

COMPACT TORIC EINSTEIN FOUR-MANIFOLDS

MINGYANG LI^{*} AND SONG SUN[†]

ABSTRACT. This is a continuation of our previous paper on toric gravitational instantons. We develop a systematic theory of compact simply-connected Einstein four-manifolds with toric symmetry. Applications include

- The construction of simply connected positive Einstein four-manifolds with arbitrarily large b_2 . Our method is based on local gluing combined with a global virtual counting argument. The key idea involves a particular design of rod structures to enable a gluing construction and restrict possible degenerations.
- A topological characterization of algebraically special toric Einstein metrics. This uses special curvature identities for W^+ in the toric setting.
- Uniqueness, up to scaling and isometry, on toric four-manifolds admitting algebraically special Einstein metrics. The proofs uses a variational study of the Einstein–Hilbert functional.
- Diffeomorphism classification of compact toric Einstein four-manifolds in terms of an improved Hitchin–Thorpe inequality $\chi \geq 3|\tau|$.

In particular, our construction recovers all the classical toric Einstein metrics, including the Page and Chen–LeBrun–Weber metrics, starting from the round sphere through moduli spaces of conical Einstein metrics.

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1. INTRODUCTION

In this paper, we construct new four-dimensional simply-connected compact Einstein manifolds with positive Ricci curvature. Previously, the known examples are given in the following list

- Round $\mathbb{S}^4$.
- Kähler–Einstein metrics on del Pezzo surfaces. The underlying manifolds are diffeomorphic to $\mathbb{CP}^2$, $\mathbb{CP}^1 \times \mathbb{CP}^1$, or $\mathbb{CP}^2 \# k\mathbb{CP}^2$ for $3 \leq k \leq 8$.
- The Page metric on $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$.

^{*} Simons Center for Geometry and Physics, Stony Brook University; mingyang.li@scgp.stonybrook.edu.

[†] Institute for Advanced Study in Mathematics, Zhejiang University; songsun@zju.edu.cn.

- The Chen–LeBrun–Weber metric on $\mathbb{CP}^2 \# \overline{2\mathbb{CP}^2}$.

All these examples have $b_2 \leq 9$ and occur on only finitely many diffeomorphism types. They are also *algebraically special*: for a suitable choice of orientation, the self-dual Weyl curvature W^+ , viewed as an endomorphism of Λ^+ , has at most two distinct eigenvalues at every point. For a compact, simply connected Einstein four-manifold, Derdziński’s theorem implies that this condition forces the metric to be either anti-self-dual or Hermitian with respect to an orientation-compatible complex structure [23]. Combining this with the classification results of Hitchin and LeBrun [28, 32], the examples listed above exhaust, up to scaling and isometry, the compact, simply connected, algebraically special Einstein four-manifolds with positive Ricci curvature. Complex geometry, through twistor theory or the theory of complex surfaces, plays a crucial role in both the construction and the classification of these metrics.

Among the examples above, the toric Einstein metrics, that is, those admitting an effective isometric $\mathbb{T}^2$ -action, are the following six, up to scaling and isometry:

- The round metric on $\mathbb{S}^4$.
- The Fubini–Study metric on $\mathbb{CP}^2$.
- The round product metric on $\mathbb{S}^2 \times \mathbb{S}^2$.
- The Page metric on $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$.
- The Chen–LeBrun–Weber metric on $\mathbb{CP}^2 \# \overline{2\mathbb{CP}^2}$.
- The dP_3 metric: the Kähler–Einstein metric on $\mathbb{CP}^2 \# \overline{3\mathbb{CP}^2}$.

These metrics were constructed by different methods. The first three are explicit and homogeneous. The Page metric is also explicit; its cohomogeneity-one symmetry reduces the Einstein equation to a system of ordinary differential equations [46]. The last two have cohomogeneity two, and no closed formulas are known. The Chen–LeBrun–Weber metric was obtained by constructing an extremal Kähler metric and making a conformal change [17], whereas the existence of the dP_3 metric follows from the theory of the complex Monge–Ampère equation [47, 48]. For numerical approximations and computations of geometric quantities, see [26, 11, 27]. See also [12] for a computer-assisted approximation of the dP_3 metric with rigorous error estimates.

The following are the existence results we prove in this paper

Theorem 1.1 (Existence Theorem). *The following manifolds admit toric Einstein metrics:*

- $\#r(\mathbb{S}^2 \times \mathbb{S}^2)$ ($r \geq 0$) (when $r = 0$, this is understood as $\mathbb{S}^4$);
- $(\mathbb{CP}^2 \# k\overline{\mathbb{CP}^2}) \# r(\mathbb{S}^2 \times \mathbb{S}^2)$, for $0 \leq k \leq 3$ and $r \geq 0$.
- $M \# N$, for either of the following choices:

$$\begin{aligned} M &\in \left\{ \mathbb{S}^4, \mathbb{S}^2 \times \mathbb{S}^2, \mathbb{CP}^2 \# k\overline{\mathbb{CP}^2} \ (1 \leq k \leq 3) \right\}, \\ N &\in \left\{ \mathbb{CP}^2, \mathbb{CP}^2 \# \overline{\mathbb{CP}^2}, 2\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}, 3\mathbb{CP}^2 \# \overline{\mathbb{CP}^2} \right\}; \end{aligned}$$

or

$$\begin{aligned} M &\in \left\{ \mathbb{S}^4, \mathbb{S}^2 \times \mathbb{S}^2, \mathbb{CP}^2 \# k\overline{\mathbb{CP}^2} \ (0 \leq k \leq 3) \right\}, \\ N &\in \left\{ \overline{\mathbb{CP}^2}, \mathbb{CP}^2 \# \overline{\mathbb{CP}^2}, \mathbb{CP}^2 \# \overline{2\mathbb{CP}^2}, \mathbb{CP}^2 \# \overline{3\mathbb{CP}^2} \right\}. \end{aligned}$$

Remark 1.2. *Independently of the present work, Ozuch [44] constructs positive toric Einstein metrics on $\#k(\mathbb{S}^2 \times \mathbb{S}^2)$ and $k\mathbb{CP}^2 \# (2k+1)\overline{\mathbb{CP}^2}$ for $k \geq 2$. His construction varies a cone angle along a generic torus orbit and obtains smooth metrics by changing the cone angle along the torus. Our construction uses gluing at torus fixed points followed by a degree argument for cone angles along rod spheres.*

A mild extension of our strategy, developed by AI, gives the following diffeomorphism classification.

Theorem 1.3 (=Theorem C.1, Diffeomorphism classification). *Let X be a closed simply connected smooth four-manifold admitting an effective $\mathbb{T}^2$ action. Then X admits a toric*

Einstein metric if and only if

$$(1.1) \quad \chi(X) \geq 3|\tau(X)|.$$

Remark 1.4. In this theorem, the Einstein metric may be invariant under a different $\mathbb{T}^2$ -action.

Some manifolds occur more than once in the list in Theorem 1.1, up to orientation reversal, because they arise from different constructions. The following uniqueness theorem shows that the resulting metrics agree up to scaling and isometry in the cases below. It is a key ingredient in proving Theorem 1.1. We also characterize algebraic speciality topologically. For a fixed orientation, a toric Einstein metric is anti-self-dual precisely when $b_2^+ = 0$, and conformally Kähler precisely when $b_2^+ = 1$; see Theorem 7.1.

Theorem 1.5 (=Corollary 7.3, Uniqueness). *On each of the manifolds*

$$\mathbb{S}^4, \quad \mathbb{S}^2 \times \mathbb{S}^2, \quad \mathbb{CP}^2 \# k \overline{\mathbb{CP}^2} \quad (0 \leq k \leq 3),$$

there is exactly one toric Einstein metric, up to scaling and isometry.

In particular, starting from the round metric on $\mathbb{S}^4$, our constructions recover the other five algebraically special toric Einstein metrics through moduli spaces of conical Einstein metrics. This provides a uniform construction of these examples.

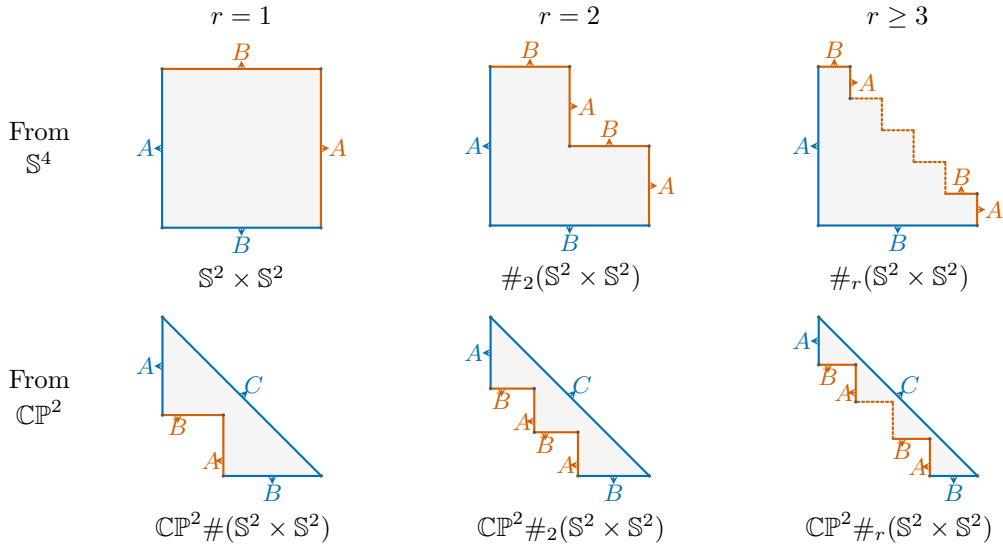


FIGURE 1. Rod structure for two infinite families in Theorem 1.1.

Our construction uses toric metrics. We obtain the examples above by choosing suitable torus actions. Their orbit spaces are topological disks: a circle subgroup collapses along each boundary edge, and the vertices correspond to torus fixed points. We use only the three circle subgroups with primitive generators

$$A = 2\pi e_1, \quad B = 2\pi e_2, \quad C = A + B,$$

where e_1, e_2 are the standard basis vectors of $\mathbb{R}^2$ and the torus lattice is $2\pi\mathbb{Z}^2$. The cyclic list of boundary labels describes the action; its precise formulation as a *rod structure* is given in Section 2. Any two distinct labels form a basis of the torus lattice, so distinct labels on adjacent edges give smooth local models at the fixed points.

The first construction replaces the two consecutive labels at a fixed point by a longer alternating sequence:

$$(1.2) \quad B, A \quad \longmapsto \quad B, A, \underbrace{B, \dots, A}_{2n \text{ new edges}}, B, A, \quad n \geq 1.$$

Topologically, this replaces M by $M \# n(\mathbb{S}^2 \times \mathbb{S}^2)$. Applying the alternating construction to the six classical toric Einstein metrics gives the first two families in Theorem 1.1. Figure 1 illustrates the cases starting from $\mathbb{S}^4$ and $\mathbb{CP}^2$.

For the finite constructions, we use the four sequences

$$(1.3) \quad \mathcal{W} = \{BCA, BACA, BCACA, BCABCA\}.$$

Each replaces the pair B, A , with its first and last labels identified with the original edges. The intermediate edges correspond to one, two, three, or four invariant spheres. These four replacements give the connected-sum summands in the third part of Theorem 1.1, with the orientation determined by the lattice identification.

For either construction, the end circles are identified with the two circles at the chosen fixed point by a torus automorphism. We call the replacement *admissible* if the resulting cyclic list still uses only $\{A, B, C\}$ after one global change of lattice basis.

The existence result Theorem 1.1 follows from a more general dichotomy theorem below.

Definition 1.6 (Toric nondegeneracy). A smooth toric Einstein metric g is *torically nondegenerate* if its space of $\mathbb{T}^2$ -invariant transverse–traceless infinitesimal Einstein deformations is trivial:

$$(1.4) \quad K_g^{\mathbb{T}^2} = \left\{ k \in C^\infty(S^2 T^* M)^{\mathbb{T}^2} : \delta_g k = 0, \quad \text{tr}_g k = 0, \quad (\nabla_g^* \nabla_g - 2\hat{R}_g)k = 0 \right\} = \{0\}.$$

Here the transverse–traceless conditions are $\delta_g k = \text{tr}_g k = 0$, $\delta_g = -\text{div}_g$, and $\hat{R}_g$ is the curvature action on symmetric two-tensors. The Einstein constant is fixed.

Theorem 1.7 (Dichotomy Theorem). *Let (M, g_0) be a compact toric Einstein four-manifold. Suppose that g_0 is torically nondegenerate and its rod directions belong to $\{A, B, C\}$ after one global change of lattice basis. Fix a torus fixed point p_0 . Then at least one of the following holds:*

- (1) *Every toric manifold obtained by an admissible replacement at p_0 of the form (1.2) or (1.3) admits a toric Einstein metric.*
- (2) *M admits a toric Einstein metric which is not isometric to any positive constant multiple of g_0 .*

The six toric Einstein metrics listed above are torically nondegenerate by Theorem 7.6, and their standard torus actions have rod directions in $\{A, B, C\}$ after a change of lattice basis. Theorem 1.5 excludes the second alternative for each of these metrics. Applying the first alternative to the admissible replacements described above yields Theorem 1.1.

Our proof is motivated by the construction of toric gravitational instantons by our previous work [34]. In the Hermitian setting, there is a striking parallel between compact Einstein four-manifolds and AF or ALF gravitational instantons [9, 33]. It is natural to ask whether such a relationship extends beyond the Hermitian setting.

For toric gravitational instantons, the Ricci-flat equation is an integrable system and reduces to an axisymmetric harmonic map equation. General harmonic map theory provides solutions with prescribed rod data, but the resulting metrics usually have conical singularities. In [34] we choose suitable rod structures and vary the rod lengths, using harmonic map bubbling and degree theory to make all cone angles equal to 2π . In the compact case, the quotient metric and its conformal structure are also unknown. Although the Einstein–Hilbert functional admits a useful variational formulation, this does not directly give an existence theory for conical Einstein metrics. See Appendix D for some discussion.

The use of varying cone angles also recalls the solution of the Kähler–Einstein problem by Chen–Donaldson–Sun [16]. In the Kähler setting, the moment map interpretation and geodesic convexity provide a clearer variational picture, accompanied by uniqueness modulo automorphisms and local continuation in the cone angle. A central difficulty there is to control degenerations without toric symmetry and use stability to identify the limits. For the general Einstein equation, these structures are unavailable even under toric symmetry, so we work directly with the moduli space of conical Einstein metrics.

We prescribe cone angle 2π along some invariant spheres and a common variable angle $2\pi\beta$ along the others. After normalization, this moduli space has expected dimension one. We construct a regular end by gluing a conical asymptotically Euclidean Ricci-flat metric to g_0 at a fixed point, with β tending to a value $\beta_* > 1$. The essential property we need about the gravitational instantons is that, at fixed cone angles, its decaying $\mathbb{T}^2$ -invariant infinitesimal deformations consist only of scaling. Prescribing total area removes this freedom, while varying the common cone angle removes the corresponding obstruction. Together with toric nondegeneracy of g_0 , this gives the required gluing family.

Our choice of the sequences W is motivated by explicit descriptions of the corresponding toric conical AE gravitational instantons. For the alternating sequence, the two rod directions force the Gram matrix to be diagonal, reducing the harmonic-map equation to a scalar axisymmetric Laplace equation. For the four sequences in (1.3), the gravitational instantons are Hermitian for a suitable orientation and admit a description by the Tod ansatz. These descriptions allow us to analyze the cone-angle map and prove the uniqueness and deformation properties needed for gluing; see Appendix B.

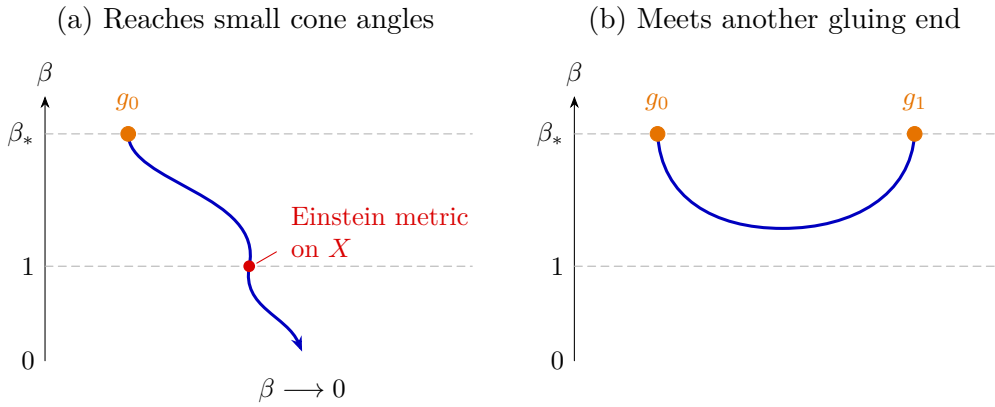


FIGURE 2. Two possible scenario for the moduli space

The global argument requires diameter bounds, noncollapsing estimates, and a precise analysis of curvature concentration when β stays in a compact subset of $(0, \infty)$. For the chosen rod structures, the relevant degenerations are described by reverse gluing near a nondegenerate compact limit. We also prove an upper bound for β , using Ricci-flat limits and curvature-energy identities. Thus the moduli space cannot escape through unbounded cone angles.

If the relevant component were a regular curve, the picture would be that it either reaches arbitrarily small positive angles, crossing $\beta = 1$, or meets another gluing end; see Figure 2. The actual moduli space can be singular and have several ends. We therefore construct an oriented Kuranishi structure and compare the boundary signs at the gluing ends. All ends arising from a compact metric isometric to g_0 have the same sign, so they cannot constitute the entire boundary. Consequently, unless M admits another Einstein metric, the moduli space contains a metric for every $\beta \in (0, 1]$. At $\beta = 1$ all conical singularities disappear. The uniqueness results for algebraically special toric Einstein metrics then give the existence results in Theorem 1.1.

The proof of Theorem 1.5, given in Section 7, combines curvature identities with a variational argument. Maximum principles and an integrated identity for the Weyl curvature force anti-self-duality when $b_2^+ = 0$ and conformal Kähler geometry when $b_2^+ = 1$. Hitchin's classification [28] gives the round metric in the spherical case. In the Hermitian cases, the integrated boundary fluxes, which are proportional to the areas of the invariant spheres, determine the moment polygon and the affine conformal factor. After fixing the total flux to remove scaling, the square root of the volume is a strictly concave function of these fluxes. This follows by expressing the volume as a minimum of quadratic forms of Lorentzian signature. The Einstein equation makes

this function stationary, so the fluxes are uniquely determined. With the polygon and conformal factor fixed, an integration-by-parts identity then identifies the metrics, proving uniqueness up to scaling and isometry.

The proof of Theorem 1.3 is a mild extension of the strategy used in the proof of Theorem 1.1. The main additional technical step is to carry out gluing over a finite-dimensional stabilization of the compact moduli space, removing the need to assume toric nondegeneracy of the intermediate compact metrics. More precisely, we use a count modulo 2 of normalized toric Einstein metrics with prescribed rod data. When the rod directions belong to $\{A, B, C\}$, our compactness results and Ricci-flat rigidity already give a compact moduli space of smooth toric Einstein metrics with a fixed rod structure. Finite-dimensional stabilization then defines the count even in the presence of degenerate metrics. We choose the perturbing tensors to be divergence-free and supported away from the chosen fixed point, allowing gluing over the resulting smooth solution space. Gluing identifies the small-area boundary count with the count on the contracted manifold, while compactness and the cone-angle bound give a cobordism to the smooth boundary. Thus the count is preserved under the prescribed corner replacements. Every cyclic rod structure with directions in $\{A, B, C\}$ reduces to a bigon or triangle, whose count is one. Counting all equivalence classes also gives the advantage of avoiding any symmetry assumption on the intermediate compact metrics, which is needed if one wants to prove Theorem 1.1 using mod 2 count. Finally, the topological classification realizes every manifold satisfying $\chi \geq 3|\tau|$ by such a rod structure, while the improved Hitchin–Thorpe inequality gives necessity.

The paper is organized as follows. Section 2 introduces compact rod structures, toric conical metrics, and the reduced Einstein equations. Section 3 establishes the geometric estimates and in particular, the a priori diameter bound and the angle control implies non-collapsing theorem. Section 4 develops the local deformation theory, the orientation of the moduli space, and the finite-dimensional perturbation argument. Section 5 studies degeneration and bubbling, proves compactness, and establishes the gluing construction and its converse. Section 6 gives an upper bound for the common cone angle. Section 7 proves the classification, uniqueness, and toric nondegeneracy results for algebraically special metrics, together with the improved Hitchin–Thorpe inequality. Section 8 determines the gluing signs and completes the proofs of the existence and dichotomy theorems. Section 9 discusses open problems. Appendices A and B establish the asymptotic classification and the uniqueness and deformation results for the gravitational instantons used in the argument. Appendix C extends the counting argument to allow degenerate compact metrics and proves the general existence criterion. Appendix D develops the reduced Einstein–Hilbert functional, proves uniqueness for a prescribed conformal quotient, and discusses a (so far unsuccessful) variational approach to existence.

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We are grateful to Dr. Tianyue Liu for explaining his interesting paper on toric Einstein four-manifolds to us in the summer of 2025, and to Christoph Böhm and Jeff Viaclovsky for pushing us to extend our work on toric gravitational instantons to the compact setting. The second author thanks Ramiro Lafuente for related discussions on toric Einstein metrics during the AMSI Winter School in Brisbane in July 2026. In particular, Lafuente informed him that he and K. Broder had a proof of the uniqueness of toric Einstein metrics on $\mathbb{S}^4$. The first author thanks the IASM for its hospitality during the summer of 2026. Parts of this work were carried out during these visits.

AI Usage Statement.

This project began with the attempt to extend our previous work [34] to the compact setting. In particular, we were interested in studying the variational problem, before clever (advanced)

AI tools became available. We could not construct conical Einstein metrics directly through this approach, by prescribing the rod fluxes, and regarded the problem as a project for the years ahead. When AI tools became available, we tried the problem as an experiment in mathematical collaboration with AI, but these initial attempts did not produce an existence argument. They did, however, reveal difficulties with prescribing the rod fluxes that led us to set this approach aside.

The authors then had the idea of studying the moduli space of conical Einstein metrics directly, without any extrinsic parametrization. This is inspired by gauge-theoretic moduli-space methods, particularly the use of compactness and boundary counts in Donaldson theory. Our initial idea was to glue a double Eguchi–Hanson gravitational instanton to the Chen–LeBrun–Weber metric and vary a single cone angle to obtain a new Einstein metric on $2\mathbb{CP}^2 \# 2\overline{\mathbb{CP}^2}$. We identified three essential ingredients: gluing, weak compactness and noncollapsing for conical toric Einstein metrics, and control of bubbling, so that the only possible degenerations in the range of cone angles under consideration would reverse the gluing construction. This led to the continuation picture in Figure 2. If successful, the argument would give either an Einstein metric on $2\mathbb{CP}^2 \# 2\overline{\mathbb{CP}^2}$ or another toric Einstein metric on $\mathbb{CP}^2 \# 2\overline{\mathbb{CP}^2}$, not isometric to a positive multiple of the Chen–LeBrun–Weber metric. Numerical experiments carried out with AI suggested that the first possibility should hold.

At this stage, we expected the gluing analysis should follow from the work of Biquard and Ozuch, and the singular moduli space to be treated using standard Kuranishi methods, as in the moduli-space arguments underlying Donaldson’s famous diagonalization theorem. However, weak compactness and noncollapsing required substantially more work than we initially anticipated. Being aware of the increasing power of AI, we tried to seek further assistance from AI. Although the individual calculations were not especially difficult, finding the differential inequalities in Section 3.1 involved considerable trial and error. We regard AI’s contribution to finding these inequalities as its most significant contribution to this project.

AI also contributed substantially to the uniqueness argument. The authors were less familiar with calculations involving W^+ . After repeated corrections and challenges from the authors, AI found the identities used in the uniqueness proof. Uniqueness excludes the second alternative above and gives the desired existence result. From the curvature identities developed in this argument, the authors subsequently observed the improved Hitchin–Thorpe inequality and formulated a conjecture describing the diffeomorphism types of compact four-manifolds admitting toric Einstein metrics.

The authors also designed the rod structures for the more general existence results in Theorem 1.1. We chose structures with at most three rod directions to control the possible degenerations and to use explicit gravitational instantons in the gluing construction. The particular choice of the replacement sequence W is guided by our own understanding of toric gravitational instantons. In particular, there are two classes of explicit families of toric gravitational instantons suitable for gluing, one from the case with two rod directions (this gives the infinite families) and one from Hermitian AE gravitational instantons (this gives the finite families). The proofs of rigidity and infinitesimal rigidity for these gravitational instantons, though fairly elementary computations, were developed by AI.

Finally, the authors asked AI to extend this strategy to prove the conjectured classification of diffeomorphism types. The proof of this final extension was developed entirely by AI, using the results and strategy established above.

Overall, this paper was developed with substantial AI assistance and represents our first active experiment in working with AI. Beyond language editing, mathematical writing, and figure preparation, AI acted as a “super assistant”, responding continually to our ideas and helping us test and develop them. We believe the project could in principle have been completed without AI, but would have required years of sustained effort and experimentation.

2. COMPACT ROD STRUCTURES AND TORIC CONICAL EINSTEIN METRICS

2.1. Compact rod structures and topology. We will always fix the standard lattice $\Lambda = (2\pi\mathbb{Z})^2 \subset \mathbb{R}^2$ and denote $\mathbb{T}^2 = \mathbb{R}^2/\Lambda$. There are standard angular coordinates φ_1, φ_2 on $\mathbb{T}^2$ with primitive period 2π . In particular,

$$(2.1) \quad \int_{\mathbb{T}^2} d\varphi_1 d\varphi_2 = (2\pi)^2.$$

Definition 2.1. A compact rod structure consists $\mathfrak{R}$ of the following data

- an oriented closed disk B ;
- $m \geq 2$ distinct points $\mathbf{p}_1, \dots, \mathbf{p}_m \in \partial B$ in positive cyclic order, with $\mathbf{p}_{m+1} := \mathbf{p}_1$;
- a primitive vector $\mathbf{v}_i \in \Lambda/\{\pm 1\}$ on each positively oriented boundary arc $\mathfrak{I}_i = [\mathbf{p}_i, \mathbf{p}_{i+1}]$,

such that for $i = 1, \dots, m$, $\mathbf{v}_i$ and $\mathbf{v}_{i+1}$ form a basis of Λ .

Each $\mathbf{p}_i$ is called a *turning point* (sometimes also referred as a *vertex*), each $\mathfrak{I}_i$ is called a *rod* and each $\mathbf{v}_i$ is called a *rod vector*.

Two compact rod structures are equivalent if, after a cyclic relabelling, there are an orientation-preserving homeomorphism $f : B \rightarrow B'$ and an orientation-preserving lattice automorphism $\mathfrak{L} : \Lambda \rightarrow \Lambda$ such that

$$f(\mathfrak{I}_i) = \mathfrak{I}'_i, \quad \mathfrak{L}(\mathbf{v}_i) = \mathbf{v}'_i \quad \text{in } \Lambda/\{\pm 1\}.$$

Remark 2.2. This definition is the counterpart of the rod structure defined in [34] for studying gravitational instantons. For our geometric purpose in this paper, we choose a simplified version than [34]. In particular, we fix the lattice Λ and avoid discussing enhancements etc.

Given a compact rod structure $\mathfrak{R}$, we set

$$X_{\mathfrak{R}} = (B \times \mathbb{T}^2) / \sim,$$

where over the interior of $\mathfrak{I}_i$ the circle generated by $\mathbf{v}_i$ is collapsed and over every $\mathbf{p}_i$ the full torus is collapsed. The local standard models are $\mathbb{R}^2 \times \mathbb{T}^2$ over the interior, $\mathbb{C} \times S^1 \times \mathbb{R}$ over a rod, and $\mathbb{C}^2$ over a turning point. Then $X_{\mathfrak{R}}$ is a smooth four-manifold endowed with the obvious $\mathbb{T}^2$ -action, and the obvious quotient map $\pi : X_{\mathfrak{R}} \rightarrow B$. We denote

$$X_{\mathfrak{R}}^{\circ} := \pi^{-1}(B^{\circ}), \quad \Sigma_i := \pi^{-1}(\mathfrak{I}_i), \quad p_i := \pi^{-1}(\mathbf{p}_i).$$

Then

$$X_{\mathfrak{R}} = X_{\mathfrak{R}}^{\circ} \sqcup \left(\bigcup_{i=1}^m \Sigma_i \right).$$

and each $\Sigma_i = \pi^{-1}(\mathfrak{I}_i)$ is an embedded two-sphere. Each turning point $\mathbf{p}_i \in \partial B$ corresponds to a unique torus fixed point $p_i \in X$, with $\pi(p_i) = \mathbf{p}_i$.

We orient $X_{\mathfrak{R}}$ by requiring that $\pi^* \nu \wedge d\varphi_1 \wedge d\varphi_2$ to be positive on $X_{\mathfrak{R}}^{\circ}$ for every positive local volume form ν on B . Notice that if $\mathfrak{R}'$ is equivalent to $\mathfrak{R}$, then there is an orientation-preserving diffeomorphism between $X_{\mathfrak{R}}$ and $X_{\mathfrak{R}'}$, which intertwines the torus actions through a lattice automorphism.

We now describe the local models for the $\mathbb{T}^2$ -action on $X_{\mathfrak{R}}$. Over a point of B° , a $\mathbb{T}^2$ -invariant neighborhood has coordinates

$$U \simeq D_b^2 \times \mathbb{T}_{\psi_1, \psi_2}^2, \quad (\varphi_1, \varphi_2) \cdot (b, \psi_1, \psi_2) = (b, \psi_1 + \varphi_1, \psi_2 + \varphi_2).$$

Over an interior point of $\mathfrak{I}_i$, choose a signed representative $\mathbf{v}_i$ and complete it to a lattice basis $(\mathbf{v}_i, \mathbf{w}_i)$. Let θ_1, θ_2 be the corresponding angular coordinates of period 2π , so that

$$\partial_{\theta_1} = \sum_{\alpha=1}^2 \frac{(\mathbf{v}_i)_{\alpha}}{2\pi} \partial_{\varphi_{\alpha}}, \quad \partial_{\theta_2} = \sum_{\alpha=1}^2 \frac{(\mathbf{w}_i)_{\alpha}}{2\pi} \partial_{\varphi_{\alpha}}.$$

A $\mathbb{T}^2$ -invariant neighborhood of the corresponding circle orbit has coordinates

$$U \simeq D_z^2 \times (-\epsilon, \epsilon)_s \times S_{\psi}^1, \quad (\theta_1, \theta_2) \cdot (z, s, \psi) = (e^{\sqrt{-1}\theta_1} z, s, \psi + \theta_2),$$

with $\Sigma_i \cap U = \{z = 0\}$.

At p_i , choose signed representatives $(\mathbf{v}_{i-1}, \mathbf{v}_i)$ forming a lattice basis, and use its angular coordinates θ_1, θ_2 of period 2π . An invariant neighborhood has coordinates

$$U \simeq D_{z_1}^2 \times D_{z_2}^2, \quad (\theta_1, \theta_2) \cdot (z_1, z_2) = (e^{\sqrt{-1}\theta_1} z_1, e^{\sqrt{-1}\theta_2} z_2),$$

with

$$\Sigma_{i-1} \cap U = \{z_1 = 0\}, \quad \Sigma_i \cap U = \{z_2 = 0\}.$$

Conversely, every closed, simply connected, oriented smooth four-manifold with an effective smooth $\mathbb{T}^2$ -action is equivariantly diffeomorphic to $X_{\mathfrak{R}}$ for some compact rod structure $\mathfrak{R}$, unique up to equivalence, with the orientation of B chosen to agree with the given orientation of the manifold. The rod data determines the action up to equivariant diffeomorphism [41]. Moreover, the underlying oriented smooth manifold is either $\mathbb{S}^4$ or a connected sum of copies of $\mathbb{S}^2 \times \mathbb{S}^2$, $\mathbb{CP}^2$, and $\overline{\mathbb{CP}}^2$.

We collect some easy topological facts about $X_{\mathfrak{R}}$. Recall that we fix the lattice to be $\Lambda = (2\pi\mathbb{Z})^2$.

- $\pi_1(X_{\mathfrak{R}}) = \{1\}$;
- $\chi(X_{\mathfrak{R}}) = m$;
- For each i , the self-intersection e_i of Σ_i is given by

$$(2\pi)^{-6} \det(\mathbf{v}_{i-1}, \mathbf{v}_i) \det(\mathbf{v}_i, \mathbf{v}_{i+1}) \det(\mathbf{v}_{i-1}, \mathbf{v}_{i+1}).$$

- $3\tau(X_{\mathfrak{R}}) = \sum_i e_i$.

The last formula is a consequence of the signature theorem. When $m = 2$ it holds trivially. For $m \geq 3$, the stable tangent-bundle splitting [20, Theorem 6.6] gives

$$p_1(TX) = \sum_{i=1}^m \text{PD}[\Sigma_i]^2.$$

2.2. Compact toric conical metrics. Roughly speaking, Riemannian metrics with cone singularities along a smooth codimension-two submanifold S are locally modelled, to leading order, on the product of a Euclidean space and a two-dimensional flat cone. The regularity required beyond this model depends on the particular setting. In Kähler geometry, a by-now standard analytic formulation for cone singularities along a smooth divisor is given by Donaldson [25]. Conical Kähler–Einstein metrics played a central role in the solution of the Kähler–Einstein problem for Fano manifolds [14, 15, 16]. For the toric metrics considered here, we use the formulation of conical regularity in terms of the rod structure developed in [34].

Definition 2.3. Given a compact rod structure $\mathfrak{R}$, a *toric conical metric* on $X_{\mathfrak{R}}$ is a $\mathbb{T}^2$ -invariant Riemannian metric g on $X_{\mathfrak{R}}^\circ$ whose metric completion is naturally identified with $X_{\mathfrak{R}}$ and which satisfies the following conditions for constants $\beta_i > 0$.

In the adapted local coordinates, lift each collapsing angular coordinate to $\mathbb{R}$. We require the following

- Near an interior point of $\mathfrak{J}_i$, the lifted metric is the pullback of a smooth Riemannian metric across $\{z = 0\}$ under

$$(r, \theta, s, \psi) \mapsto (re^{\sqrt{-1}\beta_i\theta}, s, \psi).$$

- Near p_i , the lifted metric is the pullback of a smooth Riemannian metric across both coordinate axes under

$$(r_1, \theta_1, r_2, \theta_2) \mapsto (r_1 e^{\sqrt{-1}\beta_{i-1}\theta_1}, r_2 e^{\sqrt{-1}\beta_i\theta_2}).$$

These smooth extensions are called the *local developments* of g . The cone angle along Σ_i is $2\pi\beta_i$.

The cone angle along the rod $\mathfrak{J}_i$ is $2\pi\beta_i$. When $\beta_i = 1$ for all i , then the metric is an ordinary smooth metric. The conical singularities we defined here are very mild—they arise from an identification with an incorrect period, similar to orbifold singularities. In particular, the curvature is bounded across the conical singularities.

A toric conical Einstein metric is by definition a toric conical metric which satisfies the Einstein equation

$$\text{Ric} = \lambda g$$

on $X_{\mathfrak{R}}^\circ$.

The following is well-known [19, 35]. For simplicity we give simple explanation following [34]

Lemma 2.4. *A toric conical Einstein metric on $X_{\mathfrak{R}}$ can be globally written over $X_{\mathfrak{R}}^\circ$ as*

$$(2.2) \quad g = \underline{g} + G_{\alpha\beta} d\varphi_\alpha d\varphi_\beta,$$

where $G = \rho\Phi$, with $\rho > 0$ on B° and

$$\Phi : B^\circ \longrightarrow \mathcal{H} = SL(2, \mathbb{R})/SO(2).$$

We identify $\mathcal{H}$ with the space of positive definite symmetric 2×2 matrices of determinant one.

Proof. We follow the derivation in [34]. Let $X_j = \partial_{\varphi_j}$, $j = 1, 2$, be the Killing fields of period 2π , and set $\eta_j = g(X_j, \cdot)$. The Killing identities and the Einstein equation imply that $*(\eta_j \wedge d\eta_j)$ is closed. Thus locally there are twist potentials $\mathcal{E}_j$ satisfying

$$d\mathcal{E}_j = *(\eta_j \wedge d\eta_j).$$

Set

$$\Theta_j = *(\eta_1 \wedge \eta_2 \wedge d\eta_j).$$

As in [34], $\Theta_1 = \mathcal{L}_{X_2}\mathcal{E}_1$

$$d\Theta_1 = d\mathcal{L}_{X_2}\mathcal{E}_1 = \mathcal{L}_{X_2}*(\eta_1 \wedge d\eta_1) = 0,$$

and similarly, $d\Theta_2 = 0$.

Now the $\mathbb{T}^2$ -action has at least one fixed point, at which both Θ_1 and Θ_2 tend to zero. It follows that $\Theta_1 \equiv \Theta_2 \equiv 0$. This implies that the horizontal distribution is integrable. Since B° is simply connected, its flat connection has a global horizontal section, giving the trivialization in (2.2). Finally, set $\rho = \sqrt{\det G}$ and $\Phi = \rho^{-1}G$. $\square$

In this paper we will endow $\mathcal{H}$ with the standard Riemannian metric with constant negative curvature -2 :

$$g_{\mathcal{H}} = \frac{1}{4} \text{tr}(\Phi^{-1} d\Phi)^2.$$

2.3. Reduction of the Einstein equation. For compactly supported variations over B° , consider the Einstein-Hilbert functional

$$(2.3) \quad \text{EH}_\lambda(g) = \int_X (\text{Scal}_g - 2\lambda) \text{dvol}_g.$$

It is straightforward to compute the scalar curvature

$$(2.4) \quad \text{Scal}_g = \text{Scal}_{\underline{g}} - 2\rho^{-1}\Delta_{\underline{g}}\rho + \frac{1}{2}\rho^{-2}|d\rho|_{\underline{g}}^2 - \frac{1}{4}\text{tr}(\Phi^{-1}d\Phi)_{\underline{g}}^2.$$

Proposition 2.5. *For a metric of the form (2.2), the Einstein equation $\text{Ric}(g) = \lambda g$ is equivalent on B° to the following system:*

$$(2.5) \quad \Delta_{\underline{g}}\rho = -2\lambda\rho, \quad (\text{fiber trace}),$$

$$(2.6) \quad d_{\underline{g}}^*(\rho\Phi^{-1}d\Phi) = 0, \quad (\text{fiber trace-free}),$$

$$(2.7) \quad K_{\underline{g}} = -\frac{1}{2}|d \log \rho|_{\underline{g}}^2 + \frac{1}{8}\text{tr}((G^{-1}dG)^2)_{\underline{g}}, \quad (\text{base trace}),$$

$$(2.8) \quad \left(\text{Hess}_{\underline{g}} \log \rho + \frac{1}{4}\text{tr}(G^{-1}dG \otimes G^{-1}dG) \right)^\circ = 0, \quad (\text{base trace-free}).$$

Here $K_{\underline{g}}$ denotes the Gaussian curvature of $\underline{g}$.

Proof. Variation of g in the conformal direction gives (2.5). Variation of Φ gives (2.6). For a compactly supported variation of ρ , its coefficient is

$$2K_{\underline{g}} - \rho^{-1} \Delta_{\underline{g}} \rho + \frac{|\mathrm{d}\rho|_{\underline{g}}^2}{2\rho^2} - \frac{1}{4} \mathrm{tr}((\Phi^{-1} \mathrm{d}\Phi)^2)_{\underline{g}} - 2\lambda.$$

Using (2.5) and the fact

$$\mathrm{tr}((G^{-1} \mathrm{d}G)^2)_{\underline{g}} = 2|\mathrm{d} \log \rho|_{\underline{g}}^2 + \mathrm{tr}((\Phi^{-1} \mathrm{d}\Phi)^2)_{\underline{g}},$$

we get (2.7).

Finally, a compactly supported trace-free variation of $\underline{g}$ gives

$$\left(\mathrm{Hess}_{\underline{g}} \rho - \frac{\mathrm{d}\rho \otimes \mathrm{d}\rho}{2\rho} + \frac{\rho}{4} \mathrm{tr}(\Phi^{-1} \mathrm{d}\Phi \otimes \Phi^{-1} \mathrm{d}\Phi) \right)^{\circ} = 0.$$

Here the Hessian term comes from integrating by parts in the variation of $\int_B \rho R_{\underline{g}} \mathrm{d}A_{\underline{g}}$. Dividing by ρ , and using

$$\rho^{-1} \mathrm{Hess}_{\underline{g}} \rho = \mathrm{Hess}_{\underline{g}} \log \rho + \mathrm{d} \log \rho \otimes \mathrm{d} \log \rho,$$

gives (2.8). The mixed horizontal-vertical Ricci components vanish for (2.2), so the four equations together are equivalent to the Einstein equation. $\square$

Remark 2.6. In [34], for the Ricci-flat case (i.e., $\lambda = 0$), (2.5) implies that ρ is a harmonic function on the base, so $\rho + \sqrt{-1}z$, for a harmonic conjugate z , gives a natural local conformal coordinate on the base. This fact allows us to write the whole system in terms of an axisymmetric harmonic map from a domain in $\mathbb{R}^3$ to $\mathcal{H}$.

When $\lambda \neq 0$, Φ is still a weighted harmonic map, and ρ is an eigenfunction for $\Delta_{\underline{g}}$. So we no longer have natural local conformal coordinates. This is a main difficulty to directly solve the above PDE system.

2.4. Rod asymptotics. We now derive the rod asymptotics for toric conical Einstein metrics. We first specify the coordinates used near rods and turning points.

Whenever the quotient metric $\underline{g}$ is smooth up to the interior of a rod $\mathfrak{I}_i$, choose an arclength coordinate s along the rod and extend it by keeping it constant along the inward unit-speed geodesics perpendicular to the rod. For sufficiently small $r \geq 0$, where r is the distance to the rod, this gives local Fermi coordinates (r, s) satisfying

$$\underline{g} = \mathrm{d}r^2 + h(r, s)^2 \mathrm{d}s^2, \quad h(0, s) = 1.$$

At a turning point $\mathfrak{p}_i$, choose signed representatives $(\mathfrak{v}_{i-1}, \mathfrak{v}_i)$ forming a basis of Λ . The local model for the torus action gives smooth equivariant coordinates (z_1, z_2) near the corresponding fixed point p_i , in which the two circle factors act by separate rotations and $\Sigma_{i-1} = \{z_1 = 0\}$, $\Sigma_i = \{z_2 = 0\}$. Writing

$$z_1 = x e^{\sqrt{-1}\theta_1}, \quad z_2 = y e^{\sqrt{-1}\theta_2}, \quad x, y \geq 0,$$

gives local quotient coordinates (x, y) , with the two rods corresponding to the coordinate axes.

In the following local expansions, we express the Gram matrix in the adapted angular coordinates:

$$G_{ab} = g(\partial_{\theta_a}, \partial_{\theta_b}).$$

The function $\rho = \sqrt{\det G}$ is unchanged by these integral changes of angular basis.

Lemma 2.7. *Given a compact rod structure $\mathfrak{R}$, and data $(\underline{g}, \rho, \Phi)$ which solves (2.5)–(2.8), then it determines a toric conical Einstein metric with cone angle $2\pi\beta_i$ along Σ_i , if and only if the pair $(\underline{g}, \rho\Phi)$ satisfies the following boundary conditions.*

- (1) *The metric $\underline{g}$ extends smoothly across the interior of each $\mathfrak{I}_i$.*

(2) At an interior point of $\mathfrak{J}_i$, choose the basis $(\mathbf{v}_i, \mathbf{w}_i)$ of Λ and let (r, s) be local Fermi coordinates on B . Then we have

$$\underline{g} = dr^2 + f(r^2, s)^2 ds^2$$

for a smooth nonzero function f with $f(0, s) = 1$. Moreover, choose an equivariant chart adapted to these coordinates,

$$(z, s, \theta_2), \quad z = re^{\sqrt{-1}\theta_1},$$

such that $\Sigma_i = \{z = 0\}$, then we require that the Gram matrix in these angular coordinates satisfies

$$\begin{aligned} G_{11} &= r^2(\beta_i^2 + r^2 a(r^2, s)), \\ G_{12} &= r^2 b(r^2, s), \\ G_{22} &= c(r^2, s), \end{aligned}$$

Here a, b, c are all local smooth functions and $c(0, s) > 0$.

(3) At a fixed point p_i , choose the basis $(\mathbf{v}_{i-1}, \mathbf{v}_i)$ of Λ and local equivariant coordinates

$$z_1 = xe^{\sqrt{-1}\theta_1}, \quad z_2 = ye^{\sqrt{-1}\theta_2}, \quad x, y \geq 0,$$

so that $\Sigma_{i-1} = \{z_1 = 0\}$ and $\Sigma_i = \{z_2 = 0\}$. Then we require $\underline{g}$ and G to take the form

$$\begin{aligned} \underline{g} &= a(x^2, y^2)dx^2 + 2xyb(x^2, y^2)dxdy + c(x^2, y^2)dy^2, \\ (2.9) \quad G &= \begin{pmatrix} x^2 p(x^2, y^2) & x^2 y^2 q(x^2, y^2) \\ x^2 y^2 q(x^2, y^2) & y^2 r(x^2, y^2) \end{pmatrix}, \\ p(0, y^2) &= \beta_{i-1}^2 a(0, y^2), \quad r(x^2, 0) = \beta_i^2 c(x^2, 0), \end{aligned}$$

where a, b, c, p, q, r are all smooth local functions, and $a(0, 0) > 0, c(0, 0) > 0$.

Proof. For a point in the interior of a rod $\mathfrak{J}_i$, and let (r, s) be the Fermi coordinates on B . Consider an adapted equivariant chart $(r, \theta_1, s, \theta_2)$, if we develop the collapsing S^1 factor by setting

$$\vartheta = \beta_i \theta_1,$$

and denote

$$X = r \cos \vartheta, \quad Y = r \sin \vartheta, \quad \alpha = X dY - Y dX = r^2 d\vartheta.$$

Then the smoothness and rotational invariance of the new metric precisely mean that it can be written as

$$\tilde{g} = dX^2 + dY^2 + q(r^2, s)^2 ds^2 + A(r^2, s)\alpha^2 + 2B(r^2, s)\alpha d\theta_2 + C(r^2, s)d\theta_2^2,$$

where q, A, B, C are smooth, $q(0, s) = 1$, and $C(0, s) > 0$. Substituting $\alpha = \beta_i r^2 d\theta_1$ gives

$$\underline{g} = dr^2 + q(r^2, s)^2 ds^2, \quad G_{11} = r^2(\beta_i^2 + \beta_i^2 r^2 A(r^2, s)), \quad G_{12} = \beta_i r^2 B(r^2, s), \quad G_{22} = C(r^2, s).$$

The discussion at a fixed point is similar and we omit it. $\square$

As a consequence, for a toric conical Einstein metric, the quotient metric $\underline{g}$ extends smoothly to B as a Riemannian metric on a surface with corners. The interior of each rod is a geodesic boundary arc, and adjacent arcs meet at right angles at each turning point $\mathfrak{p}_j$. Moreover, we have the boundary asymptotics for ρ as a function on B

$$(2.10) \quad \begin{cases} \rho = r\kappa(s) + O(r^3), & \kappa(s) = \beta_i \sqrt{c(0, s)} > 0, & \text{along an open rod,} \\ \rho = xy\tau(x^2, y^2), & \tau(0, 0) > 0, & \text{at a turning point.} \end{cases}$$

where κ and τ are smooth local functions.

A conformal identification of quotients is understood to match the corresponding rods and their lattice directions. If the identification changes the lattice basis, equivariance is understood with respect to the induced automorphism of $\mathbb{T}^2$.

Lemma 2.8. *A conformal identification of toric quotient disks matching the rod structures lifts to a smooth equivariant diffeomorphism.*

Proof. Using the quotient description above, combine the given map on the disk with the induced torus automorphism. Reflection extends the conformal map across each open rod. At a turning point, its expression in signed normal coordinates is

$$(x, y) \mapsto (xa(x^2, y^2), yb(x^2, y^2)),$$

with a, b smooth and positive. The lift in complex normal coordinates is therefore

$$(z_1, z_2) \mapsto (a(|z_1|^2, |z_2|^2)z_1, b(|z_1|^2, |z_2|^2)z_2).$$

It is smooth, and the inverse has the same form. The argument along an open rod uses just one normal coordinate. $\square$

The induced metric on Σ_i is given by $ds^2 + \beta_i^{-2}\kappa^2 d\theta_2^2$, with Gaussian curvature

$$K_{\Sigma_i} = -\frac{\kappa''}{\kappa}.$$

In the local smooth redeveloping model, Σ_i is a fixed-point surface and is therefore totally geodesic. By Lemma 2.4 the horizontal section is also totally geodesic. Thus the sectional curvature of a plane spanned by ∂_s and a normal direction equals $K_{\underline{g}}$. Taking the Ricci curvature in the unit ∂_s direction yields

$$(2.11) \quad \lambda = K_{\Sigma_i} + 2K_{\underline{g}} = -\frac{\kappa''}{\kappa} + 2K_{\underline{g}}.$$

The area of each Σ_i can be expressed as the rod flux of ρ :

$$(2.12) \quad \text{Area}_g(\Sigma_i) = -\frac{2\pi}{\beta_i} \int_{\mathcal{I}_i} \partial_n \rho ds_{\underline{g}}$$

where n is the outward unit normal to the rod in $\underline{g}$. Indeed, in Fermi coordinates $n = -\partial_r$ and $-\partial_n \rho = \beta_i \sqrt{c(0, s)}$, whereas

$$\text{Area}_g(\Sigma_i) = 2\pi \int_{\mathcal{I}_i} \sqrt{c(0, s)} ds_{\underline{g}}.$$

The total volume of (X, g) is given by

$$(2.13) \quad \text{Vol}_g(X) = \lambda^{-1} \pi \sum_i \beta_i \text{Area}_g(\Sigma_i).$$

This follows from the divergence theorem and (2.5):

$$\lambda \text{Vol}_g(X) = (2\pi)^2 \lambda \int_B \rho d\mu_{\underline{g}} = -2\pi^2 \int_B \Delta_{\underline{g}} \rho d\mu_{\underline{g}} = -2\pi^2 \sum_i \int_{\mathcal{I}_i} \partial_n \rho ds_{\underline{g}} = \pi \sum_i \beta_i \text{Area}_g(\Sigma_i).$$

We also have a version of Bishop-Gromov volume comparison

$$(2.14) \quad \text{Vol}_g(X) \leq \frac{24\pi^2}{\lambda^2} \min_{1 \leq i \leq m} \beta_i \beta_{i+1}, \quad \beta_{m+1} = \beta_1.$$

Indeed, by scaling the lattice, we can arrange that all the cone angles are below 2π , so the convexity argument of [34, Lemma 4.43] permits the usual Bonnet-Myers and Bishop-Gromov comparison. At the torus fixed point $\Sigma_i \cap \Sigma_{i+1}$, the volume density is $\beta_i \beta_{i+1}$, while its total volume is $\text{Vol}_g(X)$.

3. A PRIORI ESTIMATES AND GEOMETRIC BOUNDS

Throughout this section we will work with a compact toric conical Einstein metric $(X_{\mathfrak{R}}, g)$ with Einstein constant $\lambda > 0$. We will derive some a priori estimates and geometric bounds. All the quantities on B are calculated using the quotient metric $\underline{g}$. For simplicity of discussion we introduce some notations

$$u = \log \rho, \quad U = \frac{1}{2}|du|^2, \quad K = K_{\underline{g}}, \quad T_{ij} = \frac{1}{4} \operatorname{tr}(\Phi^{-1} \Phi_i \Phi^{-1} \Phi_j).$$

So T is non-negative. Denote

$$e = \frac{1}{2} \operatorname{tr} T = \frac{1}{8} \operatorname{tr}(\Phi^{-1} d\Phi)^2 \geq 0$$

and

$$\det T := \frac{\det(T_{ij})}{\det(\underline{g}_{ij})} \geq 0.$$

Notice that

$$(3.1) \quad 2 \det T + |T|^2 = (\operatorname{tr} T)^2.$$

Denote the weighted Laplacian operator

$$\mathcal{L} = \Delta_{\underline{g}} + \langle du, d\cdot \rangle.$$

The reduced system in Proposition 2.5 implies that

$$(3.2) \quad \Delta u = -2\lambda - 2U,$$

$$(3.3) \quad 2e = 2K + U,$$

$$(3.4) \quad \operatorname{Hess} u = (K - \lambda)\underline{g} - \frac{1}{2} du \otimes du - T,$$

$$(3.5) \quad \tau(\Phi) + d\Phi(\nabla u) = 0.$$

Here $\tau(\Phi) := \operatorname{tr}(\nabla d\Phi)$ is the tension field of the map Φ .

3.1. A differential inequality. We now prove a key differential inequality. One motivation is Lu's interior curvature estimate for special Kähler metrics [36]. Locally, these correspond to the above system with constant ρ and $\lambda = 0$. Then in this situation (3.4) gives $T = K\underline{g}$, thus Φ is conformal and harmonic. On $\{K > 0\}$, the metric $K\underline{g} = \Phi^* g_{\mathcal{H}}$ has curvature -2 , and hence

$$\Delta_{\underline{g}} \log K = 6K.$$

Such an equation gives a priori interior estimates on K , without extra assumptions. In our Einstein setting, we have a more complicated differential inequality.

Lemma 3.1. *For a toric conical Einstein metric, at a point in B° where $d\Phi \neq 0$, we have*

$$(3.6) \quad \Delta K + \frac{3U + 2\lambda}{2e} \langle du, dK \rangle \geq \frac{|dK|^2}{e} + \frac{K(3K - \lambda)(2U + K + \lambda)}{e} + 3 \det T.$$

Proof. First, the weighted Bochner formula for a function gives

$$\mathcal{L}U = |\operatorname{Hess} u|^2 + \langle \nabla u, \nabla \mathcal{L}u \rangle + (\operatorname{Ric} - \operatorname{Hess} u)(\nabla u, \nabla u).$$

Since $\mathcal{L}u = -2\lambda$, and the weighted Ricci curvature is

$$(3.7) \quad \operatorname{Ric}_{\underline{g}} - \operatorname{Hess} u = \lambda \underline{g} + \frac{1}{2} du \otimes du + T,$$

we obtain

$$(3.8) \quad \mathcal{L}U = |\operatorname{Hess} u|^2 + 2\lambda U + 2U^2 + T(\nabla u, \nabla u).$$

Similarly, the weighted Bochner formula for the density of Φ gives, using (3.5) and $g_{\mathcal{H}}$ has curvature -2 ,

$$(3.9) \quad \mathcal{L}e = |\nabla d\Phi|^2 + \langle \text{Ric}_{\underline{g}} - \text{Hess}_{\underline{g}} u, T \rangle_{\underline{g}} + 2(\text{tr } T)^2 - 2|T|^2$$

$$(3.10) \quad = |\nabla d\Phi|^2 + \lambda \text{tr } T + \frac{1}{2}T(\nabla u, \nabla u) + 2(\text{tr } T)^2 - |T|^2.$$

Using (3.3), (3.8), we obtain

$$(3.11) \quad \mathcal{L}K = |\nabla d\Phi|^2 - \frac{1}{2}|\text{Hess } u|^2 + 2\lambda K + 2(\text{tr } T)^2 - |T|^2 - U^2.$$

By (3.4) we have

$$dU = (K - \lambda - U)du - T(\nabla u, \cdot).$$

Together with (3.3) and (3.5), this gives

$$dK + \frac{1}{2}(K - \lambda - U)du = \sum_{j=1}^2 \langle (\nabla d\Phi)^\circ(\cdot, e_j), d\Phi(e_j) \rangle,$$

where e_1, e_2 is any local orthonormal frame and $(\nabla d\Phi)^\circ$ denotes the trace-free part of $\nabla d\Phi$.

Now we recall an elementary fact. For a symmetric trace-free (2×2) matrix A , we have

$$A^2 = \frac{1}{2}\|A\|^2 \text{Id}$$

so

$$|Av|^2 = \frac{1}{2}\|A\|^2|v|^2.$$

Apply this to each component of $(\nabla d\Phi)^\circ$ in an orthonormal frame of $\Phi^*T\mathcal{H}$, and then use Cauchy–Schwarz in the target index, we obtain

$$\begin{aligned} \left| dK + \frac{1}{2}(K - \lambda - U)du \right|^2 &= \left| \sum_{j=1}^2 \langle (\nabla d\Phi)^\circ(\cdot, e_j), d\Phi(e_j) \rangle \right|^2 \\ &\leq \frac{1}{2}|(\nabla d\Phi)^\circ|^2 |d\Phi|^2 = e|(\nabla d\Phi)^\circ|^2. \end{aligned}$$

Finally,

$$|\nabla d\Phi|^2 = |(\nabla d\Phi)^\circ|^2 + \frac{1}{2}|\tau(\Phi)|^2 = |(\nabla d\Phi)^\circ|^2 + \frac{1}{2}T(\nabla u, \nabla u).$$

Consequently, on $\{e > 0\}$, we have

$$(3.12) \quad |\nabla d\Phi|^2 \geq \frac{1}{e} \left| \nabla K + \frac{1}{2}(K - \lambda - U)\nabla u \right|^2 + \frac{1}{2}T(\nabla u, \nabla u).$$

On the other hand, by (3.4)

$$|\text{Hess } u|^2 = 2(K - \lambda)^2 - 2(K - \lambda)U - 2(K - \lambda)\text{tr } T + U^2 + |T|^2 + T(\nabla u, \nabla u).$$

Inserting these into (3.11), by (3.1) and (3.3) and a straightforward manipulation we obtain (3.6). $\square$

A first consequence of the differential inequality is a curvature upper bound on B .

Proposition 3.2. *We have*

$$(3.13) \quad K \leq \frac{\lambda}{3}.$$

Proof. Let K_0 be the maximum of K on B . Suppose that $K_0 > \lambda/3$. Then $e = K + U/2 > 0$ at every maximum. An interior maximum contradicts (3.6), since $dK = 0$, $\Delta_{\underline{g}}K \leq 0$, and its right-hand side is strictly positive. It remains to bound a boundary maximum. Since $e = K + U/2$,

$$\frac{3U + 2\lambda}{2e} = 3 + \frac{\lambda - 3K}{e}.$$

At an interior point of a rod, by (2.10) we have $u_r = r^{-1} + O(r)$, $u_s = O(1)$, and $K = O(1)$. Thus $U = (2r^2)^{-1} + O(1)$, and

$$e = K + \frac{U}{2} = (4r^2)^{-1} + O(1).$$

Since K is even in r , at a maximum $(0, s_0)$ we have

$$\lim_{r \rightarrow 0} \left(\Delta_{\underline{g}} K + \frac{3U + 2\lambda}{2e} \langle du, dK \rangle_{\underline{g}} \right) (r, s_0) = 4K_{rr}(0, s_0) + K_{ss}(0, s_0) \leq 0.$$

On the other hand, the second term on the right-hand side of (3.6) tends to $4K_0(3K_0 - \lambda) > 0$, while the other terms are nonnegative. This is a contradiction.

At a vertex, choose the coordinates (x, y) as in Lemma 2.7, with

$$\rho = xya(x^2, y^2), \quad \underline{g}(0) = dx^2 + dy^2.$$

Along $x = y = t \rightarrow 0$, we have

$$u_x = t^{-1} + O(t), \quad u_y = t^{-1} + O(t), \quad U = t^{-2} + O(1), \quad e = (2t^2)^{-1} + O(1).$$

Since K is even in both variables,

$$K_x = tK_{xx}(0) + O(t^3), \quad K_y = tK_{yy}(0) + O(t^3).$$

Consequently,

$$\Delta_{\underline{g}} K + \frac{3U + 2\lambda}{2e} \langle du, dK \rangle_{\underline{g}} \rightarrow 4(K_{xx}(0) + K_{yy}(0)) \leq 0.$$

The second term again tends to $4K_0(3K_0 - \lambda) > 0$, and the remaining terms are nonnegative. This contradiction proves $K \leq \lambda/3$ on B . $\square$

By (2.11), it follows that

Corollary 3.3. *Along the interior of a rod, we have*

$$(3.14) \quad \kappa'' \leq -\frac{\lambda}{3} \kappa < 0.$$

A consequence of the weighted Bochner formula gives

Proposition 3.4. *For $\mathfrak{p} \in B^\circ$, we have*

$$(3.15) \quad U(\mathfrak{p}) \leq \frac{12}{d(\mathfrak{p}, \partial B)^2}.$$

Proof. Fix $\mathfrak{p} \in B^\circ$ and $R < d_{\underline{g}}(\mathfrak{p}, \partial B)$, and let $r = d_{\underline{g}}(\mathfrak{p}, \cdot)$. At a smooth point of r , the Bochner formula applying to the equation $\mathcal{L}(|\nabla r|^2) = 0$ and (3.7) give

$$\begin{aligned} \partial_r(\mathcal{L}r) &= -|\text{Hess}_{\underline{g}} r|^2 - (\text{Ric}_{\underline{g}} - \text{Hess}_{\underline{g}} u)(\partial_r, \partial_r) \\ &\leq -(\Delta_{\underline{g}} r)^2 - \frac{1}{2}(u_r)^2 \leq -\frac{1}{3}(\mathcal{L}r)^2. \end{aligned}$$

Since $\mathcal{L}r = r^{-1} + O(1)$ near $\mathfrak{p}$, a scalar Riccati comparison gives $\mathcal{L}r \leq 3r^{-1}$. So

$$\mathcal{L}(r^2) \leq 8.$$

Set

$$F = (R^2 - r^2)^2 U.$$

The weighted Bochner formula (3.8) implies that

$$\mathcal{L}U \geq 2U^2.$$

If F has a positive maximum at a smooth point of r^2 , then we have

$$0 \geq \mathcal{L}F \geq 2(R^2 - r^2)^2 U^2 - 16(R^2 - r^2)U - 24r^2 U.$$

So

$$F \leq 8(R^2 - r^2) + 12r^2 \leq 12R^2.$$

If the maximum lies in the cut locus, the same estimate holds by a standard barrier argument.

Now evaluate at $\mathfrak{p}$ and letting R increase to $d_g(\mathfrak{p}, \partial B)$ we get the conclusion $\square$

By (3.3) we get an interior lower bound on K and an upper bound on the map density e .

Corollary 3.5. *We have*

$$(3.16) \quad K \geq -\frac{6}{d(\mathfrak{p}, \partial B)^2}, \quad e \leq \frac{\lambda}{3} + \frac{6}{d(\mathfrak{p}, \partial B)^2}.$$

Together with Proposition 3.2, we obtain uniform interior bound on K . Using standard elliptic theory for the reduced system we also obtain a uniform bound on the curvature of g away from the rods. Later we will prove a stronger bound including the interior of the rods (see Proposition 3.11).

3.2. Diameter and injectivity radius estimate.

Proposition 3.6. *For a toric conical Einstein metric $(X_{\mathfrak{A}}, g)$ with $\lambda > 0$ we have the uniform estimate*

$$(3.17) \quad \frac{\pi}{2\sqrt{2\lambda}} \leq \text{diam}(X_{\mathfrak{A}}, g) \leq 2\pi\sqrt{\frac{3}{\lambda}}.$$

Remark 3.7. *The upper bound is expected from Myers's theorem in the smooth setting. The proof below is a more direct argument without dealing the conical singularities, but gives slightly weaker bound which suffices for our purpose.*

Denote

$$M := \max_B \rho.$$

Lemma 3.8 (Gradient estimate). *We have a uniform bound*

$$(3.18) \quad |\mathrm{d}\rho|^2 + 2\lambda\rho^2 \leq 2\lambda M^2.$$

Proof. Denote $P = |\mathrm{d}\rho|^2 + 2\lambda\rho^2$. By the Bochner formula, we have

$$\frac{1}{2}\Delta P = |\text{Hess } \rho|^2 + K|\mathrm{d}\rho|^2 - 4\lambda^2\rho^2.$$

Suppose otherwise, then at a maximum $\mathfrak{q} \in B$ we have $P > 2\lambda M^2$. At a corner point $\mathfrak{p}_i$ we have $\rho = xya(x^2, y^2)$, so $P = 0$. In particular, $\mathfrak{q}$ can not be a corner point.

If $\mathfrak{q}$ is in B° , then it follows that $\mathrm{d}\rho \neq 0$ at $\mathfrak{q}$. So we can use an orthonormal frame (e_1, e_2) at $\mathfrak{q}$, with $e_1 = \nabla\rho/|\mathrm{d}\rho|$. At $\mathfrak{q}$ we have $\mathrm{d}P = 0$. For any tangent vector X , differentiating P gives

$$\mathrm{d}P(X) = 2\text{Hess } \rho(\nabla\rho, X) + 4\lambda\rho\mathrm{d}\rho(X).$$

Then the fact that $\Delta\rho = -2\lambda\rho$ gives

$$\text{Hess } \rho(e_1, e_1) = -2\lambda\rho, \quad \text{Hess } \rho(e_1, e_2) = \text{Hess } \rho(e_2, e_2) = 0.$$

In particular, $\text{Hess } \rho(e_2, e_2) = 0$ and it follows from (3.4) that $K \geq \lambda > 0$ at $\mathfrak{q}$. It follows that $\Delta P > 0$ at $\mathfrak{q}$. Contradiction.

Suppose now that $\mathfrak{q}$ is in the interior of a rod. We work with a local Fermi coordinates. Along the rod we have $P(0, s) = \kappa(s)^2$. So $\kappa' = 0$, $\kappa'' \leq 0$ at $\mathfrak{q}$. The equation $\Delta\rho = -2\lambda\rho$ gives

$$\rho_{rrr} - K\kappa + \kappa'' = -2\lambda\kappa.$$

Differentiating P and then using (2.11) gives that at $\mathfrak{q}$

$$P_r = 0, \quad P_{rr} = 2\kappa\rho_{rrr} + 2(\kappa')^2 + 4\lambda\kappa^2 = \lambda\kappa^2 - \kappa\kappa'' > 0.$$

Contradiction. $\square$

Proof of Proposition 3.6. For $\epsilon > 0$, (3.18) implies

$$|\mathrm{d} \arcsin(\frac{\rho}{M+\epsilon})| \leq \sqrt{2\lambda}.$$

Integrate from a maximum point $\mathfrak{q}$ of ρ to the boundary and let $\epsilon \rightarrow 0$. This gives

$$d(\mathfrak{q}, \partial B) \geq \frac{\pi}{2\sqrt{2\lambda}}$$

and hence the lower diameter bound.

For the upper bound, we first notice that

$$(3.19) \quad \mathrm{Hess} \sqrt{\rho} = \frac{\sqrt{\rho}}{2} [(K - \lambda)\underline{g} - T] \leq -\frac{\lambda}{3} \sqrt{\rho} \underline{g}.$$

Along a minimizing geodesic in B° from a maximum point of $\sqrt{\rho}$, from the Hessian bound we see that

$$\sqrt{\rho(t)} \leq \sqrt{\rho(0)} \cos(\sqrt{\lambda/3}t).$$

This implies that

$$\mathrm{diam} B \leq \pi \sqrt{3/\lambda}.$$

On $X_{\mathfrak{R}}$, the distance to a torus fixed point equals distance to its image in the quotient. Joining two points through any fixed point therefore gives $\mathrm{diam}(X_{\mathfrak{R}}, g) \leq 2 \mathrm{diam} B$. $\square$

Lemma 3.9. *We have*

$$\mathrm{inj}(\mathfrak{p}) \geq d(\mathfrak{p}, \partial B), \quad \sup_{\mathfrak{p} \in B^\circ} \mathrm{inj}(\mathfrak{p}) \geq \frac{\pi}{2\sqrt{2\lambda}}.$$

Moreover,

$$(3.20) \quad \mathrm{diam}(\partial B) \geq \frac{\pi}{\sqrt{2\lambda}}.$$

Proof. Arguing as in the proof of Proposition 3.6 we know that every geodesic starting in B° reaches ∂B before time $\pi\sqrt{\frac{3}{\lambda}}$. Fix $\mathfrak{p} \in B^\circ$, and let $D_{\mathfrak{p}} \subset T_{\mathfrak{p}}B$ be the star-shaped domain of vectors whose radial geodesics remain in B° . Since $K \leq \frac{\lambda}{3}$, Jacobi field comparison shows that these geodesics have no conjugate points before reaching the boundary. Hence $\exp_{\mathfrak{p}} : D_{\mathfrak{p}} \rightarrow B^\circ$ is a local diffeomorphism. Since the rods are totally geodesic and the vertices have angle $\pi/2$, a geodesic starting in B° cannot touch ∂B and then continue within B . By continuous dependence on initial data, a sequence approaching $\partial D_{\mathfrak{p}}$ therefore has its endpoints approaching ∂B . Since $D_{\mathfrak{p}}$ is bounded, $\exp_{\mathfrak{p}} : D_{\mathfrak{p}} \rightarrow B^\circ$ is proper. It is also onto, since a minimizing curve between two interior points stays in B° . Here we use that the rods are totally geodesic and the vertices have angle $\pi/2$. As $D_{\mathfrak{p}}$ is connected and B° is simply connected, $\exp_{\mathfrak{p}}$ is a diffeomorphism. In particular, every interior geodesic segment minimizes, and

$$\mathrm{inj}(\mathfrak{p}) \geq d(\mathfrak{p}, \partial B).$$

Again as in the proof of Proposition 3.6 we have for a maximum point $\mathfrak{q}$ of ρ ,

$$d(\mathfrak{q}, \partial B) \geq \frac{\pi}{2\sqrt{2\lambda}}.$$

Extend a geodesic through $\mathfrak{q}$ in both directions until it reaches boundary points $\mathfrak{q}_1$ and $\mathfrak{q}_2$. Its interior segments minimize, so, by continuity of distance, the entire segment from $\mathfrak{q}_1$ to $\mathfrak{q}_2$ minimizes. Both halves have length at least $d(\mathfrak{q}, \partial B)$. Therefore

$$\mathrm{diam}(\partial B) \geq d(\mathfrak{q}_1, \mathfrak{q}_2) \geq 2d(\mathfrak{q}, \partial B) \geq \frac{\pi}{\sqrt{2\lambda}}.$$

$\square$

The proof of Lemma 3.9 shows that B° is geodesically convex and has no cut or conjugate points. Since $K \leq \lambda/3$, the usual Rauch comparison gives $\mathrm{CAT}(\lambda/3)$ triangle comparison on B° . Passing to its metric completion B we get

Corollary 3.10. *The quotient (B, d_g) is a CAT($\lambda/3$) space.*

3.3. Curvature estimate away from fixed points. Let $\mathcal{V}$ denote the set of torus fixed points in M . The following provides a uniform curvature bound for the curvature of conical Einstein metrics, away from $\mathcal{V}$.

Proposition 3.11. *There is a universal constant C such that for a compact toric conical Einstein manifold (M, g) and $x \notin \mathcal{V}$ we have*

$$(3.21) \quad |\text{Rm}_g|(x) \leq C \max \left\{ \frac{\lambda}{3}, d_g(x, \mathcal{V})^{-2} \right\}.$$

Here the curvature at a point in Σ_i is to be understood in terms of local smooth development, as in the definition of toric conical metrics Definition 2.3.

We first prove a lemma. Recall that the interior of each rod is a totally geodesic boundary arc of $(B, \underline{g})$. For $\mathbf{q} \in I_i^\circ$, let $\nu_{\mathbf{q}}$ denote the inward unit normal and define

$$D_r^+(\mathbf{q}) := \left\{ v \in T_{\mathbf{q}}B : |v|_{\underline{g}} < r, \quad \underline{g}(v, \nu_{\mathbf{q}}) \geq 0 \right\}.$$

We define the *injectivity radius* at $\mathbf{q}$, denoted by $\text{inj}_{B, \underline{g}}(\mathbf{q})$, to be the supremum of all $r > 0$ such that the exponential map

$$\exp_{\mathbf{q}} : D_r^+(\mathbf{q}) \longrightarrow B_{\underline{g}}(\mathbf{q}, r)$$

is well-defined and is a diffeomorphism of manifolds with boundary. Notice that the ball $B_{\underline{g}}(\mathbf{q}, r)$ does not intersect any other rod for $r < \text{inj}_{B, \underline{g}}(\mathbf{q})$. We will also use the double of $(B, \underline{g})$, obtained by gluing two copies of B by the identity along ∂B . By Lemma 2.7, near the interior of each rod the quotient metric has the form

$$\underline{g} = dr^2 + f(r^2, s)^2 ds^2.$$

Allowing r to take both signs therefore extends the metric smoothly across the rod. At each turning point, the corner expansion extends smoothly under both coordinate reflections, and the doubled neighborhood is the quotient of this smooth extension by $(x, y) \mapsto (-x, -y)$. Thus the double is a compact surface without boundary, smooth away from the turning points, with cone angle π at each turning point. We call this the even double of $(B, \underline{g})$.

Lemma 3.12. *Suppose a point $\mathbf{q}$ lies in the interior of a rod $\mathfrak{I}_i$, and suppose that $\overline{B}_{\underline{g}}(\mathbf{q}, 2R)$ contains no turning points and that*

$$|\text{Rm}_g| \leq c \quad \text{on} \quad \pi^{-1}(B_{\underline{g}}(\mathbf{q}, 2R) \cap B^\circ),$$

Then we have

$$\text{inj}_{B, \underline{g}}(\mathbf{q}) \geq \min \left\{ R, \frac{\pi}{2\sqrt{c}} \right\},$$

Proof. Denote $r = \min \left\{ R, \frac{\pi}{2\sqrt{c}} \right\}$. Suppose $\text{inj}_{B, \underline{g}}(\mathbf{q}) < r$, then one of the following must occur

- There is a unit-speed geodesic segment $\gamma : [0, \ell] \rightarrow B$, for $\ell < r$, going from $\mathbf{q}$ and reaches the interior of another rod orthogonally.
- There is a unit-speed geodesic segment $\gamma : [0, \ell] \rightarrow B$, for $\ell < 2r$, going from $\mathbf{q}$ back to $\mathfrak{I}_i$.

In both cases, we have $\gamma((0, \ell)) \subset B^\circ$.

To see this, if another rod meets $B_{\underline{g}}(\mathbf{q}, r)$, a shortest segment to the other rods gives the first case. Otherwise, we pass to the even base double, where the curvature bound excludes conjugate points before r . A closest cut point at distance $s < r$ produces a geodesic loop of length $2s$, contained in the doubled radius- s ball. The loop cannot remain on the rod, which is an interval; its first excursion into one copy of B therefore returns to $\mathfrak{I}_i$, giving the second case.

Let $\tilde{\gamma} : [0, \ell] \rightarrow X_{\mathfrak{R}}$ be a horizontal lift of γ , so that $\pi \circ \tilde{\gamma} = \gamma$. Since the horizontal sections are totally geodesic, $\tilde{\gamma}$ is a unit-speed geodesic. Let J be the Killing Jacobi field along $\tilde{\gamma}$ generated by the circle vanishing on $\mathfrak{I}_i$. Then $J(0) = 0$, and J is nonzero in the interior. At $t = 0$, this is understood in the smooth local development. We claim that at $t = \ell$, $\langle D_t J, J \rangle = 0$. Again

this is understood in the smooth local development. In the second case, the claim follows from the fact that $J(\ell) = 0$. In the first case, notice that at the other end point, the Killing field is tangent to the corresponding totally geodesic fixed surface, whereas the lifted geodesic is normal to it. The claim then follows from $\langle D_t J, J \rangle = -\langle \nabla_J J, \dot{\gamma} \rangle = 0$.

Integration of the Jacobi equation gives

$$\int_0^\ell |D_t J|^2 dt \leq c \int_0^\ell |J|^2 dt.$$

Standard Jacobi field comparison gives

$$\ell \geq \frac{\pi}{2\sqrt{c}}$$

in the first case, and the second case, the stronger endpoint condition $J(0) = J(\ell) = 0$ gives

$$\ell \geq \frac{\pi}{\sqrt{c}}.$$

In both cases we are lead to a contradiction. $\square$

Proof of Proposition 3.11. Suppose that (3.21) fails. Then there is a sequence of toric conical Einstein metrics (X_j, g_j) , with $\text{Ric}_{g_j} = \lambda_j g_j$, such that

$$M_j := \max_{x \in X_j} a_j(x)^2 |\text{Rm}_{g_j}|(x) \longrightarrow \infty, \quad a_j(x) := \min \left\{ d_{g_j}(x, V_j), \sqrt{3/\lambda_j} \right\}.$$

Choose a maximizing point x_j , and set

$$Q_j := |\text{Rm}_{g_j}|(x_j), \quad \hat{g}_j := Q_j g_j, \quad R_j := \frac{1}{2} \sqrt{Q_j} a_j(x_j) = \frac{1}{2} \sqrt{M_j} \rightarrow \infty.$$

Since a_j is 1-Lipschitz, we have

$$|\text{Rm}_{\hat{g}_j}|(x_j) = 1, \quad \sup_{B_{\hat{g}_j}(x_j, R_j)} |\text{Rm}_{\hat{g}_j}| \leq 4.$$

Moreover,

$$d_{\hat{g}_j}(x_j, \mathcal{V}_j) \geq 2R_j \rightarrow \infty, \quad \text{Ric}_{\hat{g}_j} = \hat{\lambda}_j \hat{g}_j, \quad \hat{\lambda}_j := \frac{\lambda_j}{Q_j} \leq \frac{3}{M_j} \longrightarrow 0.$$

Now we divide into two cases:

Case (1). All the fixed surfaces escape to infinity.

In this case the limiting procedure does not see conical singularities. For sufficiently small $\delta > 0$, independent of j , the balls $B_{\hat{g}_j}(x_j, 2\delta)$ are smooth. The local covering consequence of [13, Theorem 1.3 and conditions (1.1.4)–(1.1.5)] gives a uniform volume lower bound on the universal covers:

$$\text{Vol}_{\tilde{g}_j} B_{\tilde{g}_j}(\tilde{x}_j, \delta) \geq c\delta^4.$$

Here $\tilde{g}_j$ is the lift of $\hat{g}_j$. The curvature bound and interior Einstein regularity give smooth convergence (after passing to a subsequence) on smaller lifted balls to a Ricci-flat metric $\tilde{g}_\infty$, with

$$|\text{Rm}_{\tilde{g}_\infty}|(\tilde{x}_\infty) = 1.$$

We now show that $\tilde{g}_\infty$ is flat. Here U_j, K_j, e_j, T_j are associated with the rescaled metric $\hat{g}_j$. For every fixed $A > 0$, the distance from $B_{\hat{g}_j}(x_j, A)$ to the union of the fixed surfaces tends to infinity. Thus (3.15) and (3.16), together with $\hat{\lambda}_j \rightarrow 0$, give

$$U_j \longrightarrow 0, \quad e_j \longrightarrow 0$$

uniformly on this ball. By (3.3), $K_j = e_j - U_j/2 \rightarrow 0$ uniformly as well. Finally, since T_j is nonnegative,

$$|T_j|_{\hat{g}_j} \leq \text{tr}_{\hat{g}_j} T_j = 2e_j \longrightarrow 0.$$

Normalize the constant real angular basis so that $G_j(x_j) = I$. By integrating along horizontal directions it follows that $G_j \rightarrow I$ uniformly on every fixed smaller neighborhood. Thus, in the smooth local equivariant limit, the metric becomes

$$\tilde{g}_\infty = \underline{g}_\infty + (d\theta^1)^2 + (d\theta^2)^2.$$

Since $K_j \rightarrow 0$, the base metric $\underline{g}_\infty$ is flat. So $\tilde{g}$ is also flat, contradiction.

Case (2). There exists a fixed surface Σ_j within bounded distance to x_j .

In this case away from the fixed surfaces we still have the limit of local universal cover around as above, which is Ricci-flat. But we can not deduce a contradiction directly. Instead, the flatness comes from studying the local convergence near a point in Σ_j .

We choose a point $\mathbf{q}_j \in \partial B_j$ closest to $\pi_j(x_j)$, and let $\mathfrak{I}_j$ be the rod containing $\mathbf{q}_j$. After passing to a subsequence, we have

$$d_{\underline{g}_j}(\pi_j(x_j), \mathbf{q}_j) \leq D$$

for some fixed $D < \infty$. Let q_j be the endpoint of a horizontal lift, starting at x_j , of a shortest segment from $\pi_j(x_j)$ to $\mathbf{q}_j$.

We parametrize $\mathfrak{I}_j$ by arclength s , with $s(\mathbf{q}_j) = 0$. Since all vertices escape to infinity, there are intervals $[-t_j, t_j] \subset \mathfrak{I}_j$, with $t_j \rightarrow \infty$ and $t_j + D < R_j$, such that on $\pi_j^{-1}([-t_j, t_j])$ the curvature of $\hat{g}_j$ is bounded by 4. Notice that by (3.14), κ_j is positive and concave. So we have

$$1 - \frac{|s|}{t_j} \leq \frac{\kappa_j(s)}{\kappa_j(0)} \leq 1 + \frac{|s|}{t_j}, \quad \frac{|\kappa_j'(0)|}{\kappa_j(0)} \leq \frac{1}{t_j}.$$

Thus $\kappa_j(s)/\kappa_j(0) \rightarrow 1$ uniformly on each fixed arclength interval.

By Lemma 3.12, a quotient half-disk of radius $\delta > 0$ about $\mathbf{q}_j$, with δ independent of j , meets no rod other than $\mathfrak{I}_j$. Let g_j^* be the smooth angular development across this rod. We normalize the vanishing Killing field to have unit normal rotation and a complementary Killing field to have unit length at q_j . Subtract a constant multiple of the vanishing field from the complementary field so that its derivative has zero normal rotational component at q_j . This does not change its value there. Its remaining derivative is controlled by $|\kappa_j'(0)|/\kappa_j(0)$.

The developed metrics g_j^* have uniformly bounded curvature. Applying the local covering argument from Case (1), we obtain smooth convergence on smaller lifted balls to a Ricci-flat metric g_∞ . The normalized values and derivatives of the Killing fields, together with $\nabla^2 V = \text{Rm} * V$, give smooth convergence of both fields. Thus the limit satisfies the reduced Einstein equations with $\lambda = 0$. In this angular normalization, the boundary coefficient of ρ is $\kappa_j(s)/\kappa_j(0)$, so $\kappa = 1$ along the limiting rod. Moreover, the curvature upper bound and (2.11) give

$$K \leq 0, \quad 0 = 2K - \frac{\kappa''}{\kappa} \quad \text{on the rod.}$$

So we must have $K = 0$ along the rod.

We claim that K vanishes in a collar of the rod. Notice that we have (3.6) for $\lambda = 0$:

$$(3.22) \quad \Delta_{\underline{g}_\infty} K + \frac{3U}{2e} \langle du, dK \rangle_{\underline{g}_\infty} \geq \frac{|dK|^2}{e} + \frac{3K^2(2U + K)}{e} + 3 \det T.$$

Since $e \geq 0$ and $2e = 2K + U$, we have $2U + K \geq 0$. By the rod expansion,

$$U = \frac{1}{2r^2} + O(1), \quad e = K + \frac{U}{2} = \frac{1}{4r^2} + O(1),$$

so $e > 0$ in a sufficiently small punctured collar.

To apply the strong maximum principle across the rod, we use Fermi coordinates (r, s) , where $r \geq 0$ is distance to the rod. The smooth rod expansions give

$$\underline{g}_\infty = dr^2 + f(r^2, s)^2 ds^2, \quad \rho = r + O(r^3), \quad U = \frac{1}{2r^2} + O(1), \quad e = \frac{1}{4r^2} + O(1).$$

Consequently, the operator on the left of (3.22) can be written as

$$\mathcal{A} = \partial_{rr} + \frac{3}{r}\partial_r + a(r^2, s)r\partial_r + b(r^2, s)\partial_{ss} + c(r^2, s)\partial_s,$$

where a, b, c are smooth and $b > 0$. Regard r as the radius $|x|$ in $\mathbb{R}^4$. Since K is smooth and even in r , the function $\widehat{K}(x, s) := K(|x|, s)$ is smooth, and $\mathcal{A}$ becomes the smooth uniformly elliptic operator

$$\widehat{\mathcal{A}} = \Delta_{\mathbb{R}^4} + a(|x|^2, s)x \cdot \nabla_x + b(|x|^2, s)\partial_{ss} + c(|x|^2, s)\partial_s.$$

The inequality $\widehat{\mathcal{A}}(\widehat{K}) \geq 0$ extends across $x = 0$ by continuity. Since $\widehat{K} \leq 0$ and $\widehat{K}(0, s) = 0$, the strong maximum principle implies $K \equiv 0$ in a collar of the rod.

We now show directly that the four-dimensional limit metric g_∞ is flat there. Since the base is flat and the rod is totally geodesic, its Fermi coordinates satisfy

$$\underline{g}_\infty = dr^2 + ds^2.$$

The function ρ satisfies

$$\Delta_{\underline{g}_\infty} \rho = 0, \quad \rho(0, s) = 0, \quad \partial_r \rho(0, s) = 1.$$

Uniqueness for these harmonic Cauchy data gives $\rho = r$, hence $u = \log r$.

The reduced base equation is now

$$\text{Hess}_{\underline{g}_\infty} u = -\frac{1}{2} du \otimes du - T.$$

Evaluating on ∂_s gives $T_{ss} = 0$. Since $T = \Phi^* g_{\mathcal{H}}$, this implies $\partial_s \Phi = 0$. Together with $\partial_s \rho = 0$ and $G = \rho \Phi$, we obtain $\partial_s G = 0$. This means that we can write

$$g_\infty = ds^2 + h, \quad h = dr^2 + G_{ab}(r) d\theta^a d\theta^b.$$

Since g_∞ is Ricci-flat, the three-dimensional metric h is Ricci-flat and therefore flat. Thus g_∞ is flat in the collar.

Let γ_j be the shortest quotient geodesic from $\mathbf{q}_j$ to $\pi_j(x_j)$, parametrized by arclength from $\mathbf{q}_j$. Since $\mathbf{q}_j$ is a closest boundary point,

$$d_{\widehat{g}_j}(\gamma_j(t), \partial B_j) = t.$$

Its horizontal lift has length at most D . Near q_j use the smooth angular development above. Outside a fixed small initial segment, the displayed identity gives a uniform positive distance from the fixed surfaces, so the local covering argument applies on balls of uniform radius.

Cover the lifted segment by a uniformly bounded number of overlapping neighborhoods. Choose lifts successively along the segment. On smaller neighborhoods of the overlap points, the lifted metrics are related by local isometries. The curvature bounds, local noncollapse, and Einstein regularity allow both the metrics and these local isometries to converge after passing to a subsequence.

The first limiting neighborhood contains the flat collar and is therefore flat by analyticity. Each successive Ricci-flat limiting neighborhood contains an open subset locally isometric to the preceding one. Analyticity therefore propagates flatness along the finite chain, including to the limit of x_j . This contradicts $|\text{Rm}_{\widehat{g}_j}|(x_j) = 1$. $\square$

3.4. Noncollapsing under angle bound. In this subsection, we prove a volume noncollapsing theorem.

Theorem 3.13. *Fix a compact rod structure $\mathfrak{R}$ and fix $0 < \beta_- \leq \beta_+ < \infty$. There exists $\sigma > 0$ such that any compact toric conical Einstein metric g with all cone angles contained in $[\beta_-, \beta_+]$ satisfies*

$$\text{Vol}(X_{\mathfrak{R}}, g) \geq \sigma \lambda^{-2}.$$

We will need the following elementary fact.

Lemma 3.14. *Denote*

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad A_i = (J\mathbf{v}_i)(J\mathbf{v}_i)^\top.$$

Then at every point $\mathbf{p} \in B^\circ$, we can write

$$(3.23) \quad G(\mathbf{p}) = \sum_i c_i(\mathbf{p}) A_i, \quad c_i(\mathbf{p}) \geq 0.$$

Moreover,

$$(3.24) \quad \det G = \sum_{i < j} c_i c_j \det(\mathbf{v}_i, \mathbf{v}_j)^2, \quad G(v, v) = \sum_i c_i \det(v, \mathbf{v}_i)^2.$$

Proof. With our definition, $A_i \mathbf{v}_i = 0$. Notice that Φ is a weighted harmonic map into $\mathcal{H}$:

$$d^*(\rho \Phi^{-1} d\Phi) = 0.$$

Since $\det \Phi = 1$, the matrix-valued one-form $\Phi^{-1} d\Phi$ is traceless. We apply $\mathcal{L}$ to matrix-valued functions entrywise in the fixed torus basis. Thus

$$(\mathcal{L}\Phi)_{\alpha\beta} = \Delta_{\underline{g}} \Phi_{\alpha\beta} + \langle du, d\Phi_{\alpha\beta} \rangle_{\underline{g}}.$$

Choose a local $\underline{g}$ -orthonormal frame (e_1, e_2) and write $\Phi_i = d\Phi(e_i)$. Expanding the above equation gives

$$\Phi^{-1} \mathcal{L}\Phi = \sum_i (\Phi^{-1} \Phi_i)^2.$$

By the Cayley-Hamilton identity for traceless 2×2 matrices,

$$A^2 = \frac{1}{2} \operatorname{tr}(A^2) \operatorname{Id},$$

and hence

$$\mathcal{L}\Phi = \frac{1}{2} \operatorname{tr}((\Phi^{-1} d\Phi)^2) \Phi = 4e\Phi.$$

Let ℓ be a linear functional on the space of symmetric matrices such that $\ell(A_i) \geq 0$ for every i . Then

$$(\mathcal{L} - 4e)\ell(\Phi) = 0.$$

At the interior of a rod I_i , the rod asymptotics in Lemma 2.7 gives

$$G = c(s)A_i + O(r^2), \quad \rho \sim r,$$

so $\ell(\Phi)$ has nonnegative lower limit. At a vertex between I_i and I_j , the corner expansion gives

$$G = r_1^2 a A_j + r_2^2 b A_i + O(r_1^2 r_2^2), \quad a, b > 0,$$

and hence $\ell(\Phi) \geq -Cr_1 r_2$. Thus a negative minimum of $\ell(\Phi)$ would be attained in B° , contradicting the maximum principle. Therefore Φ , and hence G , lies in the closed cone generated by the A_i at each point, proving (3.23).

Now we have

$$G(v, v) = \sum_i c_i (v^\top J \mathbf{v}_i)^2 = \sum_i c_i \det(v, \mathbf{v}_i)^2.$$

Write $J\mathbf{v}_i = (a_i, b_i)^\top$. Then

$$G = \sum_i c_i (J\mathbf{v}_i)(J\mathbf{v}_i)^\top = \begin{pmatrix} \sum_i c_i a_i^2 & \sum_i c_i a_i b_i \\ \sum_i c_i a_i b_i & \sum_i c_i b_i^2 \end{pmatrix}.$$

Hence

$$\begin{aligned}
\det G &= \left(\sum_i c_i a_i^2 \right) \left(\sum_j c_j b_j^2 \right) - \left(\sum_i c_i a_i b_i \right)^2 \\
&= \sum_{i < j} c_i c_j \left(a_i^2 b_j^2 + a_j^2 b_i^2 - 2a_i b_i a_j b_j \right) \\
&= \sum_{i < j} c_i c_j (a_i b_j - a_j b_i)^2 \\
&= \sum_{i < j} c_i c_j \det(\mathbf{v}_i, \mathbf{v}_j)^2.
\end{aligned}$$

□

Proof of Theorem 3.13. The proof is by contradiction. The rod directions identify a fixed collapsing circle. After unwrapping this circle, its normalized length converges to a positive function satisfying the equation (3.26). The rod expansions and the volume estimate near the limiting vertices justify an integration by parts that contradicts this equation.

After scaling, we may assume $\lambda = 3$. Choose a fixed integer N sufficiently large and quotient by the subgroup $\mathbb{T}^2[N] \cong (\mathbb{Z}/N\mathbb{Z})^2$. This divides all cone angles by N and the volume by N^2 , while leaving the quotient metric on the base unchanged. Relabeling the lattice, it suffices to prove the theorem when all cone angles are strictly less than 2π , with a uniform positive lower bound. The local convexity argument in [34, Lemma 4.43] then applies. Together with Bishop–Gromov comparison and the diameter bound (3.17), this gives uniform covering bounds for the spaces $(X_{\mathfrak{R}}, g)$ and their torus quotients.

Suppose the conclusion fails. There is then a sequence g_j such that $\text{Vol}(X_{\mathfrak{R}}, g_j) \rightarrow 0$. After passing to a subsequence, the bases converge in the Gromov–Hausdorff sense,

$$B_j \longrightarrow B_{\infty},$$

and the cone factors on each rod converge to positive limits. Let $\mathcal{S}$ be the limit of the vertex sets $\pi_j(\mathcal{V}_j)$, and let $\mathcal{F}$ be the limit of the union of rods. The estimates in Lemma 3.9, Proposition 3.11, and Lemma 3.12 imply that $B_{\infty} \setminus \mathcal{S}$ is a smooth surface with geodesic boundary $\mathcal{F} \setminus \mathcal{S}$, and the convergence is locally smooth up to this boundary. The boundary diameter lower bound (3.20) implies that $\mathcal{F} \setminus \mathcal{S} \neq \emptyset$.

Choose an interior point $\mathfrak{s}_{\infty}$ of a surviving rod, with rod vector $\mathbf{v}_1$, and let Q be the connected component of $B_{\infty} \setminus \mathcal{S}$ containing $\mathfrak{s}_{\infty}$. Write $Q^{\circ} = Q \setminus \mathcal{F}$. Moving a fixed small distance normally inward from $\mathfrak{s}_{\infty}$, choose $\mathfrak{q}_{\infty} \in Q^{\circ}$ and corresponding points $\mathfrak{q}_j \in B_j$. The Killing field generated by $\mathbf{v}_1$ vanishes on the rod, and its normal derivative there is controlled above and below by the cone angle bounds. The curvature bound and the Jacobi equation therefore give

$$0 < c \leq G_j(\mathbf{v}_1, \mathbf{v}_1)(\mathfrak{q}_j) \leq C.$$

Thus the $\mathbf{v}_1$ direction does not collapse at $\mathfrak{q}_j$.

On the other hand, $\rho_j = \sqrt{\det G_j}$ satisfies $-\Delta_{g_j} \rho_j = 6\rho_j$. On a fixed small neighborhood U_j of $\mathfrak{q}_j$, the base metrics converge smoothly and the Harnack inequality gives

$$\rho_j(\mathfrak{q}_j) \leq C \int_{U_j} \rho_j dA_{g_j} \leq C \text{Vol}(X_{\mathfrak{R}}, g_j) \longrightarrow 0.$$

We now use Lemma 3.14 to identify a fixed collapsing direction. Write

$$G_j = \sum_i c_{ij} A_i, \quad c_{ij} \geq 0,$$

combining the terms corresponding to parallel rod vectors. At $\mathfrak{q}_j$, Equation (3.24) gives

$$c_{ij} \det(\mathbf{v}_1, \mathbf{v}_i)^2 \leq G_j(\mathbf{v}_1, \mathbf{v}_1) \leq C$$

for every $\mathbf{v}_i$ not parallel to $\mathbf{v}_1$. These coefficients are therefore uniformly bounded. For the remaining coefficient, denoted by c_{1j} , the same equation gives

$$\det G_j \geq c_{1j} G_j(\mathbf{v}_1, \mathbf{v}_1),$$

so $c_{1j}(\mathbf{q}_j) \rightarrow 0$. Consequently, after passing to a subsequence, $G_j(\mathbf{q}_j)$ converges to a nonzero rank one matrix G_* in the cone generated by the A_i . The determinant identity shows that two generators with nonparallel rod vectors cannot both have positive coefficients in G_* . Hence

$$G_* = aA_{i_0}, \quad a > 0.$$

Set $\mathbf{w} = \mathbf{v}_{i_0}$. Since $G_*(\mathbf{v}_1, \mathbf{v}_1) > 0$, the vectors $\mathbf{w}$ and $\mathbf{v}_1$ are not parallel.

Complete $\mathbf{w}$ to a fixed integral basis $(\mathbf{w}, \mathbf{u})$ of Λ . Using the period-one angular parameters generated by this basis, write

$$(3.25) \quad g_j = \underline{g}_j + u_j^2(d\theta + a_j d\psi)^2 + f_j^2 d\psi^2,$$

where

$$u_j^2 = G_j(\mathbf{w}, \mathbf{w}), \quad a_j = \frac{G_j(\mathbf{w}, \mathbf{u})}{G_j(\mathbf{w}, \mathbf{w})}, \quad f_j^2 = G_j(\mathbf{u}, \mathbf{u}) - \frac{G_j(\mathbf{w}, \mathbf{u})^2}{G_j(\mathbf{w}, \mathbf{w})}.$$

In particular, if $D = |\det(\mathbf{w}, \mathbf{u})|$, then $u_j f_j = D\rho_j$. Here D is fixed and only records the angular normalization.

Write $\mathbf{v}_i = p_i \mathbf{w} + q_i \mathbf{u}$. The decomposition into the A_i gives

$$a_j = -\frac{\sum_{q_i \neq 0} c_{ij} p_i q_i}{\sum_{q_i \neq 0} c_{ij} q_i^2}, \quad |a_j| \leq \max_{q_i \neq 0} \left| \frac{p_i}{q_i} \right|.$$

Set

$$\varepsilon_j = u_j(\mathbf{q}_j) \rightarrow 0.$$

Since $\mathbf{v}_1 = p_1 \mathbf{w} + q_1 \mathbf{u}$ with $q_1 \neq 0$, we have

$$G_j(\mathbf{v}_1, \mathbf{v}_1)(\mathbf{q}_j) = \varepsilon_j^2(p_1 + q_1 a_j(\mathbf{q}_j))^2 + q_1^2 f_j(\mathbf{q}_j)^2.$$

The first term tends to zero, so

$$C^{-1} \leq f_j(\mathbf{q}_j) \leq C.$$

We next propagate these estimates through Q° . The interior estimates leading to (3.16), together with the equation for ρ_j , give uniform bounds for the logarithmic variation of the quadratic forms G_j on compact subsets of Q° . Integrating along fixed paths from $\mathbf{q}_\infty$, and using

$$u_j^2 = G_j(\mathbf{w}, \mathbf{w}), \quad f_j^2 = \inf_{t \in \mathbb{R}} G_j(t\mathbf{w} + \mathbf{u}, t\mathbf{w} + \mathbf{u}),$$

we obtain, for every compact subset $K \subset Q^\circ$,

$$C_K^{-1} \leq \varepsilon_j^{-1} u_j \leq C_K, \quad C_K^{-1} \leq f_j \leq C_K.$$

Thus the same fixed direction $\mathbf{w}$ collapses throughout the interior, whereas the quotient direction remains noncollapsed.

Lift the θ coordinate locally to $\mathbb{R}$ and set $\theta = \varepsilon_j^{-1} \tau$. With $V_j = \varepsilon_j^{-1} u_j$, the developed metric is

$$\underline{g}_j + V_j^2(d\tau + \varepsilon_j a_j d\psi)^2 + f_j^2 d\psi^2.$$

The preceding bounds give local noncollapse on every compact subset of Q° . The curvature estimates and Einstein regularity therefore give smooth local convergence after passing to a subsequence. Since a_j is uniformly bounded, the limit takes the diagonal form

$$g_\infty = \underline{g} + V^2 d\tau^2 + f^2 d\psi^2,$$

where $V, f > 0$ are defined throughout Q° . These developments are local, but the functions V and f agree on overlaps because they are limits of the globally defined functions V_j and f_j .

The developed metrics still satisfy $\text{Ric} = 3g$. Equations (2.5) and (2.6) therefore give

$$\Delta_{\underline{g}}(Vf) = -6Vf, \quad \text{div}_{\underline{g}}(V\nabla f - f\nabla V) = 0.$$

Combining these equations, we obtain

$$(3.26) \quad -\operatorname{div}_{\underline{g}}(f\nabla V) = 3fV.$$

We will integrate this equation against a cutoff times V^{-1} . For this purpose, we first show that V remains positive along every surviving rod in Q , and then control the contribution near $\mathcal{S}$.

Fix an arbitrary compact segment of a surviving rod in ∂Q , with rod vector

$$\mathbf{v} = p\mathbf{w} + q\mathbf{u}.$$

We claim that $q \neq 0$. Otherwise, since both vectors are primitive, $\mathbf{v} = \pm\mathbf{w}$. Choose points a fixed small normal distance $r_0 > 0$ from this rod. The Jacobi equation and the positive lower bound for the cone angles would give

$$u_j \geq cr_0$$

at these points. But the interior estimates already proved give $u_j \leq C\varepsilon_j \rightarrow 0$ there. This is a contradiction. Thus $\mathbf{w}$ is not parallel to any surviving rod vector in Q .

We now prove that near this rod,

$$(3.27) \quad V = V_0(s) + O(r^2), \quad V_0(s) > 0, \quad f = \frac{2\pi\beta}{|q|}r + O(r^3),$$

where r is the normal distance to the rod and $2\pi\beta$ is its limiting cone angle.

Fix a compact segment of a surviving rod, with rod vector $\mathbf{v} = p\mathbf{w} + q\mathbf{u}$, where $q \neq 0$. Normalize its normal cone angle using $\beta_j^{-1}\mathbf{v}$ and unwrap the $\mathbf{w}$ direction as above, with $\theta = \varepsilon_j^{-1}\tau$. The curvature bound and Lemma 3.12 give a developed collar of uniform size. The interior noncollapse and volume comparison give noncollapse up to the rod, so Einstein regularity gives smooth convergence across the fixed surface.

Let $K_{\mathbf{w},j}$ and $K_{\mathbf{v},j}$ denote the corresponding Killing fields. Their interior convergence and the Killing equation give smooth convergence across the fixed surface,

$$\varepsilon_j^{-1}K_{\mathbf{w},j} \longrightarrow K_{\mathbf{w},\infty}, \quad K_{\mathbf{v},j} \longrightarrow K_{\mathbf{v},\infty}.$$

On the interior, $K_{\mathbf{w},\infty} = \partial_\tau$ and $K_{\mathbf{v},\infty} = q\partial_\psi$. These fields commute, and the diagonal limit metric makes them orthogonal.

We claim that $K_{\mathbf{w},\infty}$ does not vanish on the fixed surface. Otherwise, at a zero, commutation with the rod rotation implies that its derivative preserves the normal plane and acts there as a rotation. Orthogonality to $K_{\mathbf{v},\infty}$ forces this normal rotation to vanish. Its flow would then fix the inward normal geodesic pointwise, contradicting $|K_{\mathbf{w},\infty}| = V > 0$ in the interior.

Thus $V = |K_{\mathbf{w},\infty}|$ is smooth, positive, and rotationally invariant near the fixed surface. Consequently,

$$(3.28) \quad V = V_0(s) + O(r^2), \quad V_0(s) > 0, \quad \partial_r V = O(r).$$

The rod rotation satisfies $|K_{\mathbf{v},\infty}| = 2\pi\beta r + O(r^3)$, where $2\pi\beta$ is the limiting cone angle. Since $|K_{\mathbf{v},\infty}| = |q|f$, we also obtain

$$(3.29) \quad f = \frac{2\pi\beta}{|q|}r + O(r^3).$$

These expansions hold on every compact segment of every surviving rod in Q .

It remains to control the neighborhoods of $\mathcal{S}$. On B_j , set

$$\nu_j = f_j dA_{\underline{g}_j}, \quad \mathbf{r}_j = d_{\underline{g}_j}(\cdot, \pi_j(\mathcal{V}_j)).$$

We claim that

$$(3.30) \quad \nu_j\{\mathbf{r}_j < r\} \leq Cr^3, \quad \nu_j(B_j) \leq C.$$

Consider the metric quotient $Y_j = X_j/S_{\mathbf{w}}^1$. On its regular part, its metric is

$$\hat{g}_j = \underline{g}_j + f_j^2 d\psi^2.$$

Since ψ has period one, the volume of the inverse image of a measurable subset of B_j equals its ν_j measure. Let $P_j \subset Y_j$ be the images of the torus fixed points. The orbit over each vertex is a single point, so the distance to P_j in Y_j equals the distance to $\pi_j(\mathcal{V}_j)$ in B_j .

The spaces Y_j inherit uniform metric doubling bounds from X_j . Since the number of vertices is fixed, each annulus

$$\{y \in Y_j : r/2 \leq d(y, P_j) < r\}$$

can therefore be covered by a uniformly bounded number of balls of radius $r/16$. On slightly enlarged balls, the curvature estimate (3.21) and O'Neill's formula give a sectional curvature lower bound $-Cr^{-2}$, with the uniform diameter bound absorbed into C . Away from P_j , a rod parallel to $\mathfrak{w}$ gives a smooth even boundary of Y_j , while a rod $\mathfrak{v} = p\mathfrak{w} + q\mathfrak{u}$ with $q \neq 0$ gives a conical curve of angle $2\pi\beta_j/|q| < 2\pi$. Reflection across the smooth boundary and local volume comparison across these conical curves show that each covering ball has volume at most Cr^3 . Consequently, the annulus has volume at most Cr^3 . Summing over dyadic annuli proves the first estimate in (3.30). Taking r larger than the uniformly bounded diameter gives the second.

Let $\mathcal{N}_r$ be the r -neighborhood of $\mathcal{S}$ intersected with Q . Smooth convergence on compact subsets of Q° , followed by exhaustion, now gives

$$\nu = f dA_{\underline{g}}, \quad 0 < \nu(Q) < \infty, \quad \nu(\mathcal{N}_r) \leq Cr^3.$$

Here positivity follows from $f > 0$ on the nonempty open set Q° .

We can now integrate (3.26). Choose a Lipschitz cutoff χ_r which vanishes on $\mathcal{N}_r$, equals one on $Q \setminus \mathcal{N}_{2r}$, and satisfies $|\mathrm{d}\chi_r| \leq Cr^{-1}$. Along each surviving rod, (3.28) and (3.29) give

$$\frac{f}{V} \partial_r V = O(r^2).$$

Thus there is no boundary contribution from the rods when we multiply (3.26) by χ_r^2/V and integrate by parts. The cutoff removes the limiting vertices. We obtain

$$\begin{aligned} (3.31) \quad 3 \int_Q f \chi_r^2 dA_{\underline{g}} &= \int_Q f \left\langle \mathrm{d}V, \mathrm{d} \left(\frac{\chi_r^2}{V} \right) \right\rangle dA_{\underline{g}} \\ &= \int_Q f \left(|\mathrm{d}\chi_r|^2 - \left| \mathrm{d}\chi_r - \frac{\chi_r}{V} \mathrm{d}V \right|^2 \right) dA_{\underline{g}} \\ &\leq \int_Q f |\mathrm{d}\chi_r|^2 dA_{\underline{g}} \\ &\leq Cr^{-2} \nu(\mathcal{N}_{2r}) \leq Cr. \end{aligned}$$

As $r \rightarrow 0$, the right-hand side tends to zero, whereas the left-hand side tends to $3\nu(Q) > 0$. This contradiction proves the volume lower bound for $\lambda = 3$. Rescaling gives the stated bound. $\square$

3.5. Hitchin-Thorpe inequality. For compact Einstein four-manifolds, a well-known topological restriction is given by the Hitchin-Thorpe inequality. This was generalized in [4, Theorem A] to Einstein four-manifolds with a cone singularity along a smooth surface in a suitable sense. In our toric conical setting, we will prove a version below.

Let $(X_{\mathfrak{R}}, g)$ be a compact toric conical Einstein four-manifold with $\mathrm{Ric}_g = \lambda g$. Let $\Sigma_1, \dots, \Sigma_m$ be associated 2-spheres along which g has cone angle $2\pi\beta_i$, and set $e_i = [\Sigma_i]^2$.

Define the modified Euler characteristic and signature by

$$(3.32) \quad \chi_c(g) = \chi(X) - 2 \sum_{i=1}^m (1 - \beta_i) + \sum_{i=1}^m (1 - \beta_i)(1 - \beta_{i+1}) = \sum_{i=1}^m \beta_i \beta_{i+1},$$

$$(3.33) \quad \tau_c(g) = \tau(X) - \frac{1}{3} \sum_{i=1}^m (1 - \beta_i^2) e_i = \frac{1}{3} \sum_{i=1}^m e_i \beta_i^2.$$

Set

$$I_{\pm}(g) = \int_{X^\circ} |W_g^{\pm}|^2 \mathrm{d} \mathrm{Vol}_g.$$

Proposition 3.15. *We have*

$$(3.34) \quad I_{\pm}(g) = 2\pi^2(2\chi_c(g) \pm 3\tau_c(g)) - \frac{\lambda^2}{3} \text{Vol}_g(X).$$

In particular, if $\lambda \neq 0$, then

$$(3.35) \quad 2 \sum_{i=1}^m \beta_i \beta_{i+1} > \left| \sum_{i=1}^m e_i \beta_i^2 \right|.$$

Proof. Let $\mathcal{E}(g)$ and $p_1(g)$ denote the normalized Euler and first Pontryagin forms, and set $\delta_i = 1 - \beta_i$. We compute their integrals by smoothing the conical singularities and applying the ordinary Chern–Gauss–Bonnet and signature formulas.

We first arrange that the metric is exactly a product of flat cones near each torus fixed point. In equivariant normal coordinates in a local development, the metric differs from its Euclidean tangent metric by $O(r^2)$, with the corresponding derivative bounds. A cutoff at scale η therefore gives a conical metric h_η which agrees with the flat product model within distance η of each torus fixed point and agrees with g outside distance 2η . These metrics have the same cone angles and uniformly bounded curvature on the regular locus. Since the modified regions have volume $O(\eta^4)$, the regular Euler and Pontryagin integrals of h_η converge to those of g as $\eta \rightarrow 0$.

Fix η and write $h = h_\eta$. Smooth the normal cones at scale ε , using product smoothings near the torus fixed points, to obtain smooth metrics h_ε on X . These agree with h outside shrinking neighborhoods of the singular surfaces. The ordinary characteristic number formulas give

$$\int_X \mathcal{E}(h_\varepsilon) = \chi(X), \quad \int_X p_1(h_\varepsilon) = 3\tau(X).$$

As $\varepsilon \rightarrow 0$, the contributions away from the singular surfaces converge to the corresponding integrals over X° . We now compute the additional contributions concentrated along the surface interiors and at their intersections.

Consider first the interior of Σ_i . Smoothing a normal cone of angle $2\pi\beta_i$ gives a smooth disk N_ε which agrees with the cone near its boundary. The boundary has total geodesic curvature $2\pi\beta_i$, so the two-dimensional Gauss–Bonnet formula gives

$$\frac{1}{2\pi} \int_{N_\varepsilon} K_N dA_N = 1 - \frac{1}{2\pi} \int_{\partial N_\varepsilon} k_N ds = 1 - \beta_i = \delta_i.$$

For a product of this disk with a surface patch $U \subset \Sigma_i^\circ$, the Euler form is the product of the two normalized curvature forms. Hence its integral is

$$\left(\frac{1}{2\pi} \int_{N_\varepsilon} K_N dA_N \right) \left(\frac{1}{2\pi} \int_U K_{\Sigma_i, h} dA_h \right) = \frac{\delta_i}{2\pi} \int_U K_{\Sigma_i, h} dA_h.$$

The local edge-cone calculation in [4, Section 2, proof of Theorem 2.1] gives the same limiting contribution for h . Here we first work away from the torus fixed points and then exhaust Σ_i° .

The induced metric on Σ_i has intrinsic cone angles $2\pi\beta_{i-1}$ and $2\pi\beta_{i+1}$ at its two poles, since these are the intersections with the neighboring singular surfaces. Thus

$$\frac{1}{2\pi} \int_{\Sigma_i^\circ} K_{\Sigma_i, h} dA_h = 2 - \delta_{i-1} - \delta_{i+1}.$$

The Euler contribution concentrated along the interior of Σ_i is therefore

$$\delta_i(2 - \delta_{i-1} - \delta_{i+1}).$$

At the torus fixed point $\Sigma_i \cap \Sigma_{i+1}$, the metric h is an exact product of two flat cones. The product smoothing has normalized curvature integrals δ_i and δ_{i+1} in the two factors. Its Euler integral is consequently $\delta_i \delta_{i+1}$. Combining the regular integral with the surface and fixed-point contributions gives

$$\chi(X) = \int_{X^\circ} \mathcal{E}(h) + \sum_i \delta_i(2 - \delta_{i-1} - \delta_{i+1}) + \sum_i \delta_i \delta_{i+1}.$$

Letting $\eta \rightarrow 0$ and using $\chi(X) = m$, we obtain

$$\begin{aligned} \int_{X^\circ} \mathcal{E}(g) &= \chi(X) - \sum_i \delta_i (2 - \delta_{i-1} - \delta_{i+1}) - \sum_i \delta_i \delta_{i+1} \\ &= \chi(X) - 2 \sum_i \delta_i + \sum_i \delta_i \delta_{i+1} \\ &= \sum_i \beta_i \beta_{i+1} = \chi_c(g). \end{aligned}$$

For the first Pontryagin form, the local normal-bundle calculation in [4, Theorem 2.2 and its proof] gives the contribution $(1 - \beta_i^2)e_i$ along Σ_i . Here $e_i = [\Sigma_i]^2$ is the Euler number of its oriented normal bundle. There is no additional fixed-point contribution: for the product smoothing, the first Pontryagin form is the sum of the squares of the two normalized curvature forms, and each square vanishes because it is pulled back from a two-dimensional factor. Taking the smoothing limit and then letting $\eta \rightarrow 0$ therefore gives

$$3\tau(X) = \int_{X^\circ} p_1(g) + \sum_i (1 - \beta_i^2)e_i.$$

Using the topological identity $3\tau(X) = \sum_i e_i$, we conclude that

$$\int_{X^\circ} p_1(g) = 3\tau(X) - \sum_i (1 - \beta_i^2)e_i = \sum_i e_i \beta_i^2 = 3\tau_c(g).$$

Finally, since $\text{Ric}_g = \lambda g$, the scalar curvature is 4λ . With the Weyl norms taken to be the Hilbert–Schmidt norms of the curvature operators on $\Lambda^\pm$, the Chern–Weil formulas give

$$\begin{aligned} 8\pi^2 \chi_c(g) &= I_+(g) + I_-(g) + \frac{2\lambda^2}{3} \text{Vol}_g(X), \\ 12\pi^2 \tau_c(g) &= I_+(g) - I_-(g). \end{aligned}$$

Taking half the sum and difference yields

$$I_\pm(g) = 2\pi^2 (2\chi_c(g) \pm 3\tau_c(g)) - \frac{\lambda^2}{3} \text{Vol}_g(X),$$

which proves (3.34). Since $I_\pm(g) \geq 0$, we have

$$2\chi_c(g) \pm 3\tau_c(g) \geq \frac{\lambda^2}{6\pi^2} \text{Vol}_g(X).$$

For $\lambda \neq 0$, the right-hand side is positive. Substituting the expressions for $\chi_c(g)$ and $\tau_c(g)$ gives (3.35). $\square$

4. LOCAL STRUCTURE OF THE MODULI SPACE

From now on in this paper we work with the setting of Theorem 1.7. Let $\mathfrak{R}_0$ be a compact rod structure with m_0 rods and a chosen turning point $\mathbf{p}$. After a global integral change of lattice basis, possibly reversing its orientation, assume that the incident rod directions are B, A in boundary order and all rod directions belong to $\{A, B, C\}$, with A, B, C as defined in the introduction.

Replace the corner (B, A) by one of the following sequences of rod directions:

$$(4.1) \quad W \in \left\{ \begin{array}{ll} (B, (A, B)^n, A) & (n \geq 1), \quad (B, C, A), \\ (B, A, C, A), & (B, C, B, A), \\ (B, C, A, C, A), & (B, C, B, C, A), \\ (B, C, A, B, C, A) \end{array} \right\}.$$

Set

$$\mathfrak{R} = \mathfrak{R}_0 \#_{\mathbf{p}} W, \quad M = X_{\mathfrak{R}_0}, \quad X = X_{\mathfrak{R}}.$$

Number the rods of $\mathfrak{R}$ in positive cyclic order so that $\mathfrak{J}_1, \dots, \mathfrak{J}_q$ correspond to the intermediate directions of W . There are $m = m_0 + q$ rods altogether. As in Section 2, write

$$\pi : X \longrightarrow B, \quad \Sigma_i = \pi^{-1}(\mathfrak{J}_i), \quad e_i = [\Sigma_i]^2,$$

with $\mathfrak{p}_i \in B$ and $p_i = \pi^{-1}(\mathfrak{p}_i)$ denoting the turning points and the corresponding torus fixed points, respectively. Set

$$\mathcal{S} = \bigcup_{\alpha=1}^q \Sigma_\alpha, \quad \sigma = \sum_{\alpha=1}^q e_\alpha.$$

The orientations of M and X are determined by the oriented quotient and the normalized angular coordinates, as in Section 2. All connected-sum identifications and self-intersection numbers below use these orientations. If the lattice normalization reverses the original lattice orientation, then these four-manifold orientations are opposite to the original ones. To express the connected-sum identifications in the original orientation of M , reverse the orientation of each listed summand. The self-intersection numbers and signatures then change sign.

4.1. Topology of the toric manifolds. It is easy to check that with the above choice of rod structures, we get exactly the smooth toric four-manifolds listed in Theorem 1.1.

Indeed, the connected-sum identifications follow directly from the neighborhoods of the corresponding spheres. For the (AB) alternating sequence, two consecutive spheres have self-intersection zero and meet transversely once. Their regular neighborhood is $(\mathbb{S}^2 \times \mathbb{S}^2) \setminus B^4$, as seen from the two coordinate spheres in $\mathbb{S}^2 \times \mathbb{S}^2$. Replacing this neighborhood by a four-ball removes one pair (A, B) from the sequence. Repeating this operation gives

$$X \cong M \# n(\mathbb{S}^2 \times \mathbb{S}^2).$$

For each remaining sequence, the self-intersection formula gives successive deletions of rods with self-intersection ± 1 , reducing the sequence to (B, A) . Each $+1$ deletion contributes a $\mathbb{CP}^2$ summand, and each -1 deletion contributes an $\overline{\mathbb{CP}^2}$ summand. Counting these deletions gives the connected-sum identifications below.

W	connected sum by
(B, C, A)	$\overline{\mathbb{CP}^2}$
$(B, A, C, A), (B, C, B, A)$	$\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$
$(B, C, A, C, A), (B, C, B, C, A)$	$\mathbb{CP}^2 \# 2\overline{\mathbb{CP}^2}$
(B, C, A, B, C, A)	$\mathbb{CP}^2 \# 3\overline{\mathbb{CP}^2}$

The two sequences grouped in each row give the same oriented connected sum summand, but their insertions into a fixed rod structure may yield inequivalent $\mathbb{T}^2$ -actions. For the four-manifolds $\#r(\mathbb{S}^2 \times \mathbb{S}^2)$ and $(\mathbb{CP}^2 \# k\overline{\mathbb{CP}^2}) \# r(\mathbb{S}^2 \times \mathbb{S}^2)$ with $0 \leq k \leq 3$ in Theorem 1.1, we can obtain them by starting with $\mathbb{S}^4$ or $\mathbb{CP}^2 \# k\overline{\mathbb{CP}^2}$ with $0 \leq k \leq 3$, then insert with $W = (B, (A, B)^r, A)$. For the third situation in Theorem 1.1, we may begin with $\mathbb{S}^4$, $\mathbb{S}^2 \times \mathbb{S}^2$, or $\mathbb{CP}^2 \# k\overline{\mathbb{CP}^2}$ with either of its orientation, where $0 \leq k \leq 3$, then insert the rest of sequences.

For all the sequences we get

$$(4.2) \quad \begin{array}{c|c|c|c} W & q & (e_1, \dots, e_q) & \sigma \\ \hline (B, (A, B)^n, A) & 2n & (0, \dots, 0) & 0 \\ (B, C, A) & 1 & (-1) & -1 \\ (B, A, C, A) & 2 & (1, 0) & 1 \\ (B, C, B, A) & 2 & (0, 1) & 1 \\ (B, C, A, C, A) & 3 & (-1, 0, 0) & -1 \\ (B, C, B, C, A) & 3 & (0, 0, -1) & -1 \\ (B, C, A, B, C, A) & 4 & (-1, -1, -1, -1) & -4 \end{array}$$

In particular, for $[(\beta, g)] \in \mathcal{M}$, we have

$$(4.3) \quad \chi_c(g) = (q-1)\beta^2 + 2\beta + m_0 - 1, \quad \tau_c(g) = \tau(X) + \frac{\beta^2 - 1}{3}\sigma.$$

Remark 4.1. For $W = (B, C, A)$, we have $q = 1$ and $\sigma = -1$, so the characteristic formulas and the Hitchin–Thorpe inequality give an upper bound on β . For the remaining sequences, these inequalities alone do not exclude $\beta \rightarrow \infty$. The upper bound is more difficult to obtain and will be proved in Section 6.

4.2. Definition of the moduli space. Let g_0 be a torically non-degenerate Einstein metric with $\text{Ric}_{g_0} = 3g_0$ for rod structure $\mathfrak{R}_0$. We consider toric conical Einstein metrics with

$$\text{Ric}_g = 3g, \quad \beta_i = \begin{cases} \beta, & 1 \leq i \leq q, \\ 1, & q < i \leq m. \end{cases}$$

Let $\mathcal{Z}$ be the space of pairs (β, g) , where $\beta > 0$ and g is such a metric satisfying $\text{Ric}(g) = 3g$. We equip $\mathcal{Z}$ with the topology of convergence of cone factors and C^∞ convergence in local developments. Recording β and the developed tensor coefficients in a fixed finite collection of charts realizes $\mathcal{Z}$ as a subspace of the product of $(0, \infty)$ with finitely many separable C^∞ spaces. In particular, $\mathcal{Z}$ is metrizable and second countable. Two pairs (β, g) and (β', g') are equivalent if $\beta = \beta'$ and $g' = F^*g$ for an orientation-preserving diffeomorphism $F : X \rightarrow X$ commuting with the fixed $\mathbb{T}^2$ -action and preserving each Σ_α . These diffeomorphisms act on $\mathcal{Z}$ by pullback. We denote the resulting quotient by $\mathcal{M}$, equip it with the quotient topology, and write $\varpi : \mathcal{Z} \rightarrow \mathcal{M}$ for the quotient projection.

The stabilizer of every pair in $\mathcal{Z}$ is precisely the group of torus translations. Indeed, an equivariant orientation-preserving isometry induces an orientation-preserving isometry of the orbit space preserving each inserted rod, so it fixes the two turning points bounding such a rod individually. Its restriction to that rod is therefore the identity. Preservation of inward normal geodesics then implies that it is the identity near the interior of the rod, and hence on the whole orbit space. On the principal orbit set, an equivariant map covering the identity has the form $x \mapsto a(\pi(x)) \cdot x$, where a maps the interior of the orbit space to $\mathbb{T}^2$. Since an isometry preserves the orthogonal complements of the torus orbits, $da = 0$. Thus a is constant, and the isometry is a torus translation. In particular, the stabilizer acts trivially on invariant tensors.

We next show that the equivalence relation is closed. Suppose that

$$(\beta_j, g_j) \longrightarrow (\beta, g), \quad (\beta_j, F_j^* g_j) \longrightarrow (\beta, h),$$

where each F_j is an allowed diffeomorphism. Smooth convergence of the metrics in local developments gives uniform bounds for the first derivatives of F_j and F_j^{-1} , and the isometry equation gives bounds for all higher derivatives. Compactness of X therefore gives a subsequence for which the maps and their inverses converge smoothly in local developments to a diffeomorphism F and its inverse. The limit preserves orientation, commutes with the torus action, and preserves each Σ_α , so it is an allowed diffeomorphism. Passing to the limit gives $h = F^*g$, proving closedness.

The projection ϖ is open. Indeed, for every open set $U \subset \mathcal{Z}$,

$$\varpi^{-1}(\varpi(U)) = \bigcup_F F^*U,$$

and the right-hand side is open because each pullback map is a homeomorphism. Closedness of the equivalence relation now implies that $\mathcal{M}$ is Hausdorff: inequivalent pairs have neighborhoods containing no equivalent pair, whose images under ϖ are disjoint open neighborhoods. Finally, since $\mathcal{Z}$ is second countable and ϖ is open, the images of a countable basis for $\mathcal{Z}$ form a countable basis for $\mathcal{M}$.

4.3. The linearized equation. We first specify the analytic conventions used below. For fixed cone angles, an invariant tensor is called *smooth* if it is smooth on the regular locus and extends smoothly in each local angular development, including the simultaneous developments at the torus fixed points. All tensor spaces below consist of $\mathbb{T}^2$ -invariant tensors with this regularity. These conventions apply to arbitrary compact toric conical metrics, including the smooth case.

We use the standard divergence gauge for the Einstein equation; see, for example, [1, Remark 4.6, equation (4.25)], [7, Section 2.10]. Denote the Einstein tensor by

$$\mathcal{E}_\lambda(g) = \text{Ric}(g) - \frac{1}{2} \text{Scal}_g g + \lambda g,$$

and define its linearization in divergence gauge by

$$\mathcal{P}_g = D\mathcal{E}_{\lambda,g} + \delta_g^* \delta_g,$$

where $\delta_g = -\text{div}_g$ and δ_g^* is its formal L^2 adjoint. In dimension four, $\mathcal{E}_\lambda(g) = 0$ is equivalent to $\text{Ric}(g) = \lambda g$. The operator $\mathcal{P}_g$ is elliptic, and its coefficients are smooth. We take $\lambda = 3$ in this section.

Fix (β_0, g) representing a point of $\mathcal{M}$. For each integer $s \geq 0$, let L_s^2 be the completion of smooth $\mathbb{T}^2$ -invariant symmetric two-tensors in the norm

$$\|k\|_{L_s^2}^2 = \sum_{j=0}^s \|\nabla_g^j k\|_{L^2}^2, \quad L^2 = L_0^2.$$

Unless otherwise stated, all tensor spaces and orthogonal complements below are defined using this fixed metric g . Since g is Einstein, we have

$$2\mathcal{P}_g k = (\nabla_g^* \nabla_g - 2\mathring{R}_g)k - \nabla_g^2 \text{tr}_g k - (\nabla_g^* \nabla_g \text{tr}_g k + \delta_g \delta_g k - 3 \text{tr}_g k)g.$$

On transverse-traceless tensors, we have

$$\mathcal{P}_g = \frac{1}{2}(\nabla_g^* \nabla_g - 2\mathring{R}_g),$$

Lemma 4.2. *The operator*

$$\mathcal{P}_g : L_2^2 \subset L^2 \longrightarrow L^2$$

is self-adjoint and Fredholm of index zero. Its kernel is

$$(4.4) \quad \mathcal{K}_g = \left\{ k \in L_2^2 : \delta_g k = 0, \quad \text{tr}_g k = 0, \quad (\nabla_g^* \nabla_g - 2\mathring{R}_g)k = 0 \right\},$$

and every element of $\mathcal{K}_g$ is smooth. For every integer $s \geq 0$, the map $\mathcal{P}_g : L_{s+2}^2 \rightarrow L_s^2$ is Fredholm of index zero with the same kernel.

Proof. At an Einstein metric, $D\mathcal{E}_{3,g}$ is formally self-adjoint, and hence so is $\mathcal{P}_g = D\mathcal{E}_{3,g} + \delta_g^* \delta_g$. Let k, h be smooth invariant symmetric two-tensors. We integrate by parts on the complement of shrinking neighborhoods of the fixed surfaces and torus fixed points. Smoothness in local developments implies that k, h and their first covariant derivatives remain bounded, so the boundary terms tend to zero as these neighborhoods shrink. Therefore

$$\int_X \langle \mathcal{P}_g k, h \rangle_g dV_g = \int_X \langle k, \mathcal{P}_g h \rangle_g dV_g.$$

By density, this identity also holds for $k, h \in L_2^2$, proving that $\mathcal{P}_g$ is symmetric on this domain.

To prove self-adjointness, we must identify the domain of the adjoint $\mathcal{P}_g^*$ of the unbounded operator $\mathcal{P}_g : L_2^2 \subset L^2 \rightarrow L^2$. Let $k \in \text{Dom}(\mathcal{P}_g^*)$. By definition, there exists $f = \mathcal{P}_g^* k \in L^2$ such that

$$\langle k, \mathcal{P}_g h \rangle_{L^2} = \langle f, h \rangle_{L^2} \quad \text{for every } h \in L_2^2.$$

At this stage, k is only known to belong to L^2 . We will use this identity to prove that $k \in L_2^2$. Lift k and f to an angularly invariant local development. Given an arbitrary compactly supported smooth test tensor φ there, its angular average is smooth and invariant, descends to the corresponding conical neighborhood, and extends by zero to a global smooth invariant tensor. It can therefore be used as h in the adjoint identity. Since $\mathcal{P}_g$ commutes with angular averaging and both k and f are invariant, the same identity holds for the arbitrary test tensor φ .

Formal self-adjointness of the differential expression then means precisely that the lifted equation $\mathcal{P}_g k = f$ holds in the distributional sense throughout the development, including across the axes. Since $f \in L^2$, interior elliptic regularity gives $k \in L^2_{2,\text{loc}}$ in each development. A finite covering and invariant smooth approximation yield $k \in L^2_2$. Thus $\text{Dom}(\mathcal{P}_g^*) \subset L^2_2$, while the reverse inclusion follows from symmetry proved above. Hence $\text{Dom}(\mathcal{P}_g^*) = L^2_2$ and $\mathcal{P}_g^* = \mathcal{P}_g$ on this domain.

The local elliptic estimates give

$$\|k\|_{L^2_2} \leq C_g(\|\mathcal{P}_g k\|_{L^2} + \|k\|_{L^2}).$$

Together with the compact inclusion $L^2_2 \hookrightarrow L^2$, this proves the Fredholm property. Self-adjointness gives index zero, and elliptic regularity gives the corresponding statements at higher Sobolev orders.

The linearized contracted Bianchi identity gives

$$(4.5) \quad \delta_g \mathcal{P}_g k = \delta_g \delta_g^* \delta_g k.$$

For $k \in \ker \mathcal{P}_g$, integration by parts first gives $\delta_g^* \delta_g k = 0$ and then $\delta_g k = 0$. For a divergence-free tensor, the trace equation is

$$(4.6) \quad \text{tr}_g(\mathcal{P}_g k) = -(\nabla_g^* \nabla_g - 3) \text{tr}_g k.$$

The integrated Bochner identity is valid for invariant functions, again by integration by parts in local developments. Since $\text{Ric}(g) = 3g$ and $\dim X = 4$, the standard Lichnerowicz eigenvalue estimate gives $\lambda_1 \geq 4$, where λ_1 denotes the first positive eigenvalue of the scalar Laplacian $\nabla_g^* \nabla_g$ on invariant functions. Consequently $\nabla_g^* \nabla_g - 3$ is invertible on invariant functions, and $\text{tr}_g k = 0$. $\square$

The ellipticity statement also applies to non-Einstein reference metrics. On a compact manifold, $\mathcal{P}_g$ differs from its formal adjoint by an operator of order zero. Its symmetrization therefore has the same elliptic principal symbol and is self-adjoint on L^2_2 domain. Since the difference is compact as a map $L^2_2 \rightarrow L^2$, the Fredholm index of $\mathcal{P}_g$ is also zero.

4.4. The local structure of the moduli space. We first identify the space of tensors when β varies. Choose $\mathbb{T}^2$ -invariant rod and corner charts and a subordinate partition of unity $\{\chi_a\}$, smooth in local developments and independent of β . Recall that $\mathcal{S} = \bigcup_{\alpha=1}^q \Sigma_\alpha$ is the union of the inserted rod spheres. In each normal plane to a component of $\mathcal{S}$, define the operator $J_{\beta,a}$ by keeping the tensor coefficients and replacing

$$\beta_0 r d\theta \longmapsto \beta r d\theta$$

in both covariant factors, while keeping the tangential part unchanged. At a fixed point, make the replacement in each normal plane corresponding to a component of $\mathcal{S}$. Thus both angles vary when both incident spheres belong to $\mathcal{S}$, and only one varies otherwise. On charts away from $\mathcal{S}$ we do not make any change. Now set

$$J_\beta = \sum_a \chi_a J_{\beta,a}, \quad b_\beta = J_\beta g.$$

For β sufficiently close to β_0 , J_β is an isomorphism between the corresponding tensor spaces and b_β is a conical metric with cone factor β along $\mathcal{S}$. Both depend real-analytically on β , with $J_{\beta_0} = \text{Id}$ and $b_{\beta_0} = g$. They may be chosen to agree with the identity and g , respectively, outside a fixed neighborhood of $\mathcal{S}$.

Throughout this subsection, all Sobolev spaces L^2_s are defined using the fixed reference metric g , with cone factor β_0 along $\mathcal{S}$. In particular, the domain space L^2_8 and the target space L^2_6 below are independent of β ; the maps J_β and J_β^{-1} transport tensors between the reference spaces and the corresponding spaces for cone factor β . Write a nearby metric with cone factor β as

$$h = b_\beta + J_\beta k, \quad k \in L^2_8,$$

and define the map with fixed domain and target

$$(4.7) \quad \Psi(\beta, k) = J_\beta^{-1} \{ \mathcal{E}_3(h) - \delta_h^* \delta_h b_\beta \}.$$

It is real-analytic from a neighborhood of $(\beta_0, 0)$ in $\mathbb{R} \times L_8^2$ to L_6^2 , and

$$D_k \Psi(\beta_0, 0) = \mathcal{P}_g.$$

Indeed, differentiating $\delta_h h = 0$ at $h = g$ shows that the derivative of the gauge term is $\delta_g^* \delta_g k$.

Proposition 4.3. *A neighborhood of $[(\beta_0, g)]$ in $\mathcal{M}$ is homeomorphic to the zero set of a real-analytic map*

$$\kappa_g : U_g \subset \mathcal{K}_g \times \mathbb{R} \longrightarrow \mathcal{K}_g,$$

where U_g is a neighborhood of $(0, 0)$. Under this homeomorphism, a point $(\xi, t) \in \kappa_g^{-1}(0)$ represents a metric with common cone angle $2\pi(\beta_0 + t)$ along $\mathcal{S}$.

Proof. At a zero of Ψ , set $\omega = \delta_h b_\beta$. The contracted Bianchi identity gives $\delta_h \delta_h^* \omega = 0$. Integration by parts yields $\delta_h^* \omega = 0$ and hence

$$\|\omega\|_{L^2(h)}^2 = \langle b_\beta, \delta_h^* \omega \rangle_{L^2(h)} = 0.$$

Thus $\Psi(\beta, k) = 0$ is equivalent to $\text{Ric}(h) = 3h$ together with $\delta_h b_\beta = 0$.

To set a nearby invariant metric h in this gauge, solve

$$\delta_h(f^* b_\beta) = 0$$

for an equivariant diffeomorphism f near the identity. At $(\beta, h, f) = (\beta_0, g, \text{Id})$, the derivative in f , after identifying vector fields and one-forms using g , is $2\delta_g \delta_g^*$. Its kernel consists of the invariant Killing fields, namely the infinitesimal torus translations. The equation is automatically orthogonal to the torus Killing fields for h . Restricting to their orthogonal complement and applying the implicit function theorem therefore gives f , locally unique modulo torus translations. The metric $(f^{-1})^* h$ satisfies the required gauge condition.

Let Π_g denote the L^2 -orthogonal projection onto $\mathcal{K}_g$. By Lemma 4.2,

$$\mathcal{P}_g : L_8^2 \cap \mathcal{K}_g^\perp \longrightarrow L_6^2 \cap \mathcal{K}_g^\perp$$

is an isomorphism. Writing $k = \xi + v$, with $\xi \in \mathcal{K}_g$ and $v \perp \mathcal{K}_g$, the implicit function theorem solves

$$(1 - \Pi_g) \Psi(\beta_0 + t, \xi + v) = 0$$

uniquely as $v = v(\xi, t)$. Set

$$\kappa_g(\xi, t) = \Pi_g \Psi(\beta_0 + t, \xi + v(\xi, t)).$$

The remaining equation is precisely $\kappa_g = 0$. As observed above, the stabilizer consists of torus translations, which act trivially on invariant tensors. Thus the local moduli space is the zero set of κ_g itself, with no further quotient. Elliptic regularity upgrades the solutions and their convergence to C^∞ . $\square$

Proposition 4.3 identifies neighborhoods in $\mathcal{M}$ with finite-dimensional real analytic zero sets. Such sets are locally connected, so every connected component of $\mathcal{M}$ is open. Extending Definition 1.6 to conical metrics, we call g *torically nondegenerate* if $\mathcal{K}_g = 0$. At such a point, the local model in Proposition 4.3 is an interval, and the moduli space is a one-dimensional manifold near $[(\beta, g)]$, with local coordinate β .

4.5. **Orientation of the moduli space.** Let $\mathcal{D}_g$ be the full linearization of (4.7) at $(\beta_0, 0)$:

$$\mathcal{D}_g : \mathbb{R} \oplus L_8^2 \longrightarrow L_6^2, \quad \mathcal{D}_g(t, k) = tB_g + \mathcal{P}_g k, \quad B_g = \partial_\beta \Psi(\beta_0, 0).$$

It is Fredholm of index one. Its cokernel is represented, using L^2 pairings, by

$$(4.8) \quad \{\tau \in \mathcal{K}_g : \langle B_g, \tau \rangle_{L^2} = 0\}.$$

In particular, every cokernel element is represented by a smooth invariant transverse–traceless tensor. We call a point *regular* if $\mathcal{D}_g$ is surjective. Every nondegenerate point is regular, but regularity alone does not assert that β is a local coordinate.

The determinant lines

$$\det \mathcal{D}_g = \Lambda^{\max} \ker \mathcal{D}_g \otimes (\Lambda^{\max} \operatorname{coker} \mathcal{D}_g)^*$$

form a line bundle over $\mathcal{M}$. To describe its local topology, choose a finite-dimensional subspace $E \subset L_6^2$ such that the stabilization

$$\mathcal{D}_{h,E}(v, e) = \mathcal{D}_h v + e$$

is surjective for h near g in a fixed chart. Its kernels form a vector bundle, and stabilization gives

$$\det \mathcal{D}_h \cong \Lambda^{\max} \ker \mathcal{D}_{h,E} \otimes (\Lambda^{\max} E)^*.$$

The standard transition isomorphisms make these descriptions independent of the stabilization and reference gauge; see [50]. They descend to the moduli space because torus translations act trivially on invariant tensors. At a regular point, the kernel of $\mathcal{D}_g$ is the tangent space of the local one-dimensional moduli space, and hence

$$\det \mathcal{D}_g \cong T_{[(\beta_0, g)]} \mathcal{M}.$$

Let L_g be the restriction of $\mathcal{P}_g$ to invariant transverse–traceless tensors. Thus

$$L_g = \frac{1}{2}(\nabla_g^* \nabla_g - 2\mathring{R}_g).$$

This operator is self-adjoint and bounded below, with compact resolvent. Let $\mu(g)$ be the number of its negative eigenvalues, counted with multiplicity.

Lemma 4.4. *The determinant line bundle $\det \mathcal{D} \rightarrow \mathcal{M}$ has a natural orientation, determined by the convention that increasing β orients the angle parameter. At a nondegenerate metric g , the induced orientation is $(-1)^{\mu(g)}$ times the orientation defined by increasing β .*

Proof. Suppose $\mathcal{P}_g k = \gamma k$. By (4.5),

$$\delta_g \delta_g^* (\delta_g k) = \gamma \delta_g k.$$

The form $\delta_g k$ is orthogonal to the $\mathbb{T}^2$ -invariant Killing fields. On this orthogonal complement, $\delta_g \delta_g^*$ has a positive spectral gap. Hence $\delta_g k = 0$ when $|\gamma|$ is sufficiently small. The trace equation (4.6) then gives

$$(\nabla_g^* \nabla_g - 3 + \gamma) \operatorname{tr}_g k = 0.$$

The spectrum of $\nabla_g^* \nabla_g$ on $\mathbb{T}^2$ -invariant functions is contained in $\{0\} \cup [4, \infty)$, so $\operatorname{tr}_g k = 0$ whenever $|\gamma| < 1$. Thus every eigentensor of $\mathcal{P}_g$ with eigenvalue sufficiently close to zero is transverse–traceless. These spectral gaps persist locally. Choose $\varepsilon > 0$ sufficiently small that every eigentensor of $\mathcal{P}_g$ with eigenvalue in $(-\varepsilon, \varepsilon)$ is transverse–traceless and neither endpoint lies in the spectrum. After shrinking the neighborhood of g , the same properties hold for nearby Einstein metrics h in the chosen chart. Let V_h be the sum of the eigenspaces in this interval. These spaces are the same for $\mathcal{P}_h$ and L_h . Their spectral projections vary continuously, and hence the spaces V_h form a finite-dimensional vector bundle of constant rank. Eigenvalues may cross zero, changing the kernel, but no eigenvalue crosses the endpoints of the chosen interval. The operators $\mathcal{P}_h$ and L_h have the same restriction A_h to V_h , and both are invertible on their respective complementary spectral subspaces. Thus their determinant lines both reduce to $\det A_h$. The exact sequence

$$0 \longrightarrow \ker A_h \longrightarrow V_h \xrightarrow{A_h} V_h \longrightarrow \operatorname{coker} A_h \longrightarrow 0$$

gives the natural identification

$$\det A_h \cong \Lambda^{\max} V_h \otimes (\Lambda^{\max} V_h)^*.$$

These identifications are continuous in h , including where the kernel dimension changes, and therefore identify the determinant line bundles

$$\det \mathcal{P} \cong \det L$$

over this neighborhood. The comparison is independent of the chosen spectral window, since changing the window adds the same invertible spectral blocks to both reductions.

Choose $\Lambda > 0$ outside the spectrum of L_g , and let E_g be the sum of its eigenspaces with eigenvalues less than Λ . This space is finite-dimensional, and L_g is positive and invertible on $E_g^\perp$. Finite-dimensional reduction gives

$$(4.9) \quad \Phi_{g,\Lambda} : \det L_g \cong \Lambda^{\max} E_g \otimes (\Lambda^{\max} E_g)^*.$$

In the following, $\det_{\text{num}}$ denotes the ordinary numerical determinant of a finite-dimensional endomorphism. The isomorphism $\Phi_{g,\Lambda}$ uses the restriction $A_g = L_g|_{E_g}$. When A_g is invertible, so is L_g , and its zero kernel and cokernel give $\det L_g = \Lambda^0\{0\} \otimes (\Lambda^0\{0\})^*$ with canonical generator $\mathbf{1}_g = 1 \otimes 1^*$. The isomorphism (4.9) sends this generator to

$$\Phi_{g,\Lambda}(\mathbf{1}_g) = u \otimes ((\Lambda^{\max} A_g)u)^* = (\det_{\text{num}} A_g)^{-1}(u \otimes u^*),$$

where u is any nonzero top wedge of E_g and w^* denotes the dual element taking value 1 on w . Orient the right-hand side of (4.9) by its canonical pairing. Increasing Λ adds only positive eigenspaces, so this orientation is independent of the cutoff. At a torically nondegenerate metric, its sign relative to the canonical generator of $\det L_g$ is therefore

$$\text{sgn } \det_{\text{num}}(L_g|_{E_g}) = (-1)^{\mu(g)}.$$

Finally, the homotopy

$$\mathcal{D}_{g,s}(t, k) = \mathcal{P}_g k + stB_g, \quad 0 \leq s \leq 1,$$

transports the product orientation of $\det \mathcal{P}_g \otimes \Lambda^1 \mathbb{R}$ to $\det \mathcal{D}_g$, where the last factor $\Lambda^1 \mathbb{R}$ is oriented by increasing β . Equivalently, one may use the determinant isomorphism for adding the angle parameter; see [50]. The spectral pairing is preserved by equivariant pullback, and changes of reference family give homotopic identifications. Thus these orientations descend and agree on overlapping charts. At a nondegenerate point, projection of $\ker \mathcal{D}_g$ onto the angle parameter is an isomorphism, giving the asserted sign. $\square$

4.6. The oriented boundary count. We now equip $\mathcal{M}$ with the determinant-line orientation of Lemma 4.4. Define the area function on $\mathcal{M}$

$$\mathcal{A}([\beta, g]) = \sum_{\alpha=1}^q \text{Area}_g(\Sigma_\alpha).$$

For $\eta > 0$, set

$$\mathcal{M}_\eta = \mathcal{A}^{-1}([\eta, \infty)), \quad \partial \mathcal{M}_\eta := \mathcal{A}^{-1}(\eta).$$

Proposition 4.5. *Let $\eta > 0$ be such that every point of $\partial \mathcal{M}_\eta$ is regular and $d\mathcal{A} \neq 0$ there. If $\mathcal{M}_\eta$ is compact, then $\partial \mathcal{M}_\eta$ is finite and*

$$\sum_{x \in \partial \mathcal{M}_\eta} \text{sign}_{\partial \mathcal{M}_\eta}(x) = 0.$$

The signs are the boundary signs induced by the determinant-line orientation, with outward direction $-\partial_a$ for the local coordinate $a = \mathcal{A}$. In particular, $\partial \mathcal{M}_\eta$ has even cardinality.

Proof. The set $\partial\mathcal{M}_\eta$ is closed in the compact space $\mathcal{M}_\eta$, hence compact. At each of its points, regularity makes the moduli space locally one-dimensional, and $d\mathcal{A} \neq 0$ makes $\mathcal{A}$ a local coordinate. Thus every point of $\partial\mathcal{M}_\eta$ is isolated, so this set is finite.

Choose neighborhoods of $\partial\mathcal{M}_\eta$ on which the linearization is surjective. The set of points in $\mathcal{M}_\eta$ where it is not surjective is compact and disjoint from these neighborhoods. By (4.8), its cokernels are represented by smooth invariant transverse–traceless tensors. Compactness and openness of surjectivity allow us to choose finitely many such tensors whose extensions stabilize the equation near this set.

We construct the extensions equivariantly. Let τ_ν be a smooth invariant tensor representing a selected element of $\text{coker } \mathcal{D}_{g_\nu}$ at (β_ν, g_ν) . Write $b_{\nu,\beta}$ and $J_{\nu,\beta}$ for the reference family and tensor identifications based at this pair, with $b_{\nu,\beta_\nu} = g_\nu$ and $J_{\nu,\beta_\nu} = \text{Id}$. Applying the earlier gauge-fixing argument with β as a parameter gives a local slice S_ν of pairs (β, h) near (β_ν, g_ν) , where

$$h = b_{\nu,\beta} + J_{\nu,\beta}k, \quad \delta_h b_{\nu,\beta} = 0.$$

The gauge-fixing diffeomorphisms preserve β , while the reference metric $b_{\nu,\beta}$ varies with it. For $(\beta, h) \in S_\nu$, set

$$\tilde{\sigma}_\nu(h) = J_{\nu,\beta}\tau_\nu.$$

After shrinking S_ν , compactness of isometries together with local gauge uniqueness implies that any allowed diffeomorphism relating two points of S_ν is a torus translation. Since these translations act trivially on invariant tensors, the rule

$$\tilde{\sigma}_\nu(F^*h) = F^*\tilde{\sigma}_\nu(h)$$

defines a well-defined extension to the saturation of S_ν . Choose a cutoff χ_ν on S_ν , equal to one near (β_ν, g_ν) and supported in a smaller slice neighborhood, and extend it invariantly by $\chi_\nu(F^*h) = \chi_\nu(h)$. Choose the corresponding supports in the quotient disjoint from the boundary neighborhoods.

The transported tensors $\tilde{\sigma}_\nu(h)$ need not be divergence-free with respect to h . To ensure this property, for an invariant symmetric tensor τ define

$$\mathbf{Q}_h\tau = \tau - \delta_h^*(\delta_h\delta_h^*)^{-1}_\perp \delta_h\tau.$$

Here the inverse is taken on invariant one-forms orthogonal to the torus Killing fields, with orthogonality defined using h . Since $\delta_h\tau$ is automatically orthogonal to these fields, the inverse is well defined. The operator $\mathbf{Q}_h$ is the $L^2(h)$ -orthogonal projection onto $\ker \delta_h$; in particular, it fixes every divergence-free tensor. Set

$$\sigma_\nu(h) = \chi_\nu(h)\mathbf{Q}_h\tilde{\sigma}_\nu(h),$$

and extend this section by zero outside its support. For each fixed h , the cutoff $\chi_\nu(h)$ is a scalar constant on X . Thus multiplication by the cutoff preserves divergence-freeness. Naturality of $\mathbf{Q}_h$ under pullback, together with the equivariance of $\tilde{\sigma}_\nu$ and invariance of χ_ν , also preserves equivariance. Consequently,

$$\delta_h\sigma_\nu(h) = 0, \quad \sigma_\nu(F^*h) = F^*\sigma_\nu(h).$$

These identities hold throughout the domain, including where the cutoff varies.

For the required differentiability, take the reference gauges and cutoffs on L_4^2 neighborhoods. On L_8^2 metrics the gauge-fixing diffeomorphisms depend smoothly with values in L_9^2 , and changes of representative are C^2 with values in L_6^2 . The cutoffs are consequently C^2 in every reference gauge. Pullback of the smooth selected tensors and the elliptic projection above give C^2 sections $\sigma_\nu : L_8^2 \rightarrow L_6^2$.

We use these sections to perturb the Einstein equation. To verify local surjectivity near (β_ν, g_ν) , first consider the perturbation directions associated with this reference pair. For $h = b_{\nu,\beta} + J_{\nu,\beta}k$,

the perturbed equation in the reference gauge is

$$J_{\nu,\beta}^{-1} \left\{ \mathcal{E}_3(h) - \delta_h^* \delta_h b_{\nu,\beta} - \sum_{j=1}^{d_\nu} u_j \sigma_{\nu,j}(h) \right\} = 0, \quad u \in \mathbb{R}^{d_\nu}.$$

Write $\mathcal{D}_h$ for the full linearization of the unperturbed gauged equation in this reference chart. At $u = 0$, the linearization of the displayed perturbed equation, with variations (t, ℓ) in (β, k) and v in u , is

$$(t, \ell, v) \mapsto \mathcal{D}_h(t, \ell) - \sum_{j=1}^{d_\nu} v_j J_{\nu,\beta}^{-1} \sigma_{\nu,j}(h).$$

At (β_ν, g_ν) , we have $J_{\nu,\beta_\nu} = \text{Id}$ and $\sigma_{\nu,j}(g_\nu) = \tau_{\nu,j}$, so this becomes

$$(t, \ell, v) \mapsto \mathcal{D}_{g_\nu}(t, \ell) - \sum_{j=1}^{d_\nu} v_j \tau_{\nu,j}.$$

This operator is surjective because the classes of the $\tau_{\nu,j}$ span $\text{coker } \mathcal{D}_{g_\nu}$. By openness of surjectivity, the augmented linearization remains surjective for nearby (β, h) . Equivalently,

$$L_6^2 = \text{im } \mathcal{D}_h + \text{span} \left\{ J_{\nu,\beta}^{-1} \sigma_{\nu,j}(h) : 1 \leq j \leq d_\nu \right\}.$$

Thus the classes of these tensors span $\text{coker } \mathcal{D}_h$ throughout a sufficiently small neighborhood of (β_ν, g_ν) .

Combining the perturbations from all reference neighborhoods gives the equivariant equation

$$\mathcal{E}_3(h) = \sum_{\nu} \sum_{j=1}^{d_\nu} u_{\nu,j} \sigma_{\nu,j}(h), \quad u \in \mathbb{R}^N, \quad N = \sum_{\nu} d_\nu,$$

with the common cone factor allowed to vary. In any reference gauge, writing b_β and J_β for its reference data, the gauged equation is

$$J_\beta^{-1} \left\{ \mathcal{E}_3(h) - \delta_h^* \delta_h b_\beta - \sum_{\nu} \sum_{j=1}^{d_\nu} u_{\nu,j} \sigma_{\nu,j}(h) \right\} = 0.$$

Since the perturbations are divergence-free, the Bianchi identity removes the gauge term exactly as in Proposition 4.3. Equivariance therefore ensures that solutions in different reference gauges represent the same geometric equation.

At every unperturbed solution $([(\beta, g)], 0)$ with $[(\beta, g)] \in \mathcal{M}_\eta$, the augmented linearization is surjective of index $N + 1$. By openness of surjectivity, it remains surjective in sufficiently small neighborhoods of these points. In each fixed reference gauge, the implicit function theorem therefore gives a local C^2 solution manifold of dimension $N + 1$.

We next check compatibility under changes of reference gauge. On an overlap, let $T_{\mu\nu}$ denote the change of representative from gauge ν to gauge μ , preserving the angle and perturbation parameters. By the previously established estimates, this map is C^2 with tensor coordinates in L_6^2 . On each local solution manifold, choose coordinates from linear functions of the angle and perturbation parameters and L^2 pairings with smooth invariant tensors, with differentials independent on the augmented kernel. These coordinate functions extend to continuous affine linear functionals on the L_6^2 coordinate space, so $T_{\mu\nu}$ induces C^2 coordinate transitions. Slice uniqueness modulo torus translations makes these transitions well defined and consistent on triple overlaps. Thus the local solution spaces glue to a C^2 manifold W near $\mathcal{M}_\eta \times \{0\}$, with the C^2 parameter projection

$$\pi : W \longrightarrow \mathbb{R}^N, \quad \pi([(\beta, h)], u) = u.$$

Near each point of $\partial \mathcal{M}_\eta$, all perturbations vanish, so the solution space is a product of the original regular solution interval and a parameter ball. On this interval, $a = \mathcal{A}$ is a local

coordinate. Restricting to $a \geq \eta$ and shrinking the neighborhood and parameter ball gives a C^2 manifold with boundary, still denoted by W , with the natural identifications

$$\pi^{-1}(0) = \mathcal{M}_\eta \times \{0\}, \quad \partial W = \partial \mathcal{M}_\eta \times B_\rho(0).$$

Under the second identification, $\pi|_{\partial W}$ is projection onto the parameter ball.

Along the zero fibre, stabilization gives

$$\Lambda^{N+1} T_{(h,0)} W \cong \det \mathcal{D}_h \otimes \Lambda^N \mathbb{R}^N.$$

The determinant-line orientation and the standard orientation of $\mathbb{R}^N$ orient W along the zero fibre. Since an orientation of a real vector bundle along a closed subset of a manifold extends to a neighborhood of that subset, this orients W after shrinking around the zero fibre and induces orientations on the regular fibres of π .

Choose a relatively open neighborhood V of the zero fibre with compact closure in W . The relative boundary $\bar{V} \setminus V$ is compact and disjoint from the zero fibre. By compactness, there is $0 < \varepsilon < \rho$ such that

$$|\pi| > \varepsilon \quad \text{on } \bar{V} \setminus V, \quad \partial \mathcal{M}_\eta \times B_\varepsilon(0) \subset V.$$

Choose a regular value $u \in B_\varepsilon(0)$ by Sard's theorem. Then

$$\pi^{-1}(u) \cap \bar{V}$$

is a compact oriented one-manifold whose boundary is naturally identified with $\partial \mathcal{M}_\eta$. The product structure near the cuts preserves their boundary signs. Since the signed boundary count of a compact oriented one-manifold is zero, the proposition follows. $\square$

Remark 4.6. *The local moduli and orientation arguments also apply when the cone angles vary independently. Fix an ordering of all invariant spheres $\Sigma_1, \dots, \Sigma_m$, preserve this order in the equivalence relation, and write $\beta = (\beta_1, \dots, \beta_m) \in (0, \infty)^m$. The fixed-angle analysis is unchanged, and the identifications J_β are constructed by making the angular replacement independently in each normal plane, including at corners. The local Kuranishi model is then*

$$\kappa_g : U_g \subset \mathcal{K}_g \times \mathbb{R}^m \longrightarrow \mathcal{K}_g.$$

The full linearization $\mathcal{D}_g$ has Fredholm index m , and the determinant-line construction gives

$$\det \mathcal{D}_g \cong \det \mathcal{P}_g \otimes \Lambda^m \mathbb{R}^m.$$

Orienting the parameter space by $d\beta_1 \wedge \dots \wedge d\beta_m$ therefore orients the determinant line bundle. At a nondegenerate metric, the cone factors are local coordinates on the moduli space, and the induced orientation is $(-1)^{\mu(g)}$ times their standard orientation.

5. END STRUCTURE OF THE MODULI SPACE

In this section the goal is to understand the end structure of the moduli space $\mathcal{M}$. We will consider a sequence $[(\beta_j, g_j)] \in \mathcal{M}$ satisfying the bound

$$1 \leq \beta_j \leq \beta_+ < \infty.$$

So passing to a subsequence, we assume $\beta_j \rightarrow \beta_\infty$.

In this section we study the limiting behavior of g_j under the above angle bound. To describe convergence near conical singularities, we use the angular developments of Definition 2.3. We say that g_j converges *smoothly in local developments* if the cone factors converge and the developed metrics converge in C^∞ on fixed smaller coordinate neighborhoods, including where the radial coordinates vanish.

The main compact Theorem 5.7 says that in our setting, there are only two alternatives passing to a subsequence

- g_j converges smoothly to a limit toric conical Einstein metric g_∞ on $X_{\mathfrak{A}}$;
- g_j converges smoothly, away from a point, to a smooth toric Einstein metric on $X_{\mathfrak{A}_0}$, and the only non-flat bubble is a AE toric conical gravitational instanton, whose rod structure is determined by W .

Our design of the particular rod structure is crucial in obtaining this.

For our purpose we also need a more refined convergence in the second alternative, to match the gluing construction in Section 5.4.

5.1. Preliminary results. Write $\mathcal{V} = \{p_1, \dots, p_m\}$ for the fixed points. Proposition 3.6, Theorem 3.13, and Proposition 3.11 give the following geometric bounds

$$(5.1) \quad \text{diam}(X_{\mathfrak{R}}, g_j) \leq D, \quad \text{Vol}_{g_j}(X_{\mathfrak{R}}) \geq v > 0, \quad |\text{Rm}_{g_j}|(x) \leq C \max\{1, d_{g_j}(x, \mathcal{V})^{-2}\}.$$

The Euler form computation in the proof of Proposition 3.15 also gives

$$(5.2) \quad \int_{X_{\mathfrak{R}}} |\text{Rm}_{g_j}|^2 \text{dvol}_{g_j} \leq C \sum_{\alpha=1}^m \beta_{\alpha,j} \beta_{\alpha+1,j} \leq C m \beta_+^2.$$

As in the proof of the angle bound in Section 3, pass to a fixed finite quotient with cone angles below 2π . Then we have volume comparison and the Sobolev inequality on the quotient. Lifting back, we get the uniform volume ratio bound

$$(5.3) \quad cr^4 \leq \text{Vol}_{g_j} B_{g_j}(x, r) \leq Cr^4 \quad (0 < r \leq 1)$$

and Sobolev inequality

$$(5.4) \quad \|f\|_{L^4}^2 \leq C(\|df\|_{L^2}^2 + \|f\|_{L^2}^2).$$

Proposition 5.1. *If*

$$\sup_j \max_{X_{\mathfrak{R}}} |\text{Rm}_{g_j}| < \infty,$$

then, after passing to a subsequence and pulling back by $\mathbb{T}^2$ -equivariant diffeomorphisms preserving each Σ_α and p_α , the metrics g_j converge smoothly in local developments to a toric conical Einstein metric g_∞ with the same rod structure $\mathfrak{R}$ and cone angle $2\pi\beta_\infty$ along $\mathcal{S}$.

Proof. Lemma 3.12 gives uniform neighborhoods along the rods away from turning points. The same Killing Jacobi field argument separates turning points from one another and from nonincident rods. Together with Lemma 3.9 and the angle bounds, this gives smooth developments of uniform size with uniform noncollapsing estimates.

Standard Einstein regularity and equivariant compactness now give smooth subsequential convergence in these developments. Retaining the fixed angular identifications preserves the rod vectors, while the separation above preserves all turning points and rods. Thus the limit has the same rod structure $\mathfrak{R}$. $\square$

The following is a generalization of the standard ϵ -regularity theorem to the toric conical setting.

Lemma 5.2. *There are $\varepsilon_0, C > 0$ such that, for $0 < r \leq \frac{1}{2}$,*

$$\int_{B_{g_j}(x, 2r)} |\text{Rm}_{g_j}|^2 \text{dvol}_{g_j} < \varepsilon_0 \implies \sup_{B_{g_j}(x, r)} |\text{Rm}_{g_j}| \leq Cr^{-2} \left(\int_{B_{g_j}(x, 2r)} |\text{Rm}_{g_j}|^2 \text{dvol}_{g_j} \right)^{1/2}.$$

Proof. The Einstein curvature equation gives away from the conical singularities the differential inequality

$$\Delta_{g_j} |\text{Rm}_{g_j}| \geq -C |\text{Rm}_{g_j}|^2.$$

We claim that this inequality holds weakly globally on X . This follows from the fact that $|\text{Rm}_{g_j}|$ is bounded on X for each fixed j and the conical locus $\mathcal{S}$ has zero $W^{1,2}$ -capacity.

Then the uniform bound on $|\text{Rm}_{g_j}|$ follows from the Sobolev inequality and the usual Moser iteration. $\square$

5.2. Bubble classification. Now suppose that

$$Q_j := \max_X |\mathrm{Rm}_{g_j}| \longrightarrow \infty, \quad \ell_j := Q_j^{-1/2}.$$

Choose a maximum point x_j . By (5.1), after passing to a subsequence there is a fixed point p_i , independent of j , such that

$$d_{g_j}(x_j, p_i) \leq C\ell_j.$$

Set $\hat{g}_j = \ell_j^{-2}g_j$ and use p_i as base point. Under this rescaling, the reduced data $\underline{g}_j$, G_j , and ρ_j are multiplied by ℓ_j^{-2} , while Φ_j is unchanged.

By (5.3) and scaling,

$$cR^4 \leq \mathrm{Vol}_{\hat{g}_j} B_{\hat{g}_j}(x, R) \leq CR^4, \quad 0 < R \leq \ell_j^{-1}.$$

Thus the rescaled sequence is uniformly noncollapsed and satisfies $|\mathrm{Rm}_{\hat{g}_j}| \leq 1$. After passing to a subsequence, $(X_{\mathfrak{X}}, \hat{g}_j, p_i)$ converges in the pointed Gromov–Hausdorff sense to (Y, d_Y, p_∞) . The local equivariant compactness argument of Proposition 5.1 gives smooth convergence in local developments on bounded subsets, retaining the fixed torus action and its primitive isotropy weights. Since the rescaled Einstein constants are $3\ell_j^2 \rightarrow 0$, the distance d_Y is induced by a complete toric conical Ricci-flat metric h .

The points x_j remain at bounded rescaled distance from p_i , so their subsequential limit satisfies $|\mathrm{Rm}_h| = 1$. Thus $\max_Y |\mathrm{Rm}_h| = 1$, and h is nonflat. The uniform curvature energy and volume bounds pass to the limit, giving finite curvature energy and Euclidean volume growth.

Moreover, Y has one end. Otherwise, since the connected torus preserves each end, the fixed finite quotient of Y would also have more than one end. The RCD(0, 4) splitting theorem gives a line factor [38, Theorem 2.16]. On the regular set, this is a local Riemannian product with a three-dimensional Ricci-flat factor, which is flat. Thus $\mathrm{Rm}_h = 0$ on the regular set and, by continuity, in every local development, contradicting $\max_Y |\mathrm{Rm}_h| = 1$.

Write $(N, \underline{h}) = Y/\mathbb{T}^2$. Equivariant convergence induces convergence of the rescaled quotient spaces to N , smoothly on their interiors. The proof of Lemma 3.9 shows that every interior geodesic segment in $(B_j, \underline{g}_j)$ is minimizing, and this property passes to N° . Minimizing geodesics between interior points remain in the interior. Moreover, $K_{\underline{h}} \leq 0$ by Proposition 3.2, so there are no conjugate points. Hence the exponential map from any interior point is a diffeomorphism from its star-shaped domain of interior geodesics onto N° . Consequently, N is contractible.

Since N is simply connected and ρ is harmonic, choose a global harmonic conjugate z satisfying $dz = *_h d\rho$.

Lemma 5.3. *The limit (Y, h) is proper in the sense of Definition A.2.*

Proof. Finite curvature energy and Lemma 5.2 give quadratic curvature decay. The limiting form of (3.19) gives

$$\mathrm{Hess}_{\underline{h}} \sqrt{\rho} \leq 0.$$

A critical point of ρ would therefore be a local maximum, contradicting the strong maximum principle. Thus $d\rho \neq 0$ on N° , and $F = (\rho, z)$ is a local conformal diffeomorphism.

Set $\nu = -\log |d\rho|_{\underline{h}}$, so that locally

$$\underline{h} = e^{2\nu}(d\rho^2 + dz^2).$$

The inequality $\mathrm{Hess}_{\underline{h}} \sqrt{\rho} \leq 0$ becomes

$$\begin{pmatrix} \nu_\rho + \frac{1}{2\rho} & \nu_z \\ \nu_z & -\nu_\rho \end{pmatrix} \geq 0.$$

Consequently,

$$-\frac{1}{2\rho} \leq \nu_\rho \leq 0, \quad \nu_\rho^2 + \nu_z^2 \leq \frac{1}{4\rho^2}.$$

Along a lift of any finite length path contained in a compact subset of $\mathbb{H}^\circ$, these estimates bound ν and hence the length of the lift. Completeness and the positive lower bound for ρ ensure that the lift extends to its endpoint in N° . Thus $F : N^\circ \rightarrow \mathbb{H}^\circ$ is a covering map and hence a diffeomorphism.

The rod asymptotics as in Section 2 show that F extends as a local homeomorphism at every boundary point. For any $z_0 \in \mathbb{R}$, the bound on ν_ρ gives

$$e^{\nu(\rho, z_0)} \leq C(z_0)\rho^{-1/2}, \quad 0 < \rho \leq 1.$$

Therefore $\rho \mapsto F^{-1}(\rho, z_0)$ has finite length as $\rho \rightarrow 0$ and converges to a boundary point mapping to $(0, z_0)$. Boundary injectivity follows from interior injectivity and the local homeomorphism property. Hence $F : N \rightarrow \mathbb{H}$ is a homeomorphism. $\square$

Proposition 5.4. *The limit (Y, h) is a proper toric conical gravitational instanton with distinct infinite rod directions. Its end is AE.*

Proof. By Lemma 5.3, (Y, h) is a proper toric conical gravitational instanton. The Euclidean volume growth case of Appendix A gives distinct infinite rod directions $\mathbf{v}_-, \mathbf{v}_+$ and ALE asymptotic. In our case, their primitive vectors form a basis of Λ , so the end is actually AE. $\square$

Proposition 5.5. *The limit (Y, h) is AE and $\beta_\infty > 1$. Its finite rods are exactly the inserted rods $\mathfrak{I}_1, \dots, \mathfrak{I}_q$ appeared in W .*

Proof. Let s, t be the original indices of the two infinite rods, and write $\beta_{\alpha, \infty} = \lim_j \beta_{\alpha, j}$. Then $\beta_{\alpha, \infty}$ is either 1 or β_∞ .

By Proposition 5.4, the volume density at infinity of Y is $\beta_{s, \infty}\beta_{t, \infty}$. Bishop–Gromov comparison (as in [34]) therefore gives, at every fixed point p_α in Y ,

$$(5.5) \quad \beta_{\alpha-1, \infty}\beta_{\alpha, \infty} > \beta_{s, \infty}\beta_{t, \infty}.$$

The inequality is strict since Y is non-flat. In particular, $\beta_\infty > 1$.

If both infinite rods had cone angle $2\pi\beta_\infty$, the right-hand side of (5.5) would be β_∞^2 , but the left-hand side is at most β_∞^2 . Contradiction. If exactly one had angle $2\pi\beta_\infty$, the fixed point incident to the smooth infinite rod would have left-hand side at most β_∞ , again contradicting (5.5). Thus both infinite rods are smooth. Moreover, (5.5) also excludes the possibility of two adjacent smooth rods.

The rods of Y occur consecutively in $\mathfrak{R}$. The finite rod adjacent to either infinite rod must therefore be inserted. Since the inserted rods form a single consecutive block, the chain inherited on Y must be of the form

$$\mathfrak{I}_m, \mathfrak{I}_1, \dots, \mathfrak{I}_q, \mathfrak{I}_{q+1}.$$

$\square$

5.3. The compact limit. We return to the original sequence g_j . The additional rescalings considered below either send the conical locus to infinity or collapse it to a point. We will show that the resulting limits are complete toric Ricci-flat spaces with smooth rod interiors and at most isolated orbifold singularities. Their finite curvature energy and Euclidean volume growth give ALE asymptotics. The restriction of the rod directions to $\{A, B, C\}$ makes the group at infinity trivial, since any two distinct directions among A, B, C are represented by primitive vectors forming a basis of Λ . Thus these limits are AE. The following simple rigidity statement then excludes any additional non-flat limit.

Lemma 5.6. *A complete toric Ricci-flat ALE four-orbifold with isolated singularities and rod directions in $\{A, B, C\}$ is Euclidean.*

Proof. If $\mathbf{v}_-, \mathbf{v}_+$ are the primitive infinite rod vectors, the group at infinity is

$$\Lambda/(\mathbb{Z}\mathbf{v}_- + \mathbb{Z}\mathbf{v}_+).$$

Its finiteness implies that the two directions are distinct. The constraint on the rod vectors therefore makes this group trivial, so the end is AE. Bishop–Gromov comparison then implies that the orbifold is Euclidean. $\square$

Recall that $\mathcal{S} = \bigcup_{\alpha=1}^q \Sigma_\alpha \subset X_{\mathfrak{R}}$ is the union of the inserted totally geodesic two-spheres, and that $p \in M$ is the original torus fixed point replaced by this chain.

Theorem 5.7. *After passing to a subsequence, either g_j converges smoothly in local developments on $X_{\mathfrak{R}}$, or $\mathcal{S}$ shrinks to p and the compact Gromov–Hausdorff limit is a smooth toric Einstein metric on M , with smooth convergence away from p . In the latter case, the limit (Y, h) in Proposition 5.5 is the only non-flat bubble.*

Proof. The bounded curvature case follows from Proposition 5.1. Otherwise, Proposition 5.5 gives the AE limit (Y, h) at scale ℓ_j , containing all the inserted spheres. Write $q \in \mathcal{S} \subset X_{\mathfrak{R}}$ for the torus fixed point used as the base point for this rescaled convergence. In particular,

$$\text{diam}_{g_j} \mathcal{S} = O(\ell_j).$$

We first claim that

$$(5.6) \quad d_{g_j}(x, \mathcal{S})^2 |\text{Rm}_{g_j}|(x) \leq C.$$

If this failed, point selection and rescaling by curvature would give a complete smooth nonflat Ricci-flat limit: the distance to $\mathcal{S}$ tends to infinity in this rescaling. Proposition 3.11 supplies a torus fixed point at bounded rescaled distance. Local equivariant compactness therefore retains the original primitive rod directions. The limit has finite curvature energy and Euclidean volume growth, so it is ALE by [6]. Its rod vectors belong to $\{A, B, C\}$, contradicting Lemma 5.6.

The geometric bounds give a compact pointed Gromov–Hausdorff limit

$$(X_{\mathfrak{R}}, g_j, q) \longrightarrow (X_\infty, p).$$

Here we use p for the limiting point, which will be identified with the original fixed point in M . Since $\text{diam}_{g_j} \mathcal{S} \rightarrow 0$, the whole chain $\mathcal{S}$ converges to p . Estimate (5.6) gives smooth Einstein convergence to a metric g_∞ away from p . Finite curvature energy and Lemma 5.2 imply

$$r^2 \sup_{B(p, 2r) \setminus B(p, r)} |\text{Rm}_{g_\infty}| \longrightarrow 0 \quad \text{as } r \rightarrow 0.$$

Volume comparison on the fixed finite quotient, followed by lifting, gives flat quotient tangent cones. Punctured neighborhoods are connected by the local Poincaré inequality and the zero capacity of a point. Comparing overlapping annuli identifies the quotient group across scales. The argument of [24, proof of Theorem 5.7] implies that g_∞ extends across p as an Einstein orbifold metric.

We next show that no fixed point outside $\mathcal{S}$ converges to p . Otherwise, after passing to a subsequence, some fixed point $p_\alpha \notin \mathcal{S}$ satisfies

$$s_j = d_{g_j}(q, p_\alpha) \longrightarrow 0.$$

Proposition 5.5 implies $s_j/\ell_j \rightarrow \infty$. Rescaling by s_j^{-2} , (5.6) gives smooth convergence away from the point to which $\mathcal{S}$ collapses. The same removal argument and [6] give a complete Ricci-flat ALE orbifold. The curvature bound near p_α retains its faithful torus representation and the original rod directions. Lemma 5.6 makes this limit Euclidean. However, its torus action has two distinct fixed points, given by the limits of q and p_α , which are at distance one. This is impossible for an effective $\mathbb{T}^2$ -action on Euclidean four-space.

Thus every remaining rod and turning point persists in the limit. The two rods incident to p have primitive vectors which form a basis of Λ , so the local orbifold group at p is trivial. The compact limit is therefore smooth, and its rod structure is $\mathfrak{R}_0$. The construction in Section 2 identifies (X_∞, p) with (M, p) . $\square$

For fixed sufficiently large R and sufficiently small $\delta > 0$, choose two invariant three-spheres in $X_{\mathfrak{R}}$ using the convergence embeddings. The inner sphere encloses $\mathcal{S}$ and, in the rescaled metrics $\ell_j^{-2} g_j$, converges smoothly together with a collar to an invariant radius- R coordinate sphere in the AE end of Y . The outer sphere, in the original metrics g_j , converges smoothly together with a collar to $\partial B_{g_\infty}(p, \delta) \subset M$. For sufficiently large j , let $A_j(R, \delta) \subset X_{\mathfrak{R}}$ be the open region between these two spheres. This region has no torus fixed points, and its two boundary rods

have directions B, A . The rod construction identifies $A_j(R, \delta)$ equivariantly with $S^3 \times (0, 1)$, and its closure with $S^3 \times [0, 1]$.

Notice that, in the above proof Chern–Gauss–Bonnet theorem on $A_j(R, \delta)$ gives

$$(5.7) \quad \lim_{\delta \rightarrow 0} \lim_{R \rightarrow \infty} \limsup_{j \rightarrow \infty} \int_{A_j(R, \delta)} |\text{Rm}_{g_j}|^2 \text{dvol}_{g_j} = 0.$$

Indeed, the annulus has Euler characteristic zero, and smooth convergence of the boundary collars makes their boundary terms approach the canceling Euclidean terms. Hence no additional nonflat bubble occurs. The same energy estimate and ε -regularity give flat Euclidean limits on intermediate annuli. Their overlapping charts provide the initial annular coordinates used later.

In particular, if $\beta_\infty = 1$, then by Proposition 5.5 we are in the first alternative and we obtain a smooth toric Einstein metric on X .

5.4. A gluing construction. Recall that g_0 is a smooth compact toric Einstein metric on M satisfying $\text{Ric}_{g_0} = 3g_0$.

Let (Y, h) be the AE toric conical Ricci-flat gravitational instanton associated with W , as constructed in Appendix B. Its two infinite rods are smooth, and its finite rods have a common cone angle $2\pi\beta_*$, where $\beta_* > 1$. In equivariant coordinates at infinity,

$$(5.8) \quad h = g_{\text{euc}} + O_\infty(r^{-2}).$$

We also denote by Σ_α the corresponding invariant spheres in Y . Fix a scale for h and set $A_* = A(h)$.

Theorem 5.8. *Assume that g_0 is torically nondegenerate. Then, for sufficiently small $a > 0$, there is a smooth family of toric conical Einstein metrics g_a on X , with common cone angle $2\pi\beta(a)$ along the inserted spheres and cone angle 2π along the remaining rods, such that*

$$\text{Ric}(g_a) = 3g_a, \quad A(g_a) = a, \quad \beta(a) \rightarrow \beta_* \quad \text{as } a \rightarrow 0.$$

*Under the gluing identifications, g_a converges smoothly to g_0 on compact subsets of $M \setminus \{p\}$, and $a^{-1}A_*g_a$ converges to h on compact subsets of Y , smoothly in local developments.*

Choose equivariant normal coordinates at p , compatible with the identification of the AE end. Then

$$(5.9) \quad g_0 = g_{\text{euc}} + O_\infty(r^2).$$

By assumption and Lemma 4.2, the $\mathbb{T}^2$ -invariant kernel of $\mathcal{P}_{g_0}$ is zero. Hence the compact linear problem is nondegenerate.

For the AE model, we use Propositions B.5 and B.7. With the Euclidean end and torus lattice fixed, every decaying invariant infinitesimal Ricci-flat deformation with variable finite cone angles is, modulo equivariant gauge transformations, obtained by varying the finite Weyl rod lengths. When the cone angles are fixed, the kernel is generated by the gauged scale deformation. Allowing the common cone angle to vary and adding the total-area equation gives an invertible augmented operator in the weighted spaces of Proposition B.7.

Set $a = \varepsilon^2 A_*$ and $r_* = \varepsilon^{2/3}$. Identify the normal coordinates x near p with the AE coordinates y by $x = \varepsilon y$. Using a smooth radial cutoff on $r_* \leq |x| \leq 2r_*$, interpolate between $\varepsilon^2 h$ and g_0 , and denote the resulting metric by b_ε . It equals the scaled model on the inner region and g_0 on the outer region. In particular, all the components of $\mathcal{S}$ lie in the region where $b_\varepsilon = \varepsilon^2 h$, so

$$A(b_\varepsilon) = \varepsilon^2 A_*.$$

Apply the tensor identifications of Section 4 with reference angle $2\pi\beta_*$. Choose the charts and partition of unity on a fixed compact neighborhood U of $\mathcal{S} \subset Y$, and transfer them to its rescaled image $U_\varepsilon \subset X$. Write J_β for the resulting identifications, suppressing their dependence on ε , and set

$$b_{\varepsilon, \beta} = J_\beta b_\varepsilon, \quad J_{\beta_*} = \text{Id}.$$

The family $b_{\varepsilon, \beta}$ depends real-analytically on β and has cone angle $2\pi\beta$ along $\mathcal{S}$. Both J_β and $b_{\varepsilon, \beta}$ are independent of β outside U_ε .

Set

$$\sigma = \varepsilon(\beta - \beta_*), \quad g = b_{\varepsilon, \beta} + J_\beta k,$$

where k is a $\mathbb{T}^2$ -invariant symmetric two-tensor, expressed in the local developments of b_ε . We solve

$$(5.10) \quad \mathcal{F}_\varepsilon(k, \sigma) = \left(J_\beta^{-1} \{ \mathcal{E}_3(g) - \delta_g^* \delta_g b_{\varepsilon, \beta} \}, \varepsilon^{-1} (A(g) - \varepsilon^2 A_*) \right) = 0.$$

The reference metric $b_{\varepsilon, \beta}$ varies with β . The map J_β^{-1} expresses the metric equation in the local developments of b_ε .

Choose a smooth positive weight w_ε satisfying

$$C^{-1}(\varepsilon + r) \leq w_\varepsilon \leq C(\varepsilon + r) \quad \text{near } p, \quad C^{-1} \leq w_\varepsilon \leq C \quad \text{away from } p,$$

where C is independent of ε . Here r is the common radial coordinate on the overlap, extended smoothly with $0 \leq r \leq C\varepsilon$ on the rescaled compact part of Y .

For a fixed $0 < \alpha < 1$, let

$$X_\varepsilon = C_{-1}^{2, \alpha}, \quad Y_\varepsilon = C_{-3}^{0, \alpha}$$

be the spaces of $\mathbb{T}^2$ -invariant symmetric two-tensors whose norms consist, respectively, of

$$\sum_{j=0}^2 \sup_X w_\varepsilon^{j+1} |\nabla^j k|, \quad \sup_X w_\varepsilon^3 |f|,$$

together with the corresponding scaled Hölder seminorms. All norms and covariant derivatives are defined using b_ε , and regularity is measured in its local developments, including over rods and turning points. Then $\mathcal{F}_\varepsilon$ maps a neighborhood of $(0, 0)$ in $X_\varepsilon \times \mathbb{R}$ to $Y_\varepsilon \times \mathbb{R}$.

Its derivative at $(0, 0)$ is

$$(5.11) \quad \mathcal{L}_\varepsilon(k, \sigma) := D\mathcal{F}_\varepsilon(0, 0)(k, \sigma) = (\mathcal{P}_{b_\varepsilon} k + \sigma B_\varepsilon, a_\varepsilon(k) + c_\varepsilon \sigma),$$

where

$$B_\varepsilon = \varepsilon^{-1} \frac{\partial}{\partial \beta} (J_\beta^{-1} \mathcal{E}_3(b_{\varepsilon, \beta})) \Big|_{\beta=\beta_*},$$

and

$$a_\varepsilon(k) = \varepsilon^{-1} dA_{b_\varepsilon}(k), \quad c_\varepsilon = \varepsilon^{-2} \partial_\beta A(b_{\varepsilon, \beta})|_{\beta=\beta_*}.$$

Indeed, since $\delta_g g = 0$, we have

$$-\delta_g^* \delta_g b_{\varepsilon, \beta} = \delta_g^* \delta_g (J_\beta k).$$

Thus the gauge term vanishes identically when $k = 0$, and its derivative in k at $(0, 0)$ is $\delta_{b_\varepsilon}^* \delta_{b_\varepsilon} k$.

Proposition 5.9. *For sufficiently small ε , $\mathcal{L}_\varepsilon$ is invertible, with inverse bounded independently of ε .*

Proof. Local Schauder estimates, applied after rescaling, give

$$(5.12) \quad \|k\|_{X_\varepsilon} \leq C \left(\|\mathcal{P}_{b_\varepsilon} k\|_{Y_\varepsilon} + \sup_X w_\varepsilon |k|_{b_\varepsilon} \right),$$

with C independent of ε . Indeed, the rescaled metrics have uniformly controlled coefficients on a fixed compact part of Y , on annuli in the overlap, and on the remaining compact part of M . Over rods and turning points we apply the estimates in local developments. This is the standard rescaling argument (see, for example, [8, p. 26, equation (54)]). The same rescaling on the support of the angle variation gives $\|B_\varepsilon\|_{Y_\varepsilon} \leq C$.

We claim that

$$\|k\|_{X_\varepsilon} + |\sigma| \leq C \|\mathcal{L}_\varepsilon(k, \sigma)\|_{Y_\varepsilon \times \mathbb{R}}.$$

If this estimate fails, there are $\varepsilon_i \rightarrow 0$ and pairs (k_i, σ_i) such that

$$\|k_i\|_{X_{\varepsilon_i}} + |\sigma_i| = 1, \quad \mathcal{L}_{\varepsilon_i}(k_i, \sigma_i) = (f_i, t_i), \quad \|f_i\|_{Y_{\varepsilon_i}} + |t_i| \rightarrow 0.$$

Pass to a subsequence with $\sigma_i \rightarrow \sigma_\infty$.

We first consider the limit on Y . Let J_β^Y be the tensor identifications on Y used to construct J_β after rescaling, and set

$$b_\beta^Y = J_\beta^Y h, \quad \dot{b} = \partial_\beta b_\beta^Y \Big|_{\beta=\beta_*}, \quad c_0 = \partial_\beta A(b_\beta^Y) \Big|_{\beta=\beta_*}.$$

Thus $\dot{b}$ is the derivative of the reference family on Y , supported in the fixed neighborhood of $\mathcal{S}$. It varies the cone angle; it is not required to be smooth in the developments for the fixed angle $2\pi\beta_*$. However,

$$D\mathcal{E}_{0,h}(\dot{b}) = \partial_\beta \left((J_\beta^Y)^{-1} \mathcal{E}_0(b_\beta^Y) \right) \Big|_{\beta=\beta_*}$$

is a smooth tensor in those developments, since $\mathcal{E}_0(h) = 0$.

Let ι_i denote the gluing embeddings from expanding subsets of Y into X , and set

$$\ell_i = \varepsilon_i^{-1} \iota_i^* k_i.$$

On each fixed compact subset of Y , for all large i ,

$$\iota_i^* b_{\varepsilon_i, \beta} = \varepsilon_i^2 b_\beta^Y.$$

The weighted bounds and local compactness give, after passing to a subsequence, convergence of ℓ_i in C^2 on compact subsets to a tensor $\ell \in C_{-1}^{2,\alpha}(Y)$. Rescaling the metric equation and the area equation gives

$$\mathcal{P}_h \ell + \sigma_\infty D\mathcal{E}_{0,h}(\dot{b}) = 0, \quad dA_h(\ell) + c_0 \sigma_\infty = 0,$$

where $\mathcal{P}_h = D\mathcal{E}_{0,h} + \delta_h^* \delta_h$. The powers of ε_i follow from

$$\varepsilon_i \iota_i^* B_{\varepsilon_i} = D\mathcal{E}_{0,h}(\dot{b}), \quad a_{\varepsilon_i}(k_i) = dA_h(\ell_i), \quad c_{\varepsilon_i} = c_0.$$

The first identity uses $b_\beta^Y = J_\beta^Y h$, so the angle derivative of $(J_\beta^Y)^{-1} (3\varepsilon_i^2 b_\beta^Y)$ vanishes. Proposition B.7 gives $\ell = 0$ and $\sigma_\infty = 0$: the angle-scale pairing forces $\sigma_\infty = 0$, and the area equation removes the remaining scale deformation.

It follows that $\sigma_i \rightarrow 0$ and $\|\mathcal{P}_{b_{\varepsilon_i}} k_i\|_{Y_{\varepsilon_i}} \rightarrow 0$. By (5.12), there are points $x_i \in X$ and a constant $c > 0$ such that

$$w_{\varepsilon_i}(x_i) |k_i(x_i)|_{b_{\varepsilon_i}} \geq c.$$

We analyze these points by a rescaling argument. The convergence on Y excludes points remaining in a rescaled fixed compact subset of Y .

If x_i stays away from p , a subsequence of k_i converges on compact subsets of $M \setminus \{p\}$ to a nonzero invariant tensor k_0 satisfying

$$\mathcal{P}_{g_0} k_0 = 0, \quad |k_0| \leq Cr^{-1}.$$

Local elliptic estimates give $|\nabla k_0| \leq Cr^{-2}$. The boundary terms in the weak equation on a sphere of radius r are therefore $O(r)$ and tend to zero. Thus the equation extends distributionally across p , and elliptic regularity makes k_0 smooth there. Toric nondegeneracy of g_0 , together with Lemma 4.2, gives $k_0 = 0$, a contradiction.

The remaining case has $\rho_i = r(x_i) \rightarrow 0$ and $\varepsilon_i/\rho_i \rightarrow 0$. In the common coordinates, let $d_i(z) = \rho_i z$. On compact subsets of $\mathbb{R}^4 \setminus \{0\}$,

$$\rho_i^{-2} d_i^* b_{\varepsilon_i} \longrightarrow g_{\text{euc}}, \quad \rho_i^{-1} d_i^* k_i \longrightarrow u,$$

where u is nonzero at some point of the unit sphere and

$$\mathcal{P}_0 u = 0, \quad |u(z)| \leq C|z|^{-1}.$$

Here $\mathcal{P}_0$ is the operator at g_{euc} with $\lambda = 0$. The same argument removes the singularity at the origin. Since $\mathcal{P}_0$ has constant coefficients and invertible symbol away from zero, the Fourier transform of u is supported at the origin. Thus u is a polynomial, and its decay at infinity forces $u = 0$, again a contradiction. These cases prove the uniform estimate.

For each fixed $\varepsilon > 0$, the weights give norms equivalent to the usual Hölder norms. The compact elliptic theory of Section 4, together with elliptic regularity, therefore shows that $\mathcal{P}_{b_\varepsilon} : X_\varepsilon \rightarrow Y_\varepsilon$ is Fredholm of index zero. Adding one scalar unknown and one scalar equation

preserves the index. Hence $\mathcal{L}_\varepsilon$ has index zero, and the uniform estimate proves both invertibility and the asserted bound. $\square$

Proof of Theorem 5.8. Since $A(b_\varepsilon) = \varepsilon^2 A_*$, the area component of $\mathcal{F}_\varepsilon(0,0)$ vanishes. The expansions (5.9) and (5.8), together with the derivative bounds for the cutoff function, give

$$\|\mathcal{F}_\varepsilon(0,0)\|_{Y_\varepsilon \times \mathbb{R}} \leq C(\varepsilon^3 + \varepsilon^2 r_*^{-1} + r_*^3) \leq C\varepsilon^{4/3},$$

where $r_* = \varepsilon^{2/3}$.

For $z = (k, \sigma)$, write

$$N_\varepsilon(z) = \|k\|_{X_\varepsilon} + |\sigma|, \quad Q_\varepsilon(z) = \mathcal{F}_\varepsilon(z) - \mathcal{F}_\varepsilon(0) - \mathcal{L}_\varepsilon z.$$

There are constants $c, C > 0$, independent of ε , such that, whenever $N_\varepsilon(z), N_\varepsilon(z') \leq c\varepsilon$,

$$\|Q_\varepsilon(z) - Q_\varepsilon(z')\|_{Y_\varepsilon \times \mathbb{R}} \leq C\varepsilon^{-1}(N_\varepsilon(z) + N_\varepsilon(z'))N_\varepsilon(z - z').$$

For the metric equation, this follows from the local $C^{2,\alpha}$ estimates after rescaling the developments to unit size, using $w_\varepsilon \geq C^{-1}\varepsilon$ and $\beta - \beta_* = \varepsilon^{-1}\sigma$. For the normalized area equation, the same estimate follows by rescaling to Y , where area scales by ε^2 .

By Proposition 5.9, the map

$$z \mapsto -\mathcal{L}_\varepsilon^{-1}(\mathcal{F}_\varepsilon(0) + Q_\varepsilon(z))$$

preserves the ball $N_\varepsilon(z) \leq R\varepsilon^{4/3}$ for a fixed sufficiently large R and all sufficiently small ε . Its Lipschitz constant is bounded by $CR\varepsilon^{1/3}$, so it is a contraction. Consequently, there is a unique solution $z_\varepsilon = (k_\varepsilon, \sigma_\varepsilon)$ in this ball, satisfying

$$\|k_\varepsilon\|_{X_\varepsilon} + |\sigma_\varepsilon| \leq C\varepsilon^{4/3}, \quad |\beta_\varepsilon - \beta_*| \leq C\varepsilon^{1/3}, \quad \beta_\varepsilon = \beta_* + \varepsilon^{-1}\sigma_\varepsilon.$$

Set

$$b = b_{\varepsilon, \beta_\varepsilon}, \quad g_\varepsilon = b + J_{\beta_\varepsilon} k_\varepsilon.$$

Elliptic regularity for the gauged equation gives smoothness in local developments. To remove the gauge term, set $\omega = \delta_{g_\varepsilon} b$. The metric equation reads

$$\mathcal{E}_3(g_\varepsilon) - \delta_{g_\varepsilon}^* \omega = 0.$$

The contracted Bianchi identity therefore gives $\delta_{g_\varepsilon} \delta_{g_\varepsilon}^* \omega = 0$. Integration by parts yields

$$0 = \langle \delta_{g_\varepsilon} \delta_{g_\varepsilon}^* \omega, \omega \rangle_{L^2(g_\varepsilon)} = \|\delta_{g_\varepsilon}^* \omega\|_{L^2(g_\varepsilon)}^2.$$

Hence $\mathcal{E}_3(g_\varepsilon) = 0$, which in dimension four is equivalent to $\text{Ric}(g_\varepsilon) = 3g_\varepsilon$. Moreover,

$$\|\omega\|_{L^2(g_\varepsilon)}^2 = \langle \delta_{g_\varepsilon} b, \omega \rangle_{L^2(g_\varepsilon)} = \langle b, \delta_{g_\varepsilon}^* \omega \rangle_{L^2(g_\varepsilon)} = 0.$$

Thus the gauge condition also holds. These integrations by parts are justified by excising shrinking neighborhoods of the fixed surfaces and torus fixed points; smoothness in local developments makes all boundary terms tend to zero.

On compact subsets of $M \setminus \{p\}$, the reference metric equals g_0 for sufficiently small ε , and the weighted estimate gives $g_\varepsilon \rightarrow g_0$ in $C^{2,\alpha}$. Let ι_ε denote the gluing embeddings from expanding subsets of Y . For every compact $K \subset Y$, the same estimate gives

$$\|\varepsilon^{-2} \iota_\varepsilon^* k_\varepsilon\|_{C^{2,\alpha}(K,h)} \leq C_K \varepsilon^{-1} \|k_\varepsilon\|_{X_\varepsilon} \leq C_K \varepsilon^{1/3}.$$

Together with $\beta_\varepsilon \rightarrow \beta_*$ and the construction of $b_{\varepsilon, \beta}$, this proves that $\varepsilon^{-2} \iota_\varepsilon^* g_\varepsilon \rightarrow h$ in $C^{2,\alpha}$ in local developments. Elliptic estimates upgrade both convergences to smooth convergence.

Finally, the nonlinear estimate implies

$$\|\mathcal{L}_\varepsilon^{-1}(D\mathcal{F}_\varepsilon(z_\varepsilon) - \mathcal{L}_\varepsilon)\|_{\text{op}} \leq C\varepsilon^{1/3}.$$

Thus the augmented derivative remains invertible. The implicit-function theorem gives smooth dependence on $\varepsilon > 0$ and local uniqueness in the chosen gauge for each prescribed area. Since $A(g_\varepsilon) = \varepsilon^2 A_*$, the change of parameter $a = \varepsilon^2 A_*$ gives the asserted family g_a . $\square$

Remark 5.10. *Gluing Einstein metrics in dimension four generally involves finite-dimensional obstructions. In our setting, two ingredients remove these obstructions: toric symmetry and the freedom to vary the cone angle.*

In Biquard’s desingularization of A_1 Einstein orbifold singularities using Eguchi–Hanson metrics, the decaying model kernel is three-dimensional, and the leading obstruction imposes the curvature condition $\det R_{g_0}^+(p) = 0$; see [8]. Under the standard torus action, however, only the scale direction is invariant. The corresponding one-dimensional obstruction can be removed by allowing the cone angle along the exceptional sphere to vary.

For the AE models considered here, Proposition B.5 gives the corresponding rigidity statement: at fixed cone angles, every decaying invariant infinitesimal Ricci-flat deformation is a scale deformation modulo equivariant coordinate changes. The nonzero angle–scale pairing of Lemma B.6 shows that varying the common cone angle supplies the missing cokernel direction, while prescribing total area removes the scale kernel. Together, these yield the augmented inverse of Proposition B.7.

Consequently, under toric nondegeneracy of g_0 , this conical gluing construction requires no additional pointwise curvature condition at p . The cone angle is determined as part of the solution.

5.5. Degeneration versus gluing. Consider a degenerating sequence g_j as in Theorem 5.7, with compact limit (M, g_0) . Throughout this subsection, we assume g_0 is torically nondegenerate.

We use the notation of Section 5.4 and set

$$(5.13) \quad \varepsilon_j^2 = \frac{A(g_j)}{A_*}.$$

Since the rescaled convergence includes every Σ_α , this scale is comparable to the curvature scale. The uniqueness up to scaling in Appendix B therefore gives

$$\varepsilon_j^{-2} g_j \longrightarrow h \quad \text{on compact subsets of } Y, \quad \beta_j \longrightarrow \beta_*,$$

with convergence smooth in local developments.

The proof of the following proposition is a mild adaptation of Ozuch’s arguments [42, 43] to our toric conical setting.

Proposition 5.11. *For all sufficiently large j , the metric g_j is equivariantly isometric to $g_{A(g_j)}$ from Theorem 5.8, by a diffeomorphism preserving each Σ_α .*

Write $|u|_{m,\alpha;w_\varepsilon}$ for the local $C^{m,\alpha}$ norm at scale w_ε , with derivatives of order ℓ multiplied by w_ε^ℓ , and set

$$H_\varepsilon(k) = \sup_X |k|_{2,\alpha;w_\varepsilon}.$$

We use b_ε and its local developments to measure these norms. The maps J_β identify tensors at the varying cone angles, as in Section 5.4.

Lemma 5.12. *After equivariant pullbacks preserving each Σ_α , one can write*

$$g_j = b_{\varepsilon_j, \beta_j} + J_{\beta_j} k_j, \quad \delta_{g_j} b_{\varepsilon_j, \beta_j} = 0, \quad H_{\varepsilon_j}(k_j) \longrightarrow 0.$$

Proof. Let $A_j(R, s)$ be the annulus between the images of the radius- R sphere in Y and the radius- s sphere about p . Theorem 5.7 identifies this annulus equivariantly with $S^3 \times (0, 1)$ and gives flat limits at every intermediate scale. The Gauss–Bonnet argument following that theorem, together with (5.9) and (5.8), gives the quantitative bound

$$(5.14) \quad \limsup_{j \rightarrow \infty} \int_{A_j(R, s)} |\text{Rm}_{g_j}|^2 \, \text{dvol}_{g_j} \leq C(s^4 + R^{-4}).$$

We apply Ozuch’s annular coordinate estimates on this smooth region. The simultaneous half-turn in the two circle directions A, B acts freely there, so its quotient has flat model $\mathbb{R}^4/\{\pm \text{Id}\}$. Thus the nontrivial-group hypothesis in [42, Propositions 7–8 and Remark 16] is satisfied. Apply those local estimates on the quotient and lift the resulting coordinates to $A_j(R, s)$.

More precisely, choose $L_j \rightarrow \infty$ sufficiently slowly that $\varepsilon_j L_j^2 \rightarrow 0$ and convergence on the two limiting regions, through six derivatives and including the boundary terms in (5.14), has error at most L_j^{-M} for a fixed sufficiently large M . With $R = L_j$ and $s = L_j^{-1}$, the lifted estimate gives, on a slightly smaller annulus,

$$(5.15) \quad |g_j - g_{\text{euc}}|_{4,\alpha;r} \leq C L_j^{-2} \left\{ \left(\frac{\varepsilon_j L_j}{r} \right)^\gamma + (r L_j)^\gamma \right\}$$

for some $\gamma > 0$. Its endpoint radii lie between fixed positive multiples of $\varepsilon_j L_j$ and L_j^{-1} , respectively. Here the prefactor follows from (5.14); the decay along the annulus is the conclusion of the cited coordinate estimate.

These coordinates can be chosen equivariantly. The constant-mean-curvature leaves are invariant by uniqueness. On a lifted central leaf, the spectral subspace near the first nonzero eigenvalue of the round sphere gives an equivariant identification with S^3 ; extend it along the normal flow. The rod directions fix the two rotation planes. The remaining rotations extend to isometries of g_0 and h , so the coordinates can be matched to the converging coordinates on the two end collars.

Fix $0 < \nu < \min\{\gamma, \frac{1}{2}\}$ and set

$$\omega_\varepsilon = w_\varepsilon^\nu + (\varepsilon/w_\varepsilon)^\nu.$$

For the gauge argument we use the weighted norms of [43, Definitions 2.5–2.6, p. 30], restricted to invariant tensors and vector fields. In our notation the tensor norm is

$$\|u\|_{m,*} = \inf_T \left\{ |T| + \sup_X \omega_\varepsilon^{-1} |u - \chi T|_{m,\alpha;w_\varepsilon} \right\}.$$

Here χ equals one on the middle annulus and has transitions at a fixed large radius in Y and a fixed small radius in M ; T ranges over invariant constant symmetric two-tensors in the annular coordinates. We include the trace direction, since no volume constraint is imposed in this gauge argument.

After division by ω_{ε_j} , (5.15) is bounded by $C L_j^{\nu-2}$. The reference metric satisfies

$$\frac{r^2 + (\varepsilon_j/r)^2}{\omega_{\varepsilon_j}} \leq C \left\{ r^{2-\nu} + (\varepsilon_j/r)^{2-\nu} \right\}.$$

Together with convergence on the two ends and the collar identifications, these estimates give

$$(5.16) \quad \left\| J_{\beta_j}^{-1}(g_j - b_{\varepsilon_j,\beta_j}) \right\|_{4,*} \rightarrow 0.$$

We next verify the model conditions for the uniform gauge estimate. On (M, g_0) , an invariant Killing field vanishes at p , and its derivative commutes with the two normal rotations. Subtracting a Killing field generated by $\text{Lie}(\mathbb{T}^2)$ therefore gives a Killing field with zero first jet, hence the zero field. The same argument applies to every reference metric. In each case the invariant Killing fields are precisely generated by $\text{Lie}(\mathbb{T}^2)$.

On (Y, h) there is no nonzero invariant solution of

$$\delta_h \delta_h^* v = 0, \quad |v| = O(r^{1-\nu}).$$

Indeed, the order-two asymptotic expansion and elliptic estimates give $\delta_{\text{euc}} \delta_{\text{euc}}^* v = O(r^{-3-\nu})$. The Euclidean fundamental solution improves the decay to $O(r^{-1-\nu})$, modulo a constant vector, which torus invariance excludes. The boundary term in Green's formula is then $O(r^{-2\nu})$ at infinity. The boundary terms along the conical strata vanish in local developments. It follows that $\delta_h^* v = 0$, so v is a decaying invariant Killing field, hence a Killing field generated by an element in $\text{Lie}(\mathbb{T}^2)$ by the same fixed-point argument. Its decay forces it to vanish.

For vector fields, the norm uses $w_\varepsilon \omega_\varepsilon$, with cutoffs of invariant linear fields retained separately; the image norm uses $\omega_\varepsilon/w_\varepsilon$. The flat annular estimate of [43, Proposition 4] applies after the same half-turn quotient. The preceding model checks therefore allow the proof of [43, Lemma 3.4] to be carried out in local developments. In that proof the compact estimate controls the linear coefficients on the annulus; after these are subtracted, the estimate on Y uses precisely

the sublinear class considered above. The annular estimate controls the remaining part. The errors from our order-two model decay are bounded by $r^{2-\nu} + (\varepsilon/r)^{2-\nu}$, so the same absorption argument applies. This gives a uniform inverse for $\delta_b \delta_b^*$ on the complement of the torus fields, where $b = b_{\varepsilon, \beta_*}$. Fredholm theory and integration by parts are taken in local developments. In local developments, changing the cone angle changes the rescaled coefficients by $O(|\beta - \beta_*|)$, so the estimate persists for $b = b_{\varepsilon, \beta}$.

To obtain the gauge in (5.10), apply the fixed-point argument of [43, proof of Proposition 10] to

$$v \mapsto \delta_g((\exp_v)^* b), \quad b = b_{\varepsilon_j, \beta_j}, \quad g = g_j.$$

Here the exponential map is taken with respect to b . At $g = b$ the derivative at zero is $2\delta_b \delta_b^*$, with vector fields identified with one-forms using b . For g satisfying (5.16), this derivative remains invertible on the torus complement. The nonlinear estimates in the cited proof hold in the rescaled developments and allow the invariant constant tensors retained above. Moreover, the image is orthogonal to the torus fields: they are exact Killing fields of g , so integration by parts gives $\int_X (\delta_g u)(K) d\text{vol}_g = 0$ for every Killing field K generated by an element in $\text{Lie}(\mathbb{T}^2)$ and every developed symmetric tensor u . Consequently the projected equation is the full equation. We obtain an equivariant diffeomorphism ψ_j such that

$$\delta_{g_j}(\psi_j^* b_{\varepsilon_j, \beta_j}) = 0,$$

and the resulting change of metric tends to zero in the $\|\cdot\|_{2,*}$ norm. Pulling back by ψ_j^{-1} gives $\delta_{g_j} b_{\varepsilon_j, \beta_j} = 0$ and the asserted estimate. The generating fields are invariant, so their flows preserve each Σ_α . $\square$

Proof of Proposition 5.11. Use the representatives supplied by Lemma 5.12 and set

$$z_j = (k_j, \sigma_j), \quad \sigma_j = \varepsilon_j(\beta_j - \beta_*).$$

The gauge condition and (5.13) imply $\mathcal{F}_{\varepsilon_j}(z_j) = 0$ for the map (5.10). The pullbacks preserve the total area because they preserve each Σ_α .

The local estimates used in the proof of Theorem 5.8 also give

$$(5.17) \quad \|(D\mathcal{F}_\varepsilon(z) - \mathcal{L}_\varepsilon)\dot{z}\| \leq C(H_\varepsilon(k) + |\beta - \beta_*|)(\|\dot{k}\|_{X_\varepsilon} + |\dot{\sigma}|),$$

whenever $H_\varepsilon(k) + |\beta - \beta_*|$ is sufficiently small, where $z = (k, \varepsilon(\beta - \beta_*))$ and the target norm is that of $Y_\varepsilon \times \mathbb{R}$. Indeed, at scale w_ε the metric coefficients and their first two derivatives are controlled by this quantity. The angle variation is supported where $C^{-1}\varepsilon \leq w_\varepsilon \leq C\varepsilon$, so the normalization of σ gives the same bound for its derivative. Rescaling the area equation to Y gives the corresponding estimate for its derivatives as well.

By Lemma 5.12, the coefficient on the right of (5.17) tends to zero along the segment from 0 to z_j . Taylor's formula, Proposition 5.9, and the reference error estimate in the proof of Theorem 5.8 give

$$\|k_j\|_{X_{\varepsilon_j}} + |\sigma_j| \leq C\varepsilon_j^{4/3} + o(1)(\|k_j\|_{X_{\varepsilon_j}} + |\sigma_j|).$$

Absorbing the last term yields

$$\|k_j\|_{X_{\varepsilon_j}} + \varepsilon_j|\beta_j - \beta_*| \leq C\varepsilon_j^{4/3}.$$

The solution constructed in Theorem 5.8 with total area $\varepsilon_j^2 A_*$ satisfies the same bound. Applying (5.17) along the segment between the two solutions and using the uniform inverse once more shows that their difference is zero. This proves the proposition. $\square$

6. AN UPPER BOUND ON THE CONE ANGLE

We continue to adopt the setting in Section 4. In this section, the main goal is

Theorem 6.1. *There is a finite constant $\beta_{\max}$ depending only on the rod structures $\mathfrak{R}$ such that for any $[(\beta, g)] \in \mathcal{M}$, we have*

$$\beta \leq \beta_{\max}$$

Before we give a rigorous proof let us give some heuristic arguments. The angle bound comes from comparing the curvature required by the Ricci-flat bubbles with the curvature available on the compact manifold. If $\beta \rightarrow \infty$, changing the angular periods by β^{-1} makes the designated cone angles equal to 2π , while the original cone angles and the total volume tend to zero, so we are in a collapsing situation. The internal torus fixed points are smooth, and rescaling around them produces nonflat AF or ALF gravitational instantons. The conical corrections account for the disappearing contribution of the original rods to the Euler formula. Gauss–Bonnet then shows that the Ricci-flat bubbles use all the available total Weyl curvature energy. The signature formula imposes a further constraint: the twisting of the circle fibration at each ALF end contributes a sign, which must balance together with the signatures of the limits. Our choice of rod structures make this balance impossible. For the alternating sequence, the odd number of internal fixed points forces an odd number of ALF ends, whose signs cannot cancel, although the limiting signature integral is zero. The other chosen sequences give analogous contradictions through their intersection forms. Splitting into several limits therefore cannot resolve the obstruction. Below we give the rigorous details.

Given $[(\beta, g)] \in \mathcal{M}$. To prove the above theorem we always assume $\beta > 1$. As in Remark 4.1, the upper bound is automatic if $q = 1$, so we will assume $q \geq 2$ throughout this section.

To apply Riemannian convergence theory, it is helpful to change the angular periods by defining

$$\hat{g} = \underline{g} + \beta^{-2} G_{ab} d\varphi_a d\varphi_b.$$

Then $\hat{g}$ is again a toric conical Einstein metric, and each cone angle is multiplied by β^{-1} . Thus the rods $\mathfrak{I}_1, \dots, \mathfrak{I}_q$ become smooth, while the remaining rods have cone angle $2\pi\beta^{-1} < 2\pi$. Moreover we have

$$(6.1) \quad \text{Vol}_{\hat{g}}(X) = \beta^{-2} \text{Vol}_g(X), \quad I_{\pm}(\hat{g}) = \beta^{-2} I_{\pm}(g).$$

Suppose $\beta_j \rightarrow \infty$. Then by (2.14) and (4.3) we have

$$(6.2) \quad \text{Vol}_{\hat{g}_j}(X) \leq \frac{8\pi^2}{3\beta_j^2} \rightarrow 0,$$

$$(6.3) \quad \chi_c(\hat{g}_j) = q - 1 + \frac{2}{\beta_j} + \frac{m_0 - 1}{\beta_j^2},$$

$$(6.3) \quad \tau_c(\hat{g}_j) = \frac{\sigma}{3} + \frac{1}{\beta_j^2} \left(\tau(X) - \frac{\sigma}{3} \right),$$

$$(6.4) \quad I_{\pm}(\hat{g}_j) \rightarrow 2\pi^2(2(q-1) \pm \sigma).$$

In particular, the L^2 curvature energies of $\hat{g}_j$ are uniformly bounded.

Let $\mathcal{S}_0$ be the union of the spheres corresponding to the original rods in $\mathfrak{R}_0$, and denote the internal fixed points by $p_j^1, \dots, p_j^{q-1}$. These points are smooth for $\hat{g}_j$. Write ω_4 for the volume of the Euclidean unit ball and, for each $1 \leq a \leq q-1$, choose $r_j^a > 0$ such that

$$(6.5) \quad \text{Vol}_{\hat{g}_j} B_{\hat{g}_j}(p_j^a, r_j^a) = \frac{3}{4} \omega_4 (r_j^a)^4.$$

Such a radius exists because Bishop–Gromov volume comparison holds for $\hat{g}_j$ (since the cone angles are all smaller than 2π) and the total volume goes to zero. Moreover, (6.2) implies $r_j^a \leq C\beta_j^{-1/2} \rightarrow 0$. Set

$$h_j^a = (r_j^a)^{-2} \hat{g}_j.$$

Lemma 6.2. *After passing to a subsequence, each (X, h_j^a, p_j^a) converges smoothly and equivariantly to a complete noncompact nonflat toric Ricci-flat four-manifold with finite curvature energy.*

Proof. Fix a and write $p_j = p_j^a$, $r_j = r_j^a$, and $h_j = h_j^a$. For $z \in \mathcal{S}_0$, its volume density is at most β_j^{-1} . Bishop–Gromov comparison therefore gives

$$\frac{3}{4}\omega_4 r_j^4 \leq \text{Vol}_{g_j} B_{g_j}(z, d_{g_j}(p_j, z) + r_j) \leq \frac{\omega_4}{\beta_j} (d_{g_j}(p_j, z) + r_j)^4.$$

Consequently,

$$(6.6) \quad d_{h_j}(p_j, \mathcal{S}_0) \geq \left(\frac{3\beta_j}{4}\right)^{1/4} - 1 \longrightarrow \infty.$$

Thus every fixed-radius ball about p_j is smooth for large j . Comparison also gives

$$(6.7) \quad \text{Vol}_{h_j} B_{h_j}(x, t) \geq \frac{3\omega_4}{4(R+1)^4} t^4, \quad x \in B_{h_j}(p_j, R), \quad 0 < t \leq R+1.$$

If curvature were unbounded on a bounded ball, the point-selection argument of the proof of Theorem 5.7 would produce a complete smooth nonflat Ricci-flat limit with finite curvature energy and Euclidean volume growth. The fixed-point curvature estimate 3.11 places a smooth torus fixed point at bounded distance in this further rescaling. The equivariant convergence argument therefore gives an effective limiting torus action with rod directions in $\{A, B, C\}$. By [6, Theorem 1.5], this limit is ALE, contradicting the ALE rigidity Lemma 5.6.

Consequently, curvature is uniformly bounded on every bounded ball. Smooth equivariant compactness gives a complete toric Ricci-flat limit (Y, h, p_∞) , with finite curvature energy by lower semicontinuity. The limiting torus generators vanish at p_∞ and have the standard rotational derivatives there. Thus Y cannot be compact. If Y were flat, this torus action would force it to be Euclidean, contrary to

$$\text{Vol}_h B_h(p_\infty, 1) = \frac{3}{4}\omega_4.$$

□

Equivariant convergence identifies stabilizers and their fixed sets on compact subsets. Thus the limiting torus fixed points are precisely the limits of internal fixed points at bounded rescaled distance. The local uniqueness of the surface fixed by each stabilizer circle, together with the equivariant charts at the torus fixed points, identifies every finite limiting rod with one designated rod and its original endpoints. Consequently, every finite chain in the limit follows a consecutive part of the designated chain.

Let (Y, h) be a limit from Lemma 6.2, and write

$$(B_\infty, \underline{h}) = Y/\mathbb{T}^2, \quad \mathcal{V}_\infty = Y^{\mathbb{T}^2}.$$

The set $\mathcal{V}_\infty$ is finite. Moreover, the rods are smooth and the rod vectors belong to $\{A, B, C\}$.

Proposition 6.3. *The limit (Y, h) is a proper toric gravitational instanton of AF or ALF type. The rods of $Y/\mathbb{T}^2$, in boundary order, correspond to a consecutive part of the designated chain.*

Proof. As in Corollary 3.10, the quotient $B_\infty = Y/\mathbb{T}^2$ is complete, has simply connected interior, and has no interior cut locus. Since $K_{\underline{h}} \leq 0$, the same comparison argument as in Corollary 3.10 shows that B_∞ is CAT(0). Its boundary is nonempty because the basepoint is fixed by the torus. The horizontal distribution is integrable by Lemma 2.4, and (3.21) gives quadratic curvature decay at infinity, since the torus fixed-point set is finite. Lemma A.17 therefore gives properness and identifies B_∞ with the closed half-plane.

The rod-length estimate of Corollary 3.3 and the CAT(0) property show that each original rod is minimizing. Hence each contributes at most one connected boundary piece. The correspondence of rods and endpoints established above then identifies the boundary sequence of rods with a consecutive part of the designated chain.

Theorem A.4 gives ALE, ALF or AF asymptotics. Since Y is nonflat, the ALE case is excluded by the rigidity argument used in Lemma 6.2. □

We denote

$$I_{\pm}(Y) = \int_Y |W_h^{\pm}|^2 d\text{Vol}_h.$$

Let v be the number of torus fixed points in Y . Set $\varepsilon(Y) = 0$ for an AF end; for an ALF end, let $\varepsilon(Y) = \pm 1$ be the oriented Euler number of the circle bundle at infinity.

Lemma 6.4. *We have $\chi(Y) = v$, and $\tau(Y)$ is the signature of the intersection matrix of the finite rod spheres. Moreover,*

$$(6.8) \quad I_{\pm}(Y) = 4\pi^2 \left(v \pm \left(\frac{3}{2}\tau(Y) + \varepsilon(Y) \right) \right).$$

If only two rod directions occur, then $\tau(Y) = 0$ and

$$(6.9) \quad I_{\pm}(Y) = 4\pi^2(v \pm \varepsilon(Y)), \quad \varepsilon(Y) = \begin{cases} 0, & v \text{ even}, \\ \pm 1, & v \text{ odd}. \end{cases}$$

Proof. The plumbing description gives $\chi(Y) = v$ and identifies the intersection form with that of the $v - 1$ finite rod spheres. Equal extreme directions give an AF end; distinct directions form a lattice basis and give an ALF end with Euler number ± 1 .

The AF and ALF Chern–Weil formulas give

$$(6.10) \quad I_+(Y) + I_-(Y) = 8\pi^2 v,$$

$$(6.11) \quad \frac{I_+(Y) - I_-(Y)}{12\pi^2} = \tau(Y) + \frac{2}{3}\varepsilon(Y),$$

see [18, Section 6.2, equations (6.19)–(6.25)] and [39, Section 4.1]. For a screw AF end, the signature boundary term also vanishes: equatorial reflection commutes with the screw rotation and reverses the orientation of the model cross-section. Adding and subtracting the two identities gives (6.8).

With only two directions, the rods alternate and every finite sphere has self-intersection zero. Changing the signs of alternate sphere classes negates the intersection matrix, so $\tau(Y) = 0$. The extreme directions agree exactly when v is even, proving (6.9). $\square$

Proof of Theorem 6.1. Recall that $q \geq 2$. Pass to a subsequence so that the volume radii have a fixed order and all ratios of fixed-point distances to these radii converge in $[0, \infty]$. Successively choose a fixed point with largest remaining radius, take its pointed limit, and remove all fixed points at bounded distance on that scale. Later centers escape at every earlier scale, and their radii are no larger. Thus the corresponding rescaled balls are disjoint, and no fixed point occurs in two limits.

The resulting limits $Y_1, \dots, Y_t$ therefore satisfy

$$(6.12) \quad \sum_{\alpha=1}^t v_{\alpha} = q - 1.$$

Smooth convergence, disjointness and exhaustion give

$$(6.13) \quad \sum_{\alpha=1}^t I_{\pm}(Y_{\alpha}) \leq \lim_{j \rightarrow \infty} I_{\pm}(\hat{g}_j) = 2\pi^2(2(q-1) \pm \sigma).$$

Substituting (6.8) into both inequalities yields

$$(6.14) \quad \sum_{\alpha=1}^t (3\tau(Y_{\alpha}) + 2\varepsilon(Y_{\alpha})) = \sigma.$$

For an alternating insertion, every limit uses only A, B , so $\tau(Y_{\alpha}) = 0$. Since $\sigma = 0$, (6.14) gives $\sum_{\alpha} \varepsilon(Y_{\alpha}) = 0$. However, (6.9) implies

$$\sum_{\alpha} \varepsilon(Y_{\alpha}) \equiv \sum_{\alpha} v_{\alpha} = 2n - 1 \pmod{2},$$

a contradiction.

For $W = (B, A, C, A)$ or (B, C, A, C, A) , every limit uses only A, C , so again all signatures vanish. Equation (6.14) then has an even left-hand side, whereas $\sigma = 1$ or -1 . The same argument applies to (B, C, B, A) and (B, C, B, C, A) : their limiting rods use only B, C , and $\sigma = 1$ and -1 , respectively.

Finally, let $W = (B, C, A, B, C, A)$. Here $q - 1 = 3$ and $\sigma = -4$. Each limiting rod sequence is a consecutive part of (C, A, B, C) . A limit with two or three torus fixed points has intersection matrix, respectively,

$$(-1), \quad \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix},$$

and hence signature -1 . If such a limit occurs, $\sum_{\alpha} \tau(Y_{\alpha}) = -1$, making the left-hand side of (6.14) odd, a contradiction. Otherwise there are three limits, each with one torus fixed point, signature zero and $\varepsilon(Y_{\alpha}) = \pm 1$. Equation (6.14) requires these three signs to sum to -2 , which is also impossible. $\square$

7. RIGIDITY OF TORIC EINSTEIN METRICS WITH $b_2^+ \leq 1$

Throughout this section, g is a smooth toric Einstein metric on the compact four-manifold $X_{\mathfrak{R}}$, with $\text{Ric}(g) = \lambda g$ and $\lambda > 0$. We retain the orientation convention of Section 2. We first give a topological criterion for g to be algebraically special; its proof is given in Section 7.1.

Theorem 7.1. *Let g be a toric Einstein metric on $X_{\mathfrak{R}}$. Then*

- $b_2^+(X_{\mathfrak{R}}) = 0$ if and only if g is anti-self-dual, that is, $W_g^+ = 0$.
- $b_2^+(X_{\mathfrak{R}}) = 1$ if and only if g is conformally Kähler with respect to a complex structure compatible with the orientation of $X_{\mathfrak{R}}$.

As in the case of toric gravitational instantons studied in [34], one can also formulate the criterion in terms of the *degree* of the rod structure. Recall that the rod vector on $\mathcal{I}_i$ is defined up to sign, so it determines a projective direction $[\mathbf{v}_i] \in \mathbb{RP}^1$.

Definition 7.2. Let $\mathbf{v}_1, \dots, \mathbf{v}_m$ be the rod vectors in positive boundary order, with $\mathbf{v}_{m+1} = \mathbf{v}_1$. Choose real angular lifts $\theta_1, \dots, \theta_{m+1} \in \mathbb{R}$ of these projective directions successively so that

$$0 < \theta_i - \theta_{i+1} < \pi, \quad i = 1, \dots, m.$$

The *degree* of $\mathfrak{R}$ is

$$d(\mathfrak{R}) = \frac{\theta_1 - \theta_{m+1}}{\pi} - 1.$$

Thus $d(\mathfrak{R}) + 1$ counts the clockwise turns made by joining consecutive directions in $\mathbb{RP}^1$.

The degree is a nonnegative integer, independent of the initial angular lift and invariant under orientation-preserving changes of lattice basis. With our orientation conventions,

$$(7.1) \quad d(\mathfrak{R}) = b_2^+(X_{\mathfrak{R}}).$$

To prove this identity, choose a circle subgroup generated by a primitive lattice vector ξ whose direction differs from every rod direction. Its fixed points are precisely the m torus fixed points. At the vertex between $\mathcal{I}_i$ and $\mathcal{I}_{i+1}$, choose signed representatives so that $(\mathbf{v}_i, \mathbf{v}_{i+1})$ is a positive lattice basis, and write

$$\xi = a_i \mathbf{v}_i + b_i \mathbf{v}_{i+1}, \quad a_i, b_i \in \mathbb{Z} \setminus \{0\}.$$

In the corresponding local complex coordinates, the circle action is

$$t \cdot (z_1, z_2) = (t^{a_i} z_1, t^{b_i} z_2), \quad t \in \mathbb{S}^1.$$

Thus a_i, b_i are the two rotation weights. To compare orientations, write $z_k = r_k e^{\sqrt{-1}\psi_k}$. The rods $\mathcal{I}_i$ and $\mathcal{I}_{i+1}$ are given by $r_1 = 0$ and $r_2 = 0$, respectively, so their positive boundary order makes $dr_1 \wedge dr_2$ positive on the orbit space. The positive lattice basis makes $d\psi_1 \wedge d\psi_2$ positive on the torus. Our orientation of $X_{\mathfrak{R}}$ is therefore represented by $dr_1 \wedge dr_2 \wedge d\psi_1 \wedge d\psi_2$, whereas the complex orientation is represented by

$$(r_1 dr_1 \wedge d\psi_1) \wedge (r_2 dr_2 \wedge d\psi_2) = -r_1 r_2 dr_1 \wedge dr_2 \wedge d\psi_1 \wedge d\psi_2.$$

Hence these local complex coordinates induce the orientation opposite to that of $X_{\mathfrak{R}}$.

We apply the isolated-fixed-point form of the Atiyah–Singer equivariant signature theorem [5, Theorem 6.12, p. 582]. Since the circle is connected, its action on cohomology is trivial, so the equivariant signature equals the ordinary signature $\tau(X_{\mathfrak{R}})$. The fixed-point formula therefore gives, for generic $t \in \mathbb{S}^1$,

$$\tau(X_{\mathfrak{R}}) = - \sum_{i=1}^m \frac{1+t^{a_i}}{1-t^{a_i}} \frac{1+t^{b_i}}{1-t^{b_i}}.$$

The minus sign in each summand accounts for the opposite local complex orientation. This is an identity of rational functions of t , so we may take $t \rightarrow 0$. Since

$$\lim_{t \rightarrow 0^+} \frac{1+t^a}{1-t^a} = \operatorname{sgn}(a) \quad (a \neq 0),$$

we obtain

$$\tau(X_{\mathfrak{R}}) = - \sum_{i=1}^m \operatorname{sgn}(a_i b_i).$$

Thus the contribution of the vertex between $\mathfrak{J}_i$ and $\mathfrak{J}_{i+1}$ to this signature sum is $-\operatorname{sgn}(a_i b_i)$: it is $+1$ when the two rotation weights have opposite signs and -1 when they have the same sign.

This contribution is $+1$ precisely when the chosen circle direction lies in the clockwise projective arc from $\mathbf{v}_i$ to $\mathbf{v}_{i+1}$, and is -1 otherwise. There are exactly $d(\mathfrak{R}) + 1$ such arcs containing this direction, counted with multiplicity. Therefore

$$\tau(X_{\mathfrak{R}}) = 2d(\mathfrak{R}) + 2 - m.$$

Together with $\chi(X_{\mathfrak{R}}) = m$ and $b_1(X_{\mathfrak{R}}) = 0$, this proves (7.1). Using $3\tau(X_{\mathfrak{R}}) = \sum_i [\Sigma_i]^2$, we also obtain

$$d(\mathfrak{R}) = \frac{m-2}{2} + \frac{1}{6} \sum_{i=1}^m [\Sigma_i]^2.$$

Thus the two cases in Theorem 7.1 correspond to degrees zero and one, respectively. Reversing orientation changes the degree to $m - 2 - d(\mathfrak{R}) = b_2^-(X_{\mathfrak{R}})$ and gives the corresponding criteria for W_g^- .

Together with the classification results of Hitchin and LeBrun, Theorem 7.1 gives the following classification. Section 7.3 gives a direct proof of the uniqueness assertion in the Hermitian cases.

Corollary 7.3. *If $b_2^+(X_{\mathfrak{R}}) \leq 1$, then, up to scaling and isometry, a toric Einstein metric g on $X_{\mathfrak{R}}$ is one of the following:*

- the round metric on $\mathbb{S}^4$;
- the Fubini–Study metric on $\mathbb{CP}^2$;
- the round product metric on $\mathbb{S}^2 \times \mathbb{S}^2$;
- the Page metric on $\mathbb{CP}^2 \# \mathbb{CP}^2$;
- the Chen–LeBrun–Weber metric on $\mathbb{CP}^2 \# 2\overline{\mathbb{CP}^2}$;
- the Kähler–Einstein metric on $\mathbb{CP}^2 \# 3\overline{\mathbb{CP}^2}$.

Proof. If $b_2^+(X_{\mathfrak{R}}) = 0$, Theorem 7.1 and Hitchin’s classification [28] give the round sphere or the Fubini–Study metric.

If $b_2^+(X_{\mathfrak{R}}) = 1$, the theorem makes g Hermitian. LeBrun’s classification [32, Theorem A] gives the Page metric, the Chen–LeBrun–Weber metric, or a Kähler–Einstein metric. The torus preserves the simple eigenline of W^+ and its associated Kähler form: connectedness excludes a change of sign. Thus its action is holomorphic for the resulting complex structure.

In the Kähler–Einstein case, the classification of toric del Pezzo surfaces and Matsushima’s obstruction leave

$$\mathbb{CP}^2, \quad \mathbb{CP}^1 \times \mathbb{CP}^1, \quad \mathbb{CP}^2 \# 3\overline{\mathbb{CP}^2}.$$

The Bando–Mabuchi uniqueness theorem identifies the metric, up to scaling and isometry, with the corresponding metric in the list; for the last case, see also [32, Theorem C]. $\square$

In particular, we obtain the following nonexistence result.

Corollary 7.4. *For $k \geq 4$, the manifold $\mathbb{CP}^2 \# k\overline{\mathbb{CP}^2}$ admits no toric Einstein metric.*

Remark 7.5. *For positive Einstein metrics, the Hitchin–Thorpe inequality only excludes $k \geq 9$ in the above corollary. Indeed,*

$$\chi(\mathbb{CP}^2 \# k\overline{\mathbb{CP}^2}) = 3 + k, \quad \tau(\mathbb{CP}^2 \# k\overline{\mathbb{CP}^2}) = 1 - k,$$

so $2\chi > 3|\tau|$ gives $k < 9$. For $4 \leq k \leq 8$, these manifolds admit Kähler–Einstein metrics, but none is toric.

In Section 7.3, we also prove infinitesimal rigidity for algebraically special toric Einstein metrics.

Theorem 7.6. *All the metrics listed in Corollary 7.3 are torically nondegenerate.*

7.1. Proof of Theorem 7.1. We use the notation of Sections 2 and 3. Recall that

$$\begin{aligned} u &= \log \rho, & U &= \frac{1}{2} |du|_{\underline{g}}^2, & K &= K_{\underline{g}}, \\ T &= \Phi^* g_{\mathcal{H}}, & e &= \frac{1}{2} \operatorname{tr}_{\underline{g}} T = K + \frac{U}{2}. \end{aligned}$$

We identify the upper half-plane with the space of positive-definite symmetric matrices of determinant one by

$$(X, Y) \mapsto \begin{pmatrix} X + X^{-1}Y^2 & X^{-1}Y \\ X^{-1}Y & X^{-1} \end{pmatrix}, \quad X > 0.$$

In these coordinates, the target metric $\frac{1}{4} \operatorname{tr}(\Phi^{-1} d\Phi)^2$ is

$$g_{\mathcal{H}} = \frac{dX^2 + dY^2}{2X^2},$$

and $\Phi = \rho^{-1}G$ has coordinates

$$X = \frac{\rho}{G_{22}}, \quad Y = \frac{G_{12}}{G_{22}}.$$

Choose a positively oriented orthonormal frame e_1, e_2, e_3, e_4 locally on $X_{\mathfrak{H}}$, with e_1, e_2 horizontal and e_3, e_4 vertical. Set

$$J = R_{1234}.$$

Lemma 7.7. *The curvature component J is the negative Jacobian of Φ . More precisely,*

$$(7.2) \quad J dA_{\underline{g}} = d \left(\frac{G_{22}}{2\rho} d \frac{G_{12}}{G_{22}} \right) = - \frac{dX \wedge dY}{2X^2}.$$

Proof. Let K_1, K_2 be the positively oriented angular generators. Choose the orthonormal vertical frame

$$e_3 = \frac{\sqrt{G_{22}}}{\rho} (K_1 - Y K_2), \quad e_4 = \frac{K_2}{\sqrt{G_{22}}}.$$

For a horizontal vector ξ , the Koszul formula gives $2\langle \nabla_{\xi} K_a, K_b \rangle = \xi(G_{ab})$. Thus the vertical connection form along the base is

$$\alpha = \langle \nabla e_3, e_4 \rangle = - \frac{dG_{12} - Y dG_{22}}{2\rho} = - \frac{dY}{2X}.$$

Covariant differentiation in horizontal directions preserves the vertical distribution, so

$$\nabla_{\xi} e_3 = \alpha(\xi) e_4, \quad \nabla_{\xi} e_4 = -\alpha(\xi) e_3.$$

The curvature equation, with our convention $R_{ijkl} = \langle R(e_i, e_j)e_l, e_k \rangle$, therefore yields

$$J dA_{\underline{g}} = -d\alpha = - \frac{dX \wedge dY}{2X^2}.$$

□

The integral of J is determined by the signature of $X_{\mathfrak{H}}$.

Lemma 7.8. *We have*

$$(7.3) \quad \int_B J dA_{\underline{g}} = -\frac{\pi}{2} \tau(X_{\mathfrak{R}}),$$

and

$$(7.4) \quad \int_B (K + J) dA_{\underline{g}} = \pi(1 - b_2^+(X_{\mathfrak{R}})).$$

Proof. Choose a positively oriented angular basis whose second circle is different from every rod circle. Along each open rod, G_{12}/G_{22} has constant boundary value and an error of order r^2 , whereas ρ/G_{22} vanishes linearly. The boundary integral in (7.2) therefore vanishes along open rods.

At a turning point, choose radial coordinates x, y with $dx \wedge dy$ positive, and signs for the adjacent primitive generators so that they form a positive lattice basis. Write the two global generators in this local basis as (a, b) and (c, d) , where $ad - bc = 1$ and $cd \neq 0$. On the arc $x = r \cos t$, $y = r \sin t$, the leading orbit matrix is the transform of $\text{diag}(x^2, y^2)$. Consequently,

$$-\frac{G_{22}}{2\rho} d \frac{G_{12}}{G_{22}} \Big|_{\text{arc}} = \left(\frac{cd}{c^2 \cos^2 t + d^2 \sin^2 t} + O(r^2) \right) dt.$$

The error is uniform up to the endpoints by the corner expansions. The boundary of the excised domain traverses this arc from $\pi/2$ to 0, so its limiting contribution is $-\pi \text{sgn}(cd)/2$.

Apply Stokes' theorem on domains cut off near the rods and vertices. First let the offsets from the open rods tend to zero with the corner radii fixed, and then let the corner radii tend to zero. The boundary contributions computed above give (7.3).

The rods are geodesic and the corner angles are $\pi/2$. Since their number is $\chi(X_{\mathfrak{R}})$, Gauss–Bonnet gives

$$\int_B K dA_{\underline{g}} = 2\pi - \frac{\pi}{2} \chi(X_{\mathfrak{R}}).$$

The second identity follows from $\chi(X_{\mathfrak{R}}) + \tau(X_{\mathfrak{R}}) = 2 + 2b_2^+(X_{\mathfrak{R}})$. $\square$

Lemma 7.7 gives

$$J^2 = \det_{\underline{g}} T, \quad |J| \leq \frac{1}{2} \text{tr}_{\underline{g}} T = e.$$

Consider the orthonormal basis of Λ^+ given by

$$\omega_0 = \frac{e^{12} + e^{34}}{\sqrt{2}}, \quad \omega_1 = \frac{e^{13} - e^{24}}{\sqrt{2}}, \quad \omega_2 = \frac{e^{14} + e^{23}}{\sqrt{2}}.$$

In this basis, the local reflection $\varphi \mapsto -\varphi$ acts as $\text{diag}(1, -1, -1)$. Since W^+ commutes with this reflection, ω_0 is an eigenform of W^+ . The Einstein condition gives $R_{3434} = R_{1212} = K$, and hence

$$\langle \text{Rm}_g \omega_0, \omega_0 \rangle = \frac{1}{2} (R_{1212} + R_{3434} + 2R_{1234}) = K + J.$$

On the other hand,

$$\text{Rm}_g|_{\Lambda^+} = W^+ + \frac{\text{Scal}_g}{12} \text{Id} = W^+ + \frac{\lambda}{3} \text{Id}.$$

Thus, defining real coefficient functions a, b in the chosen frame, we may write

$$(7.5) \quad W^+ = \begin{pmatrix} -w & 0 & 0 \\ 0 & w/2 + a & b \\ 0 & b & w/2 - a \end{pmatrix},$$

where

$$w = \frac{\lambda}{3} - K - J, \quad q = a + \sqrt{-1}b.$$

The coefficient q depends on the frame, but its squared norm

$$(7.6) \quad |q|^2 = \frac{1}{2} \left(|W^+|^2 - \frac{3}{2} w^2 \right)$$

is well defined.

The boundary expansions of Lemma 2.7 show that $K, J, w, |q|^2$ are smooth and even in the normal variables at rods and turning points. On an open rod, choose the limiting frame so that e_1, e_3 span the normal plane to the fixed surface. The stabilizer circle fixes ω_1 and rotates $\text{span}\{\omega_0, \omega_2\}$. Since W^+ is self-adjoint and commutes with this action, it acts as $-w \text{Id}$ on this two-plane. Trace freeness therefore gives eigenvalues $-w, -w, 2w$, so

$$(7.7) \quad |q|^2 = \frac{1}{2} \left(6w^2 - \frac{3}{2}w^2 \right) = \frac{9}{4}w^2 \quad \text{on } \partial B.$$

The tensor W^+ satisfies the contracted Bianchi identity

$$(7.8) \quad \delta_g W^+ = 0$$

and the Weyl Weitzenböck formula [23, Section 2],

$$(7.9) \quad \Delta_g W^+ = 2\lambda W^+ - 6(W^+)^2 + 2|W^+|^2 \text{Id}.$$

Here $\Delta_g = -\nabla_g^* \nabla_g$ on tensors. These identities give the following equations for w and q .

Lemma 7.9. *On B° , we have*

$$(7.10) \quad \left| dw + \frac{3}{2}wdu \right|_g^2 = 4(e + J)|q|^2,$$

$$(7.11) \quad \Delta_g w + 3\langle du, dw \rangle_g = 4\lambda w - 3w^2 - 4|q|^2.$$

Proof. Use a $\mathbb{T}^2$ -invariant orthonormal frame and write

$$(\langle \nabla_{e_\alpha} e_i, e_\beta \rangle)_{\alpha, \beta=3,4} = \begin{pmatrix} u_i/2 + p_i & t_i \\ t_i & u_i/2 - p_i \end{pmatrix}, \quad i = 1, 2.$$

Here e_1, e_2 are horizontal, e_3, e_4 are vertical, and $u_i = e_i(u)$. The functions p_i, t_i describe the trace-free part of the matrix. Set

$$\zeta = 2(p_1 - t_2 + \sqrt{-1}(p_2 + t_1)).$$

We first establish the identities

$$e = p_1^2 + p_2^2 + t_1^2 + t_2^2, \quad J = -2(p_1 t_2 - p_2 t_1), \quad |\zeta|^2 = 4(e + J).$$

Take constant linear combinations of the angular Killing fields that agree with e_3, e_4 at the chosen point. Their Gram matrix satisfies $G = \text{Id}$ and $\rho = 1$ there, so

$$e_i(G) = \begin{pmatrix} u_i + 2p_i & 2t_i \\ 2t_i & u_i - 2p_i \end{pmatrix},$$

$$e_i(\Phi) = e_i(G) - u_i \text{Id} = 2 \begin{pmatrix} p_i & t_i \\ t_i & -p_i \end{pmatrix}.$$

Hence the definition of the target metric gives

$$e = \frac{1}{8} \sum_{i=1}^2 \text{tr}((\Phi^{-1} e_i(\Phi))^2) = p_1^2 + p_2^2 + t_1^2 + t_2^2.$$

In the upper half-plane coordinates, we obtain

$$e_i(X) = e_i(\rho) - e_i(G_{22}) = 2p_i, \quad e_i(Y) = 2t_i.$$

Since $X = 1$ at the chosen point, (7.2) gives

$$J = -\frac{1}{2}(\text{d}X \wedge \text{d}Y)(e_1, e_2) = -2(p_1 t_2 - p_2 t_1).$$

Finally, recalling that

$$\zeta = 2(p_1 - t_2 + \sqrt{-1}(p_2 + t_1)),$$

we obtain

$$|\zeta|^2 = 4((p_1 - t_2)^2 + (p_2 + t_1)^2) = 4(e + J).$$

For a horizontal vector ξ , define

$$\vartheta(\xi) = 2(\langle \nabla_\xi e_1, e_2 \rangle + \langle \nabla_\xi e_3, e_4 \rangle).$$

The two summands are the horizontal and vertical connection forms along the base. From the definitions of ω_1, ω_2 ,

$$\nabla_\xi \omega_1 = \frac{\vartheta(\xi)}{2} \omega_2, \quad \nabla_\xi \omega_2 = -\frac{\vartheta(\xi)}{2} \omega_1.$$

The induced covariant derivative of the trace-free symmetric block represented by $q = a + \sqrt{-1}b$ is therefore

$$Dq = dq + \sqrt{-1}\vartheta q = (da - \vartheta b) + \sqrt{-1}(db + \vartheta a).$$

The factor two arises because the components of a trace-free symmetric endomorphism rotate through twice the angle of the frame. With our curvature convention, the horizontal and vertical connection forms have curvatures $-KdA_{\underline{g}}$ and $-JdA_{\underline{g}}$, respectively. Hence

$$(7.12) \quad d\vartheta = -2(K + J)dA_{\underline{g}}.$$

At the chosen point, rotate the horizontal and vertical frames so that their connection forms vanish in horizontal directions. Then $D_i q = a_i + \sqrt{-1}b_i$ there, where $a_i = e_i(a)$ and $b_i = e_i(b)$. Although the coefficients of W^+ are $\mathbb{T}^2$ -invariant, their covariant derivatives in vertical directions include the derivatives of the frame.

The shape coefficients give

$$\begin{aligned} \nabla_{e_3} \omega_0 &= \left(\frac{u_2}{2} + p_2 + t_1 \right) \omega_1 - \left(\frac{u_1}{2} + p_1 - t_2 \right) \omega_2, \\ \nabla_{e_4} \omega_0 &= \left(\frac{u_1}{2} - p_1 + t_2 \right) \omega_1 + \left(\frac{u_2}{2} - p_2 - t_1 \right) \omega_2. \end{aligned}$$

Differentiating $W^+ \omega_0 = -w \omega_0$ and pairing with ω_j , for $j = 1, 2$, yields

$$\langle (\nabla_{e_\alpha} W^+) \omega_0, \omega_j \rangle = -\langle (W^+ + w \text{Id}) \nabla_{e_\alpha} \omega_0, \omega_j \rangle.$$

Now we use the contracted Bianchi identity $\delta_g W^+ = 0$. Its e_2 -component, paired with ω_0 , reads

$$0 = w_1 + \langle (\nabla_{e_3} W^+) \omega_0, \omega_2 \rangle - \langle (\nabla_{e_4} W^+) \omega_0, \omega_1 \rangle.$$

Substituting the preceding expressions, and computing the e_1 -component in the same way, gives

$$(7.13) \quad \begin{aligned} w_1 + \frac{3}{2} w u_1 &= 2a(p_1 - t_2) + 2b(p_2 + t_1), \\ w_2 + \frac{3}{2} w u_2 &= 2b(p_1 - t_2) - 2a(p_2 + t_1). \end{aligned}$$

Squaring and adding gives

$$\begin{aligned} \left| dw + \frac{3}{2} w du \right|_{\underline{g}}^2 &= 4(a^2 + b^2)((p_1 - t_2)^2 + (p_2 + t_1)^2) \\ &= 4(e + J)|q|^2. \end{aligned}$$

This proves (7.10).

For later use, we also derive an equation for q . Recall that, for a horizontal vector ξ ,

$$\vartheta(\xi) = 2(\langle \nabla_\xi e_1, e_2 \rangle + \langle \nabla_\xi e_3, e_4 \rangle), \quad Dq = dq + \sqrt{-1}\vartheta q.$$

Indeed,

$$\nabla_\xi \omega_1 = \frac{\vartheta(\xi)}{2} \omega_2, \quad \nabla_\xi \omega_2 = -\frac{\vartheta(\xi)}{2} \omega_1,$$

so Dq represents the covariant derivative of the trace-free symmetric block of W^+ . The horizontal and vertical connection forms have curvatures $-KdA_{\underline{g}}$ and $-JdA_{\underline{g}}$, respectively. Hence

$$(7.14) \quad d\vartheta = -2(K + J)dA_{\underline{g}}.$$

At a chosen point, choose the horizontal and vertical frames so that their connection forms vanish in horizontal directions. Then $D_i q = a_i + \sqrt{-1}b_i$, where $a_i = e_i(a)$ and $b_i = e_i(b)$. Pairing the e_3 -component of $\delta_g W^+ = 0$ with ω_1, ω_2 gives

$$\begin{aligned} 0 &= a_1 + b_2 + \frac{1}{2}w_1 - \langle (\nabla_{e_4} W^+) \omega_0, \omega_1 \rangle, \\ 0 &= b_1 - a_2 + \frac{1}{2}w_2 - \langle (\nabla_{e_4} W^+) \omega_0, \omega_2 \rangle. \end{aligned}$$

Using the preceding formulas for $\nabla_{e_4} \omega_0$ and $(\nabla_{e_4} W^+) \omega_0$, these become

$$\begin{aligned} 0 &= a_1 + b_2 + \frac{1}{2}w_1 + \left(\frac{3}{2}w + a\right) \left(\frac{1}{2}u_1 - p_1 + t_2\right) + b \left(\frac{1}{2}u_2 - p_2 - t_1\right), \\ 0 &= b_1 - a_2 + \frac{1}{2}w_2 + b \left(\frac{1}{2}u_1 - p_1 + t_2\right) + \left(\frac{3}{2}w - a\right) \left(\frac{1}{2}u_2 - p_2 - t_1\right). \end{aligned}$$

Substituting (7.13) yields

$$\begin{aligned} a_1 + b_2 &= \frac{3}{2}w(p_1 - t_2) - \frac{1}{2}(au_1 + bu_2), \\ b_1 - a_2 &= \frac{3}{2}w(p_2 + t_1) - \frac{1}{2}(bu_1 - au_2). \end{aligned}$$

With $\zeta = 2(p_1 - t_2 + \sqrt{-1}(p_2 + t_1))$, these combine into

$$(7.15) \quad D_1 q - \sqrt{-1}D_2 q = \frac{3}{4}w\zeta - \frac{1}{2}q(u_1 - \sqrt{-1}u_2).$$

The use of covariant derivatives makes this identity independent of the choice of oriented frames.

To prove (7.11), we use the Weyl Weitzenböck formula. Differentiating $W^+ \omega_0 = -w\omega_0$ twice gives

$$(7.16) \quad \Delta_g w = -\langle (\Delta_g W^+) \omega_0, \omega_0 \rangle + 2 \sum_{\alpha=1}^4 \langle (w \text{Id} + W^+) \nabla_{e_\alpha} \omega_0, \nabla_{e_\alpha} \omega_0 \rangle.$$

The horizontal derivatives of ω_0 vanish, while

$$\begin{aligned} \nabla_{e_3} \omega_0 &= \frac{1}{2}(u_2 + \text{Im } \zeta) \omega_1 - \frac{1}{2}(u_1 + \text{Re } \zeta) \omega_2, \\ \nabla_{e_4} \omega_0 &= \frac{1}{2}(u_1 - \text{Re } \zeta) \omega_1 + \frac{1}{2}(u_2 - \text{Im } \zeta) \omega_2. \end{aligned}$$

Substituting these and (7.13) into the sum in (7.16) gives

$$\begin{aligned} &2 \sum_{\alpha=1}^4 \langle (w \text{Id} + W^+) \nabla_{e_\alpha} \omega_0, \nabla_{e_\alpha} \omega_0 \rangle \\ &= 6w(K + J) - 2\langle du, dw \rangle_g. \end{aligned}$$

The first term on the right of (7.16) is $2\lambda w + 3w^2 - 4|q|^2$. Equation (7.11) now follows from $K + J = \lambda/3 - w$ and

$$\Delta_g w = \Delta_{\underline{g}} w + \langle du, dw \rangle_{\underline{g}}.$$

□

The scalar equation implies the following positivity statement.

Proposition 7.10. *Either $W^+ = 0$ everywhere, or $w > 0$ on B .*

Proof. At an open rod, the operator on the left of (7.11) has the form

$$\partial_{rr} + \frac{3}{r}\partial_r + f(r^2, t)^{-2}\partial_{tt} + rb_1(r^2, t)\partial_r + b_2(r^2, t)\partial_t.$$

On even functions it is the restriction of a smooth uniformly elliptic operator obtained by replacing r with the radial variable in $\mathbb{R}^4$. At a turning point it similarly has the form

$$a \left(\partial_{xx} + \frac{3}{x}\partial_x \right) + c \left(\partial_{yy} + \frac{3}{y}\partial_y \right) + 2xyb\partial_{xy} + xd_1\partial_x + yd_2\partial_y,$$

where the coefficients are smooth in x^2, y^2 . Writing $x = |\mathbf{x}|$ and $y = |\mathbf{y}|$, with $\mathbf{x}, \mathbf{y} \in \mathbb{R}^4$, gives a smooth uniformly elliptic operator; its mixed principal coefficients are $b\mathbf{x}_i\mathbf{y}_j$. The ordinary maximum principle therefore also applies at boundary extrema of w .

At a negative minimum, the left-hand side of (7.11) is nonnegative and the right-hand side is negative. Thus $w \geq 0$. Moreover,

$$\Delta_{\underline{g}}w + 3\langle du, dw \rangle_{\underline{g}} - 4\lambda w = -3w^2 - 4|q|^2 \leq 0.$$

The strong maximum principle gives either $w > 0$ everywhere or $w = 0$ everywhere. In the latter case (7.11) gives $q = 0$, hence $W^+ = 0$. $\square$

Suppose now that $W^+ \neq 0$. By Proposition 7.10, $w > 0$ on B . Using $\Delta_{\underline{g}}u = -2\lambda - 2U$ and $e = K + U/2$, equations (7.10)–(7.11) give

$$(7.17) \quad \Delta_{\underline{g}} \log w = 3(K + J) + \left(\frac{4\lambda}{3} + 2U\right) \left(\frac{9}{4} - \frac{|q|^2}{w^2}\right),$$

$$(7.18) \quad \Delta_{\underline{g}} \log(\rho^{3/2}w) = 3(e + J) - 4\left(\frac{\lambda}{3} + \frac{U}{2}\right) \frac{|q|^2}{w^2}.$$

Lemma 7.11. *On B , one has $|q| \leq 3w/2$.*

Proof. We first derive an equation for $|q|/w$ on $\{q \neq 0\}$. Write $q = |q|e^{\sqrt{-1}\theta}$ locally on this set. Then

$$\frac{D_i q}{q} = e_i(\log |q|) + \sqrt{-1}(d\theta + \vartheta)(e_i).$$

Dividing (7.15) by q and using the conjugate of (7.13), we obtain

$$\frac{D_1 q - \sqrt{-1}D_2 q}{q} = \frac{3w^2}{4|q|^2}(e_1 - \sqrt{-1}e_2) \log(\rho^{3/2}w) - \frac{1}{2}(u_1 - \sqrt{-1}u_2).$$

Taking real and imaginary parts gives

$$\begin{aligned} (d\theta + \vartheta)(e_1) &= e_2(\log |q|) + \frac{1}{2}u_2 - \frac{3w^2}{4|q|^2}e_2 \log(\rho^{3/2}w), \\ (d\theta + \vartheta)(e_2) &= -e_1(\log |q|) - \frac{1}{2}u_1 + \frac{3w^2}{4|q|^2}e_1 \log(\rho^{3/2}w). \end{aligned}$$

Let $\underline{*}$ denote the Hodge star of $\underline{g}$. Since $\underline{*}e^1 = e^2$ and $\underline{*}e^2 = -e^1$, these identities are equivalent to

$$(7.19) \quad d\theta + \vartheta = -\underline{*}d \log |q| - \frac{1}{2}\underline{*}du + \frac{3w^2}{4|q|^2}\underline{*}d \log(\rho^{3/2}w).$$

Taking exterior derivatives and applying (7.12) gives

$$\begin{aligned} \Delta_{\underline{g}} \log \frac{|q|}{w} - \operatorname{div}_{\underline{g}} \left[\left(\frac{3w^2}{4|q|^2} - \frac{1}{3} \right) \nabla_{\underline{g}} \log(\rho^{3/2}w) \right] \\ = 2(K + J) - \frac{2}{3}\Delta_{\underline{g}} \log w \\ = -\frac{8}{3} \left(\frac{\lambda}{3} + \frac{U}{2} \right) \left(\frac{9}{4} - \frac{|q|^2}{w^2} \right). \end{aligned}$$

Expanding the divergence and using (7.18) gives

$$\begin{aligned} (7.20) \quad \Delta_{\underline{g}} \log \frac{|q|}{w} + \frac{3w^2}{2|q|^2} \left\langle d \log(\rho^{3/2}w), d \log \frac{|q|}{w} \right\rangle_{\underline{g}} \\ = \left(\frac{9}{4} - \frac{|q|^2}{w^2} \right) \left(\frac{(e + J)w^2}{|q|^2} - 4 \left(\frac{\lambda}{3} + \frac{U}{2} \right) \right). \end{aligned}$$

By (7.7), the ratio equals $3/2$ on ∂B . If it had a larger maximum, that maximum would occur in $B^\circ \cap \{q \neq 0\}$. The left-hand side of (7.20) would be nonpositive there, whereas both factors on the right would be negative. For the second factor, use $e + J \geq 0$ and

$$\frac{(e + J)w^2}{|q|^2} - 4\left(\frac{\lambda}{3} + \frac{U}{2}\right) = (e + J)\left(\frac{w^2}{|q|^2} - 4\right) - 4w < 0.$$

This is a contradiction. $\square$

Proof of Theorem 7.1. Suppose first that $W^+ \not\equiv 0$. The function w is positive and smooth on the closed quotient. Its evenness gives

$$\int_B \Delta_{\underline{g}} \log w dA_{\underline{g}} = 0.$$

Indeed, the normal derivative vanishes on open rods, and the flux over a corner arc of radius r is $O(r^2)$. Integrating (7.17) therefore gives

$$(7.21) \quad \int_B \left(\frac{4\lambda}{3} + 2U\right) \left(\frac{9}{4} - \frac{|q|^2}{w^2}\right) dA_{\underline{g}} = 3\pi(b_2^+(X_{\mathfrak{R}}) - 1).$$

The integrand is nonnegative by Lemma 7.11. It is also integrable: its second factor is $O(r^2)$ on a rod and $x^2 y^2$ times a smooth function at a turning point, while $U = O(x^{-2} + y^{-2})$ there. If $b_2^+(X_{\mathfrak{R}}) = 0$, (7.21) is impossible. Hence $W^+ = 0$ in that case.

If $b_2^+(X_{\mathfrak{R}}) = 1$, the case $W^+ = 0$ is excluded by the Weitzenböck formula for harmonic self-dual two-forms,

$$0 = \nabla_g^* \nabla_g \eta + \frac{4\lambda}{3} \eta - 2W^+(\eta).$$

Thus (7.21) applies and forces $|q| = 3w/2$. By (7.5),

$$\text{spec}(W^+) = \{-w, -w, 2w\}, \quad w > 0.$$

Derdziński's theorem [23, Proposition 5(i), pp. 420–421] implies that $w^{2/3}g$ is locally Kähler. Since $X_{\mathfrak{R}}$ is simply connected, the simple eigenline of W^+ admits a global unit section, and the local Kähler forms agree after a choice of sign. This proves the global conclusion. $\square$

7.2. An improved Hitchin–Thorpe inequality and topological constraint. As an application of the preceding results, we obtain the following improvement of the Hitchin–Thorpe inequality and the restriction on self-intersection numbers for compact toric Einstein metrics.

Theorem 7.12 (Improved Hitchin–Thorpe inequality). *Let $(X_{\mathfrak{R}}, g)$ be a compact toric Einstein four-manifold. Then*

$$(7.22) \quad \chi(X_{\mathfrak{R}}) \geq 3|\tau(X_{\mathfrak{R}})|.$$

Equality holds if and only if all rod self-intersections have the same value $\varepsilon \in \{-1, 1\}$. Furthermore, this inequality is sharp.

We rescale g so that $\text{Ric}(g) = 3g$. We first constrain the self-intersection numbers of the rod spheres.

Lemma 7.13. *For every rod sphere Σ_i , write $e_i = [\Sigma_i]^2$. Then*

$$(7.23) \quad |e_i| + \frac{\text{Area}_g(\Sigma_i)}{2\pi} \leq 2.$$

In particular, $e_i \in \{-1, 0, 1\}$.

Proof. Let w_+ be the function w of Section 7.1, and let w_- be its counterpart for the opposite orientation. Proposition 7.10, applied in both orientations, gives $w_{\pm} \geq 0$. This includes the half-conformally flat cases, when the corresponding function vanishes identically.

The sphere Σ_i is totally geodesic. At a point over the interior of its rod, let σ and ν be the unit tangent and normal area forms, with $*\sigma = \nu$. Then $(\sigma \pm \nu)/\sqrt{2}$ are eigenforms of $W^{\pm}$ with

eigenvalues $2w_{\pm}$, respectively. Since the Einstein curvature operator on $\Lambda^{\pm}$ is $W^{\pm} + \text{Id}$, these identities give

$$\text{Rm}_g(\sigma) = (1 + w_+ + w_-)\sigma + (w_+ - w_-)\nu.$$

Denote by K_i the intrinsic Gaussian curvature of Σ_i and by $K_i^{\perp}$ the scalar curvature of its oriented normal connection. The Gauss and Ricci equations give

$$(7.24) \quad K_i = 1 + w_+ + w_-, \quad |K_i^{\perp}| = |w_+ - w_-| \leq K_i - 1.$$

The normal Euler-number formula and Gauss–Bonnet therefore yield

$$\begin{aligned} 2\pi|e_i| &= \left| \int_{\Sigma_i} K_i^{\perp} dA_g \right| \\ &\leq \int_{\Sigma_i} |K_i^{\perp}| dA_g \\ &\leq 4\pi - \text{Area}_g(\Sigma_i). \end{aligned}$$

This proves (7.23). The area is positive, so $|e_i| < 2$, and the integrality of e_i gives the last assertion. $\square$

Proof of Theorem 7.12. Recall that

$$\chi(X_{\mathfrak{R}}) = m, \quad 3\tau(X_{\mathfrak{R}}) = \sum_{i=1}^m e_i.$$

By Lemma 7.13, each e_i belongs to $\{-1, 0, 1\}$. Consequently,

$$3|\tau(X_{\mathfrak{R}})| = \left| \sum_{i=1}^m e_i \right| \leq \sum_{i=1}^m |e_i| \leq m = \chi(X_{\mathfrak{R}}).$$

Equality in the last inequality requires $|e_i| = 1$ for every i , while equality in the triangle inequality then requires all e_i to have the same sign. Conversely, these conditions imply equality. Finally, the Fubini–Study metric on $\mathbb{CP}^2$ is toric Einstein and satisfies $\chi(\mathbb{CP}^2) = 3$ and $\tau(\mathbb{CP}^2) = 1$, proving sharpness. $\square$

Corollary 7.14. *The connected sum*

$$X_{\mathfrak{R}} = p\mathbb{CP}^2 \#_q \overline{\mathbb{CP}^2} \#_r (\mathbb{S}^2 \times \mathbb{S}^2)$$

admits no toric Einstein metric if

$$\max\{p, q\} > 2\min\{p, q\} + r + 1.$$

Here p, q, r are nonnegative integers.

Proof. Since

$$\chi(X_{\mathfrak{R}}) = 2 + p + q + 2r, \quad \tau(X_{\mathfrak{R}}) = p - q,$$

the improved Hitchin–Thorpe inequality (7.22) requires

$$2 + p + q + 2r \geq 3|p - q|,$$

or equivalently $\max\{p, q\} \leq 2\min\{p, q\} + r + 1$. $\square$

Remark 7.15. *The individual self-intersection bounds give a stronger obstruction for a prescribed rod structure than the improved Hitchin–Thorpe inequality. For example, the standard toric action on the Hirzebruch surface $\mathbb{F}_2$ has rod self-intersections $(2, 0, -2, 0)$. Although $\chi = 4$ and $\tau = 0$ satisfy the improved Hitchin–Thorpe inequality, this action admits no invariant Einstein metric. Its underlying manifold is $\mathbb{S}^2 \times \mathbb{S}^2$, which admits the round product Einstein metric with a different torus action.*

Corollary 7.16. *If a compact rod structure $\mathfrak{R}$ has a rod sphere with $|e_i| \geq 2$, then $X_{\mathfrak{R}}$ admits no Einstein metric invariant under its prescribed $\mathbb{T}^2$ action. Moreover, the manifold $p\mathbb{CP}^2 \# q\overline{\mathbb{CP}^2}$ admits no toric Einstein metric whenever*

$$\max\{p, q\} > 2\min\{p, q\} + 1.$$

In particular, $\mathbb{CP}^2 \# k\overline{\mathbb{CP}^2}$ admits no toric Einstein metric for $k \geq 4$.

Proof. The first assertion follows from Lemma 7.13. For the second, the identities

$$\chi = 2 + p + q, \quad \tau = p - q$$

and Theorem 7.12 give

$$2 + p + q \geq 3|p - q|,$$

which is equivalent to $\max\{p, q\} \leq 2\min\{p, q\} + 1$. $\square$

7.3. Rigidity in the Hermitian case. We apply the Einstein–Hilbert functional (2.3) to conformally Kähler metrics. The integrated boundary fluxes determine the moment polygon and the conformal factor. Uniqueness follows from strict concavity of the square root of the volume as a function of these fluxes.

Proposition 7.17. *Let $X_{\mathfrak{R}}$ be one of the complex surfaces $\mathbb{CP}^2 \# k\overline{\mathbb{CP}^2}$, $0 \leq k \leq 3$, or $\mathbb{S}^2 \times \mathbb{S}^2$, with its standard complex structure $\mathcal{J}$ and $\mathbb{T}^2$ -action. A toric Einstein metric on $X_{\mathfrak{R}}$ that is Hermitian with respect to $\mathcal{J}$ is unique up to scaling and isometry.*

By the conformally Kähler reduction in [32, Proposition 1], we can write $g = f^{-2}h$, where h is a $\mathbb{T}^2$ -invariant extremal Kähler metric in the sense of Calabi. Its scalar curvature is affine in moment coordinates. In the non-Kähler case, f is a positive constant multiple of Scal_h , and hence is a positive affine function on the moment polygon P . In the Kähler case, take $h = g$ and $f = 1$.

Choose angular coordinates of period 2π and write

$$\begin{aligned} h &= H_{ij} dx_i dx_j + H^{ij} d\theta_i d\theta_j, \\ H_{ij} &= \partial_i \partial_j \psi, \end{aligned}$$

where $(H^{ij}) = H^{-1}$ and ψ is a symplectic potential.

Let E_i be the edge of P corresponding to $\mathcal{J}_i$, with endpoints z_i, z_{i+1} and primitive outward normal n_i . Write

$$P = \{x : \langle n_i, x \rangle \leq b_i\}, \quad z_{i+1} - z_i = L_i t_i,$$

where t_i is the primitive tangent in the positive boundary direction. The affine boundary measure $d\sigma$ satisfies $\int_{E_i} d\sigma = L_i$.

Using the notation of Section 2, define the integrated boundary fluxes and normalized volume by

$$\begin{aligned} M_i &= - \int_{\mathcal{J}_i} \partial_n \rho ds_g = \frac{\text{Area}_g(\Sigma_i)}{2\pi} \\ &= \int_{E_i} f^{-2} d\sigma = \frac{L_i}{f(z_i)f(z_{i+1})}, \\ M_{\text{tot}} &= \sum_i M_i, \\ V &= \int_P f^{-4} dx = \frac{\text{Vol}_g(X_{\mathfrak{R}})}{(2\pi)^2}. \end{aligned} \tag{7.25}$$

Here dx denotes the affine area measure on P , n is the outward g -unit normal, and the area identity follows from (2.12). We identify (h, f) with $(s^2 h, s f)$ for $s > 0$, since they define the same metric $g = f^{-2}h$.

We now show that the integrated boundary fluxes give smooth coordinates for the pairs (P, f) .

Lemma 7.18. *With the primitive outward normals n_i fixed, the positive vectors $\mathbf{M} = (M_1, \dots, M_m)$ parametrize the pairs (P, f) , up to translation and the preceding rescaling. This parametrization and $V(\mathbf{M})$ are smooth.*

Proof. The divergence theorem gives

$$(7.26) \quad \sum_i M_i n_i = -2 \nabla f \int_P f^{-3} dx.$$

If the sum vanishes, f is constant. Taking $f = 1$ gives $L_i = M_i$, which determines a convex polygon up to translation. Otherwise the sum determines the direction and sign of ∇f . Use the rescaling to fix its magnitude, and choose oriented affine coordinates in which $f = y$. Set $r_i = 1/y(z_i)$. Then

$$r_{i+1} - r_i = -M_i(t_i)_y, \quad r_i = r + c_i,$$

and we have

$$L_i(r) = \frac{M_i}{(r + c_i)(r + c_{i+1})},$$

where the c_i are determined by $\mathbf{M}$. The sum of the edge displacements in the y -direction vanishes by telescoping. To obtain a closed polygon, we must also require that their sum in the x -direction vanish:

$$F(r) = \sum_i \frac{M_i(t_i)_x}{(r + c_i)(r + c_{i+1})} = 0.$$

At every zero, the positive edge lengths give a convex polygon, and

$$\begin{aligned} F'(r) &= - \sum_i L_i(t_i)_x (y(z_i) + y(z_{i+1})) \\ &= -2 \int_{\partial P} y dx = 2 \text{Area}(P) > 0. \end{aligned}$$

Thus there is at most one zero. With the direction fixed by (7.26),

$$\sum_i M_i(t_i)_x > 0,$$

so $F(r) > 0$ for large r . At the other endpoint, a horizontal upper edge contributes a negative term of order $(r + \min c_i)^{-2}$. If there is no such edge, the two edges meeting the upper vertex contribute a term of order $(r + \min c_i)^{-1}$ with coefficient

$$\frac{(t_-)_x}{(t_-)_y} - \frac{(t_+)_x}{(t_+)_y} = \frac{\det(t_-, t_+)}{(t_-)_y(t_+)_y} < 0.$$

Hence a zero exists. This also proves smooth dependence when $\nabla f \neq 0$.

At $f = 1$, fix one vertex at the origin and normalize $f(0) = 1$. For $\dot{f} = \xi \cdot x$, the differential of the flux map is

$$\begin{aligned} \dot{M}_i &= \dot{L}_i - L_i \xi \cdot (z_i + z_{i+1}), \\ \sum_i \dot{M}_i n_i &= -2 \text{Area}(P) \xi. \end{aligned}$$

These equations determine ξ and then every $\dot{L}_i$. The inverse function theorem therefore gives smoothness also at constant f . $\square$

We consider polygons with the prescribed outward normals in each of the five families above. We first express the volume functional in terms of quadratic forms in the fluxes.

Fix a polygon P , translate its centroid to the origin, and set $A = \text{Area}(P)$. Write L_i for its edge coefficients and t_i for the primitive tangents in positive boundary order. Thus

$$z_{i+1} - z_i = L_i t_i, \quad \sum_i L_i n_i = \sum_i L_i t_i = 0.$$

A general flux vector $\mathbf{M}$ does not satisfy this closure condition. We first separate the part that does. For a constant vector $\xi \in \mathbb{R}^2$, define

$$(T_P \xi)_i = -2 \int_{E_i} \xi \cdot x d\sigma.$$

The divergence theorem gives

$$\sum_i (T_P \xi)_i n_i = -2 \int_P \nabla(\xi \cdot x) dx = -2A\xi.$$

Consequently, there is a unique decomposition

$$\mathbf{M} = \eta + T_P \xi, \quad \sum_i \eta_i n_i = 0,$$

given explicitly by

$$\xi = -\frac{1}{2A} \sum_i M_i n_i, \quad \eta = \mathbf{M} - T_P \xi.$$

For fixed P , both η and ξ depend linearly on $\mathbf{M}$.

We now define the area associated with η . Since $\sum_i \eta_i t_i = 0$, the vectors $\eta_i t_i$, placed successively, form a closed polygon. Its signed area is

$$A(\eta) = \frac{1}{2} \sum_{i < j} \eta_i \eta_j \det(t_i, t_j).$$

Indeed, with the first vertex at the origin, the i -th vertex is $\sum_{j < i} \eta_j t_j$; summing the signed triangle areas gives the displayed formula. When all $\eta_i > 0$, this is the ordinary area of the convex polygon with these edges. The formula extends it quadratically to all vectors satisfying the closure condition.

For $L = (L_i)$, we have $A(L) = A$. Define the mixed area by

$$A(P, \eta) = \frac{1}{2} (A(L + \eta) - A(L) - A(\eta)).$$

If the edge equations of P are $\langle n_i, x \rangle = b_i$, then

$$2A(P, \eta) = \sum_i b_i \eta_i.$$

To see this, let P_s have edge coefficients $L_i + s\eta_i$ and support numbers $b_i(s)$. The first variation of area is

$$\left. \frac{d}{ds} \text{Area}(P_s) \right|_{s=0} = \sum_i L_i \dot{b}_i.$$

Differentiating therefore shows that this same derivative equals $\sum_i b_i \eta_i$. By polarization, it is also $2A(P, \eta)$.

For the decomposition above, define

$$(7.27) \quad Q_P(\mathbf{M}) = 2A(\eta) + \frac{4A(P, \eta)^2}{A} - 12 \int_P (\xi \cdot x)^2 dx.$$

For fixed P , this is a quadratic form in $\mathbf{M}$. The centroid condition removes the choice of translation, and positive homotheties of P leave Q_P unchanged.

Lemma 7.19. *For every vector $\mathbf{M}$ with $M_i > 0$, we have*

$$(7.28) \quad 6V(\mathbf{M}) = \min_P Q_P(\mathbf{M}).$$

The minimum is attained at the polygon associated with $\mathbf{M}$ by Lemma 7.18. Moreover, each Q_P has signature $(1, m-1)$ and is positive whenever $M_i > 0$ for all i .

Proof. We first evaluate Q_P when $\mathbf{M}$ is the flux vector of a pair (P, f) :

$$M_i = \int_{E_i} f^{-2} d\sigma, \quad V = \int_P f^{-4} dx.$$

Suppose first that f is nonconstant. Rotate coordinates so that $f = c + py$, with $p \neq 0$, and set

$$I_3 = \int_P f^{-3} dx dy, \quad \kappa = \frac{pI_3}{A}.$$

Since

$$\sum_i M_i n_i = \int_P \nabla(f^{-2}) dx dy = -2I_3 \nabla f,$$

the preceding decomposition gives

$$\xi = \kappa \partial_y, \quad \eta_i = \int_{E_i} (f^{-2} + 2\kappa y) d\sigma.$$

Set

$$a_0(y) = f^{-2} + 2\kappa y, \quad a_1(y) = -\frac{1}{pf} + \kappa y^2.$$

Then $a'_1 = a_0$. Integrating $a_0(y)(dx, dy)$ along ∂P gives a closed curve with edge vectors $\eta_i t_i$. Each edge is straight, and its vertical coordinate is $a_1(y)$ up to an additive constant. Its signed area is therefore

$$A(\eta) = - \int_{\partial P} a_1(y) a_0(y) dx.$$

Green's formula and direct differentiation give

$$\begin{aligned} A(\eta) &= \int_P (a_0^2 + a_1 a'_0) dx dy \\ &= \int_P \left(3f^{-4} - \frac{2\kappa c^2}{p} f^{-3} + 6\kappa^2 y^2 \right) dx dy \\ &= 3V - \frac{2c^2 I_3^2}{A} + 6\kappa^2 \int_P y^2 dx dy. \end{aligned}$$

The mixed area is equally explicit. Using the support number formula and then the divergence theorem,

$$\begin{aligned} 2A(P, \eta) &= \int_P \operatorname{div}(a_0(y)(x, y)) dx dy \\ &= 2cI_3 + 6\kappa \int_P y dx dy = 2cI_3, \end{aligned}$$

where the last equality uses the centroid condition.

Substituting these two identities into (7.27) gives

$$(7.29) \quad Q_P(\mathbf{M}) = 6 \int_P f^{-4} dx$$

when $\mathbf{M}$ is the flux vector of (P, f) . If $f = c$ is constant, then $\xi = 0$ and $\eta = c^{-2}L$, so the same identity follows immediately. Thus equality holds at the polygon associated with $\mathbf{M}$. To prove the minimum formula, we must compare this value with those of the other polygons. This step uses the five polygon families in the statement.

For the hexagon, choose the translate

$$\begin{aligned} P &= \operatorname{conv}\{(c, 0), (a, 0), (a, b-d), (a-d, b), (0, b), (0, c)\}, \\ \min\{a, b\} &> c, d > 0. \end{aligned}$$

Number its edges from 0 to 5. Then

$$\begin{aligned} A &= ab - \frac{c^2 + d^2}{2}, \\ (L_0, \dots, L_5) &= (a-c, b-d, d, a-d, b-c, c). \end{aligned}$$

Although this polygon need not have centroid zero, we compute Q_P after centering, as above.

We evaluate (7.27) using

$$\int_{E_i} x d\sigma = \frac{L_i}{2}(z_i + z_{i+1}).$$

The interior moments are obtained from the rectangle $[0, a] \times [0, b]$ by subtracting the two corner triangles with leg lengths c and d . Substitution gives

$$(7.30) \quad \begin{aligned} Q_P(\mathbf{M}) &= Q_0(\mathbf{M}) + \frac{N_{\mathbf{M}}(a, b, c, d)}{A}, \\ Q_0(\mathbf{M}) &= \sum_{i < j} M_i M_j - M_0 M_3 - M_1 M_4 - M_2 M_5, \end{aligned}$$

where

$$(7.31) \quad \begin{aligned} N_{\mathbf{M}} &= (b-d)^2 M_0 M_2 + b^2 M_0 M_3 + c^2 M_0 M_4 + d^2 M_1 M_3 + a^2 M_1 M_4 \\ &+ (a-c)^2 M_1 M_5 + (a-d)^2 M_2 M_4 + (a+b-c-d)^2 M_2 M_5 + (b-c)^2 M_3 M_5. \end{aligned}$$

For fixed $\mathbf{M}$, the term $Q_0(\mathbf{M})$ is independent of the polygon. Thus minimizing Q_P amounts to minimizing $N_{\mathbf{M}}/A$.

At the polygon associated with $\mathbf{M}$, this quotient is stationary:

$$(7.32) \quad d_{(a,b,c,d)} N_{\mathbf{M}} = \frac{N_{\mathbf{M}}}{A} d_{(a,b,c,d)} A.$$

Here $\mathbf{M}$ is held fixed when differentiating. The identity can be checked directly from (7.31). For example,

$$\frac{1}{2} \partial_a N_{\mathbf{M}} = a M_1 M_4 + (a-c) M_1 M_5 + (a-d) M_2 M_4 + (a+b-c-d) M_2 M_5.$$

Substitute

$$M_i = \frac{L_i}{f(z_i) f(z_{i+1})}, \quad f(x, y) = r + sx + ty.$$

This gives $\partial_a N_{\mathbf{M}} = b N_{\mathbf{M}}/A$. The other three derivatives give

$$\begin{aligned} \partial_b N_{\mathbf{M}} &= \frac{a N_{\mathbf{M}}}{A}, \\ \partial_c N_{\mathbf{M}} &= -\frac{c N_{\mathbf{M}}}{A}, \\ \partial_d N_{\mathbf{M}} &= -\frac{d N_{\mathbf{M}}}{A}. \end{aligned}$$

These identities are verified by multiplying by $\prod_{i=0}^5 f(z_i)$ and expanding; every denominator then cancels.

We next show that this stationary point is the global minimum. For $M_i > 0$, the quadratic form $N_{\mathbf{M}}$ is positive definite: its displayed terms include positive multiples of a^2, b^2, c^2, d^2 . In contrast,

$$A = \left(\frac{a+b}{2}\right)^2 - \left(\frac{a-b}{2}\right)^2 - \frac{c^2 + d^2}{2}$$

has one positive and three negative directions. Since the quotient is unchanged by scaling, normalize $N_{\mathbf{M}} = 1$. Minimizing $N_{\mathbf{M}}/A$ for $A > 0$ is then equivalent to maximizing A on this ellipsoid. After a linear change of coordinates, the ellipsoid is a unit sphere and the quadratic form A still has exactly one positive eigenvalue. Hence it has only one positive critical value, which is its global maximum. There is therefore a unique minimizing ray in the component containing the polygons.

Equation (7.32) identifies this ray with the polygon associated with $\mathbf{M}$. Its value is $6V(\mathbf{M})$ by (7.29), proving (7.28) for the hexagon.

The other four polygon families are obtained by the following restrictions. The flux of every omitted edge is set to zero.

Polygon	Restrictions	Area	Omitted edges
Pentagon	$d = 0$	$ab - c^2/2$	2
Rectangle	$c = d = 0$	ab	2, 5
Trapezoid	$d = 0, b = c$	$ab - b^2/2$	2, 4
Triangle	$d = 0, a = b = c$	$a^2/2$	0, 2, 4

The stationarity identity restricts to each of these families. For the pentagon, the numerator still contains positive multiples of a^2, b^2, c^2 . For the rectangle and trapezoid, respectively, it becomes

$$b^2 M_0 M_3 + a^2 M_1 M_4, \quad b^2 M_0 M_3 + (a - b)^2 M_1 M_5.$$

These are positive definite in their two independent parameters. The corresponding area forms have one positive direction and all remaining directions negative, so the same minimization argument applies. For the triangle, the numerator vanishes and

$$Q_P(\mathbf{M}) = \sum_{i < j} M_i M_j$$

is independent of the polygon scale. Thus (7.28) holds in all five cases.

Finally, we check the signature and positivity of Q_P . The edge coefficients are linear coordinates subject to the two closure conditions. The area formulas above therefore show that $A(\eta)$ has signature $(1, m - 3)$ on this $(m - 2)$ -dimensional space. Decompose

$$\eta = \frac{A(P, \eta)}{A} L + \eta_0, \quad A(P, \eta_0) = 0.$$

Since $A(L) = A > 0$, the area form is negative definite on the subspace containing η_0 . Formula (7.27) becomes

$$Q_P(\mathbf{M}) = \frac{6A(P, \eta)^2}{A} + 2A(\eta_0) - 12 \int_P (\xi \cdot x)^2 dx.$$

The last term is negative for every nonzero ξ . Thus Q_P has exactly one positive and $m - 1$ negative directions.

In (7.30), every coefficient of $M_i M_j$, $i < j$, is positive. Hence the vectors $\mathbf{M}$ with $M_i > 0$ lie in one component of $\{Q_P > 0\}$. $\square$

Proposition 7.20. *The function $\sqrt{V(\mathbf{M})}$ is strictly concave on*

$$\{M_i > 0 : \sum_i M_i = 1\}.$$

Proof. By Lemma 7.19 and the reverse Cauchy–Schwarz inequality, each $\sqrt{Q_P}$ is concave on the set $M_i > 0$, with strict inequality between nonproportional vectors.

Let $\mathbf{M}_0, \mathbf{M}_1$ be distinct vectors in the displayed simplex, and set $\mathbf{M}_t = (1 - t)\mathbf{M}_0 + t\mathbf{M}_1$, where $0 < t < 1$. Choose a polygon P attaining the minimum for $\mathbf{M}_t$. Since $\mathbf{M}_0$ and $\mathbf{M}_1$ are not proportional,

$$\begin{aligned} \sqrt{6V(\mathbf{M}_t)} &= \sqrt{Q_P(\mathbf{M}_t)} \\ &> (1 - t)\sqrt{Q_P(\mathbf{M}_0)} + t\sqrt{Q_P(\mathbf{M}_1)} \\ &\geq (1 - t)\sqrt{6V(\mathbf{M}_0)} + t\sqrt{6V(\mathbf{M}_1)}. \end{aligned}$$

This proves the proposition. $\square$

Proof of Proposition 7.17. The conformal scalar-curvature formula, with f affine, is

$$(7.33) \quad \text{Scal}_g = -f^5 \partial_i \partial_j (f^{-3} H^{ij}).$$

At a facet $\ell_i = b_i - \langle n_i, x \rangle = 0$, the potential has singular part $\frac{1}{2} \ell_i \log \ell_i$. Integration by parts therefore gives

$$(7.34) \quad \frac{1}{(2\pi)^2} \int_{X_{\text{gr}}} \text{Scal}_g \, d\text{vol}_g = 2M_{\text{tot}}.$$

At an Einstein metric, $M_{\text{tot}} = 2\lambda V$, and the first variation of the Einstein–Hilbert functional along the metrics $f^{-2}h$ gives

$$d(2M_{\text{tot}} - 2\lambda V) = 0, \quad d\left(\frac{M_{\text{tot}}^2}{V}\right) = 0.$$

These variations allow arbitrary nearby polygons and positive affine functions. By Lemma 7.18, they allow every flux variation. After scaling so that $M_{\text{tot}} = 1$, each Einstein metric is therefore a critical point of $\sqrt{V}$. Proposition 7.20 shows that this critical point is unique.

The two metrics now have the same flux vector, so Lemma 7.18 identifies their pairs (P, f) . Equation (7.34) also identifies their Einstein constants. Let H_0, H_1 be the Hessians of their symplectic potentials. Subtract (7.33), multiply by $\psi_1 - \psi_0$, and integrate twice by parts. The difference of the potentials is smooth on P , and the common facet singularities cancel the boundary terms. We obtain

$$\begin{aligned} 0 &= \int_P f^{-3}(H_1 - H_0) : (H_1^{-1} - H_0^{-1}) dx \\ &= - \int_P f^{-3} |H_1^{-1/2}(H_1 - H_0)H_0^{-1/2}|^2 dx. \end{aligned}$$

Thus $H_1 = H_0$, and the metrics agree. $\square$

We now prove infinitesimal rigidity.

Proof of Theorem 7.6. Let $k \in K_g^{\mathbb{T}^2}$. For the round sphere,

$$0 = \int_{X_{\mathfrak{R}}} \left(|\nabla_g k|_g^2 + \frac{2\lambda}{3} |k|_g^2 \right) d\text{vol}_g,$$

so $k = 0$. We consider the remaining metrics in their complex orientations, and write $\mathcal{J}$ for the corresponding complex structure.

We first represent k by a variation of (P, f, ψ) . Applied to $g + tk$, [2, Lemma 3.1] shows that the conformally Kähler condition holds to first order: the divergence terms integrate to zero on $X_{\mathfrak{R}}$, and the remaining terms are nonnegative squares. Since $H^1(X_{\mathfrak{R}}, T_{\mathcal{J}}^{1,0} X_{\mathfrak{R}}) = 0$, an equivariant infinitesimal diffeomorphism removes the complex-structure variation. The anti-Hermitian part of the linearized conformal Ricci equation then makes $\mathcal{J}\nabla_h f$ Killing for h to first order, which in moment coordinates says that f is affine to first order. Thus, modulo equivariant diffeomorphisms, k is represented by a curve (P_t, f_t, ψ_t) of the data used above.

Let $v = \dot{\mathbf{M}}$ be its flux variation. By (7.34), the Einstein–Hilbert functional with fixed λ is

$$\frac{1}{(2\pi)^2} \int_{X_{\mathfrak{R}}} (\text{Scal}_g - 2\lambda) d\text{vol}_g = 2M_{\text{tot}} - 2\lambda V.$$

Every flux variation is realized by Lemma 7.18, so $2M_{\text{tot}} - 2\lambda V$ is critical at $\mathbf{M}$. Its Hessian vanishes on the variation induced by k , because k solves the linearized Einstein equation. Since $M_{\text{tot}} = \sum_i M_i$ is linear,

$$0 = D^2(2M_{\text{tot}} - 2\lambda V)_{\mathbf{M}}(v, v) = -2\lambda D^2 V_{\mathbf{M}}(v, v).$$

Also, $\text{tr}_g k = 0$ gives $DV_{\mathbf{M}}[v] = 0$.

Fix the polygon P associated with $\mathbf{M}$. By (7.28) and (7.29),

$$Q_P(\mathbf{N}) \geq 6V(\mathbf{N}), \quad Q_P(\mathbf{M}) = 6V(\mathbf{M}) > 0.$$

Thus $Q_P - 6V$ has a minimum at $\mathbf{M}$, so $DQ_P(\mathbf{M})[v] = 6DV_{\mathbf{M}}[v] = 0$: the vector v is orthogonal to $\mathbf{M}$ for the bilinear form associated with Q_P . Since Q_P has signature $(1, m-1)$, it is negative definite on this orthogonal complement. On the other hand, differentiating the preceding inequality twice gives

$$0 \leq D^2(Q_P - 6V)_{\mathbf{M}}(v, v) = 2Q_P(v).$$

These two facts force $v = 0$.

By Lemma 7.18, we may now choose representatives with $\dot{P} = \dot{f} = 0$. The linearized scalar-curvature equation becomes

$$\partial_i \partial_j \left(f^{-3} H^{ia} \dot{\psi}_{ab} H^{bj} \right) = 0.$$

Multiplying by $\dot{\psi}$ and integrating twice by parts gives

$$0 = \int_P f^{-3} H^{ia} H^{bj} \dot{\psi}_{ij} \dot{\psi}_{ab} dx.$$

Here $\dot{\psi}$ is smooth up to the boundary, and the boundary terms vanish by the same fixed-facet calculation used in the uniqueness proof. Positivity implies $D^2 \dot{\psi} = 0$, so this variation does not change the metric.

Consequently, the original deformation is $k = \mathcal{L}_Z g$. Since $\delta_g k = 0$,

$$\int_{X_{\mathfrak{R}}} |k|_g^2 d\text{vol}_g = 2 \int_{X_{\mathfrak{R}}} \langle \delta_g k, Z \rangle_g d\text{vol}_g = 0.$$

Thus $k = 0$. □

8. PROOF OF THEOREM 1.7 AND THEOREM 1.1

For a gluing family g_a from Theorem 5.8, define its sign to be the boundary sign of $g_\eta \in \partial \mathcal{M}_\eta$, for sufficiently small $\eta > 0$. The outward direction is $-\partial_a$.

Lemma 8.1. *There is a sign $\epsilon_W \in \{\pm 1\}$, depending only on W , such that for every torically nondegenerate compact Einstein metric g_0 , the metric g_η constructed in Theorem 5.8 satisfies*

$$\text{sign}_{\partial \mathcal{M}_\eta}(g_\eta) = \epsilon_W (-1)^{\mu(g_0)}$$

for all sufficiently small $\eta > 0$.

Proof. We compute the orientation of the area coordinate by adding the area equation to the linearized Einstein equation, as in (5.11). The estimates in the proof of Theorem 5.8 show that, at the glued solution, $D\mathcal{F}_\varepsilon$ differs from $\mathcal{L}_\varepsilon$ by $O(\varepsilon^{1/3})$ in operator norm. Proposition 5.9 therefore joins these operators through invertible augmented operators. It suffices to compute the sign using $\mathcal{L}_\varepsilon$.

We compare the operator on (X, b_ε) with the operators on the compact Einstein manifold (M, g_0) and the complete Ricci-flat space (Y, h) . All model operators use the weighted spaces of Proposition 5.9.

Let E_0 be the sum of the negative eigenspaces of L_{g_0} , so that $\dim E_0 = \mu(g_0)$, and let ζ be the decaying infinitesimal deformation obtained by rescaling h , expressed in divergence gauge and normalized by

$$dA_h(\zeta) = 2A_*.$$

By Proposition B.7, $\ker \mathcal{P}_h = \mathbb{R}\zeta$ in the weighted space used here. Set

$$B_h = D\mathcal{E}_{0,h}(\dot{b}).$$

By Lemma B.6, B_h has nonzero pairing with the model cokernel. Consequently the operators

$$T_0(u, e) = \mathcal{P}_{g_0} u + e, \quad e \in E_0, \quad T_h(v, t) = \mathcal{P}_h v + tB_h, \quad t \in \mathbb{R},$$

are surjective, with

$$K_0 := \ker T_0 = \{(u, -L_{g_0} u) : u \in E_0\}, \quad K_h := \ker T_h = \mathbb{R}(\zeta, 0).$$

Choose the right inverse of T_0 by requiring $u \perp_{L^2(g_0)} E_0$, and the right inverse of T_h by requiring $dA_h(v) = 0$.

Using the gluing identifications, cut off and rescale the tensors in E_0 and the tensor B_h to obtain tensors on X . Denote the resulting linear map by

$$j_\varepsilon : E_0 \oplus \mathbb{R} \longrightarrow Y_\varepsilon.$$

and define

$$T_\varepsilon(k, e, t) = \mathcal{P}_{b_\varepsilon} k + j_\varepsilon(e, t), \quad K_\varepsilon = \ker T_\varepsilon.$$

Patching the two right inverses with logarithmic cutoffs gives an approximate right inverse Q_ε^0 satisfying

$$T_\varepsilon Q_\varepsilon^0 = I + \mathcal{R}_\varepsilon.$$

The first and second scaled derivatives of the cutoffs are bounded by $C/\log R$. The weighted estimates of Proposition 5.9, together with (5.9) and (5.8), therefore give

$$\|\mathcal{R}_\varepsilon\| \leq C((\log R)^{-1} + R^{-2}) + o_\varepsilon(1).$$

Choose R large and then ε small so that this norm is less than one. The convergent geometric series

$$Q_\varepsilon = Q_\varepsilon^0 \sum_{m=0}^{\infty} (-\mathcal{R}_\varepsilon)^m$$

then satisfies $T_\varepsilon Q_\varepsilon = I$.

Let C_ε be the map that cuts off and transfers elements of $K_0 \oplus K_h$ to the domain of T_ε . The actual comparison of kernels is

$$\Phi_\varepsilon = (I - Q_\varepsilon T_\varepsilon) C_\varepsilon : K_0 \oplus K_h \longrightarrow K_\varepsilon.$$

The same estimates show that Φ_ε is a small perturbation of the injective cutoff map C_ε , and hence is injective. Since $\mathcal{P}_{b_\varepsilon}$ has index zero and T_ε is surjective,

$$\dim K_\varepsilon = \dim E_0 + 1 = \dim(K_0 \oplus K_h).$$

Thus Φ_ε is an isomorphism. The absence of additional modes in the intermediate region is the weight -1 removability and Liouville argument in Proposition 5.9.

Under this identification, adding the angle variable and the area equation gives a finite-dimensional reduction arbitrarily close to the direct sum of the compact spectral reduction and the model block computed below. The small error can be deformed to zero through invertible reduced operators. We use the naturality and direct-sum properties of the determinant-line identifications in [50, Theorem 1] to compare their orientations.

The compact contribution is $(-1)^{\mu(g_0)}$ by Lemma 4.4. The complementary model operator contributes a fixed sign. To compute the remaining block, let s_T be the gauged scale tensor of Proposition B.7, and use $\dot{b}$ and c_0 from the proof of Proposition 5.9. Lemma B.6 and the gauge correction in Proposition B.7 give

$$dA_h(s_T) = 2A_*, \quad \int_Y \left\langle D\mathcal{E}_{0,h}(\dot{b}), s_T \right\rangle_h d\text{vol}_h = -8\pi A_*.$$

Represent the model cokernel by pairing with s_T , and choose $\mathbb{R}D\mathcal{E}_{0,h}(\dot{b})$ as a complement to $\text{Im } \mathcal{P}_h$. After eliminating the invertible model part, the obstruction and area equations, in the ordered scale and angle coordinates, have matrix

$$\begin{pmatrix} 0 & -8\pi A_* \\ 2A_* & c_0 \end{pmatrix}, \quad \det = 16\pi A_*^2 > 0.$$

Thus the angle derivative of area does not affect the sign.

All rescalings in (5.11) are positive. Combining the fixed model contribution with the boundary convention $-\partial_a$ gives a sign ϵ_W independent of g_0 and η , proving the formula.

Finally, equivariant pullback conjugates the tensor operators and preserves the spectral pairing of Lemma 4.4, as well as increasing scale, cone angle, and area. Hence the comparison is independent of the end coordinates. The same argument applies to orientation-reversing isometries that identify the prescribed gluing data. $\square$

Proof of Theorem 1.7. Rescale g_0 so that $\text{Ric}(g_0) = 3g_0$, and suppose that the second alternative does not hold. Fix an admissible sequence W in (4.1), and use the notation of Section 4. Let $\mathcal{M}_0$ be the connected component of $\mathcal{M}$ containing the family g_a of Theorem 5.8 for sufficiently small $a > 0$.

Suppose that $\mathcal{M}_0$ contains no point with $\beta = 1$. Since $\beta(g_a) \rightarrow \beta_* > 1$, connectedness gives $\beta > 1$ throughout $\mathcal{M}_0$. We let $\mathcal{M}_{0,\eta} = \mathcal{M} \cap \mathcal{M}_\eta$. For every $\eta > 0$, Theorems 6.1 and 5.7 imply that $\mathcal{M}_{0,\eta}$ is compact. Indeed, the cone factors are bounded above and below, and the only degeneration in Theorem 5.7 has $\mathcal{A} \rightarrow 0$. Every other limit belongs to $\mathcal{M}_{0,\eta}$, since a connected component is closed.

By our assumption, every compact Einstein metric arising in such a degeneration is isometric to g_0 . The given two-tori are maximal in the respective isometry groups, so conjugacy of maximal tori makes these isometries weakly equivariant. Consequently, all these limits are torically nondegenerate and have the same value $\mu(g_0)$.

Theorem 5.8 and Proposition 5.11 therefore imply that, for sufficiently small $\eta > 0$, every point of $\partial\mathcal{M}_{0,\eta}$ is regular and $\mathcal{A}$ is a local coordinate there. This boundary is nonempty, since it contains the class of g_η . Proposition 4.5 and Lemma 8.1 now give

$$0 = \sum_{x \in \partial\mathcal{M}_{0,\eta}} \text{sign}_{\partial\mathcal{M}_{0,\eta}}(x) = \epsilon_W(-1)^{\mu(g_0)} \# \partial\mathcal{M}_{0,\eta} \neq 0,$$

a contradiction.

Thus $\mathcal{M}_0$ contains a metric with $\beta = 1$. All its cone angles are 2π , so it is a smooth toric Einstein metric on X . Since W was arbitrary, the first alternative follows. $\square$

Proof of Theorem 1.1. Equip

$$\mathbb{S}^4, \quad \mathbb{S}^2 \times \mathbb{S}^2, \quad \mathbb{CP}^2 \# k \overline{\mathbb{CP}^2} \quad (0 \leq k \leq 3)$$

with their classical Einstein metrics and standard torus actions, normalized by $\text{Ric}(g) = 3g$. Their rod directions belong to $\{A, B, C\}$ after a global change of lattice basis. Corollary 7.3 gives uniqueness up to isometry, and Theorem 7.6 gives toric nondegeneracy. Thus the second alternative of Theorem 1.7 is excluded, and every admissible replacement produces a toric Einstein metric.

By the topological calculations in Section 4, the alternating sequences applied to $\mathbb{S}^4$ and $\mathbb{CP}^2 \# k \overline{\mathbb{CP}^2}$ give

$$\#_r(\mathbb{S}^2 \times \mathbb{S}^2), \quad (\mathbb{CP}^2 \# k \overline{\mathbb{CP}^2}) \#_r(\mathbb{S}^2 \times \mathbb{S}^2), \quad r \geq 1, \quad 0 \leq k \leq 3.$$

The cases $r = 0$ are the classical metrics themselves.

The four sequences in (1.3), applied to the six manifolds above, give the stated finite list of connected sums $X \# Y$. The admissible choices of corner and lattice identification give precisely the listed choices of X and Y , with the indicated orientations. $\square$

Remark 8.2. *The possible multiplicity of boundary points is a symmetry issue. Even when every compact limit is isometric to g_0 , different contraction points with their ordered incident rods need not be related by an isometry compatible with the torus action, even allowing a change of lattice basis. Thus $\partial\mathcal{M}_{0,\eta}$ may a priori contain several points. The oriented count handles this multiplicity: Lemma 8.1 assigns every such boundary point the same sign $\epsilon_W(-1)^{\mu(g_0)}$.*

9. FURTHER QUESTIONS

Our results suggest several future directions for exploration.

Question 9.1. For a fixed compact rod structure,

- (Uniqueness) Is a toric Einstein metric unique up to scaling and equivariant isometry?
- (Non-degeneracy) Is every compact toric Einstein metric torically nondegenerate?
- (Stability) Which compact toric Einstein metrics are stable?

It would be more interesting the answers to these question are NO. As suggested by AI, the smooth compactness and virtual counting results above developed in Appendix C can extend to arbitrary compact rod structures: for each fixed $\mathfrak{R}$, the normalized smooth toric Einstein moduli space should be compact and admit a well-defined virtual count mod 2 (or an oriented count).

Then we will have an interesting virtual counting number, even when the moduli space is not discrete. We leave the proposed extension to arbitrary rod structures for future investigation.

The toric Einstein theory developed in this paper are likely useful in studying toric shrinking Ricci solitons in dimensions. In particular, a natural question is

Question 9.2. Does there exist a compact simply connected toric gradient shrinking Ricci soliton that is neither Einstein nor Kähler?

There is also a natural Ricci-flow problem. Liu proved that integrability of the horizontal distribution is preserved under smooth Ricci flow on closed simply connected four-manifolds; see [35, Theorem 8.1]. Thus metrics with this property can be studied through their evolving quotient metrics and orbit matrices.

Question 9.3. Can we have a theory of Ricci flow with surgery for $\mathbb{T}^2$ -invariant Ricci flow with integrable horizontal distribution?

This could serve as an experiment towards understanding of general singularity formation of 4d Ricci flow.

APPENDIX A. ASYMPTOTICS OF TORIC GRAVITATIONAL INSTANTONS

In this appendix we establish the classification of asymptotic geometries for toric conical gravitational instantons and discuss some applications.

A.1. Definitions and statements. As in [34], a gravitational instanton is a complete connected oriented Ricci-flat four-manifold with quadratic curvature decay

$$(A.1) \quad |\mathrm{Rm}_g|_g(p) \leq C(1 + d_g(o, p))^{-2},$$

where o is a fixed base point.

We first recall relevant notations in [34]. We say a Ricci-flat four-manifold (M, g) is *toric*, if every point on M has a connected stabilizer group. Denote the quotient by (N, g) , which is a smooth manifold with boundary, and the interior of N by N° . On M there is the horizontal distribution that is orthogonal to the $\mathbb{T}^2$ -orbits. The toric Ricci-flat four-manifold is called *integrable* if the horizontal distribution is integrable, which automatically holds when the $\mathbb{T}^2$ -action has a fixed point. After fixing a set of generators $\{\partial_{\phi_1}, \partial_{\phi_2}\}$, there is the *Gram matrix*

$$G_{ij} := g(\partial_{\phi_i}, \partial_{\phi_j}).$$

Write $\rho = \sqrt{\det G}$ which is a function on the quotient N and define

$$\Phi := \rho^{-1}G,$$

which is a map from N° into the space of symmetric positive (2×2) matrices with determinant one. The function ρ is smoothly defined on N° and continuously extends to N and vanishes on ∂N . We say (M, g) is *regular* if $d\rho \neq 0$ on N° .

Definition A.1. A toric gravitational instanton is a complete Ricci-flat oriented four-manifold (M, g) that is integrable and regular, with quadratic curvature decay (A.1).

For a toric gravitational instanton, the function ρ on (N, g) is harmonic on N° , and locally there is the harmonic conjugate z that satisfies $dz = \ast d\rho$.

Definition A.2. A toric gravitational instanton is proper if there is a globally defined harmonic conjugate z of ρ on N such that the map (ρ, z) induces a homeomorphism from N to $\mathbb{H}$.

Here $\mathbb{H}$ refers to the two-dimensional half-plane.

Remark A.3. The properness assumption is a genuine restriction. The periodic quotients of the four-dimensional examples in [29, Theorems 2 and 4] are complete toric gravitational instantons with Kasner asymptotics; see also [45, Section 2.3]. Their orbit spaces are half-cylinders $[0, \infty) \times (\mathbb{R}/L\mathbb{Z})$, where $L > 0$, and the local harmonic conjugate z is periodic. Consequently, $\ast d\rho$

has a nonzero period around the cylinder and admits no globally defined real-valued primitive. These gravitational instantons are therefore not proper.

The corresponding infinite periodic covers have half-plane orbit spaces, but fail the quadratic curvature decay condition (A.1), since nonzero curvature recurs arbitrarily far from a fixed base point under the periodic isometries.

For a proper toric gravitational instanton, we have the global coordinates (ρ, z) for N , and the Ricci-flat metric can be written as

$$(A.2) \quad g = e^{2\nu} (d\rho^2 + dz^2) + \rho \Phi_{ij} d\phi_i d\phi_j.$$

It has been shown in [34] that the Ricci-flat equation for a metric in the form of (A.2) is equivalent to the following system:

$$(A.3) \quad (\rho \Phi^{-1} \Phi_\rho)_\rho + (\rho \Phi^{-1} \Phi_z)_z = 0,$$

$$(A.4) \quad \partial_z \nu = \frac{1}{4\rho} \operatorname{tr}(\mathcal{U}\mathcal{V}),$$

$$(A.5) \quad \partial_\rho \nu = -\frac{1}{2\rho} + \frac{1}{8\rho} \operatorname{tr}(\mathcal{U}^2 - \mathcal{V}^2),$$

where

$$(A.6) \quad \mathcal{U} = \operatorname{Id} + \rho \Phi^{-1} \Phi_\rho, \quad \mathcal{V} = \rho \Phi^{-1} \Phi_z.$$

The target of Φ , which is the space of symmetric positive (2×2) matrices with determinant one, can be identified with the two-dimensional hyperbolic upper-half-plane with curvature -2

$$\mathcal{H} = \{(X, Y) \mid X > 0\}, \quad g_{\mathcal{H}} = \frac{dX^2 + dY^2}{2X^2}$$

via

$$(A.7) \quad (X, Y) \mapsto \begin{pmatrix} X + X^{-1}Y^2 & X^{-1}Y \\ X^{-1}Y & X^{-1} \end{pmatrix}.$$

In the Euclidean $\mathbb{R}^3$, denote the z -axis as Γ . Let ρ be the distance function to Γ in $\mathbb{R}^3$. Then (A.3) is equivalent to

$$\Phi : \mathbb{R}^3 \setminus \Gamma \rightarrow \mathcal{H}$$

is an axisymmetric harmonic map. For each toric gravitational instanton (M, g) , there is an associated *rod structure*

$$\mathfrak{R} = (\{z_j\}_{j=1}^n, \{\mathbf{v}_j\}_{j=0}^n),$$

which consists of

- A collection of *turning points* $0 = z_1 < \dots < z_n$ on $\Gamma = \mathbb{R}$. We also write $z_0 = -\infty$ and $z_{n+1} = \infty$. For $0 \leq j \leq n$, the open interval $\mathfrak{I}_j := (z_j, z_{j+1}) \subset \Gamma$ is called a rod.
- A collection of *rod vectors* $\mathbf{v}_0, \dots, \mathbf{v}_n \in \mathbb{RP}^1$ with $\mathbf{v}_j \neq \mathbf{v}_{j+1}$.

Note that the definition of rod structures we use in this appendix is actually the *untyped rod structure* under the terminology of [34]. Under the coordinate (ρ, z) on N , the boundary ∂N is identified with Γ . Each fixed point of $\mathbb{T}^2$ in M descends to a point in ∂N , and corresponds to a turning point z_j on Γ . Each rod $\mathfrak{I}_j$ corresponds to a totally geodesic $\mathbb{S}^2$ inside M , whose isotropy group is S^1 . Under our choice of generator $\{\partial_{\phi_1}, \partial_{\phi_2}\}$ where we identify them as $(1, 0)^\top$ and $(0, 1)^\top$, the isotropy of the rod $\mathfrak{I}_j$ is described by the rod vector $\mathbf{v}_j$. Under our terminology of [34], the axisymmetric harmonic map Φ is locally tamed by the rod structure $\mathfrak{R}$. Along each rod, there is the notion of *cone angles*. See [34, Definition 4.19]. In the smooth case the cone angles are all 2π . For toric *conical* gravitational instantons, cone angles are allowed to be different from 2π . Throughout this appendix, we take the lattice to be $\Lambda = (2\pi\mathbb{Z})^2$, which means the periods of ϕ_1, ϕ_2 are 2π .

In this appendix, we work with proper toric conical gravitational instantons, which also corresponds to axisymmetric harmonic maps as above. The only difference is the cone angles along each rod is allowed to be different from 2π . For a precise formulation, see [34]. We prove

Theorem A.4. *A proper toric conical gravitational instanton (M, g) with finite rods is of Asymptotic Type ALE, $\text{ALF}^\pm$, or AF_β . In the $\text{ALF}^\pm$ and AF_β cases, there are a corresponding model end (E_0, g_0) of Asymptotic Type ALE, $\text{ALF}^\pm$, or AF_β , and a diffeomorphism $\varpi : E_0 \rightarrow M \setminus K$ such that*

$$(A.8) \quad |\nabla_{g_0}^k(\varpi^*g - g_0)|_{g_0} = O(\mathfrak{r}^{-1-k}), \quad k \geq 0,$$

where $\mathfrak{r}$ is the distance to a fixed base point.

A.2. Maximal volume growth. In the maximal volume growth case, the gravitational instanton is ALE.

Proposition A.5. *Let (M, g) be a proper toric conical gravitational instanton with maximal volume growth. Then the two infinite rods have distinct directions. After changing the angular lattice on the complement of a compact set, the resulting smooth Ricci-flat end is asymptotically Euclidean. With the original angular lattice, the end is asymptotic to the corresponding flat toric cone, allowing conical singularities along its two infinite rods and a finite angular quotient.*

Proof. As before, by enlarging the lattice, we can arrange all cone angles to be strictly less than 2π . We continue to denote it by (M, g) . The Bishop-Gromov comparison applies, as in [34]. In particular, for a constant $v > 0$,

$$vt^4 \leq \text{Vol}_g B_g(p, t) \leq \omega_4 t^4$$

for every $p \in M$ and $t > 0$, where ω_4 is the volume of the Euclidean unit ball.

Now let $R_j \rightarrow \infty$. Pointed compactness gives a subsequential blow-down limit. If v_∞ is the asymptotic volume ratio, volume convergence gives

$$\text{Vol}_{g_{R_j}} B_{g_{R_j}}(o, t) \rightarrow v_\infty \omega_4 t^4$$

for every $t > 0$. The volume-cone rigidity theorem [21] identifies the limit with a metric cone

$$(C(Y), g_C), \quad g_C = dr^2 + r^2 h.$$

The smooth convergence on regular annuli makes g_C Ricci-flat there. Consequently $\text{Ric}_h = 2h$ on the regular part of the three-dimensional link. In dimension three this is equivalent to sectional curvature one. Hence g_C is flat on its regular part, and carries the effective torus action. In a flat development, this action is generated by rotations in two orthogonal two-planes. Consequently, up to a finite angular quotient, the cone is a product $\mathbb{R}_{\alpha_-}^2 \times \mathbb{R}_{\alpha_+}^2$, where $\mathbb{R}_\alpha^2$ denotes the flat two-dimensional cone of angle α . Its two rods correspond to the collapsing circles in the two factors and therefore have distinct rod directions.

If the two original infinite rod directions coincided, the same primitive generator X would vanish on both rods. Smooth convergence in the local developments would therefore give a nonzero limiting Killing field vanishing on both limiting rods. This contradicts the two-plane description of g_C . The original infinite rod directions are consequently distinct as well.

We now return to the original manifold. Choose primitive period vectors $\ell_-, \ell_+ \in \Lambda$ representing the two infinite rod directions, and denote their cone angles by $2\pi\beta_-$ and $2\pi\beta_+$. Here $\ell_\pm$ are period vectors, so the corresponding 2π -periodic infinitesimal generators are $\ell_\pm/(2\pi)$. On an end containing no finite rods, replace Λ by

$$\Lambda_{\text{sm}} = \mathbb{Z} \frac{\ell_-}{\beta_-} + \mathbb{Z} \frac{\ell_+}{\beta_+}.$$

The two vectors are independent and form a basis of this lattice. Both infinite rods now have cone angle 2π , giving a smooth Ricci-flat end $(E_{\text{sm}}, g_{\text{sm}})$ with cross-section S^3 . This operation need not extend over the compact region. Here g_{sm} still has quadratic curvature decay and maximal volume growth. The exterior argument of Bando–Kasue–Nakajima [6] now gives ALE asymptotics; its applicability to a Ricci-flat end without a prescribed Ricci-flat filling is also explained in [34, proof of Proposition 4.38]. Since the cross-section is S^3 , the asymptotic group is trivial and the modified end is AE. As in that proof, the asymptotic identification can be chosen equivariantly. Restoring the original angular periods then gives the corresponding flat

conical asymptotic model for the original end, with the differentiated decay furnished by the exterior estimates. This proves the maximal volume growth case. $\square$

A.3. Estimates on the quotient. In the following we consider the case where the gravitational instanton has non-maximal volume growth. For a proper toric gravitational instanton (M, g) , the quotient $(N, \underline{g})$ is a smooth manifold with boundary and corners. The rods are the boundaries and turning points are the corners. The rods $\mathfrak{I}_j$ are totally geodesic in N .

Lemma A.6. *The quotient $(N, \underline{g})$ has non-positive sectional curvature $K_{\underline{g}} \leq 0$ and quadratic curvature decay. Moreover, $(N, \underline{g})$ is a CAT(0) space.*

Proof. Because (M, g) is integrable, the sectional curvature of $(N, \underline{g})$ is equal to the sectional curvature of (M, g) along $\partial_\rho \wedge \partial_z$, which therefore decays quadratically. In particular $K_{\underline{g}} \rightarrow 0$ at infinity, so any positive global maximum is attained. The proof of Lemma 3.1 and Proposition 3.2 go through as well for the Ricci-flat case, hence we know there is no interior maximum. It follows $K_{\underline{g}} \leq 0$. Since it has geodesic boundaries and right angle corners, it is CAT(0) [10, Theorem II.4.1]. $\square$

In $(N, \underline{g})$, we set $o \in \partial N$ to be the turning point z_1 . If there is no turning point in the rod structure, we just set o to be an arbitrary base point in ∂N and normalize z to be zero at o . Write $r = \sqrt{\rho^2 + z^2}$ and $d = d_{\underline{g}}(o, \cdot)$. Denote

$$A_d(a, b) = \{p \in N \mid a < d(p) < b\}, \quad A_r(a, b) = \{p \in N \mid a < r(p) < b\}.$$

By the next several lemmas, we prove as $R \rightarrow \infty$, the scaled down sequence $(A_d(aR, bR), R^{-2}g)$ is noncollapsed. We denote the metric annulus in (M, g) centered at o as $A_g(a, b)$, which covers $A_d(a, b)$.

Consider any point q that is on the two infinite rods that are sufficiently far away from o . Write $O = \mathbb{T}^2 \cdot q$ as the orbits at q , which is an S^1 , and F as the fixed surface corresponding to the infinite rod that contains q . Set $R = d(q)$ and the length of orbit O at q as $\sigma(q) = |O|$. Let X be a primitive Killing vector field with period 2π that vanishes along F .

Lemma A.7. *There are constants $\epsilon, \delta > 0$, such that the normal exponential map of the orbit O is injective on its radius ϵR normal disk bundle. Moreover,*

$$(A.9) \quad \text{Vol}(B_g(o, (1 + \epsilon)R)) \geq \delta \sigma(q) R^3.$$

Proof. The fixed surface F is totally geodesic and

$$g_F = dl^2 + \sigma(l)^2 d\psi^2, \quad \sigma'' + K_F \sigma = 0,$$

where ∂_ψ is a primitive integral Killing field in $\mathbb{T}^2$ with period 2π and l is the arclength from the torus fixed point contained in F . The Ricci-flat version of (2.11) and $K_{\underline{g}} \leq 0$ give $\sigma'' = 2K_{\underline{g}}\sigma \leq 0$. When there is no turning point for $\mathfrak{R}$, the function σ is nowhere vanishing and concave, hence constant. When there are turning points, since $\sigma(0) = 0$ and σ stays positive on the whole ray, concavity gives $0 \leq \sigma' \leq \sigma/l$. Since the ray is a geodesic in $(N, \underline{g})$, so $l(q) = R + O(1)$. On the ray when $|l - l(q)| < R/4$ we consequently have

$$(A.10) \quad |\sigma'|/\sigma \leq C/R.$$

The orbit circle has second fundamental form of the norm $|\sigma'|/\sigma$. Normal Jacobi comparison gives a focal radius at least $c_0 R$. If normal injectivity radius were smaller than a sufficiently small $\epsilon R < c_0 R$, then there would be a geodesic γ from O to O , normal to both ends, of length $L < 2\epsilon R$. The Jacobi field $J = X|_\gamma$ vanishes at both ends. From the Jacobi field equation we obtain

$$\int_0^L |D_t J|^2 dt \leq C R^{-2} \int_0^L |J|^2 dt.$$

If ϵ is small enough, the Poincaré inequality forces $J = 0$. Thus γ lies in F . The velocity $\dot{\gamma}$ is orthogonal to ∂_ψ , since $\frac{d}{dt}g(\dot{\gamma}, \partial_\psi) = 0$ and γ starts orthogonally to O . So it is a radial l -geodesic, which cannot go back to O , a contradiction.

To see the lower bound on the volume, the ϵR lower bound on injectivity radius gives

$$\text{Vol}(B_g(o, (1 + \epsilon)R)) \geq \sigma(q)\delta R^3$$

for a small constant δ . Since σ is concave, this in particular implies (M, g) has at least cubic volume growth. $\square$

Lemma A.8. *For every $0 < a < b < \infty$, as $R \rightarrow \infty$, the sequence of scaled down annuli $(A_d(aR, bR), R^{-2}g)$ has uniformly bounded curvature and is noncollapsed. Each annulus can be covered by a uniform number of balls with radius ϵ .*

Proof. This is clear from Lemma A.7, where we have a uniform injectivity radius lower bound ϵR at points on the infinite rods. The scaled down annulus has uniformly bounded curvature since (N, g) has quadratic curvature decay. The number of balls to cover $A_d(aR, bR)$ can be controlled by the number of balls to cover $A_g(aR, bR)$ in the Ricci-flat (M, g) , which is controlled by volume comparison. $\square$

Next we control ν and ρ . Recall $\underline{g} = e^{2\nu}(d\rho^2 + dz^2)$. Recall that q is any point on the two infinite rods that are far away from o .

Lemma A.9. *There is a constant $C > 0$ such that, outside a compact set,*

$$(A.11) \quad re^\nu \leq Cd.$$

Moreover, on $A_d(R/2, 2R)$,

$$(A.12) \quad C^{-1}\sigma(q) \leq |d\rho|_{\underline{g}} \leq C\sigma(q), \quad C^{-1}R\sigma(q) \leq |z(q)| \leq CR\sigma(q).$$

Proof. Consider annuli $A_g(aR, bR)$ of the gravitational instanton (M, g) centered at o . Since (M, g) has quadratic curvature decay, in $g_R = R^{-2}g$ the curvature of $(A_g(aR, bR), g_R)$ is uniformly bounded. By Cheeger-Fukaya-Gromov for any $p \in (A_g(aR, bR), g_R)$ there is a small ball $(B_\delta(p), g_R)$ such that its universal cover $(\tilde{B}_\delta(p), \tilde{g}_R)$ is noncollapsed. We pick the point $p \in A_g(aR, bR)$ to be above q . Then the universal cover $(\tilde{B}_\delta(p), \tilde{g}_R)$ is noncollapsed, and by the regularity of Einstein equation, it converges smoothly to a Ricci-flat limit after passing to subsequences. By Lemma A.8, we know $(A_d(aR, bR), \underline{g}_R)$ converges smoothly to a limit $(A_{d,\infty}, \underline{g}_\infty)$ with geodesic boundaries.

As before, we let X be a primitive Killing field with period 2π that vanishes along F . Let Y be another Killing field in $\mathbb{T}^2$, normalized such that $|Y(p)|_{\underline{g}} = \sigma(p)/2\pi$. Such a Y restricts to ∂_ψ on F . We then normalize X, Y in $A_g(aR, bR)$ to

$$X, \quad V_R = \frac{R}{\sigma(q)}(Y - b_RX),$$

where we take the constant b_R to cancel the normal rotational part of Y at p . In the noncollapsed limit of the universal cover $(\tilde{B}_{g_\infty}, \tilde{g}_\infty)$, the normalized Killing fields X, V_R converge to nontrivial limit Killing fields X_∞, V_∞ . From the Killing field equation, this implies the normalized Killing fields X, V_R converge on the local noncollapsed cover for any point in $A_d(aR, bR)$. This implies

$$\rho_R := \frac{\rho}{R\sigma(q)},$$

which is the square root of the determinant of the Gram matrix formed by X, V_R in g_R , converges smoothly to a limit harmonic function ρ_∞ on $A_{d,\infty}$. The conjugate of ρ_R , which is related to z via

$$z_R = \frac{z - z(q)}{R\sigma(q)},$$

converges smoothly to a limit harmonic function z_∞ as well on $A_{d,\infty}$. The holomorphic functions $w_R = z_R + \sqrt{-1}\rho_R$ have nowhere vanishing derivative on $A_d(aR, bR)$, so Hurwitz's theorem implies that $w_\infty = z_\infty + \sqrt{-1}\rho_\infty$ has no critical points on $A_{d,\infty}$. Therefore, we have uniform upper and lower bound on $|d\rho_R|$. This proves $C^{-1}\sigma(q) \leq |d\rho|_{\underline{g}} \leq C\sigma(q)$.

From the convergence of z_R we know for any two points $q_1, q_2 \in A_d(aR, bR)$, we have

$$|z(q_1) - z(q_2)| \leq CR\sigma(q).$$

Since $A_d(aR, bR)$ contains part of the two infinite rods, on which the values of z are positive and negative respectively, there exists points in $A_d(aR, bR)$ with $z = 0$. Hence on $A_d(aR, bR)$ we obtain the bound

$$\rho + |z| \leq CR\sigma(q).$$

On a rod segment of intrinsic length cR , (A.10) gives that σ is comparable to $\sigma(q)$. Harmonic conjugacy gives the identity $|dz/dl| = a$ on the rod. Since the base point o is certainly not contained in $A_R(aR, bR)$, integration gives $|z(q)| \geq cR\sigma(q)$.

Finally, since $|d\rho|_{\underline{g}} = e^{-\nu}$ and $d(q) = R$, we have obtained $re^\nu \leq Cd$. $\square$

A.4. Limits of the rescaled harmonic maps. Now we justify the existence of scale-downs of Φ . Recall Φ is an axisymmetric harmonic map from $\mathbb{H}^\circ = \{(X, Y) \mid X > 0\}$ into $\mathcal{H}$. For each $R \gg 1$, there is the scaled down harmonic map $\Phi(R\rho, Rz)$. Set $p_* = (1, 0)$ in $\mathbb{H}$. For $R \rightarrow \infty$, we choose an isometry T_R of the hyperbolic space with $T_R\Phi(Rp_*) = (1, 0)$. As $R \rightarrow \infty$, we define

$$\Phi_R(\rho, z) = T_R\Phi(R\rho, Rz).$$

Lemma A.10. *As $R \rightarrow \infty$, after passing to subsequences, Φ_R converges smoothly on any compact subset of $\mathbb{H}^\circ$ to a limit harmonic map Φ_∞ .*

Proof. The distance function d is 1-Lipschitz for $\underline{g}$, and (A.11) therefore gives that in the Euclidean metric $d\rho^2 + dz^2$ for $\mathbb{H}$,

$$|\nabla_{\text{Eucl}} \log d| \leq C/r$$

almost everywhere. Integrating along paths of bounded length in an annulus $A_r(aR, bR)$ shows that in such an annulus the values of the distance function d are uniformly comparable. Hence, if we set $d_R = d((R, 0))$ and enlarge the annulus so that $a < 1 < b$, $A_r(aR, bR)$ is contained in $A_d(a'd_R, b'd_R)$ for some $0 < a' < b' < \infty$, independent of R .

Here R is the Weyl coordinate scale, whereas the geometric rescaling is $g_{d_R} = d_R^{-2}g$. Choose $q = (0, R)$ on the positive infinite rod and use the normalization

$$V_R = \frac{d_R}{\sigma(q)}(Y - b_RX),$$

where X, Y and b_R are chosen as in the proof of Lemma A.9. Since $d(q)$ is uniformly comparable to d_R , that proof gives smooth convergence of the normalized Killing fields X, V_R on the local noncollapsed limits around any point in $(A_g(a'd_R, b'd_R), g_{d_R})$. Hence the associated Gram matrix converges on each local noncollapsed cover. After division by the square root of its determinant, this Gram matrix differs from Φ by an isometry of the hyperbolic space via (A.7); the scalar metric rescaling cancels in this determinant normalization. Hence after a correct isometry, the rescaled harmonic map converges on any compact subset of $\mathbb{H}^\circ$. Since the isometry T_R normalizes $\Phi(Rp_*)$ to $(1, 0)$ and the stabilizer of $(1, 0)$ is compact, we know Φ_R converges to a limit harmonic map Φ_∞ after passing to a subsequence. $\square$

Lemma A.11. *Any limit harmonic map Φ_∞ has a rod structure $\mathfrak{R}_\infty$ that only consists of two infinite rods and one turning point. The two infinite rods have the same rod vector. The harmonic map Φ_∞ is locally tamed by $\mathfrak{R}_\infty$ in the interior of the infinite rods.*

Here, after applying an isometry of the hyperbolic space, we may assume the rod vectors is $[1, 0]^\top$, then Φ_∞ being locally tamed in the interior of infinite rods means that for any point in the interior of infinite rods, there is a neighborhood such that

$$d_{\mathcal{H}}(\Phi_\infty, \Phi_{[1,0]^\top}) \text{ is bounded,} \quad \Phi_{[1,0]^\top} = \begin{pmatrix} \rho & 0 \\ 0 & \rho^{-1} \end{pmatrix}.$$

Proof. We keep R as the Weyl coordinate scale and set $d_R = d((R, 0))$. We take $q = (0, R)$ on the positive infinite rod, and let F be the fixed surface in M corresponding to this infinite rod as before. Let X be a primitive Killing field with period 2π that vanishes along F , and let Y be another Killing field normalized such that $|Y(q)|_g = \sigma(q)/2\pi$, which restricts to ∂_ψ on F . From (A.9), applied at the geometric scale $d(q)$, since (M, g) has non-maximal volume growth, we know $\sigma(q)/d(q) \rightarrow 0$ along the infinite rod. This implies that $|Y(q)|_g/d(q) \rightarrow 0$.

We also pick a point q' on the negative infinite rods, such that $q, q' \in A_r(aR, bR)$ for fixed $0 < a < 1 < b$. Let F' be the fixed surface, and let X' be a primitive Killing field with period 2π that vanishes along the rod and Y' be another Killing field. Then $X = mX' + nY'$ for some constants m, n . Similarly, $|Y'(q')|_g/d(q') \rightarrow 0$. By the preceding lemma, both $d(q)/d_R$ and $d(q')/d_R$ are bounded above and below independently of R .

Now we use $g_{d_R} = d_R^{-2}g$ and $V_R = \frac{d_R}{\sigma(q)}(Y - b_RX)$, with b_R chosen as above, and consider the Gram matrix for g_{d_R} under the basis X, V_R . We express this matrix in the rescaled Weyl coordinates and denote it by

$$G_R = \begin{pmatrix} G_{R,11} & G_{R,12} \\ G_{R,21} & G_{R,22} \end{pmatrix}.$$

Since we have the smooth convergence of X, V_R in any local noncollapsed cover and $A_r(aR, bR) \subset A_d(a'd_R, b'd_R)$, G_R converges smoothly to a limit G_∞ on $\mathbb{H}^\circ$. The component $G_{R,11}$ vanishes along the positive infinite rod. On the negative infinite rod, since $X' = 0$ there, we have

$$\begin{aligned} G_{R,11}(0, z(q')/R) &= \frac{|nY'(q')|_g^2}{d_R^2} \\ &= \frac{|nY'(q')|_g^2}{d(q')^2} \frac{d(q')^2}{d_R^2} \rightarrow 0. \end{aligned}$$

This holds for any $q' \in A_r(aR, bR)$ on the other infinite rod, hence $G_{\infty,11} = 0$ along the other infinite rod as well. This indicates the limit Gram matrix G_∞ has the same one-dimensional kernel along the two infinite rods, hence the two infinite rods share the same rod vector.

Since G_R is defined using the basis X, V_R and X, V_R converge smoothly, it is clear that Φ_∞ is locally tamed in the interior of the positive infinite rod. As for the negative infinite rod, a similar normalization of Killing fields X', V'_R converge smoothly in any local noncollapsed cover so we obtain a limit harmonic map Φ'_∞ that is locally tamed in the interior of the negative infinite rod. Comparing the two normalization shows Φ_∞ is also locally tamed in the interior of the negative infinite rod. $\square$

For a limit harmonic map Φ_∞ , we always apply an isometry of the hyperbolic space so that the rod vector is $[1, 0]^\top$.

Proposition A.12. *The convergence $\Phi_R \rightarrow \Phi_\infty$ is $C^{1,\alpha}$ on any compact subset of $\mathbb{H} \setminus \{o\}$. There exists a constant matrix $P \in \text{SL}(2, \mathbb{R})$ such that*

$$\Phi_\infty = (P^{-1})^\top \Phi_{[1,0]^\top} P^{-1}.$$

For the precise definition of $C^{1,\alpha}$ convergence of harmonic maps, see [34, Definition 4.20].

Proof. We write $\rho = r \sin \theta$ and $z = r \cos \theta$. Let $X = \rho e^u$ and $Y = v$ where we parametrize $\mathcal{H}$ via (X, Y) . From the smooth convergence to Φ_∞ and the local tameness of Φ_∞ by $\mathfrak{R}_\infty$, from the regularity of [34, Proposition 4.12], we have the $C^{1,\alpha}$ convergence of Φ_R to Φ_∞ .

Now we consider a limit harmonic map $\Phi_\infty = (\rho e^{u_\infty}, v_\infty)$, and choose a sequence $R_j \rightarrow \infty$ such that $\Phi_{R_j} \rightarrow \Phi_\infty$. The annular estimates and rod regularity used above also give uniform estimates for the normalized secondary rescalings at every scale $R > 0$. Indeed, fix $R > 0$, let ξ_j be the ideal endpoint associated with the positive rod of Φ_{R_j} , and choose the orientation-preserving hyperbolic isometry $S_{j,R}$ that sends ξ_j to 0 and $\Phi_{R_j}(R, 0)$ to $(1, 0)$. Since $\xi_j \rightarrow 0$, these isometries converge to the isometry S_R satisfying

$$S_R(0) = 0, \quad S_R \Phi_\infty(R, 0) = (1, 0).$$

The map $S_{j,R}\Phi_{R_j}(R\rho, Rz)$ is a normalized rescaling of the original map at scale $R_jR \rightarrow \infty$. Consequently, its annular interior estimates and rod-adapted estimates pass to $S_R\Phi_\infty(R\rho, Rz)$. Their constants are independent of R , since they are independent of the original scale R_jR . The limiting rod values on both sides are 0 by Lemma A.11. In particular, for every $R > 0$,

$$d_{\mathcal{H}}\left(S_R\Phi_\infty(R\sin\theta, R\cos\theta), (\sin\theta, 0)\right) \leq C, \quad 0 < \theta < \pi,$$

and the family $S_R\Phi_\infty(R\rho, Rz)$ has uniform interior derivative bounds on compact subsets of $\mathbb{H}^\circ$. Writing

$$S_R\Phi_\infty(R\rho, Rz) = (\rho e^{u_{\infty,R}}, v_{\infty,R}),$$

the same rod regularity gives uniform $C^{1,\alpha}$ bounds for $u_{\infty,R}$ and the corresponding weighted derivative bounds for $v_{\infty,R}$ on compact subsets of $\mathbb{H} \setminus \{o\}$; see [34, Proposition 4.12]. Thus these estimates hold at both arbitrarily large and arbitrarily small secondary scales.

The curves $S_R^{-1}(\sin\theta, 0)$ and $(R\sin\theta, 0)$, parametrized by $-\log\sin\theta$, are both geodesics in $\mathcal{H}$ with the same constant speed $1/\sqrt{2}$, sharing the same ideal endpoint. Their distance is non-increasing in the parameter $-\log\sin\theta$ due to hyperbolic geometry, where $-\log\sin\theta \geq 0$ and $-\log\sin\theta = 0$ at $\theta = \frac{\pi}{2}$. So

$$\begin{aligned} d_{\mathcal{H}}\left(S_R^{-1}(\sin\theta, 0), (R\sin\theta, 0)\right) &\leq d_{\mathcal{H}}\left(S_R^{-1}(1, 0), (R, 0)\right) \\ &= d_{\mathcal{H}}\left(\Phi_\infty(R, 0), (R, 0)\right). \end{aligned}$$

From this we obtain that at the point $(R\sin\theta, R\cos\theta)$,

$$\begin{aligned} d_{\mathcal{H}}(\Phi_\infty, \Phi_{[1,0]^\top}) &= d_{\mathcal{H}}\left(\Phi_\infty, (R\sin\theta, 0)\right) \\ &\leq d_{\mathcal{H}}\left(\Phi_\infty, S_R^{-1}(\sin\theta, 0)\right) + d_{\mathcal{H}}\left(S_R^{-1}(\sin\theta, 0), (R\sin\theta, 0)\right) \\ &\leq C + d_{\mathcal{H}}\left(\Phi_\infty(R, 0), (R, 0)\right). \end{aligned}$$

The uniform interior derivative bound for $S_R\Phi_\infty(R\rho, Rz)$ at $(\rho, z) = (1, 0)$, valid for every $R > 0$, gives

$$|\partial_\rho\Phi_\infty(\rho, 0)|_{\mathcal{H}} \leq \frac{C}{\rho}, \quad \rho > 0.$$

Integrating from 1 to ρ , in either direction, gives

$$d_{\mathcal{H}}\left(\Phi_\infty(\rho, 0), (1, 0)\right) \leq C(|\log\rho| + 1).$$

Since the hyperbolic distance from $(\rho, 0)$ to $(1, 0)$ is $|\log\rho|/\sqrt{2}$, we conclude

$$d_{\mathcal{H}}\left(\Phi_\infty(R, 0), (R, 0)\right) \leq C(|\log R| + 1), \quad R > 0,$$

and therefore

$$(A.13) \quad d_{\mathcal{H}}(\Phi_\infty, \Phi_{[1,0]^\top}) \leq C(1 + |\log r|), \quad r > 0.$$

Finally we compare Φ_∞ to itself after a z -translation by t

$$D_t(\rho, z) := d_{\mathcal{H}}\left(\Phi_\infty(\rho, z+t), \Phi_\infty(\rho, z)\right).$$

Both are harmonic maps from $\mathbb{R}^3 \setminus \Gamma$ to $\mathcal{H}$, so D_t is subharmonic. The uniform interior and rod estimates for the secondary rescalings at every scale $R > 0$ imply the following uniform bound on $\mathbb{H}^\circ$

$$(A.14) \quad |\partial_z\Phi_\infty|_{\mathcal{H}} \leq \frac{C}{r}.$$

Indeed, writing $S_R\Phi_\infty(R\rho, Rz) = (\rho e^{u_{\infty,R}}, v_{\infty,R})$, the rod estimates give uniform bounds for $u_{\infty,R}$, $\partial_z u_{\infty,R}$ and $\rho^{-1}\partial_z v_{\infty,R}$ on the unit semicircle. Thus the hyperbolic norm of the z -derivative

of the normalized map is uniformly bounded. Rescaling and using that S_R is an isometry give (A.14) for every $r > 0$. Integration along the segment from (ρ, z) to $(\rho, z + t)$ gives

$$D_t(\rho, z) \leq C \frac{|t|}{r}, \quad \text{if } r \gg 1 + |t|.$$

The function D_t is locally bounded near the rods except possibly at $(0, 0)$ and $(0, -t)$, since Φ_∞ is locally tamed in the interior of the two rods. Hence it extends subharmonically across the rods except the two points. As $\Phi_{[1,0]^\top}$ is independent of z , the bound (A.13) gives at most logarithmic growth of D_t at these two exceptional points. For $\epsilon > 0$, apply maximum principle to

$$D_t(\rho, z) - \epsilon \left(r^{-1} + \frac{1}{\sqrt{\rho^2 + (z+t)^2}} \right)$$

on large balls with small balls about the two exceptional points deleted. The logarithmic growth makes the inner boundary values negative; the outer supremum tends to zero. For fixed t and $\epsilon > 0$, let the radii of the deleted balls tend to zero and the outer radius tend to infinity. The maximum principle gives

$$0 \leq D_t(\rho, z) \leq \epsilon \left(r^{-1} + \frac{1}{\sqrt{\rho^2 + (z+t)^2}} \right).$$

Letting $\epsilon \rightarrow 0$ at each fixed point with $\rho > 0$ gives $D_t = 0$, which holds for any t .

Thus Φ_∞ depends only on ρ . The axisymmetric harmonic map equation then says $t \mapsto \Phi_\infty(e^t, z)$ is a geodesic in $\mathcal{H}$. After an isometry $P \in \text{SL}(2, \mathbb{R})$ it is identified to the geodesic $t \mapsto (e^t, 0)$ that corresponds to the harmonic map $\Phi_{[1,0]^\top}$. This gives $\Phi_\infty = (P^{-1})^\top \Phi_{[1,0]^\top} P^{-1}$, as desired. $\square$

A.5. Convergence at infinity. So far we have shown that for the normalized harmonic map $\Phi_R(\rho, z) = T_R \Phi(R\rho, Rz)$, any subsequential limit of it is equivalent to $\Phi_{[1,0]^\top}$. In the following, let $s = \log r$. We first study the behavior of Φ along the ray $(R, 0)$ in $\mathbb{H}$ as $R \rightarrow \infty$.

Lemma A.13. *There is a constant matrix $Q_1^{-1} \in \text{SL}(2, \mathbb{R})$ and a constant b , such that for $\Phi_1 = (Q_1^{-1})^\top \Phi Q_1^{-1}$, as $R \rightarrow \infty$,*

$$Y(\Phi_1(R, 0)) \rightarrow b, \quad X(\Phi_1(R, 0)) \rightarrow 0, \quad \frac{|Y(\Phi_1(R, 0)) - b|}{X(\Phi_1(R, 0))} \rightarrow 0.$$

Proof. We first perform a Möbius transformation

$$Y + \sqrt{-1}X \mapsto -\frac{1}{Y + \sqrt{-1}X - c}.$$

This is a fixed hyperbolic isometry, which corresponds to conjugation by a matrix $Q_1^{-1} \in \text{SL}(2, \mathbb{R})$. The value of Y is a constant along the positive infinite rod, and we take c to be that value. Then the boundary value of $\Phi_1 := (Q_1^{-1})^\top \Phi Q_1^{-1}$ on the positive rod, which is a point in the ideal boundary, is infinity.

We follow the ray $(R, 0)$ in $\mathbb{H}$ to infinity and consider the values $\Phi_1(R, 0)$. For $R > 0$, define the real functions

$$f(R) = Y(\Phi_1(R, 0)), \quad h(R) = X(\Phi_1(R, 0)).$$

Let A_R be the hyperbolic isometry that is defined by

$$A_R : (X, Y) \mapsto \left(\frac{X}{h(R)}, \frac{Y - f(R)}{h(R)} \right).$$

Thus the normalized harmonic map

$$\Phi_{1,R}(\rho, z) = A_R(\Phi_1(R\rho, Rz))$$

has value $(X, Y) = (1, 0)$ at $(\rho, z) = (1, 0)$, while its positive rod value remains infinity.

Write

$$F_R = (Y + \sqrt{-1}X) \circ \Phi_{1,R},$$

so F_R is the complex coordinate expression of $\Phi_{1,R}$. The preceding blow-down result shows F_R has a limit, which corresponds to a geodesic in $\mathcal{H}$. The boundary value of $Y(\Phi_{1,R})$ is infinity on the positive rod, and $F_R(1,0) = (1,0)$. Since the limit geodesic has to pass through $(1,0)$ and diverge to infinity, it is the vertical line $(X,Y) = (\frac{1}{\rho},0)$, and

$$F_R(\rho, z) \rightarrow \frac{\sqrt{-1}}{\rho}.$$

The convergence is smooth near $(1,0)$. Differentiating in the domain variable ρ , with R fixed, therefore gives

$$\partial_\rho F_R(1,0) = \frac{Rf'(R) + \sqrt{-1}Rh'(R)}{h(R)} \rightarrow -\sqrt{-1}, \quad \text{as } R \rightarrow \infty.$$

Consequently,

$$\frac{Rf'(R)}{h(R)} = o(1), \quad \frac{Rh'(R)}{h(R)} = -1 + o(1), \quad \text{as } R \rightarrow \infty.$$

Integration of the second relation yields

$$h(R) = R^{-1+o(1)}.$$

Moreover, for all sufficiently large R , we have $h'(R)/h(R) < -\frac{1}{2R}$. So integration gives

$$\frac{h(t)}{h(R)} \leq \left(\frac{t}{R}\right)^{-1/2}, \quad \text{for } t \geq R.$$

It follows that $|f'| \leq \frac{h(t)}{t}$, so f' is absolutely integrable at infinity. Hence $f(R) \rightarrow b$ for some $b \in \mathbb{R}$, and

$$\frac{|f(R) - b|}{h(R)} \leq \left(\sup_{t \geq R} \frac{t|f'(t)|}{h(t)}\right) \int_R^\infty \left(\frac{t}{R}\right)^{-1/2} \frac{dt}{t} = 2 \sup_{t \geq R} \frac{t|f'(t)|}{h(t)} \rightarrow 0.$$

□

Lemma A.14. *There is a constant matrix $Q \in \text{SL}(2, \mathbb{R})$ and constant a , such that for*

$$(Q^{-1})^\top \Phi Q^{-1} = (\rho e^u, v),$$

as $s \rightarrow \infty$,

$$u = o(s), \quad u_s = o(1), \quad u_\theta = o(1),$$

uniformly in θ . Moreover, for every $\epsilon > 0$,

$$\frac{|v - a \cos \theta|}{\sin^2 \theta} \leq C_\epsilon e^{(1+\epsilon)s}$$

for all sufficiently large s .

Proof. Introduce the Möbius transformation

$$Y + \sqrt{-1}X \mapsto -\frac{1}{Y + \sqrt{-1}X - b},$$

which corresponds to conjugation by another matrix $Q_2^{-1} \in \text{SL}(2, \mathbb{R})$. Then we set

$$\Phi_2 := (Q_2^{-1})^\top \Phi_1 Q_2^{-1}.$$

The Möbius transformation sends $(b,0)$ to infinity and the positive rod value of Φ_1 to $(0,0)$. We claim boundary values of Φ_2 on both infinite rods are finite. For the positive rod, the value of Φ_1 is infinite, so the value of Φ_2 is $(0,0)$, which is finite. For the negative rod, if the value of Φ_2 was not finite, then Φ_1 had negative rod value b . Then for the normalized maps

$$\Phi_{1,R}(\rho, z) = A_R(\Phi_1(R\rho, Rz)),$$

the positive rod value is infinity, whereas the negative rod value is

$$\frac{b - f(R)}{h(R)} \rightarrow 0.$$

This is a contradiction since $F_R \rightarrow \frac{\sqrt{-1}}{\rho}$, where the negative rod value should diverge to infinity.

Let c_+, c_- be the positive and negative finite rod values of $Y(\Phi_2)$, so $c_+ = 0$. Define a third hyperbolic isometry via

$$Y + \sqrt{-1}X \mapsto Y + \sqrt{-1}X - \frac{c_+ + c_-}{2}, \quad a = \frac{c_+ - c_-}{2}.$$

The hyperbolic isometry corresponds to a conjugation by $Q_3^{-1} \in \text{SL}(2, \mathbb{R})$. Set

$$\Phi_3 = (Q_3^{-1})^\top \Phi_2 Q_3^{-1}.$$

Then the boundary values of Φ_3 are a and $-a$. Set

$$H_R = X(\Phi_3(R, 0)), \quad K_R = Y(\Phi_3(R, 0)), \quad \alpha_R = \frac{f(R) - b}{h(R)}.$$

Direct calculation gives

$$H_R = \frac{1}{h(R)(1 + \alpha_R^2)} = R^{1+o(1)}.$$

Define the hyperbolic isometry

$$B_R : (X, Y) \mapsto \left(\frac{X}{H_R}, \frac{Y - K_R}{H_R} \right),$$

where $B_R(\Phi_3(R\rho, Rz))$ has value $(1, 0)$ at $(\rho, z) = (1, 0)$. This indicates that $B_R(\Phi_3(R\rho, Rz))$ converges to a limit. The complex coordinate expression

$$\hat{F}_R(\rho, z) = \frac{(Y + \sqrt{-1}X)(\Phi_3(R\rho, Rz)) - K_R}{H_R}$$

satisfies

$$\hat{F}_R = \frac{\alpha_R F_R - 1}{F_R + \alpha_R}.$$

Since $\alpha_R \rightarrow 0$ and $F_R \rightarrow \sqrt{-1}/\rho$, we obtain

$$\hat{F}_R \rightarrow \sqrt{-1}\rho.$$

Define $Q = Q_3 Q_2 Q_1$ and $(Q^{-1})^\top \Phi Q^{-1} = (\rho e^u, v)$, so $\Phi_3 = (Q^{-1})^\top \Phi Q^{-1}$. Let

$$U_R(\rho, z) = \log \frac{\text{Im } \hat{F}_R}{\rho} = u(R\rho, Rz) - \log \frac{H_R}{R}, \quad V_R(\rho, z) = \text{Re } \hat{F}_R = \frac{1}{H_R}(v(R\rho, Rz) - K_R).$$

The established rod compactness gives $C^{1,\alpha}$ convergence of these functions to zero on closed annuli away from o . For each fixed R , the definition of U_R gives,

$$u(s, \theta) = U_R \left(\frac{e^s}{R} \sin \theta, \frac{e^s}{R} \cos \theta \right) + \log \frac{H_R}{R}$$

for s near $\log R$. In particular,

$$u(\log R, \theta) = U_R(\sin \theta, \cos \theta) + \log \frac{H_R}{R}.$$

Since $U_R \rightarrow 0$ uniformly on the unit semicircle and $H_R = R^{1+o(1)}$, we obtain

$$u = o(s)$$

uniformly in θ . To estimate the derivatives, keep R fixed in the preceding identity, differentiate with respect to s and θ , and then evaluate at $s = \log R$. This gives

$$\begin{aligned} u_s(\log R, \theta) &= \sin \theta \partial_\rho U_R(\sin \theta, \cos \theta) + \cos \theta \partial_z U_R(\sin \theta, \cos \theta), \\ u_\theta(\log R, \theta) &= \cos \theta \partial_\rho U_R(\sin \theta, \cos \theta) - \sin \theta \partial_z U_R(\sin \theta, \cos \theta). \end{aligned}$$

The C^1 convergence of U_R therefore implies

$$u_s = o(1), \quad u_\theta = o(1),$$

uniformly in θ .

It remains to estimate v . On the portions of the outer rods contained in the fixed annulus, the boundary values of V_R are

$$V_R(0, z) = \begin{cases} (a - K_R)/H_R, & z > 0, \\ (-a - K_R)/H_R, & z < 0. \end{cases}$$

The uniform $C^{1,\alpha}$ convergence and the regularity of [34, Proposition 4.12] consequently gives

$$\left| V_R(\rho, z) - \frac{a - K_R}{H_R} \right| \leq C\rho^2 \quad \text{near } (0, 1),$$

and

$$\left| V_R(\rho, z) - \frac{-a - K_R}{H_R} \right| \leq C\rho^2 \quad \text{near } (0, -1),$$

with neighborhoods and constants independent of R . For θ sufficiently close to 0, we therefore have

$$\begin{aligned} \left| V_R(\sin \theta, \cos \theta) - \frac{a \cos \theta - K_R}{H_R} \right| &\leq \left| V_R(\sin \theta, \cos \theta) - \frac{a - K_R}{H_R} \right| + \frac{|a|}{H_R}(1 - \cos \theta) \\ &\leq C \sin^2 \theta. \end{aligned}$$

Here we used the boundedness of $|a|/H_R$. The analogous estimate near $\theta = \pi$ follows from the negative rod value.

On the remaining compact angular interval, $\sin \theta$ is bounded below. Moreover, V_R and its two rod values are uniformly bounded. Their interpolation

$$\frac{a \cos \theta - K_R}{H_R} = \frac{1 + \cos \theta}{2} \frac{a - K_R}{H_R} + \frac{1 - \cos \theta}{2} \frac{-a - K_R}{H_R}$$

is therefore uniformly bounded as well. Increasing C if necessary, we obtain on the entire unit semicircle

$$\left| V_R(\sin \theta, \cos \theta) - \frac{a \cos \theta - K_R}{H_R} \right| \leq C \sin^2 \theta.$$

By the definition of V_R , multiplication by H_R gives

$$|v(R \sin \theta, R \cos \theta) - a \cos \theta| \leq C H_R \sin^2 \theta.$$

Finally, $H_R = R^{1+o(1)}$ implies that, for every $\epsilon > 0$,

$$\frac{|v(s, \theta) - a \cos \theta|}{\sin^2 \theta} \leq C_\epsilon e^{(1+\epsilon)s}$$

for all sufficiently large s and $0 < \theta < \pi$. This proves the stated estimates. $\square$

Next we improve the estimate that we obtain in Lemma A.14. In the following, scalar functions are regarded as axisymmetric functions on Euclidean $\mathbb{R}^3$, and Δ and div denote its Euclidean operators. For $f = f(\rho, z)$, the gradient norm is $|\nabla f|^2 = f_\rho^2 + f_z^2$, which also equals its flat half-plane gradient norm.

Lemma A.15. *We have the following bounds on v when r is sufficiently large*

$$|v - a \cos \theta| \leq C \sin^{3/2} \theta, \quad |\nabla v| \leq \frac{C}{r} \sin^{1/2} \theta.$$

Proof. Recall that our notation is

$$X \circ \Phi_3 = \rho e^u, \quad Y \circ \Phi_3 = v,$$

and that the two outer rod values of v are a and $-a$. The harmonic map equations are

$$(A.15) \quad \Delta_{\mathbb{R}^3} u = -\rho^{-2} e^{-2u} |\nabla v|^2, \quad \text{div}_{\mathbb{R}^3} (\rho^{-2} e^{-2u} \nabla v) = 0.$$

By Lemma A.14,

$$(A.16) \quad u = o(s), \quad u_s = o(1), \quad u_\theta = o(1), \quad s = \log r,$$

as $s \rightarrow \infty$, uniformly in θ . Moreover, for every $\epsilon > 0$,

$$(A.17) \quad |v - a \cos \theta| \leq C_\epsilon e^{(1+\epsilon)s} \sin^2 \theta$$

for all sufficiently large s .

We first improve (A.17) to a bound independent of s . In logarithmic spherical coordinates, the second equation in (A.15) is $Lv = 0$, where

$$L = \partial_s^2 - (1 + 2u_s)\partial_s + \partial_\theta^2 - \cot \theta \partial_\theta - 2u_\theta \partial_\theta.$$

Consequently,

$$L(v - a \cos \theta) = -2au_\theta \sin \theta.$$

For $\lambda = 0$ and $\lambda = \frac{3}{2}$, direct computation gives

$$\frac{L(e^{\lambda s} \sin^{3/2} \theta)}{e^{\lambda s} \sin^{3/2} \theta} = \lambda^2 - \lambda - \frac{3}{2} - \frac{3}{4} \cot^2 \theta - 2\lambda u_s - 3u_\theta \cot \theta.$$

Choose s_0 sufficiently large that

$$|u_s|, |u_\theta| \leq \frac{1}{12}, \quad \text{for } s \geq s_0.$$

Using $|\cot \theta| \leq (1 + \cot^2 \theta)/2$, we obtain

$$(A.18) \quad L(e^{\lambda s} \sin^{3/2} \theta) \leq -\frac{3}{8} e^{\lambda s} \sin^{-1/2} \theta, \quad \lambda \in \left\{0, \frac{3}{2}\right\}.$$

Choose C sufficiently large to control $2au_\theta \sin \theta$ and the boundary values at $s = s_0$. For $\delta > 0$, consider

$$\pm(v - a \cos \theta) - C \sin^{3/2} \theta - \delta e^{3s/2} \sin^{3/2} \theta$$

on $[s_0, T] \times [0, \pi]$. By (A.18), these functions have nonnegative L -image and are non-positive at $s = s_0$. Fixing $0 < \epsilon < \frac{1}{2}$ in (A.17), we see that they are also non-positive at $s = T$ for all sufficiently large T . They vanish continuously at $\theta = 0, \pi$. Any positive maximum would therefore occur strictly inside the rectangle, where L is regular, contradicting the maximum principle. Letting $T \rightarrow \infty$ and then $\delta \rightarrow 0$, we obtain

$$(A.19) \quad |v - a \cos \theta| \leq C \sin^{3/2} \theta.$$

We next estimate the source in the equation for u . Away from the axis, the equation for v can be written as

$$v_{\rho\rho} + v_{zz} - (\rho^{-1} + 2u_\rho) v_\rho - 2u_z v_z = 0.$$

By (A.16),

$$|\nabla u| = \frac{\sqrt{u_s^2 + u_\theta^2}}{r} \leq \frac{C}{r}.$$

Fix $p = (\rho_0, z_0)$, set $r_0 = (\rho_0^2 + z_0^2)^{1/2}$, and consider the Euclidean disk

$$B = B_{\rho_0/8}(p) \subset \{\rho > 0\}.$$

On B , we have ρ is comparable to ρ_0 and r is comparable to r_0 . Hence

$$\rho_0 \sup_B (\rho^{-1} + 2|\nabla u|) \leq C.$$

Thus, after rescaling B to a unit disk, the equation for v has uniformly bounded first-order coefficients. The standard interior gradient estimate gives

$$|\nabla v(p)| \leq \frac{C}{\rho_0} \text{osc}_B v,$$

where C is independent of p . This follows from the interior $W^{2,q}$ estimate and Sobolev embedding with $q > 2$; only a uniform bound on the first-order coefficients is required.

To estimate the oscillation, (A.19) gives

$$\sup_B |v - a \cos \theta| \leq C \left(\frac{\rho_0}{r_0} \right)^{3/2}.$$

Moreover, since

$$|\nabla \cos \theta| = \frac{\rho}{r^2},$$

we have

$$\text{osc}_B(a \cos \theta) \leq C \frac{\rho_0^2}{r_0^2}.$$

Consequently,

$$\begin{aligned} \text{osc}_B v &\leq 2 \sup_B |v - a \cos \theta| + \text{osc}_B(a \cos \theta) \\ &\leq C \left(\frac{\rho_0}{r_0} \right)^{3/2}, \end{aligned}$$

where we used $0 < \rho_0/r_0 \leq 1$. Therefore

$$|\nabla v(p)| \leq \frac{C}{\rho_0} \left(\frac{\rho_0}{r_0} \right)^{3/2}.$$

Since p was arbitrary, we obtain

$$(A.20) \quad |\nabla v| \leq \frac{C}{\rho} \left(\frac{\rho}{r} \right)^{3/2} = \frac{C}{r} \sin^{1/2} \theta$$

for all sufficiently large r . □

Finally we have the following strong tameness.

Proposition A.16. *There exists $c > 0$ such that the harmonic model Φ_0 , defined by*

$$\Phi_0(\rho, z) = \left(\rho \left(c + \frac{|a|}{r} \right), a \cos \theta \right)$$

satisfies

$$d_{\mathcal{H}}(\Phi_3, \Phi_0) = O(r^{-1}).$$

Proof. Since $u = o(\log r)$, equations (A.15) and (A.20) imply, for every $\epsilon > 0$,

$$|\Delta_{\mathbb{R}^3} u| \leq C_\epsilon r^{-4+\epsilon} \sin^{-1} \theta.$$

The angular factor is integrable on $\mathbb{S}^2$. Moreover, $|\nabla u| \leq C/r$ implies that the flux through a shrinking cylindrical tube around the axis tends to zero on every compact exterior annulus. Thus the first equation in (A.15) holds weakly across the exterior axis. Consider the spherical mean

$$m(r) = \frac{1}{4\pi} \int_{\mathbb{S}^2} u(r, \omega) d\omega.$$

Averaging the equation for u gives

$$(r^2 m'(r))' = \frac{r^2}{4\pi} \int_{\mathbb{S}^2} \Delta_{\mathbb{R}^3} u(r, \omega) d\omega = O(r^{-2+\epsilon}).$$

Taking $0 < \epsilon < 1$, the right-hand side is integrable at infinity. Hence $r^2 m'(r)$ has a finite limit, $m'(r) = O(r^{-2})$, and $m(r)$ converges to a finite constant u_∞ . Furthermore,

$$\sup_\theta |u(r, \theta) - m(r)| \leq \pi \sup_\theta |u_\theta(r, \theta)| = o(1).$$

Therefore

$$(A.21) \quad u \rightarrow u_\infty, \quad \text{uniformly in } \theta.$$

Set

$$c = e^{u_\infty} > 0, \quad H = c + \frac{|a|}{r},$$

and define Φ_0 by

$$X \circ \Phi_0 = \rho H, \quad Y \circ \Phi_0 = a \cos \theta.$$

This map is harmonic. Write

$$D = d_{\mathcal{H}}(\Phi_3, \Phi_0).$$

The hyperbolic distance formula gives

$$\cosh(\sqrt{2}D) - 1 = \frac{(e^u - H)^2}{2e^u H} + \frac{(v - a \cos \theta)^2}{2\rho^2 e^u H}.$$

The first term tends uniformly to zero by (A.21). Since e^u and H are bounded above and below for large r , estimate (A.19) bounds the second term by

$$Cr^{-2} \sin \theta.$$

Thus $D \rightarrow 0$ uniformly for $0 < \theta < \pi$.

Finally, the distance between harmonic maps into $\mathcal{H}$ is subharmonic away from the axis. Fix a sufficiently large R_0 and set

$$C = R_0 \sup_{r=R_0} D.$$

Given $\eta > 0$, take $T > R_0$ sufficiently large that $D \leq \eta$ on $r = T$. For $\delta > 0$, the function

$$D - \frac{C}{r} - \eta + \delta \log \frac{\rho}{2T}$$

is subharmonic on

$$\{R_0 < r < T, \rho > 0\},$$

because both r^{-1} and $\log \rho$ are harmonic there. It is non-positive on the spherical boundaries and tends to $-\infty$ at the axis, since D is bounded there. Applying the maximum principle, first on the domain with a sufficiently thin cylindrical tube removed, gives

$$D - \frac{C}{r} - \eta + \delta \log \frac{\rho}{2T} \leq 0.$$

Letting $\delta \rightarrow 0$, then $T \rightarrow \infty$ and $\eta \rightarrow 0$, we conclude that

$$D \leq \frac{C}{r}.$$

This proves strong tameness. □

Starting from the strong tameness of the harmonic map, it is a direct application of [34, Proposition A.1] to conclude Theorem A.4. Although [34, Proposition A.1] is stated for smooth ends, the same argument applies to conical ends by working changing the lattice for the end. Accordingly, the asymptotic model in Theorem A.4 is allowed to have cone angles different from 2π along its infinite rods.

A.6. Properness and ALE rigidity. Finally, we record a criterion for properness and a rigidity statement for toric ALE orbifolds. We continue to write $(N, \underline{g}) = M/\mathbb{T}^2$, with $G, \rho = \sqrt{\det G}$, and $\Phi = \rho^{-1}G$ as above.

Lemma A.17. *Let (M, g) be a complete connected oriented toric conical Ricci-flat four-manifold that is integrable. Assume that N° is simply connected, $\partial N \neq \emptyset$, and $K_{\underline{g}} \leq 0$. Then $d\rho \neq 0$ on N° , and there is a globally defined harmonic conjugate z , satisfying $dz = \ast d\rho$, such that*

$$F = (\rho, z) : N \longrightarrow \mathbb{H} = \{(\rho, z) \in \mathbb{R}^2 : \rho \geq 0\}$$

is a homeomorphism. Its restriction to the interiors is a conformal diffeomorphism. If g also has quadratic curvature decay (A.1), then (M, g) is a proper toric conical gravitational instanton in the sense of Definitions A.1 and A.2.

Proof. Since (M, g) is complete and the torus is compact, the quotient $(N, \underline{g})$ is complete as a metric space, including its boundary. The reduced identities (2.5) and (3.4), applied with $\lambda = 0$, give $\Delta_{\underline{g}}\rho = 0$ and

$$(A.22) \quad \text{Hess}_{\underline{g}} \sqrt{\rho} = \frac{\sqrt{\rho}}{2} (K_{\underline{g}} \underline{g} - T) \leq 0, \quad T = \frac{1}{4} \text{tr}(\Phi^{-1} d\Phi \otimes \Phi^{-1} d\Phi).$$

Here T is the nonnegative tensor introduced before (3.4). These are local identities for an integrable Einstein metric, so they apply in the present noncompact Ricci-flat setting. The

function ρ is positive on N° and vanishes on ∂N , and is therefore nonconstant. If $d\rho(p) = 0$ at some $p \in N^\circ$, then along every sufficiently short geodesic γ issuing from p ,

$$\left. \frac{d}{dt} \sqrt{\rho(\gamma(t))} \right|_{t=0} = 0, \quad \frac{d^2}{dt^2} \sqrt{\rho(\gamma(t))} \leq 0.$$

Thus p would be a local maximum of ρ , contradicting the strong maximum principle for the nonconstant harmonic function ρ . Hence $d\rho \neq 0$ on N° .

Since N° is simply connected and $d(*d\rho) = 0$, there is a global harmonic conjugate z on N° . The map $F = (\rho, z)$ is a local conformal diffeomorphism. Set $\nu = -\log |d\rho|_{\underline{g}}$, so that in each such coordinate neighborhood

$$\underline{g} = e^{2\nu}(d\rho^2 + dz^2).$$

Equation (A.22) implies

$$\begin{pmatrix} \nu_\rho + \frac{1}{2\rho} & \nu_z \\ \nu_z & -\nu_\rho \end{pmatrix} \geq 0.$$

The diagonal entries and determinant therefore give

$$(A.23) \quad -\frac{1}{2\rho} \leq \nu_\rho \leq 0, \quad \nu_\rho^2 + \nu_z^2 \leq -\frac{\nu_\rho}{2\rho} \leq \frac{1}{4\rho^2}.$$

We claim that $F : N^\circ \rightarrow \mathbb{H}^\circ$ is a diffeomorphism. Let $\gamma : [0, 1] \rightarrow \mathbb{H}^\circ$ be a piecewise smooth path starting at $F(p)$, and lift it locally starting at $p \in N^\circ$. Since $\gamma([0, 1])$ is compact, its ρ coordinate is bounded below by some $\varepsilon > 0$. Along the lift, (A.23) gives

$$\left| \frac{d}{dt} \nu(\tilde{\gamma}(t)) \right| \leq \frac{|\dot{\gamma}(t)|}{2\varepsilon}, \quad \nu(\tilde{\gamma}(t)) \leq \nu(p) + \frac{L_{\text{Euc}}(\gamma)}{2\varepsilon}.$$

Consequently the $\underline{g}$ -length of every lifted subarc is bounded by a fixed constant times the Euclidean length of the corresponding subarc of γ . If the lift stopped before reaching the endpoint, its tail would be Cauchy. Completeness would give a limit in N , and the lower bound $\rho \geq \varepsilon$ would place this limit in N° , where the local inverse extends the lift. Thus every such path lifts throughout, and in particular F is onto. It follows that F is a covering map. Since N° is connected and $\mathbb{H}^\circ$ is simply connected, F is a diffeomorphism.

We next extend F across ∂N , using the local expansions in Lemma 2.7. Near an interior point of a rod, let $t \geq 0$ be the inward distance to that rod in the quotient metric $\underline{g}$, and let s be arclength along the rod, extended constantly along the normal geodesics. Thus (t, s) are local Fermi coordinates. By (2.10),

$$\begin{aligned} \underline{g} &= dt^2 + f(t^2, s)^2 ds^2, & f(0, s) &= 1, \\ \rho &= t\kappa(s) + O(t^3), & \kappa(s) &:= \partial_t \rho(0, s) > 0. \end{aligned}$$

In particular, $\kappa(s)$ is the inward normal derivative of ρ along the rod. The one-form $*d\rho$ extends smoothly to $t = 0$, so the same is true of z . Choose the direction of s so that $*dt = ds$ on the rod. Then

$$d\rho|_{t=0} = \kappa(s)dt, \quad dz|_{t=0} = \kappa(s)ds.$$

Hence F has nonzero Jacobian there and is a local diffeomorphism of surfaces with boundary.

At a turning point, use the quadrant coordinates (x, y) of (2.9), rescaled so that $\underline{g}(0) = dx^2 + dy^2$ and ordered consistently with the orientation defining $*$. The corner expansion and $dz = *d\rho$ give

$$\begin{aligned} \rho &= cxy + O((x^2 + y^2)^2), \\ z &= z_0 + \frac{c}{2}(y^2 - x^2) + O((x^2 + y^2)^2), \quad c > 0. \end{aligned}$$

Here z again extends smoothly to the closed quadrant, since $*d\rho$ is smooth and closed there. For a sufficiently small quadrant neighborhood, z is strictly decreasing on $y = 0$ and strictly increasing on $x = 0$, with values on opposite sides of z_0 . Together with the injectivity of F on N° and the positivity of ρ away from the rods, this shows that F is injective on a small

closed quarter-disc. It is therefore a homeomorphism from the quarter-disc onto its image. The boundary of the image consists of an interval on $\{\rho = 0\}$ and a simple arc whose interior lies in $\{\rho > 0\}$. By the Jordan curve theorem, the image contains a neighborhood of $(0, z_0)$ relative to $\mathbb{H}$. Thus F is a local homeomorphism at the turning point as well.

It remains to show that the extended boundary map is bijective. For any $z_0 \in \mathbb{R}$, integrating the lower bound for ν_ρ in (A.23) gives

$$e^{\nu(\rho, z_0)} \leq e^{\nu(1, z_0)} \rho^{-1/2}, \quad 0 < \rho \leq 1.$$

The curve $\rho \mapsto F^{-1}(\rho, z_0)$ therefore has finite $\underline{g}$ -length as $\rho \rightarrow 0$. By completeness it converges to a point of ∂N , whose image is $(0, z_0)$. Thus the boundary map is onto. If two distinct boundary points had the same image, disjoint local homeomorphism neighborhoods would give two preimages of nearby interior points, contradicting interior injectivity. Hence $F : N \rightarrow \mathbb{H}$ is a bijective local homeomorphism, and therefore a homeomorphism. We have proved regularity and the existence of global (ρ, z) coordinates. With quadratic curvature decay, the final assertion follows from Definitions A.1 and A.2. $\square$

Lemma A.18. *Let (M, g) be a complete connected Ricci-flat ALE four-orbifold with at most isolated orbifold singularities. Suppose it carries an effective isometric $\mathbb{T}^2$ -action with connected stabilizers. In a fixed basis of 2π -periodic angular generators, set*

$$A = (1, 0)^\top, \quad B = (0, 1)^\top, \quad C = (1, 1)^\top.$$

If every circle stabilizer has one of the unoriented directions represented by A , B , and C , then (M, g) is isometric to Euclidean four-space.

Proof. We first show that M is smooth. Every isolated orbifold point is fixed by the connected torus, since the torus preserves the discrete singular set. Let v, w be the primitive rod vectors incident to such a point. In the local toric model, the orbifold group is the quotient of $\mathbb{Z}^2$ by the sublattice generated by v, w , and hence has order $|\det(v, w)|$; see [34, Section 3]. Any two distinct directions among A, B, C have determinant ± 1 . Thus the local group is trivial at every torus fixed point, and there are no orbifold singularities. Since the only singularities allowed in the statement were isolated orbifold points, the metric is now smooth everywhere, including along all rods.

Let Γ be the finite group at infinity. A sufficiently large invariant cross-section L of the ALE end is diffeomorphic to S^3/Γ . To obtain such a cross-section, extend the squared ALE radius smoothly over M and average it over $\mathbb{T}^2$. The averaged function has Hessian $2g + o(1)$ and is strictly increasing along radial rays sufficiently far out, so a large level has the required properties. The torus action on L is effective: an element fixing L pointwise also fixes its normal vector, and is therefore the identity isometry. Its stabilizers are connected, and it has no fixed points, since $\mathbb{T}^2$ has no faithful orthogonal representation in dimension three. Consequently $L/\mathbb{T}^2$ is a circle or an interval. The circle case would make L a principal $\mathbb{T}^2$ -bundle over a circle and give an infinite fundamental group, contrary to $\pi_1(L) \cong \Gamma$. Thus $L/\mathbb{T}^2$ is an interval.

Let $v_-, v_+ \in \{A, B, C\}$ represent the two endpoint stabilizers. The corresponding circles bound meridian discs in the two solid tori making up L , so

$$(A.24) \quad \Gamma \cong \pi_1(L) \cong \mathbb{Z}^2 / (\mathbb{Z}v_- + \mathbb{Z}v_+).$$

Here $\mathbb{Z}^2$ records integer coefficients in the angular basis, whose periods are 2π . Finiteness forces the endpoint directions to be distinct, so the same determinant calculation gives $\Gamma = \{1\}$. The end is therefore AE.

We can now apply ordinary Bishop–Gromov comparison, as in the volume comparison argument of [34, proof of Proposition 4.42]. For any $p \in M$, the function

$$R \mapsto \frac{\text{Vol}_g(B_g(p, R))}{\omega_4 R^4}$$

is nonincreasing, where ω_4 is the volume of the Euclidean unit four-ball. Its limit is one both as $R \downarrow 0$, by smoothness, and as $R \rightarrow \infty$, by the AE asymptotics. Thus it is identically one, and the equality case of Bishop–Gromov comparison gives $(M, g) \cong \mathbb{R}^4$. $\square$

Remark A.19. Lemma A.17 allows arbitrary positive cone angles. In Lemma A.18, however, all rod interiors are smooth and have cone angle 2π . The orbifold formulation is useful for the intermediate rescalings in the proof of Theorem 5.7, where the conical surfaces have collapsed to an isolated point; the determinant calculation above removes that possible singularity. The main conical bubble in Proposition 5.5 retains its finite conical rods and is instead treated by the angle comparison argument of [34, proof of Proposition 4.42].

A.7. Topological bounds for toric gravitational instantons. We prove some topological bounds for toric gravitational instantons similar to the case of compact toric Einstein metrics in Section 7.2, which may be of independent interest. For AF ends, the sharper bound below strengthens the restrictions obtained by Nilsson [39, Section 4.1 and Remark 5].

Let (M, g) be a smooth, simply connected, proper toric gravitational instanton with $n \geq 1$ turning points. The finite rods $\mathfrak{I}_j$, $1 \leq j < n$, give spheres $\Sigma_j = \pi^{-1}(\mathfrak{I}_j)$; set $e_j = [\Sigma_j]^2$. Choose representatives $\mathbf{v}_j = 2\pi u_j$ with $\det(u_{j-1}, u_j) = 1$. Then

$$(A.25) \quad e_j = -\det(u_{j-1}, u_{j+1}), \quad u_{j+1} = -e_j u_j - u_{j-1}.$$

The sign differs from the compact recurrence because increasing z is opposite to the positive boundary orientation of the half-plane. The plumbing description gives

$$(A.26) \quad \chi(M) = n, \quad \tau(M) = \text{sign } Q, \quad Q = \begin{pmatrix} e_1 & 1 & & 0 \\ 1 & e_2 & \ddots & \\ & \ddots & \ddots & 1 \\ 0 & & 1 & e_{n-1} \end{pmatrix};$$

see [39, Theorem 1]. The matrix represents the intersection form in the basis of finite rod spheres, oriented so adjacent intersections are one. Its zero eigenvalues do not contribute to the signature. For $n = 1$, the matrix is empty.

Lemma A.20. Every finite rod sphere has $|e_j| \leq 2$. Unless g is hyperkähler in one of the two orientations,

$$(A.27) \quad e_j \in \{-1, 0, 1\}.$$

Proof. Apply the Weyl identities of Section 7.1 in both orientations, with $\lambda = 0$. The functions $w_{\pm} = -K \mp J$ tend to zero at infinity by quadratic curvature decay and satisfy

$$(A.28) \quad \Delta_g w_{\pm} + 3\langle d \log \rho, dw_{\pm} \rangle_g = -3w_{\pm}^2 - 4|q_{\pm}|^2.$$

The even extensions across rods and turning points used in the proof of Proposition 7.10 allow the maximum principle there. A negative minimum is impossible. Thus each $w_{\pm}$ is either positive everywhere or identically zero; in the latter case (A.28) also gives $q_{\pm} = 0$, hence $W^{\pm} = 0$.

The rod curvature calculation in the main text, now with $\lambda = 0$, gives

$$(A.29) \quad K_j = w_+ + w_-, \quad |K_j^{\perp}| = |w_+ - w_-| \leq K_j,$$

where K_j and $K_j^{\perp}$ are the tangent and normal curvatures of Σ_j . Consequently,

$$2\pi|e_j| \leq \int_{\Sigma_j} |K_j^{\perp}| dA_g \leq \int_{\Sigma_j} K_j dA_g = 4\pi.$$

If both Weyl halves are nonzero, the last inequality is strict and integrality gives (A.27). Otherwise Ricci flatness and simple connectivity give a parallel triple in Λ^+ or Λ^- , so the metric is hyperkähler in that orientation. $\square$

Proposition A.21. *If g is not hyperkähler in either orientation, then*

$$(A.30) \quad |\tau(M)| \leq \left\lfloor \frac{\chi(M) + 1}{3} \right\rfloor.$$

For an AF end, one always has

$$(A.31) \quad 3|\tau(M)| \leq \chi(M).$$

Remark A.22. *Notice that, it is essential that we exclude hyperkähler metrics in (A.30). The ALE and ALF Gibbons–Hawking metrics with three collinear centres have $\chi(M) = 3$ and $|\tau(M)| = 2$, and hence violate this bound.*

Proof. Step 1. Bound the signature of the intersection matrix. Set $m = n - 1$; the case $m = 0$ is immediate. Let A_m have entries one immediately above and below the diagonal and zero elsewhere. Lemma A.20 gives

$$A_m - \text{Id} \leq Q \leq A_m + \text{Id}.$$

Eigenvalue monotonicity bounds the signature by those of the two outer matrices. These have opposite signatures, since conjugating by $\text{diag}(1, -1, 1, -1, \dots)$ changes A_m to $-A_m$. The eigenvalues of $A_m + \text{Id}$ are

$$1 + 2 \cos \frac{k\pi}{m+1}, \quad 1 \leq k \leq m.$$

Counting their signs gives $|\tau(M)| \leq \lfloor (m+2)/3 \rfloor$, proving (A.30).

Step 2. The strenghtening in the AF case. The two infinite rods have the same direction, so $u_n = \pm u_0$ and $n \geq 2$. The recurrence gives $\det Q = (-1)^{n-1} \det(u_0, u_n) = 0$. Since all adjacent entries are one, a vector in $\ker Q$ is determined by its first component. Thus $\dim \ker Q = 1$.

If one Weyl half vanished, the signed version of (A.29) would give $e_j = 2$ for every finite rod, or $e_j = -2$ for every finite rod. The resulting matrix $\pm 2 \text{Id} + A_m$ is definite, a contradiction. The flat case is also excluded by Gauss–Bonnet on a finite rod sphere. Hence the first bound applies. Finally,

$$\tau(M) \equiv \text{rank } Q = n - 2 \equiv \chi(M) \pmod{2}.$$

Thus $\chi(M) - 3|\tau(M)|$ is even and at least -1 , so it is nonnegative. $\square$

Remark A.23. *The self-intersection bound is proved without using the Hitchin–Thorpe inequality. It implies the signature estimate above, but also gives restrictions on individual rod spheres that neither signature inequality detects. For example, the AF chains with finite self-intersections $(a, 0, -a)$ in [39, Figure 5] have $\chi(M) = 4$ and $\tau(M) = 0$ for every integer a . Both the classical and improved signature inequalities hold for every a , whereas Lemma A.20 requires $|a| \leq 1$.*

For AF ends, (A.31) strengthens $2\chi(M) \geq 3|\tau(M)|$ to $\chi(M) \geq 3|\tau(M)|$. Nilsson observes that the former inequality excludes none of the AF rod structures with four turning points [39, Remark 5]. The latter excludes all four exceptional structures in [39, Figure 6]: their finite self-intersections are

$$\pm(1, 2, 1) \quad \text{or} \quad \pm(2, 1, 2),$$

so $\chi(M) = 4$ and $|\tau(M)| = 2$.

For ALE and ALF ends, (A.30) provides an additional restriction outside the hyperkähler case; the classical inequalities also involve boundary terms at infinity [39, Section 4.1].

For example, the AF rod sequence with primitive representatives

$$(u_0, u_1, u_2, u_3, u_4) = ((0, 1), (-1, 0), (1, -1), (-1, 2), (0, -1))$$

has finite self-intersections $(1, 2, 1)$ and intersection eigenvalues $0, 1, 3$. It has $\chi = 4$ and $\tau = 2$, so it satisfies the classical inequality but violates (A.31). Likewise, the AF chains $(a, 0, -a)$ have $\chi = 4$ and $\tau = 0$ for every integer a , whereas the individual rod bound requires $|a| \leq 1$.

APPENDIX B. TORIC AE GRAVITATIONAL INSTANTONS WITH EQUAL CONE ANGLES

In this appendix we study toric conical equal cone angle AE gravitational instantons that we need for the gluing construction. See Appendix A for the reduction of proper toric conical gravitational instantons to axisymmetric harmonic maps. Theorem A.4 has already proved that for a proper toric conical gravitational instanton, it is of Asymptotic Type ALE, $\text{ALF}^\pm$, or AF_β . As it proved in [34], given a rod structure $\mathfrak{R}$ with an Asymptotic Type, there is a unique axisymmetric harmonic map that is strongly tamed by $\mathfrak{R}$ and the Asymptotic Type. So any proper toric conical gravitational instanton corresponds to a rod structure and an Asymptotic Type, and the harmonic map is exactly the one strongly tamed by the rod structure and Asymptotic Type. Note that in [34], the definition of rod structures includes the Asymptotic Types, which is different from the notion of rod structures we use here. Here AE means that the end is ALE and has trivial group at infinity.

The sequences W used in our construction admit two explicit descriptions. For $(B, (A, B)^n, A)$, diagonality forces the harmonic map to take values in a geodesic of the hyperbolic plane. For BCA , $BACA$, $BCACA$, and $BCABCA$, the Hermitian structure allows us to use the Tod ansatz, with its description by a convex piecewise linear function on the axis. The first alternating case $BABA$ is also Hermitian. In both families, the resulting formulas give control of the cone-angle map. We use this to prove existence and uniqueness of the equal-angle metric up to homothety, and to show that its decaying $\mathbb{T}^2$ -invariant infinitesimal deformations at fixed cone angles consist only of scaling modulo coordinate changes. These properties provide the linear analysis required for gluing.

B.1. Toric conical AE gravitational instantons with two rod directions. In this section, we consider rod structures that only consist of two rod directions, $e_1 = [1, 0]^\top$ and $e_2 = [0, 1]^\top$. We prove

Theorem B.1. *For every $n \geq 1$ there is a proper toric conical AE gravitational instanton, whose rod vectors are*

$$(e_2, e_1, e_2, \dots, e_2, e_1),$$

with $2n$ finite rods, smooth cone angles along infinite rods, and the same cone angle $2\pi\beta_n$ on all finite rods. It is unique up to homothety and equivariant isometry among proper toric conical AE gravitational instantons with these rod vectors, whose finite rods all have the same cone angle and infinite rods have cone angle 2π . Moreover, $\beta_n > 1$.

These AE gravitational instantons are actually explicit. The rod lengths are invariant under reflection of z at the middle turning point, and the metric admits an orientation-reversing isometry.

We claim that the condition that there are only two rod directions forces the metric to be diagonal. For any proper toric conical AE gravitational instanton with rod directions e_1, e_2 and smooth infinite rods, let Φ be its associated harmonic map. The target isometry $\Phi \mapsto S^\top \Phi S$, where $S = \text{diag}(1, -1)$, fixes both rod lines and preserves the normalized Euclidean asymptotics. Thus $S^\top \Phi S$ is strongly tamed by the same rod structure and has the same asymptotic data. By the uniqueness theorem [34, Theorem 4.24], we have

$$S^\top \Phi S = \Phi,$$

which forces $\Phi_{12} = 0$. Consequently, the metric has the diagonal form (B.1), and the harmonic-map equation reduces to the scalar axisymmetric Laplace equation. This argument applies for arbitrary positive rod lengths, independently of the equal-angle condition.

Set $m = 2n + 1$ and choose

$$a_1 < \dots < a_m, \quad \epsilon_i = (-1)^{i+1}.$$

For $\rho > 0$, set

$$R_i = \sqrt{\rho^2 + (z - a_i)^2}, \quad \mu_i = R_i - (z - a_i), \quad E_{ij} = R_i R_j + (z - a_i)(z - a_j) + \rho^2,$$

$$U = \prod_{i=1}^m \mu_i^{\epsilon_i}, \quad V = \frac{\prod_{i < j} E_{ij}^{-\epsilon_i \epsilon_j}}{2^{n+1} \prod_i R_i}.$$

Then

$$(B.1) \quad g_a = V(d\rho^2 + dz^2) + U d\phi_1^2 + \rho^2 U^{-1} d\phi_2^2$$

is positive definite and Ricci-flat. Indeed, with $u = \frac{1}{2} \log U$ and $\nu = \frac{1}{2} \log V$, the Ricci-flat equations are

$$\begin{aligned} u_{\rho\rho} + \rho^{-1} u_\rho + u_{zz} &= 0, \\ \nu_\rho &= \rho(u_\rho^2 - u_z^2) - u_\rho, \\ \nu_z &= 2\rho u_\rho u_z - u_z. \end{aligned}$$

The first follows from the same equation for $\log \mu_i$. Differentiating the displayed product for V gives the other two. All factors are positive when $\rho > 0$.

The outer rods have cone angle 2π , and the cone angle $2\pi\alpha_k$ of (a_k, a_{k+1}) is

$$(B.2) \quad \alpha_k(a) = \prod_{i \leq k < j} (a_j - a_i)^{\epsilon_i \epsilon_j}, \quad 1 \leq k \leq 2n.$$

For example, on a finite rod the expansions

$$\mu_i = \frac{\rho^2}{2(z - a_i)}(1 + O(\rho^2)) \quad (i \leq k), \quad \mu_j = 2(a_j - z)(1 + O(\rho^2)) \quad (j > k)$$

give $\alpha_k^2 = \lim_{\rho \rightarrow 0} U/(\rho^2 V)$ for odd k , and $\alpha_k^2 = \lim_{\rho \rightarrow 0} 1/(UV)$ for even k . Substitution gives (B.2); all factors involving z cancel. The same calculation on either outer rod gives one.

At infinity let $R = \sqrt{\rho^2 + z^2}$. Since $\sum_i \epsilon_i = 1$ and $\sum_{i < j} \epsilon_i \epsilon_j = -n$,

$$U = (R - z)(1 + O_\infty(R^{-1})), \quad V = (2R)^{-1}(1 + O_\infty(R^{-1})).$$

The quotients in these formulas have the same differentiated expansions at the outer axes: one can use $\mu_i = \rho^2/(R_i + z - a_i)$ near the positive axis. Setting $x^2 = R + z$ and $y^2 = R - z$ therefore gives

$$g_a = dx^2 + x^2 d\phi_2^2 + dy^2 + y^2 d\phi_1^2 + O_\infty(r^{-2}), \quad r^2 = x^2 + y^2.$$

The error is understood in the Euclidean tensor norm, with its corresponding differentiated bounds. This is an AE end with smooth outer axes.

For every choice of turning points $a_1 < \dots < a_{2n+1}$, the formulas above define a harmonic map

$$\Phi_a = \rho^{-1} \text{diag}(U, \rho^2 U^{-1})$$

strongly tamed by the corresponding alternating rod structure of AE type. By the existence and uniqueness theorem in [34], this is precisely the harmonic map associated with these data. The regularity theorem in [34] then gives the conical extension of g_a across the fixed surfaces and torus fixed points. Since the turning points are arbitrary, the metrics g_a exhaust all proper toric conical AE gravitational instantons with these alternating rod vectors, smooth infinite rods, and the prescribed angular lattice. Here the augmentation is fixed by the Euclidean normalization at infinity.

Translation and homothety allow the normalization $a_1 = 0$, $a_m = 1$. Write

$$\ell_k = a_{k+1} - a_k > 0, \quad \sum_{k=1}^{2n} \ell_k = 1, \quad d_{ij} = a_j - a_i.$$

On the simplex of ℓ , define

$$(B.3) \quad F(\ell) = \sum_{i < j} \epsilon_i \epsilon_j d_{ij} \log d_{ij}.$$

The convention $0 \log 0 = 0$ extends F continuously to the closed simplex. The sum $s_k = \sum_{i \leq k} \epsilon_i$ is zero or one. Hence

$$\sum_{i \leq k < j} \epsilon_i \epsilon_j = s_k(1 - s_k) = 0,$$

and differentiation gives

$$(B.4) \quad dF_\ell(\dot{\ell}) = \sum_{k=1}^{2n} \log \alpha_k(\ell) \dot{\ell}_k, \quad \sum_{k=1}^{2n} \dot{\ell}_k = 0.$$

Thus equal angles are precisely the critical points of F under the normalization of total length.

Lemma B.2. *On the open simplex*

$$\left\{ \ell = (\ell_1, \dots, \ell_{2n}) \in (0, \infty)^{2n} : \sum_{k=1}^{2n} \ell_k = 1 \right\},$$

the Hessian of F is negative definite. More precisely, for every nonzero variation $\dot{\ell} = (\dot{\ell}_1, \dots, \dot{\ell}_{2n})$ satisfying $\sum_k \dot{\ell}_k = 0$, one has

$$D^2F(\ell)[\dot{\ell}, \dot{\ell}] < 0.$$

Consequently, the map

$$\ell \mapsto (\log \alpha_1(\ell), \dots, \log \alpha_{2n}(\ell))$$

has differential of rank $2n - 1$ everywhere on this simplex.

Proof. Fix ℓ and a variation $\dot{\ell}$ satisfying $\sum_k \dot{\ell}_k = 0$. Set $m = 2n + 1$ and define the corresponding variations of the turning points by

$$\dot{a}_1 = 0, \quad \dot{a}_i = \sum_{k=1}^{i-1} \dot{\ell}_k, \quad 2 \leq i \leq m.$$

Thus $\dot{a}_m = 0$, consistently with the normalization $a_1 = 0, a_m = 1$. Differentiating twice along $\ell + t\dot{\ell}$ gives

$$D^2F(\ell)[\dot{\ell}, \dot{\ell}] = \sum_{i < j} \epsilon_i \epsilon_j \frac{(\dot{a}_j - \dot{a}_i)^2}{a_j - a_i}.$$

For $\tau \in \mathbb{R}$, define

$$H(\tau) = \sum_{i < j} \epsilon_i \epsilon_j |\tau(a_j - a_i) + \dot{a}_j - \dot{a}_i|.$$

Order the numbers $\tau a_i + \dot{a}_i$ as $y_1 \leq \dots \leq y_m$, and let $q_j \in \{1, -1\}$ be the sign ϵ_i attached to y_j . Set

$$S_k = \sum_{j=1}^k q_j.$$

Since $\sum_j q_j = \sum_i \epsilon_i = 1$, writing each difference $y_j - y_i$ as a sum of consecutive gaps yields

$$H(\tau) = \sum_{k=1}^{m-1} S_k(1 - S_k)(y_{k+1} - y_k) \leq 0.$$

Indeed, S_k is an integer, so $S_k(1 - S_k) \leq 0$. For sufficiently large positive or negative τ , the signs occur in the original or reversed alternating order. In either case every S_k is zero or one, so $H(\tau) = 0$.

For $d > 0$ and $c \in \mathbb{R}$, direct integration gives

$$\int_{\mathbb{R}} (|d\tau + c| - d|\tau| - c \operatorname{sgn} \tau) d\tau = \frac{c^2}{d}.$$

The identity

$$\sum_{i \leq k < j} \epsilon_i \epsilon_j = 0$$

implies both

$$\sum_{i < j} \epsilon_i \epsilon_j (a_j - a_i) = 0, \quad \sum_{i < j} \epsilon_i \epsilon_j (\dot{a}_j - \dot{a}_i) = 0.$$

Applying the integral formula to each pair therefore gives

$$D^2 F(\ell)[\dot{\ell}, \dot{\ell}] = \int_{\mathbb{R}} H(\tau) d\tau \leq 0.$$

We now determine when equality holds. If the integral vanishes, then $H \equiv 0$, since H is continuous and non-positive. Whenever the numbers $\tau a_i + \dot{a}_i$ are distinct, all their consecutive gaps are positive. Hence every S_k must be zero or one, which forces the ordered signs to alternate, beginning and ending with $+1$.

We claim that the points $(a_i, \dot{a}_i)$ are collinear. We use Kelly's elementary minimum-distance proof of the Sylvester–Gallai theorem. Suppose otherwise. Among all positive distances from one of these points to a line through two others, choose a smallest one, realized by a point P and a line L . The line L contains exactly two of the points. Indeed, if it contained at least three, two of them, say A and B , would lie on the same closed side of the perpendicular foot from P , with B farther from that foot. Then

$$\text{dist}(A, PB) = \text{dist}(P, L) \frac{|AB|}{|PB|} < \text{dist}(P, L),$$

contradicting the choice of P and L .

Since the a_i are distinct, L is not vertical. Thus there is a value τ_0 at which the two affine functions $\tau a_i + \dot{a}_i$ corresponding to the points on L have the same value, and no third function has that value. These two functions occupy adjacent positions in the ordered list near τ_0 , and exchange positions as τ passes through τ_0 . Their signs must be opposite for the ordering to alternate on one side, but exchanging them prevents the ordering from alternating on both sides. Any simultaneous crossings at other values do not affect these two adjacent positions. This contradicts $H \equiv 0$.

The points $(a_i, \dot{a}_i)$ are therefore collinear. Since they include $(a_1, \dot{a}_1) = (0, 0)$ and $(a_m, \dot{a}_m) = (1, 0)$, we obtain $\dot{a}_i = 0$ for every i , and consequently $\dot{\ell}_k = \dot{a}_{k+1} - \dot{a}_k = 0$ for every k . Thus the Hessian is negative definite on the tangent space $\{\dot{\ell} : \sum_k \dot{\ell}_k = 0\}$.

Finally, the identity

$$dF = \sum_{k=1}^{2n} \log \alpha_k d\ell_k$$

gives

$$D^2 F(\ell)[\dot{\ell}, \dot{\ell}] = \sum_{k=1}^{2n} \dot{\ell}_k D_{\dot{\ell}} \log \alpha_k.$$

If $\dot{\ell}$ lies in the kernel of the differential of the logarithmic cone-angle map, the right-hand side vanishes. Negative definiteness then forces $\dot{\ell} = 0$. The differential is therefore injective on the $(2n - 1)$ -dimensional tangent space, proving the rank statement. $\square$

The continuous function F attains a maximum on the closed simplex. We show that no boundary point is a maximum. On the boundary of the simplex, some $\ell_k = a_{k+1} - a_k = 0$. This means some consecutive turning points coincide. We show that slightly separating the coincident turning points increases F , so such a configuration cannot maximize it.

At a boundary point, partition the turning points into maximal consecutive blocks having the same position. Since the coefficients ϵ_i alternate between $+1$ and -1 , their sum within each group is zero if the number of turning points is even, and is $+1$ or -1 if that number is odd. At least one group contains more than one turning point.

Suppose first that a group contains an even number of turning points. Divide it into two consecutive groups, each containing an odd number of turning points. The sums of their respective coefficients ϵ_i are opposite, so their product is -1 . Separate the positions of these two groups by a small distance $\delta > 0$, keeping all turning points in $[0, 1]$ and preserving $a_1 = 0$

and $a_m = 1$. The contribution to F from pairs with one turning point in each of these two groups changes from zero to $-\delta \log \delta$. Contributions involving turning points outside the original group change by $O(\delta)$, since their distances from that group remain bounded away from zero. Consequently,

$$F_{\text{new}} - F_{\text{old}} = -\delta \log \delta + O(\delta) > 0$$

for sufficiently small $\delta > 0$.

It remains to consider a group containing an odd number, at least three, of coincident turning points. Let c be their common position and define

$$\sigma = \sum_{\{i|a_i=c\}} \epsilon_i \in \{-1, 1\}.$$

Divide this group into three consecutive groups, each containing an odd number of turning points. The sums of their respective coefficients ϵ_i are $\sigma, -\sigma, \sigma$.

Suppose first that $0 < c < 1$. Move these three groups to $c - \delta, c, c + \delta$, respectively, choosing $\delta > 0$ sufficiently small to preserve the ordering of all turning points. The contribution to F from pairs within the original group increases by

$$-\delta \log \delta - \delta \log \delta + 2\delta \log(2\delta) = 2\delta \log 2.$$

For each turning point outside this group, its distances from the three new positions remain bounded away from zero. The terms linear in δ cancel between the groups at $c - \delta$ and $c + \delta$, since both have coefficient sum σ . Taylor expansion therefore shows that all contributions involving turning points outside the original group change by $O(\delta^2)$. Consequently,

$$F_{\text{new}} - F_{\text{old}} = 2\delta \log 2 + O(\delta^2) > 0$$

for sufficiently small $\delta > 0$.

Suppose next that $c = 0$. Place the three groups at $0, p\delta, \delta$, respectively, where $0 < p < 1$. This preserves $a_1 = 0$. Their contribution to F increases by

$$\delta h(p), \quad h(p) = -p \log p - (1 - p) \log(1 - p).$$

For each turning point a_j outside the original group, we have $a_j > 0$, and the change in its contribution with the original group is exactly

$$\sigma \epsilon_j [(a_j - \delta) \log(a_j - \delta) - (a_j - p\delta) \log(a_j - p\delta)].$$

Choose δ smaller than half the distance from zero to every turning point outside the original group. The mean value theorem then bounds the absolute value of the sum of these changes by

$$C(1 - p)\delta,$$

where C is independent of p and δ . Thus

$$F_{\text{new}} - F_{\text{old}} \geq \delta(h(p) - C(1 - p)).$$

Since

$$\frac{h(p)}{1 - p} \longrightarrow +\infty \quad \text{as } p \longrightarrow 1,$$

we can first choose p sufficiently close to one and then choose $\delta > 0$ sufficiently small. The resulting change in F is positive. The case $c = 1$ follows by reflecting the construction about $1/2$, preserving $a_m = 1$.

All these perturbations remain in the closed simplex. Therefore no boundary point can maximize F . Its maximum lies in the open simplex and is unique by Lemma B.2. Since

$$dF = \sum_{k=1}^{2n} \log \alpha_k d\ell_k,$$

the critical-point condition under $\sum_k \ell_k = 1$ is

$$\log \alpha_1 = \cdots = \log \alpha_{2n}.$$

Hence there is a unique choice of normalized rod lengths for which all finite cone angles are equal. The identification of all metrics with the family g_a then gives uniqueness of the corresponding metric up to homothety and equivariant isometry.

Finally, after normalizing the total rod length to one, the Jacobian of the equations $\log \alpha_i - \log \alpha_{2n} = 0$, $1 \leq i \leq 2n-1$, with respect to the rod lengths is non-singular at the equal-angle solution. We finish by verifying this assertion. Consider the map

$$\left\{ \ell \in (0, \infty)^{2n} : \sum_{k=1}^{2n} \ell_k = 1 \right\} \longrightarrow \mathbb{R}^{2n-1},$$

$$\ell \longmapsto \left(\log \frac{\alpha_1(\ell)}{\alpha_{2n}(\ell)}, \dots, \log \frac{\alpha_{2n-1}(\ell)}{\alpha_{2n}(\ell)} \right).$$

At the equal-angle solution, its differential is an isomorphism from the tangent space

$$\left\{ \dot{\ell} \in \mathbb{R}^{2n} : \sum_{k=1}^{2n} \dot{\ell}_k = 0 \right\}$$

onto $\mathbb{R}^{2n-1}$. Indeed, if $\dot{\ell}$ lies in its kernel, then

$$D_{\dot{\ell}} \log \alpha_k = c$$

for some common value c and every k . Therefore

$$D^2 F(\ell)[\dot{\ell}, \dot{\ell}] = \sum_k \dot{\ell}_k D_{\dot{\ell}} \log \alpha_k = c \sum_k \dot{\ell}_k = 0.$$

Lemma B.2 implies $\dot{\ell} = 0$. The differential is thus injective, and the domain and target both have dimension $2n-1$.

Denote the common finite cone factor by β_n . The formula for the first cone factor gives the common cone angle is

$$\beta_n = \alpha_1 = \dots = \alpha_{2n} = \prod_{j=1}^n \frac{a_{2j+1} - a_1}{a_{2j} - a_1} > 1.$$

This justifies $\beta_n > 1$. For the toric conical AE gravitational instanton with equal cone angle on finite rods, the configuration of turning points on $[0, 1]$ must be invariant under the reflection $z \mapsto 1 - z$, which follows from the uniqueness of the maximum of F . The map $(\rho, z, \phi_1, \phi_2) \mapsto (\rho, z, -\phi_1, \phi_2)$ induces an isometry of the gravitational instanton, swapping its two orientations.

We also record an identity used below. Let $\dot{\ell}$ be any variation tangent to the normalized simplex, and let $\dot{\alpha}_k$ and $\dot{d}_{ij}$ denote the corresponding variations of α_k and d_{ij} . Since

$$d_{ij} = \sum_{k=i}^{j-1} \ell_k, \quad \log \alpha_k = \sum_{i \leq k < j} \epsilon_i \epsilon_j \log d_{ij},$$

we have

$$\begin{aligned} \sum_k \ell_k \frac{\dot{\alpha}_k}{\alpha_k} &= \sum_{i < j} \epsilon_i \epsilon_j \frac{\dot{d}_{ij}}{d_{ij}} \sum_{k=i}^{j-1} \ell_k \\ &= \sum_{i < j} \epsilon_i \epsilon_j \dot{d}_{ij} \\ &= \sum_k \left(\sum_{i \leq k < j} \epsilon_i \epsilon_j \right) \dot{\ell}_k = 0. \end{aligned}$$

Thus, on the normalized simplex,

$$(B.5) \quad \sum_k \ell_k d \log \alpha_k = 0, \quad \sum_k A_k d\alpha_k = 0, \quad A_k = \frac{2\pi \ell_k}{\alpha_k}.$$

Here A_k is the area of the fixed surface corresponding to the k -th finite rod. Indeed, its induced area element is $\alpha_k^{-1} dz d\phi$, where ϕ is the noncollapsing angular coordinate of period 2π . In particular, if a variation in this family satisfies $\dot{\alpha}_1 = \dots = \dot{\alpha}_{2n} = c$, then

$$0 = \sum_k A_k \dot{\alpha}_k = c \sum_k A_k.$$

Since every $A_k > 0$, necessarily $c = 0$.

B.2. Four Hermitian AE metrics. In this section we study four Hermitian toric conical AE gravitational instantons using the piecewise linear description of the Tod ansatz in [9, 3].

Set $A = e_1$, $B = e_2$, and $C = e_1 + e_2$. The rod vectors below include the two infinite rods. We write $2\pi\alpha_j$ for the cone angle along the j -th finite rod, and β_* for their common cone factor when all finite cone angles are equal. The infinite rods are smooth. Let ϕ_1, ϕ_2 be the 2π -periodic angular coordinates associated with e_1, e_2 , and set $o_0 = d\rho \wedge dz \wedge d\phi_1 \wedge d\phi_2$.

Theorem B.3. *For each sequence of rod vectors in the following table there is a unique proper toric conical AE gravitational instanton with smooth infinite rods and common finite cone angle $2\pi\beta_*$, up to homothety and equivariant isometry:*

rod vectors	β_*	rod lengths, up to scale
BCA	2	(1)
$BACA$	$1 + 2/\sqrt{3}$	(1, 2)
$BCACA$	$4/\sqrt{3}$	$(3, 2\sqrt{3} - 2, \sqrt{3} + 2)$
$BCABCA$	$1 + \sqrt{2}$	$(1, \sqrt{2} - 1, \sqrt{2} - 1, 1)$.

The first two metrics are Hermitian in orientation o_0 , and the last two are Hermitian in $-o_0$. The first is hyperkähler in $-o_0$, and the second is Hermitian in both orientations. This follows from counting the ALE topological degree of [34]. Combining this with the representation of [9, Section 3] and [3, Propositions 2.4 and 4.5], every such instanton, in the Hermitian non-Kähler orientation, takes the form of the ansatz below.

Other presentations in the theorem give equivalent metrics. For example, reversing the order of the rods and interchanging A and B takes $BACA$ to $BCBA$. This identifies the metrics after the corresponding change of angular basis, and gives the two more sequences that appear in (4.1).

B.2.1. The ansatz and its rod data. Choose $N \geq 2$, turning points $z_1 < \dots < z_N$, and constants $a_i > 0$ with $\sum_{i=1}^N a_i = 1$. The ansatz is parametrized by the positive convex piecewise linear function

$$f(z) = \sum_{i=1}^N a_i |z - z_i|.$$

Its slopes on the two infinite rods are -1 and 1 . The absence of an additive constant in this formula is the AE condition. We recall the reconstruction of the metric to fix the normalization.

The axisymmetric harmonic function determined by f is

$$(B.6) \quad \begin{aligned} \mathcal{U}(\rho, z) &= \sum_{i=1}^N a_i \mathcal{U}_0(\rho, z - z_i), \\ \mathcal{U}_0(\rho, t) &= 2\sqrt{\rho^2 + t^2} - t \log \frac{\sqrt{\rho^2 + t^2} + t}{\sqrt{\rho^2 + t^2} - t}. \end{aligned}$$

In particular, $\mathcal{U}(\rho, z) = f(z) \log \rho^2 + O(1)$ as $\rho \rightarrow 0$ on the interior of each rod. Set

$$\mu = \sum_{i=1}^N a_i z_i, \quad c = -2 \sum_{i=1}^N a_i (z_i - \mu)^2 < 0,$$

and let H satisfy

$$H_\rho = -\rho \mathcal{U}_z, \quad H_z = \rho \mathcal{U}_\rho.$$

Tod's reconstruction is

$$(B.7) \quad g_f = W^{-1}(d\tau + Fdy)^2 + W\rho^2 dy^2 + e^{2\nu}(d\rho^2 + dz^2),$$

where

$$\begin{aligned} W &= \frac{1}{2c} \left(\rho \mathcal{U}_\rho + \frac{\mathcal{U}_\rho^2 \mathcal{U}_{zz}}{\mathcal{U}_{zz}^2 + \mathcal{U}_{\rho z}^2} \right), \\ e^{2\nu} &= \frac{W\rho^2}{4} (\mathcal{U}_{zz}^2 + \mathcal{U}_{\rho z}^2), \\ F &= \frac{1}{2c} \left(\frac{\rho \mathcal{U}_\rho^2 \mathcal{U}_{\rho z}}{\mathcal{U}_{zz}^2 + \mathcal{U}_{\rho z}^2} - \rho^2 \mathcal{U}_z - 2H \right). \end{aligned}$$

Choose the additive constant in H so that $F = 0$ on the right infinite rod. The choice of c then gives $F = -1$ on the left infinite rod. These are the standard Tod formulas of [3, Proposition 2.4]; all subsequent calculations use only the function f and its rod data.

For BCA and $BACA$ take $(\tau, y) = (-\phi_2, \phi_1 - \phi_2)$, while for $BCACA$ and $BCABCA$ take $(\tau, y) = (\phi_2, \phi_1 + \phi_2)$. These choices identify the two infinite rod lines with B and A and fix the angular lattice. The construction gives the normalized Euclidean end. It depends smoothly on the parameters while the turning points remain distinct and the a_i remain positive. In the corresponding normalized end charts, its asymptotic expansion gives

$$|\nabla_{g_E}^k (g_f - g_E)|_{g_E} = O(r^{-2-k}), \quad k \geq 0,$$

where r is the Euclidean radius. The same estimates hold after differentiation in the parameters, locally uniformly in the parameter space. When the collapsing fields agree with the prescribed primitive rod lines, as verified below for the metrics and nearby families under consideration, the cited regularity results give the conical extension along the fixed surfaces and at the torus fixed points.

There are $q = N - 1$ finite rods. Write

$$\begin{aligned} I_0 &= (-\infty, z_1), & I_j &= (z_j, z_{j+1}) \quad (1 \leq j \leq q), & I_N &= (z_N, \infty), \\ s_j &= f'|_{I_j}, & f_i &= f(z_i), & \ell_j &= z_{j+1} - z_j. \end{aligned}$$

Thus $s_0 = -1$, $s_N = 1$, $a_i = (s_i - s_{i-1})/2$, and

$$f_{j+1} - f_j = s_j \ell_j.$$

The condition that f have no additive constant is equivalent to

$$(B.8) \quad f_1 + f_N = z_N - z_1 = \sum_{j=1}^q \ell_j.$$

We next recall the rod formulas from [3, Lemmas 5.1–5.2]. A normalized rod generator means a collapsing Killing field for which period 2π would give cone angle 2π . Up to sign, it is

$$u_j = \begin{cases} s_j(\partial_y - F_j \partial_\tau), & s_j \neq 0, \\ -c^{-1} f_j^2 \partial_\tau, & s_j = 0, \end{cases}$$

where F_j is the constant value of F on I_j when $s_j \neq 0$. If v_j is the primitive lattice vector of the rod, then $v_j = \pm \alpha_j u_j$, where we set $\alpha_0 = \alpha_N = 1$ for the two infinite rods. Thus its cone angle is $2\pi \alpha_j$.

If s_{i-1} and s_i are both nonzero, then

$$(B.9) \quad F_i - F_{i-1} = \frac{f_i^2}{c} \left(\frac{1}{s_i} - \frac{1}{s_{i-1}} \right).$$

If $s_j = 0$, then $f_j = f_{j+1}$ and

$$(B.10) \quad F_{j+1} - F_{j-1} = \frac{1}{c} \left[f_j^2 \left(\frac{1}{s_{j+1}} - \frac{1}{s_{j-1}} \right) - 2\ell_j f_j \right].$$

Consecutive primitive rod vectors in our four configurations have determinant of absolute value one. Taking determinants of their normalized generators gives

$$f_i^2(s_i - s_{i-1}) = \frac{|c|}{\alpha_{i-1}\alpha_i}, \quad 1 \leq i \leq N.$$

This identity also holds at an endpoint of a zero-slope rod.

B.2.2. The equal-angle metrics. For the first two configurations, choose the signed rod lists $(B, -C, A)$ and $(B, -A, -C, A)$. The coefficient of ∂_y is -1 on B , 1 on A , and 0 on C . Consequently, the slopes for BCA are $(-1, 0, 1)$. Up to translation, its function is

$$f(z) = \frac{1}{2}|z| + \frac{1}{2}|z - L|, \quad L > 0.$$

Its finite rod has height $L/2$, and $c = -L^2/2$. Since its primitive rod vector is $C = -\partial_\tau$, the zero-slope generator gives

$$\alpha_1 = \frac{|c|}{(L/2)^2} = 2.$$

There is therefore exactly one such metric up to scale.

For $BACA$, set the turning points at $0, L, L + M$, where $L, M > 0$, and write $b = \alpha_1$. The slopes are

$$(-1, -1/b, 0, 1).$$

In particular, $b > 1$. Set $h = f_2 = f_3$ and $x = L/(bh)$, so that $f_1 = h + L/b = h(1 + x)$. The zero-slope generator gives $\alpha_2 = |c|/h^2$. The jump at the first turning point, the jump across the zero-slope rod, and (B.8) give, respectively,

$$\alpha_2 = (b - 1)(1 + x)^2, \quad M = \frac{b + 1}{2}h, \quad L + M = 2h + \frac{L}{b}.$$

Hence

$$(B.11) \quad \alpha_2 = \frac{(b + 1)^2}{4(b - 1)}, \quad \frac{M}{L} = \frac{b^2 - 1}{b(3 - b)}, \quad 1 < b < 3.$$

Conversely, these formulas determine a positive convex function f with the prescribed rods for each $1 < b < 3$ and $L > 0$. The derivative of the length ratio is

$$\frac{d}{db} \left(\frac{b^2 - 1}{b(3 - b)} \right) = \frac{3b^2 - 2b + 3}{b^2(3 - b)^2} > 0,$$

and the ratio ranges from zero to infinity. Thus this family realizes every pair of positive finite rod lengths. The equal-angle equation $\alpha_2 = b$ has the unique solution

$$b = 1 + \frac{2}{\sqrt{3}}, \quad \frac{M}{L} = 2.$$

For the last two configurations, the two infinite rod lines again have slope coefficients of absolute value one. Opposite coefficients on A and B would give zero coefficient on both occurrences of C , contrary to the strict increase of the slopes. The signed rod lists are therefore

$$(-B, -C, -A, C, A), \quad (-B, -C, -A, B, C, A),$$

respectively. At common finite cone factor b , their slopes are

$$\begin{aligned} BCACA : \quad & (-1, -2/b, -1/b, 2/b, 1), \\ BCABCA : \quad & (-1, -2/b, -1/b, 1/b, 2/b, 1). \end{aligned}$$

In particular, $b > 2$. A homothety allows us to set $|c| = 1$. The determinant identity then gives the heights

$$\begin{aligned} BCACA : \quad & (f_1, f_2, f_3, f_4) = \left(\frac{1}{\sqrt{b-2}}, \frac{1}{\sqrt{b}}, \frac{1}{\sqrt{3b}}, \frac{1}{\sqrt{b-2}} \right), \\ BCABCA : \quad & (f_1, f_2, f_3, f_4, f_5) = \left(\frac{1}{\sqrt{b-2}}, \frac{1}{\sqrt{b}}, \frac{1}{\sqrt{2b}}, \frac{1}{\sqrt{b}}, \frac{1}{\sqrt{b-2}} \right). \end{aligned}$$

Recover the lengths from $\ell_j = (f_{j+1} - f_j)/s_j$. The identity (B.8) becomes, respectively,

$$\begin{aligned} 2\sqrt{b-2} + (1 - \sqrt{3})\sqrt{b} &= 0, \\ 2\sqrt{b-2} + 2(1 - \sqrt{2})\sqrt{b} &= 0. \end{aligned}$$

Their unique solutions for $b > 2$ are $4/\sqrt{3}$ and $1 + \sqrt{2}$. Substitution gives the positive length ratios in the theorem.

Conversely, these heights and lengths determine z_i up to translation and $a_i = (s_i - s_{i-1})/2 > 0$. The identity (B.8) ensures that the resulting function is $f(z) = \sum_i a_i |z - z_i|$. The rod formulas then give the required primitive rod vectors and common cone factors. The reconstruction, including the normalization of c , is also verified in the nearby-angle calculation below. This proves existence. Every competitor has the ansatz already described, so the preceding calculations force its finite lengths to have the same ratios. After fixing scale and translation, the uniqueness theorem for harmonic maps with prescribed rod data and normalized end, [34, Theorem 4.24], identifies the metrics. This proves Theorem B.3.

B.2.3. Nearby angles. We describe the nearby families for use in the next subsection. For *BCA* the finite cone factor remains 2, and the only parameter is scale. For *BACA*, equation (B.11) describes the nearby cone factors, with scale providing the second parameter.

For *BCACA*, normalize $|c| = 1$ and set

$$\begin{aligned} (s_0, s_1, s_2, s_3, s_4) &= \left(-1, -\frac{2}{\alpha_1}, -\frac{1}{\alpha_2}, \frac{2}{\alpha_3}, 1\right), \\ (d_1, d_2, d_3, d_4) &= (\alpha_1 - 2, 2\alpha_2 - \alpha_1, 2\alpha_2 + \alpha_3, \alpha_3 - 2), \quad f_i = d_i^{-1/2}. \end{aligned}$$

The lengths are

$$(B.12) \quad \ell_1 = \frac{\alpha_1}{2}(f_1 - f_2), \quad \ell_2 = \alpha_2(f_2 - f_3), \quad \ell_3 = \frac{\alpha_3}{2}(f_4 - f_3),$$

and the condition (B.8) is

$$(B.13) \quad H_5(\alpha) := \sqrt{d_1} + \sqrt{d_2} - \sqrt{d_3} + \sqrt{d_4} = 0.$$

For *BCABCA*, set

$$\begin{aligned} (s_0, s_1, s_2, s_3, s_4, s_5) &= \left(-1, -\frac{2}{\alpha_1}, -\frac{1}{\alpha_2}, \frac{1}{\alpha_3}, \frac{2}{\alpha_4}, 1\right), \\ (d_1, d_2, d_3, d_4, d_5) &= (\alpha_1 - 2, 2\alpha_2 - \alpha_1, \alpha_2 + \alpha_3, 2\alpha_3 - \alpha_4, \alpha_4 - 2), \quad f_i = d_i^{-1/2}. \end{aligned}$$

Then

$$(B.14) \quad \begin{aligned} \ell_1 &= \frac{\alpha_1}{2}(f_1 - f_2), & \ell_2 &= \alpha_2(f_2 - f_3), \\ \ell_3 &= \alpha_3(f_4 - f_3), & \ell_4 &= \frac{\alpha_4}{2}(f_5 - f_4), \end{aligned}$$

and the corresponding condition is

$$(B.15) \quad H_6(\alpha) := \sqrt{d_1} + \sqrt{d_2} - 2\sqrt{d_3} + \sqrt{d_4} + \sqrt{d_5} = 0.$$

All d_i and ℓ_j are positive near the equal-angle solutions. Conversely, a nearby solution of $H_5 = 0$ or $H_6 = 0$ reconstructs f : set $z_1 = 0$, $z_{j+1} = z_j + \ell_j$, and $a_i = (s_i - s_{i-1})/2$. The slopes are strictly increasing, and (B.8) ensures that $f(z) = \sum_i a_i |z - z_i|$ has the prescribed heights.

For completeness, the normalization of the end imposes no additional equation. Write $t_j = \alpha_j s_j$, where $\alpha_0 = \alpha_N = 1$. These are the ∂_y coefficients of the signed primitive rod vectors,

namely $(-1, -2, -1, 2, 1)$ and $(-1, -2, -1, 1, 2, 1)$. The determinant identity gives

$$\sum_{i=1}^N f_i^2 \left(\frac{1}{s_i} - \frac{1}{s_{i-1}} \right) = - \sum_{i=1}^N \frac{1}{t_{i-1} t_i} = -1,$$

$$\sum_{i=1}^N f_i^2 \left(\frac{1}{s_i} - \frac{1}{s_{i-1}} \right) = f_1^2 + f_N^2 - \sum_{j=1}^q \ell_j (f_j + f_{j+1}) = -2 \sum_{i=1}^N a_i (z_i - \mu)^2.$$

Thus the variance formula for c indeed gives $c = -1$. Moreover, each jump of F is $1/(t_{i-1} t_i)$, which is exactly the jump prescribed by the consecutive signed primitive rod vectors. Starting from $F_0 = -1$ therefore recovers every rod and gives $F_N = 0$. A common rescaling of the lengths supplies the scale parameter.

Differentiating the defining functions gives

$$(B.16) \quad \frac{\partial H_5}{\partial \alpha_j} = \frac{\ell_j}{\alpha_j} > 0 \quad (1 \leq j \leq 3), \quad \frac{\partial H_6}{\partial \alpha_j} = \frac{\ell_j}{\alpha_j} > 0 \quad (1 \leq j \leq 4).$$

For *BACA*, use

$$H_4(\alpha_1, \alpha_2) = \alpha_2 - \frac{(\alpha_1 + 1)^2}{4(\alpha_1 - 1)}.$$

Its partial derivatives satisfy

$$\frac{\partial H_4}{\partial \alpha_1} = \frac{(\alpha_1 + 1)(3 - \alpha_1)}{4(\alpha_1 - 1)^2} > 0, \quad \frac{\partial H_4}{\partial \alpha_2} = 1,$$

for $1 < \alpha_1 < 3$. Consequently, in each case the admissible cone factors form a smooth hypersurface near the equal-angle point, and the common-angle direction $(1, \dots, 1)$ is not tangent to it.

More explicitly, on this hypersurface the map

$$\alpha \mapsto (\log \alpha_1 - \log \alpha_q, \dots, \log \alpha_{q-1} - \log \alpha_q)$$

has invertible differential at the equal-angle point. Indeed, a tangent vector in its kernel has all components equal, and the positive partial derivatives of the defining function force this common component to vanish. For $q = 1$ there are no angle-difference equations. Thus, including scale, each metric belongs to a smooth q -parameter family. Proposition B.5 below proves that the finite Weyl lengths are local coordinates on this family; after fixing their total length, the same calculation gives a nonsingular Jacobian for the $q - 1$ equal-angle equations.

B.3. Infinitesimal deformations. Let (X, g) be one of the equal-angle metrics constructed above. There are q finite rods, with lengths $\ell_j = z_{j+1} - z_j$ and cone angles $2\pi\alpha_j$. The two infinite rods have cone angle 2π . We keep the torus action, the primitive rod vectors, and the normalized Euclidean end fixed.

In this section, we will prove that every decaying torus-invariant solution of the linearized Ricci-flat equation in the class specified below comes from varying the rod lengths in the families above, up to a change of coordinates. If all finite cone angles have the same first variation, that variation must be zero and only scaling remains.

We first specify the symmetry and regularity of the fixed-angle tensors, using the coordinates in Lemma 2.7. Variations of the cone angles will be specified separately. Write $(N, g) = X/\mathbb{T}^2$ and let $\pi : X \rightarrow N$ be the quotient map. All coordinates below are chosen for the background metric g and are kept fixed when describing tensor regularity. We consider $\mathbb{T}^2$ -invariant symmetric two-tensors k ; their coefficients in the quotient and angular coordinates are therefore independent of the angular variables.

Near an interior point of the j -th rod $\mathcal{I}_j$, take the coordinates of Lemma 2.7, denoted here by (ξ, η, ϕ, ψ) , with (ξ, η) the quotient coordinates and ϕ, ψ the angular coordinates of period 2π . Here the rod is given by $\xi = 0$, η parametrizes it, and ∂_ϕ is the primitive angular generator that vanishes on the rod surface $\pi^{-1}(\mathcal{I}_j)$. Thus (η, ψ) are local coordinates along that surface. If the cone angle is $2\pi\alpha_j$, set

$$x = \xi \cos(\alpha_j \phi), \quad y = \xi \sin(\alpha_j \phi).$$

We require that k , when expressed in (x, y, η, ψ) , extend as a smooth symmetric two-tensor across $x = y = 0$.

Near the torus fixed point over a turning point between rods $\mathfrak{I}_i$ and $\mathfrak{I}_j$, take the coordinates supplied by the corner description in Lemma 2.7, denoted here by $(\xi_1, \xi_2, \phi_1, \phi_2)$. The quotient is locally the quadrant $\xi_1, \xi_2 \geq 0$, with

$$\mathfrak{I}_i = \{\xi_1 = 0\}, \quad \mathfrak{I}_j = \{\xi_2 = 0\},$$

and the primitive angular generators $\partial_{\phi_1}, \partial_{\phi_2}$ collapse on the corresponding rod surfaces. No distance interpretation of ξ_1, ξ_2 is required. If the two cone angles are $2\pi\alpha_i, 2\pi\alpha_j$, set

$$\begin{aligned} x_1 &= \xi_1 \cos(\alpha_i \phi_1), & y_1 &= \xi_1 \sin(\alpha_i \phi_1), \\ x_2 &= \xi_2 \cos(\alpha_j \phi_2), & y_2 &= \xi_2 \sin(\alpha_j \phi_2). \end{aligned}$$

We require that k , when expressed in (x_1, y_1, x_2, y_2) , extend as a smooth symmetric two-tensor across both $\{x_1 = y_1 = 0\}$ and $\{x_2 = y_2 = 0\}$, including their intersection.

To allow changing cone angles, for each finite rod $\mathfrak{I}_j$ choose a smooth one-parameter family of torus-invariant conical metrics $g^{(j)}(t)$ with $g^{(j)}(0) = g$, keeping the torus action and rod vectors fixed. Such families can be constructed by modifying the Gram matrix. The family can be chosen to agree with g outside a compact neighborhood of the finite rod surfaces. Its cone angle along $\mathfrak{I}_i$ is $2\pi\alpha_i^{(j)}(t)$, where

$$\alpha_i^{(j)}(t) = \alpha_i + t\delta_{ij}, \quad 1 \leq i \leq q,$$

while both infinite cone angles remain 2π . Define the reference variations by

$$b_j = \left. \frac{d}{dt} \right|_{t=0} g^{(j)}(t).$$

Thus b_j is a compactly supported invariant symmetric two-tensor which changes the cone angle along $\mathfrak{I}_j$ at speed 2π and leaves every other cone angle unchanged.

A general variation considered here is

$$(B.17) \quad h = k + \sum_{j=1}^q c_j b_j,$$

where k is a torus-invariant fixed-angle tensor with the regularity just described and $c_j \in \mathbb{R}$. The variation of its cone angle along $\mathfrak{I}_j$ is $2\pi c_j$, equivalently $\dot{\alpha}_j(h) = c_j$, and its infinite cone angles have zero variation.

For the deformation statement below, k is smooth in these coordinates and satisfies

$$(B.18) \quad |\nabla_{g_{\text{Euc}}}^m k|_{g_{\text{Euc}}} = O(r_{\text{Euc}}^{-1-m}), \quad m \geq 0,$$

on the Euclidean end. Here g_{Euc} is the Euclidean metric and r_{Euc} is its distance to the origin; all end quantities are understood using a fixed equivariant Euclidean end chart. Since the b_j have compact support, the same decay holds for h . The equation $(D \text{Ric})_g h = 0$ is imposed on the complement of the rod surfaces. The coordinate changes we quotient out are generated by torus-invariant vector fields Y which are smooth in the same coordinates, preserve the rod surfaces and torus fixed points, and satisfy

$$(B.19) \quad |\nabla_{g_{\text{Euc}}}^m Y|_{g_{\text{Euc}}} = O(r_{\text{Euc}}^{-m}), \quad m \geq 0.$$

In particular, Y is bounded at infinity.

Scaling gives the family $e^{2t}g$, whose derivative at $t = 0$ is $2g$. This tensor is smooth in the fixed-angle coordinates described above and preserves all cone angles, but it does not decay at infinity. To obtain a decaying variation representing scaling, choose a smooth torus-invariant vector field Z_0 which vanishes near all finite rod surfaces and is equal to $r_{\text{Euc}}\partial_{r_{\text{Euc}}}$ on the end, transported to X by the fixed equivariant Euclidean end chart. Let Ψ_t be the flow of Z_0 , and define

$$(B.20) \quad h_{\text{sc}} = \left. \frac{d}{dt} \right|_{t=0} (e^{2t}\Psi_{-t}^*g) = 2g - \mathcal{L}_{Z_0}g.$$

The family $e^{2t}\Psi_{-t}^*g$ is torus-invariant and Ricci-flat, with the same cone angles as g . Consequently, h_{sc} is a smooth fixed-angle tensor satisfying $(D\text{Ric})_g h_{\text{sc}} = 0$. This tensor represents scaling and decays at infinity.

Let E_j be the compact surface corresponding to the j -th finite rod. Its area and the total rod area are

$$(B.21) \quad A_j = \text{Area}_g(E_j) = \frac{2\pi\ell_j}{\alpha_j}, \quad A_* = \sum_{j=1}^q A_j > 0.$$

The area formula follows from $\det G = \rho^2$. The induced area element on the rod is $\alpha_j^{-1}dzd\psi$, with ψ of period 2π . Scaling gives

$$\dot{\ell}_j(h_{\text{sc}}) = 2\ell_j, \quad \dot{\alpha}_j(h_{\text{sc}}) = 0, \quad DA_*(h_{\text{sc}}) = 2A_*.$$

We study tensors h satisfying $(D\text{Ric})_g h = 0$ and taking the form (B.17), with the decay (B.18). Recall that the reduction of the toric Ricci-flat metric gives

$$g = \underline{g} + \sum_{i,j=1}^2 G_{ij}d\phi_i d\phi_j.$$

We first prove that, up to a gauge term, we can remove the mixed components of h between the quotient and angular directions.

Lemma B.4. *There is a torus-invariant vector field Y' , tangent to the torus orbits and satisfying (B.19), such that*

$$h + \mathcal{L}_{Y'}g = \underline{\dot{g}} + \sum_{i,j=1}^2 \dot{G}_{ij}d\phi_i d\phi_j.$$

Proof. Let $K_i = \partial_{\phi_i}$. On N° , use the background coordinates (ρ, z) and define one-forms by

$$a^i = \sum_{j=1}^2 G^{ij} (h(\partial_\rho, K_j)d\rho + h(\partial_z, K_j)dz), \quad i = 1, 2,$$

where $(G^{ij}) = G^{-1}$. These one-forms encode precisely the mixed components:

$$h = \underline{\dot{g}} + 2 \sum_{i,j=1}^2 G_{ij}a^j d\phi_i + \sum_{i,j=1}^2 \dot{G}_{ij}d\phi_i d\phi_j.$$

Here $\underline{\dot{g}}$ and $\dot{G}$ denote the quotient and angular components of h . The reference variations b_j preserve the block form, so the mixed components come entirely from the fixed-angle tensor k . Although G^{-1} is singular at a rod, the regularity of k supplies the vanishing needed in this formula. Indeed, in the adapted coordinates of Lemma 2.7, smoothness and $\mathbb{T}^2$ invariance give

$$a^i = \xi p_i(\xi^2, \eta)d\xi + q_i(\xi^2, \eta)d\eta$$

near an interior rod point, and

$$a^i = \xi_1 p_{i1}(\xi_1^2, \xi_2^2)d\xi_1 + \xi_2 p_{i2}(\xi_1^2, \xi_2^2)d\xi_2$$

near a turning point, with smooth coefficient functions. Thus these one-forms are smooth in the variables (ξ^2, η) and (ξ_1^2, ξ_2^2) , respectively.

Linearizing the identities used in the proof of Lemma 2.4, the mixed components of $(D\text{Ric})_g h = 0$ give

$$d\left(\rho \sum_{j=1}^2 G_{ij} * \underline{\dot{g}} da^j\right) = 0, \quad i = 1, 2,$$

where $*_{\underline{\dot{g}}}$ is the Hodge star on the quotient. Each expression inside the parentheses is therefore a constant function. The regularity just described and the vanishing of ρ on the rods imply that these constants are zero. Since $\rho > 0$ and G is invertible in the interior, it follows that $da^i = 0$.

The quotient interior is simply connected, so there are functions f^i satisfying $a^i = \mathrm{d}f^i$. Set

$$Y' = - \sum_{i=1}^2 f^i K_i.$$

This vector field changes only the angular coordinates. Its Lie derivative cancels the mixed components:

$$h + \mathcal{L}_{Y'} g = \underline{\dot{g}} + \sum_{i,j=1}^2 \dot{G}_{ij} \mathrm{d}\phi_i \mathrm{d}\phi_j.$$

We check that this is an allowed coordinate change. The local forms of a^i above imply that their primitives are smooth in (ξ^2, η) near a rod and in (ξ_1^2, ξ_2^2) near a turning point. Consequently, Y' extends smoothly in the fixed-angle Cartesian coordinates and preserves the rod surfaces and torus fixed points.

For the end estimates, use the fixed equivariant Euclidean end chart. Applying the same division argument on rescaled annuli, the differentiated decay of h gives

$$a^i = O_\infty(r_{\text{Euc}}^{-2}),$$

uniformly up to the smooth infinite rods. The primitives have a common limit along the end; subtracting their constant limits gives $f^i = O_\infty(r_{\text{Euc}}^{-1})$. Since the torus generators have size $O(r_{\text{Euc}})$, we obtain

$$|\nabla_{g_{\text{Euc}}}^m Y'|_{g_{\text{Euc}}} = O(r_{\text{Euc}}^{-m}), \quad m \geq 0.$$

In particular, Y' is bounded. □

We may therefore replace h by $h + \mathcal{L}_{Y'} g$ and assume from now on that it has only quotient and angular components.

Proposition B.5. *The tensor h can be written as*

$$h = \sum_{j=1}^q \dot{\ell}_j \frac{\partial g_\ell}{\partial \ell_j} + \mathcal{L}_Y g,$$

where the coefficients $\dot{\ell}_j$ are uniquely determined and Y is a bounded vector field of the kind specified above.

At the equal-angle metric with rod-length vector $\ell = (\ell_1, \dots, \ell_q)$,

$$(B.22) \quad \ker D_\ell(\log \alpha_1, \dots, \log \alpha_q) = \mathbb{R}\ell.$$

If $c_1 = \dots = c_q$ in (B.17), then every c_j is zero and

$$h = ah_{\text{sc}} + \mathcal{L}_Y g$$

for some $a \in \mathbb{R}$. In particular, fixing all cone angles leaves only scaling modulo coordinate changes. If also $DA_*(h) = 0$, then h is pure gauge.

We prove this proposition in the rest of this section.

By the preceding lemma, we may replace h by $h + \mathcal{L}_{Y'} g$ and assume that it has no mixed components between the quotient and angular directions. Write

$$h = \underline{\dot{g}} + \sum_{i,j=1}^2 \dot{G}_{ij} \mathrm{d}\phi_i \mathrm{d}\phi_j.$$

We first prove that if all rod-length variations vanish, then h is pure gauge. We then use this fact to identify the general deformation with a tangent to the families constructed above.

Write $\ast$ for the Hodge star of the quotient metric $\underline{g}$, so that $\mathrm{d}z = \ast \mathrm{d}\rho$, and let $\dot{\ast}$ be its first variation induced by $\underline{\dot{g}}$. The first variations of $\rho = \sqrt{\det G}$ and its harmonic conjugate satisfy

$$\begin{aligned} \dot{\rho} &= \frac{\rho}{2} \text{tr}(G^{-1} \dot{G}), \\ \mathrm{d}\dot{z} &= \dot{\ast} \mathrm{d}\rho + \ast \mathrm{d}\dot{\rho}. \end{aligned}$$

Indeed, linearizing (2.5) with $\lambda = 0$ shows that the one-form in the second line is closed. Since N° is simply connected, $\dot{z}$ exists globally and is unique up to an additive constant. Its values $\dot{z}_i$ at the turning points determine the rod-length variations

$$\dot{\ell}_j = \dot{z}_{j+1} - \dot{z}_j.$$

These differences are unchanged by the allowed coordinate changes.

Suppose now that every $\dot{\ell}_j$ is zero. All the $\dot{z}_i$ are then equal, so we choose the additive constant in $\dot{z}$ to make them zero. On N° , define

$$V = -\dot{\rho}\partial_\rho - \dot{z}\partial_z,$$

and lift V horizontally to X . Replacing h by $h + \mathcal{L}_V g$ makes the first variations of both ρ and z zero. We check that V is an allowed vector field, including along the rod surfaces and at the torus fixed points.

Near a turning point, use the quotient coordinates (ξ_1, ξ_2) of Lemma 2.7. By reflecting four times, we can glue four copies of a small disc along its two rods around the turning point, and we obtain a reflected disc, where g extends as a smooth metric. The original disk in the quotient is a quadrant of the reflected disc. The reference variations b_j change only the Gram matrix, so $\underline{\dot{g}}$ is the quotient component of the fixed-angle tensor k . Its smoothness and $\mathbb{T}^2$ invariance give

$$\underline{\dot{g}} = \dot{a}d\xi_1^2 + 2\xi_1\xi_2\dot{b}d\xi_1d\xi_2 + \dot{c}d\xi_2^2,$$

where $\dot{a}, \dot{b}, \dot{c}$ are smooth functions of (ξ_1^2, ξ_2^2) . This formula extends $\underline{\dot{g}}$ smoothly to either sign of ξ_1, ξ_2 and makes it invariant under both coordinate reflections. Together with the corresponding extension of $\underline{g}$, it gives a smooth metric and a smooth symmetric two-tensor on the full reflected disc.

Choose a fixed isothermal coordinate $\zeta = x + \sqrt{-1}y$ on the reflected disc, centered at the turning point and compatible with both reflections, such that $\underline{g} = e^{2\sigma}|\mathrm{d}\zeta|^2$. Set $w = z + \sqrt{-1}\rho$ and $\dot{w} = \dot{z} + \sqrt{-1}\dot{\rho}$, so that the equations defining V become $\mathrm{d}w(V) = -\dot{w}$. Choose the complex orientation of ζ so that w is holomorphic. The map w takes the original quadrant to the Weyl half-plane, and the corner expansion gives

$$w = z_i + a_i\zeta^2 + O(\zeta^4), \quad \partial_\zeta w = 2a_i\zeta + O(\zeta^3), \quad a_i \neq 0.$$

Thus ζ is a regular coordinate at the corner, whereas w has vanishing differential there. The corner expansions of G and $\dot{G}$, together with $\rho = \sqrt{\det G}$, give

$$\dot{\rho} = \xi_1\xi_2 f(\xi_1^2, \xi_2^2) = O(|\zeta|^2)$$

for a smooth function f . In particular, $\mathrm{d}\rho$ and $\mathrm{d}\dot{\rho}$ are $O(|\zeta|)$. Differentiating $\mathrm{d}z = \ast\mathrm{d}\rho$ gives

$$\mathrm{d}\dot{z} = \ast\mathrm{d}\rho + \ast\mathrm{d}\dot{\rho}.$$

The right-hand side extends smoothly to the reflected disc and is $O(|\zeta|)$. Since $\dot{z}_i = 0$, it follows that $\dot{z} = O(|\zeta|^2)$, and hence $\dot{w} = O(|\zeta|^2)$. In the fixed coordinate ζ , the differentiated conjugacy equation reads

$$\partial_{\bar{\zeta}}\dot{w} = \mu\partial_\zeta w, \quad \mu = \frac{\dot{g}_{xx} - \dot{g}_{yy} + 2\sqrt{-1}\dot{g}_{xy}}{4e^{2\sigma}},$$

where μ is smooth on the reflected disc. Since w is holomorphic and $\mathrm{d}w(V) = -\dot{w}$, the complex component of V satisfies

$$v := \mathrm{d}\zeta(V) = -\frac{\dot{w}}{\partial_\zeta w} = O(|\zeta|), \quad \partial_{\bar{\zeta}}v = -\mu$$

away from the corner. Extending $v(0) = 0$, the latter equation holds distributionally on the full disc: integration by parts around a circle of radius ε produces a boundary term of order $O(\varepsilon^2)$. We now apply interior elliptic regularity to $\partial_{\bar{\zeta}}v = -\mu$. Thus v is smooth across the corner.

The reflection symmetries then imply

$$V = \xi_1 v_1(\xi_1^2, \xi_2^2)\partial_{\xi_1} + \xi_2 v_2(\xi_1^2, \xi_2^2)\partial_{\xi_2},$$

with smooth v_1, v_2 . Its lift is therefore smooth in the fixed-angle Cartesian coordinates and preserves both rod surfaces and the torus fixed point. At an interior rod point, $\partial_\zeta w \neq 0$, and the same conclusion follows directly from one reflection.

On the end, the decay of h gives $\dot{\rho}, \dot{z} = O_\infty(r_{\text{Euc}})$. The Weyl-coordinate differential has size comparable to r_{Euc} , so

$$|\nabla_{g_{\text{Euc}}}^m V|_{g_{\text{Euc}}} = O(r_{\text{Euc}}^{-m}), \quad m \geq 0.$$

Thus the torus-invariant lift of V is an allowed vector field. Replacing h by $h + \mathcal{L}_V g$, we may henceforth assume $\dot{\rho} = \dot{z} = 0$.

Linearizing $dz = \star d\rho$ now gives $\star d\rho = 0$. Since $\star^2 = -1$ on one-forms, we also have $\star dz = 0$. Thus the Hodge star is fixed to first order, which in two dimensions means that the quotient-metric variation is conformal. Write

$$\underline{g} = e^{2\nu}(\underline{d\rho}^2 + \underline{dz}^2), \quad \underline{\dot{g}} = 2\nu \underline{g}.$$

With the coordinates (ρ, z) fixed, linearizing (A.3) shows that $Z = \dot{\Phi}$, where $\Phi = \rho^{-1}G$, is a Jacobi field along the axisymmetric harmonic map Φ . The target $\mathcal{H} = SL(2, \mathbb{R})/SO(2)$ has negative curvature. Consequently,

$$(B.23) \quad \Delta_{\mathbb{R}^3}|Z|^2 = 2|DZ|^2 - 2 \sum_{a=1}^3 \langle \mathcal{R}(Z, d\Phi(e_a))d\Phi(e_a), Z \rangle \geq 0.$$

Here the norms and inner product use the target metric, D is the pullback connection, $\mathcal{R}$ is the target curvature, and e_a is an orthonormal frame for the Euclidean $\mathbb{R}^3$. Its axisymmetric Laplacian is $\partial_\rho^2 + \rho^{-1}\partial_\rho + \partial_z^2$.

To apply the maximum principle, we check that $|Z|$ is bounded near the auxiliary axis and tends to zero at infinity. At a turning point, rescale the rows and columns of G and $\dot{G}$ by ξ_1^{-1}, ξ_2^{-1} in the adjacent primitive rod basis. The prescribed regularity gives smooth bounded matrices after this rescaling, and the rescaled background matrix is positive definite. Hence the relative Gram variation, and therefore $|Z|$, is bounded. At an interior rod point, the same argument rescales only the row and column corresponding to the collapsing circle. On the end, $|Z| = O(r_{\text{Euc}}^{-1})$ by the decay of h .

The axis has locally zero Newtonian capacity in $\mathbb{R}^3$, so the bounded subharmonic function $|Z|^2$ extends across it. Since it tends to zero at infinity, the maximum principle gives $Z = 0$. We already have $\dot{\rho} = 0$, so $\dot{G} = 0$ as well. The remaining reduced equations for ν then give $d\nu = 0$, and the end normalization forces this constant to be zero. Thus the tensor vanishes after the allowed coordinate changes. This proves that every deformation with zero rod-length variations is pure gauge.

We next show that the rod lengths are smooth local coordinates on the families constructed in Sections B.1-B.2. For the alternating families, they are already independent parameters. For a Hermitian family, the parameters α_j determining the finite cone angles lie on the hypersurface described in the preceding subsection. Together with scale, they give q parameters. If a tangent to this family has zero rod-length variations, the result just proved makes it pure gauge. It therefore has zero variation of every cone angle, so its component tangent to the angle hypersurface is zero. The remaining scale component changes the length vector by a nonzero multiple of ℓ , unless that component is zero. Thus the length differential is injective between two q -dimensional spaces, and the inverse function theorem makes the rod lengths local coordinates.

For a general solution h , choose the unique family tangent with the same rod-length variations and subtract it. The difference has zero rod-length variations and is therefore pure gauge. This gives

$$h = \sum_{j=1}^q \dot{\ell}_j \frac{\partial g_\ell}{\partial \ell_j} + \mathcal{L}_Y g.$$

The coefficients are uniquely determined because the allowed coordinate changes do not change the rod-length variations.

It remains to consider the cone angles. In an alternating family, split a length variation into a multiple of ℓ and a tangent to the simplex of fixed total rod length. The differential of the angle differences on that simplex is nonsingular by Lemma B.2 and the calculation following it. If all finite cone angles have the same first variation, the angle differences have zero variation, so the tangent component vanishes. Only scale remains, and scale preserves all cone angles.

For a Hermitian family with $q \geq 2$, differentiating its angle relation gives

$$\sum_{j=1}^q \frac{\partial H}{\partial \alpha_j} c_j = 0,$$

where $H = H_4, H_5$ or H_6 and $c_j = \dot{\alpha}_j(h)$. All the partial derivatives are positive by (B.16) and the formula for H_4 . If the c_j have a common value, that value must be zero. The tangent component along the angle hypersurface then vanishes, leaving only scale. For BCA , the single finite cone angle is identically 4π , and the family has only the scale parameter. This proves (B.22) and gives, in the common-angle case,

$$h = ah_{\text{sc}} + \mathcal{L}_Y g.$$

Finally, the allowed coordinate changes preserve the areas of the compact rod surfaces, while scaling gives $DA_*(h_{\text{sc}}) = 2A_*$. Therefore

$$DA_*(ah_{\text{sc}} + \mathcal{L}_Y g) = 2aA_*.$$

Since $A_* > 0$, the condition $DA_*(h) = 0$ forces $a = 0$, and h is pure gauge.

B.4. Angle variation and invertibility at fixed area. We consider the equal-angle metric (X, g) and keep the fixed-angle tensor regularity from the preceding subsection. Using the notation of Section 4, with Einstein constant zero, set

$$\mathcal{E}_0(g) = \text{Ric}_g - \frac{1}{2} \text{Scal}_g g, \quad \mathcal{P}_g k = (D\mathcal{E}_0)_g k + \delta_g^* \delta_g k,$$

where $\delta_g = -\text{div}_g$ and δ_g^* is its formal adjoint. The domain of $\mathcal{P}_g$ consists of torus-invariant fixed-angle symmetric two-tensors. All components are allowed, including the mixed components between quotient and angular directions.

Let b_j be the reference variations chosen in the preceding subsection, and set

$$b = \sum_{j=1}^q b_j, \quad B = (D\mathcal{E}_0)_g b.$$

Thus $\dot{\alpha}_j(b) = 1$ for every finite rod, so b changes each finite cone angle at speed 2π . It has compact support and leaves both infinite cone angles unchanged.

Although b itself need not be smooth in the fixed-angle Cartesian coordinates, B is smooth. We explain this in the following. Let g_t be the reference family inducing b , with $g_0 = g$ and $\dot{g}_0 = b$. The quotient metric is fixed, while the Gram matrix varies.

Near an interior point of a finite rod, set $\alpha_j(t) = \alpha_j + t$ and consider, on an angular sector, the map

$$F_t(\xi, \eta, \phi, \psi) = (\xi \cos(\alpha_j(t)\phi), \xi \sin(\alpha_j(t)\phi), \eta, \psi).$$

The regularity of the reference family means that

$$g_t = F_t^* \tilde{g}_t,$$

where $\tilde{g}_t$ is a smooth family of smooth metrics on a fixed Cartesian domain with coordinates (x, y, η, ψ) , including $x = y = 0$. These Cartesian coordinates are held fixed when differentiating $\tilde{g}_t$. At $t = 0$, the map F_0 is exactly the map used to define fixed-angle regularity for the

background metric. Set $S_t = \mathcal{E}_0(\tilde{g}_t)$. This is a smooth family of smooth tensors on the Cartesian domain. Naturality of the Einstein tensor gives

$$\mathcal{E}_0(g_t) = F_t^* S_t.$$

Since g is Ricci-flat, $S_0 = 0$. Differentiating on the regular locus therefore gives

$$B = \left. \frac{d}{dt} \right|_{t=0} (F_t^* S_t) = F_0^* \dot{S}_0 + \left. \frac{d}{dt} \right|_{t=0} (F_t^* S_0) = F_0^* \dot{S}_0.$$

The term coming from the changing coordinates vanishes because $F_t^* S_0$ is identically zero. Thus the coefficients of B in the background Cartesian coordinates are precisely those of the smooth tensor $\dot{S}_0$. This proves smooth extension across the rod.

At a torus fixed point, apply the same argument using both angle-adapted Cartesian coordinate pairs from the preceding subsection. The reference metrics are jointly smooth in t and these four Cartesian variables, so the same calculation proves smoothness of B across both rod surfaces and their intersection. Hence B is a fixed-angle smooth tensor. It has compact support because b does, and differentiating the Bianchi identity at the Ricci-flat background gives $\delta_g B = 0$.

Lemma B.6. *Choose the vector field Z_0 in (B.20) to vanish on a neighborhood of $\text{supp } b$. Then*

$$h_{\text{sc}} = 2g - \mathcal{L}_{Z_0} g = O_\infty(r_{\text{Euc}}^{-2})$$

satisfies $(D\mathcal{E}_0)_g h_{\text{sc}} = 0$, $DA_(h_{\text{sc}}) = 2A_*$, and*

$$(B.24) \quad \int_X \langle B, h_{\text{sc}} \rangle_g dV_g = -8\pi A_* \neq 0.$$

Proof. The equation and the area variation follow from the scaling construction in the preceding subsection. Since Z_0 agrees with the Euclidean dilation field on the end and $g - g_{\text{Euc}} = O_\infty(r_{\text{Euc}}^{-2})$, its leading Euclidean contribution cancels that of $2g$, giving the stated decay.

We compute the pairing by integrating the scalar-curvature variation. Near an interior point of E_j , use the coordinates (ξ, η, ϕ, ψ) from the preceding subsection. Here ξ is normal distance to the rod, and the leading normal metric is

$$d\xi^2 + \alpha_j^2 \xi^2 d\phi^2.$$

The reference variation keeps the quotient metric fixed and satisfies $\dot{\alpha}_j(b) = 1$. Differentiating the angular term therefore gives the leading normal part of b :

$$2\alpha_j \xi^2 d\phi \otimes d\phi.$$

At a Ricci-flat metric, the scalar-curvature variation is

$$(D\text{Scal})_g b = \text{div}_g(J^\sharp), \quad J = \text{div}_g b - d \text{tr}_g b.$$

Here J is a one-form, and $J^\sharp$ is its metric dual. The rod expansions give

$$(\text{div}_g b)(\partial_\xi) = -\frac{2}{\alpha_j \xi} + O(1), \quad d(\text{tr}_g b)(\partial_\xi) = O(1).$$

Indeed, for the leading normal metric, $\Gamma_{\phi\xi}^\phi = \xi^{-1}$ and the angular component of b with one index raised is $2/\alpha_j$. These give the term $-2/(\alpha_j \xi)$ in the divergence. The trace of the leading normal variation is the constant $2/\alpha_j$, and the remaining terms in the rod expansions contribute only bounded errors.

Delete tubes of radius ε about the finite rod surfaces. Along the boundary near an interior rod point, the outward unit normal of the remaining region is $-\partial_\xi$. Hence

$$J(-\partial_\xi) = \frac{2}{\alpha_j \varepsilon} + O(1).$$

After integration in ϕ , whose period is 2π , the boundary measure is

$$(2\pi\alpha_j\varepsilon + O(\varepsilon^2))dA_g.$$

The limiting flux along E_j is therefore

$$\int_{E_j} \frac{2}{\alpha_j \varepsilon} (2\pi \alpha_j \varepsilon) dA_g = 4\pi A_j.$$

The boundary pieces obtained by rounding the tubes near the torus fixed points contribute $O(\varepsilon^2)$: their measure is $O(\varepsilon^3)$ and the current is $O(\varepsilon^{-1})$. There is no contribution at infinity because b has compact support. Applying the divergence theorem and letting ε tend to zero gives

$$\int_X (D \text{Scal})_g b dV_g = 4\pi A_*.$$

In dimension four, at the Ricci-flat background,

$$\text{tr}_g((D\mathcal{E}_0)_g b) = -(D \text{Scal})_g b.$$

Moreover, $\text{supp } B \subset \text{supp } b$, and Z_0 vanishes on a neighborhood of this support. Thus $h_{\text{sc}} = 2g$ on $\text{supp } B$, and consequently

$$\begin{aligned} \int_X \langle B, h_{\text{sc}} \rangle_g dV_g &= 2 \int_X \text{tr}_g B dV_g \\ &= -2 \int_X (D \text{Scal})_g b dV_g = -8\pi A_*. \end{aligned}$$

□

We now specify the weighted spaces. Choose a smooth positive torus-invariant function ω on X which equals $1 + r_{\text{Euc}}$ on the end. For an integer $m \geq 0$ and a weight $\tau \in \mathbb{R}$, set

$$(B.25) \quad \|k\|_{H_\tau^m}^2 = \sum_{j=0}^m \int_X \omega^{-2\tau-4+2j} |\nabla_g^j k|_g^2 dV_g.$$

The space H_τ^m is the completion in this norm of compactly supported torus-invariant tensors with the fixed-angle Cartesian regularity specified above. Write $L_\tau^2 = H_\tau^0$. On the compact part, the corresponding Hölder spaces are defined using the same Cartesian coordinates and an invariant partition of unity. On the end, the $C_\tau^{m,\gamma}$ norm uses $\omega^{j-\tau} |\nabla_g^j k|_g$ and the scaled Hölder seminorm on dyadic annuli, where $0 < \gamma < 1$. We use the same conventions for functions and one-forms. All spaces below are $\mathbb{T}^2$ invariant.

Proposition B.7. *The operator*

$$\mathcal{P}_g : H_{-1}^2 \longrightarrow L_{-3}^2$$

is Fredholm of index zero. Its kernel and its cokernel, the latter represented using the unweighted integral pairing, are generated by a same divergence-free, trace-free tensor s_T representing scaling variation. It satisfies $DA_(s_T) = 2A_*$. For any $c_0 \in \mathbb{R}$, the map*

$$(B.26) \quad (k, t) \longmapsto (\mathcal{P}_g k + tB, DA_*(k) + c_0 t)$$

is an isomorphism from $H_{-1}^2 \oplus \mathbb{R}$ to $L_{-3}^2 \oplus \mathbb{R}$. It is also an isomorphism from $C_{-1}^{2,\gamma} \oplus \mathbb{R}$ to $C_{-3}^{0,\gamma} \oplus \mathbb{R}$ for $0 < \gamma < 1$.

In the rest part of this section, we prove this proposition.

We first establish the local and weighted estimates. In the fixed-angle Cartesian coordinates from the preceding subsection, the metric and the operator have smooth coefficients, and the tensors under consideration are invariant under rotations in the corresponding coordinate pairs. Ordinary interior elliptic estimates therefore apply across a rod surface and across a torus fixed point.

Let $\mathcal{P}_0$ be the operator for g_{Euc} . The standard fundamental solution construction for constant-coefficient elliptic operators gives a kernel K , smooth away from the origin, satisfying

$$|\partial^j K(x)| \leq C_j |x|^{-2-j}, \quad x \neq 0, \quad j \geq 0.$$

For every smooth compactly supported symmetric two-tensor f on $\mathbb{R}^4$, the convolution $k = K * f$ satisfies $\mathcal{P}_0 k = f$. We use this kernel to establish the weighted estimates below. The kernel bound, together with rescaling on annuli and interior elliptic estimates, gives

$$\|K * f\|_{H_\tau^2(\mathbb{R}^4)} \leq C_\tau \|f\|_{L_{\tau-2}^2(\mathbb{R}^4)}, \quad -2 < \tau < 0,$$

for smooth compactly supported f . By density, convolution with K therefore defines a bounded right inverse in these weighted spaces. The homogeneous equation has no nonzero solution in H_τ^2 : applying scaled interior estimates on expanding Euclidean balls and using $\tau < 0$ forces such a solution to vanish. Consequently,

$$\mathcal{P}_0 : H_\tau^2(\mathbb{R}^4) \longrightarrow L_{\tau-2}^2(\mathbb{R}^4)$$

is an isomorphism for $-2 < \tau < 0$.

Since $g - g_{\text{Euc}} = O_\infty(r_{\text{Euc}}^{-2})$, the coefficients of $\mathcal{P}_g - \mathcal{P}_0$ in orders two, one, and zero are respectively $O(r_{\text{Euc}}^{-2})$, $O(r_{\text{Euc}}^{-3})$, and $O(r_{\text{Euc}}^{-4})$. We may therefore absorb these terms outside a sufficiently large compact set. Combining the resulting end estimate with the local estimates gives

$$(B.27) \quad \|k\|_{H_\tau^2} \leq C(\|\mathcal{P}_g k\|_{L_{\tau-2}^2} + \|k\|_{L^2(K_0)}), \quad -2 < \tau < 0,$$

where $K_0 \subset X$ is compact. Rellich compactness implies that the kernel is finite dimensional and the range is closed.

At a Ricci-flat metric, $(D\mathcal{E}_0)_g$ is, up to an overall sign, the Hessian of the Einstein–Hilbert functional. Hence $\mathcal{P}_g$ is formally self-adjoint. The Cartesian regularity makes the inner boundary terms in Green’s formula vanish. Under the unweighted integral pairing, the dual of $L_{\tau-2}^2$ is $L_{-\tau-2}^2$. If a tensor in this dual space annihilates the range, it satisfies the adjoint equation across the rod surfaces and torus fixed points as well as in the regular locus. Indeed, in a Cartesian chart, average an arbitrary smooth test tensor over the rotations. Its pairing with an invariant tensor is unchanged, and the averaged test tensor belongs to the specified domain. Distributional elliptic regularity on the whole chart therefore applies. End estimates then place the adjoint solution in $H_{-\tau-2}^2$. This proves the Fredholm property. At $\tau = -1$, the complementary weight is again -1 , so formal self-adjointness gives index zero.

We will also need a decay improvement. If $\mathcal{P}_g k = 0$ and $k \in H_\tau^2$ for some $-2 < \tau < 0$, scaled elliptic estimates first give $k = O_\infty(r_{\text{Euc}}^\tau)$. Multiply k by a cutoff supported on the end and extend the resulting tensor by zero to $\mathbb{R}^4$. Its flat equation has a right-hand side which is compactly supported plus $O_\infty(r_{\text{Euc}}^{\tau-4})$. This right-hand side is integrable because $\tau < 0$. Convolution with K gives $O(r_{\text{Euc}}^{-2})$; the difference from the cutoff tensor is a decaying entire flat solution and is therefore zero. Differentiated estimates give

$$k = O_\infty(r_{\text{Euc}}^{-2}).$$

To set the scaling variation in divergence gauge, consider

$$Q = 2\delta_g \delta_g^*$$

on invariant one-forms with the fixed-angle regularity specified above. Since g is Ricci-flat,

$$Q\vartheta = \nabla_g^* \nabla_g \vartheta + d(\delta_g \vartheta).$$

This is a formally self-adjoint elliptic operator. Its Euclidean fundamental kernel satisfies the same bounds used above. The preceding weighted estimates and decay argument therefore apply to Q : the map

$$Q : H_\tau^2 \longrightarrow L_{\tau-2}^2$$

is Fredholm for $-2 < \tau < 0$, and every element of its kernel is $O_\infty(r_{\text{Euc}}^{-2})$. If $Q\vartheta = 0$, integration by parts gives

$$0 = \int_X \langle Q\vartheta, \vartheta \rangle_g dV_g = 2 \int_X |\delta_g^* \vartheta|_g^2 dV_g.$$

The boundary terms vanish by the fixed-angle regularity at the rods and the decay at infinity. Hence $\vartheta^\sharp$ is a decaying Killing field. Ricci-flatness and the Bochner maximum principle imply that $\vartheta = 0$. By formal self-adjointness, the cokernel is represented by the kernel of Q at the complementary weight $-\tau - 2$. This weight also lies in $(-2, 0)$, so the same argument shows that the cokernel is zero. Consequently,

$$Q : H_\tau^2 \longrightarrow L_{\tau-2}^2$$

is an isomorphism for every $-2 < \tau < 0$.

Fix $0 < \epsilon < 1$ and solve

$$Q\vartheta = -\delta_g h_{\text{sc}}$$

at weight $\tau = -1 + \epsilon$. This small shift avoids the borderline weight: $\delta_g h_{\text{sc}} = O_\infty(r_{\text{Euc}}^{-3})$ belongs to $L_{-3+\epsilon}^2$. Define

$$s_T = h_{\text{sc}} + 2\delta_g^* \vartheta = h_{\text{sc}} + \mathcal{L}_{\vartheta^\sharp} g.$$

The vector field $\vartheta^\sharp$ is invariant, smooth in the fixed-angle coordinates, and tangent to the rod surfaces; it vanishes at the torus fixed points. By construction and diffeomorphism invariance,

$$\delta_g s_T = 0, \quad (D\text{Ric})_g s_T = 0.$$

Initially $s_T = O_\infty(r_{\text{Euc}}^{-2+\epsilon})$, so $s_T \in H_{-1}^2$. Its trace is a decaying harmonic function and hence is zero by the maximum principle, applied across the rods using the Cartesian coordinates. It follows that $\mathcal{P}_g s_T = 0$, and the decay improvement gives $s_T = O_\infty(r_{\text{Euc}}^{-2})$. Since the gauge field preserves every rod surface, it does not change its area. Thus $DA_*(s_T) = 2A_*$.

Now suppose $\mathcal{P}_g k = 0$ in H_{-1}^2 . The linearized Bianchi identity gives

$$Q(\delta_g k) = 0.$$

The decay improvement for k gives $\delta_g k = O_\infty(r_{\text{Euc}}^{-3})$, so the kernel calculation for Q implies $\delta_g k = 0$. Taking the trace of $\mathcal{P}_g k = 0$ then shows that $\text{tr}_g k$ is harmonic. It decays, so it vanishes. Consequently $(D\text{Ric})_g k = 0$, and Proposition B.5 applies. Since k is a fixed-angle variation, it is a multiple of h_{sc} plus an allowed bounded gauge term.

Subtracting the corresponding multiple of s_T , we obtain a tensor $\mathcal{L}_Y g$ with zero divergence and trace. The vector field Y has the regularity and differentiated bounds specified in (B.19). The divergence equation and Ricci-flatness give $\nabla_g^* \nabla_g Y = 0$. Comparing this equation with the Euclidean operator on the end gives

$$\Delta_{\mathbb{R}^4} Y = O_\infty(r_{\text{Euc}}^{-4}).$$

The Newton kernel estimate gives

$$Y = v_0 + O_\infty(r_{\text{Euc}}^{-2} \log r_{\text{Euc}})$$

for a constant vector v_0 . The standard torus action on the Euclidean end has no nonzero fixed vector, so invariance forces $v_0 = 0$. The Bochner identity and the maximum principle now give $Y = 0$. Hence the kernel and cokernel of $\mathcal{P}_g$ are both represented by $\mathbb{R}s_T$. The same conclusion holds at every weight $-2 < \tau < 0$, because every homogeneous solution in these weights has the decay just proved.

Since B is divergence-free and compactly supported, integration by parts gives

$$\int_X \langle B, 2\delta_g^* \vartheta \rangle_g dV_g = 0.$$

Lemma B.6 therefore yields

$$\int_X \langle B, s_T \rangle_g dV_g = -8\pi A_* \neq 0.$$

Given a first component f , pairing $\mathcal{P}_g k + tB = f$ with s_T determines t uniquely. The Fredholm alternative then gives k modulo $\mathbb{R}s_T$. The second equation in (B.26) determines the remaining coefficient because $DA_*(s_T) = 2A_* > 0$. The area functional is bounded on H^2 : in the Cartesian charts, the codimension-two trace theorem gives a bounded restriction to each compact rod surface. Thus the bounded inverse theorem proves the Sobolev assertion, for every $c_0 \in \mathbb{R}$.

Finally, let $f \in C_{-3}^{0,\gamma}$. It need not belong to L_{-3}^2 , but it belongs to $L_{-3+\epsilon}^2$ for every $\epsilon > 0$. Solve the augmented problem first at weight $-1 + \epsilon$, with $0 < \epsilon < 1$, using the same kernel and cokernel calculation. Local regularity and scaled estimates give $k = O(r_{\text{Euc}}^{-1+\epsilon})$ with the corresponding derivative bounds through order two. On the end, its flat equation has right-hand side

$$O(r_{\text{Euc}}^{-3}) + O(r_{\text{Euc}}^{-5+\epsilon}).$$

Convolution with K improves k to $O(r_{\text{Euc}}^{-1})$. Scaled Schauder estimates then give $k \in C_{-1}^{2,\gamma}$ and the bounded inverse estimate in these spaces. Uniqueness follows from the kernel and area calculation above. The same argument gives the corresponding higher regularity statements.

APPENDIX C. DIFFEOMORPHISM TYPES OF COMPACT TORIC EINSTEIN FOUR-MANIFOLDS

Here we prove that the improved Hitchin–Thorpe inequality is sufficient for the existence of a toric Einstein metric on the underlying manifold. The key idea is to compare the virtual counts modulo two before and after a replacement of a corner. We cut the conical moduli space at $\beta = 1$ and at a sufficiently small positive total area of the designated spheres. The resulting boundary counts correspond to smooth metrics and small-area metrics, respectively. The compactness and angle estimates proved in the main text allow us to compare these boundaries by a finite-dimensional perturbation. A gluing construction identifies the small-area count with the count on the contracted manifold. The key new difficulty is to carry out this gluing without assuming toric nondegeneracy of the compact metric, since we have established nondegeneracy only for the algebraically special metrics. We overcome this difficulty using the standard finite-dimensional stabilization procedure for Fredholm equations. We choose the perturbation terms to be divergence-free and supported away from the chosen fixed point, and carry out the gluing construction over the resulting smooth solution space of the perturbed equation.

More precisely, we use finite-dimensional stabilization, as in the gluing modulo obstructions framework of [43]. Delay’s localized divergence correction [22] allows us to choose the stabilizing tensors to vanish near the chosen fixed point. We then glue over the resulting smooth space of solutions to the stabilized equation. The angle–area inverse from Section 5.4 handles the Ricci-flat part; the stabilization handles the compact kernel. This proves that the virtual count is preserved under replacement. Every cyclic list with directions in $\{A, B, C\}$ reduces to a bigon or triangle by such operations, so its virtual count is odd.

Theorem C.1. *Let X be a simply connected smooth compact four-manifold admitting an effective $\mathbb{T}^2$ -action. Then X admits a toric Einstein metric if and only if*

$$(C.1) \quad \chi(X) \geq 3|\tau(X)|.$$

The action in this theorem is allowed to vary. For prescribed actions we prove the following stronger (virtual) counting statement.

Theorem C.2. *If the directions of a compact rod structure $\mathfrak{R}$ belong to $\{A, B, C\}$ after a global change of lattice basis, then*

$$\mathfrak{n}(\mathfrak{R}) \equiv 1 \pmod{2}.$$

In particular, $X_{\mathfrak{R}}$ admits a toric Einstein metric for its prescribed action.

We normalize the Einstein constant to be three. The allowed diffeomorphisms preserve orientation, commute with the fixed torus action, and preserve each rod sphere. When a corner is chosen, its fixed point and ordered incident rods are also fixed. All tensor spaces are $\mathbb{T}^2$ -invariant, with regularity measured in local developments. We write $\mathcal{M}_{\mathfrak{R}}^{\text{sm}}$ for the quotient of the smooth toric Einstein metrics on $X_{\mathfrak{R}}$ by these diffeomorphisms and define the count $\mathfrak{n}(\mathfrak{R})$ modulo two below.

C.1. Compactness and the invariant.

Lemma C.3. *The normalized smooth moduli space $\mathcal{M}_{\mathfrak{R}}^{\text{sm}}$ is compact whenever $\mathfrak{R}$ has directions in $\{A, B, C\}$.*

Proof. Theorem 3.13, Myers' theorem, and Proposition 3.15 give volume noncollapsing and diameter upper bound. The curvature energy is also uniformly bounded by the Gauss-Bonnet theorem. We claim the curvature is uniformly bounded, then Proposition 5.1 gives smooth convergence after orientation-preserving diffeomorphisms commuting with the fixed torus action and preserving each rod sphere.

If the maximum curvature were unbounded, rescaling at a curvature maximum would give a complete smooth nonflat Ricci-flat limit with finite curvature energy and Euclidean volume growth, by similar argument as in Theorem 5.7. There is also an effective limiting torus action with the original primitive rod directions. The limit is ALE by [6, Theorem 1.5], contradicting Lemma 5.6. $\square$

Proposition C.4. *For a compact rod structure with directions in $\{A, B, C\}$, there is a well-defined virtual count $\mathfrak{n}(\mathfrak{R}) \in \mathbb{Z}/2\mathbb{Z}$, independent of the reference gauges, stabilizing sections, and sufficiently small regular perturbation.*

Remark C.5. *The invariant $\mathfrak{n}(\mathfrak{R})$ is a virtual count modulo two: it is defined by the mod-two degree of the parameter projection from a finite-dimensional stabilization of the Einstein equation. Its definition does not require the original moduli space to be discrete. If every Einstein metric is torically nondegenerate, the moduli space is finite and $\mathfrak{n}(\mathfrak{R})$ equals its cardinality modulo two. In all cases, $\mathfrak{n}(\mathfrak{R}) \neq 0$ implies that the moduli space is nonempty.*

Proof. Set $\mathcal{M} = \mathcal{M}_{\mathfrak{R}}^{\text{sm}}$. If this space is empty, set $\mathfrak{n}(\mathfrak{R}) = 0$. Otherwise, Lemma 4.2 gives a fixed-angle Fredholm equation of index zero whose cokernel consists of smooth invariant infinitesimal Einstein deformations. By compactness and the extension construction in the proof of Proposition 4.5, finitely many C^2 sections $\tau_1, \dots, \tau_N$ span these cokernels along $\mathcal{M}$. They satisfy

$$\delta_h \tau_\nu(h) = 0, \quad \tau_\nu(F^*h) = F^* \tau_\nu(h)$$

for every allowed diffeomorphism F . For clarity, the divergence-free extension is obtained by transporting a chosen tensor in a reference gauge, applying

$$\mathbf{Q}_h \tau = \tau - \delta_h^* (\delta_h \delta_h^*)_{\perp}^{-1} \delta_h \tau,$$

and multiplying by a cutoff on the space of metrics. The inverse is taken on invariant one-forms orthogonal to the torus Killing fields. The slice and regularity arguments of Section 4 apply with the cone factors fixed.

Consider the stabilized equation

$$(C.2) \quad \mathcal{E}_3(h) = \sum_{\nu=1}^N u_\nu \tau_\nu(h), \quad u \in \mathbb{R}^N.$$

In a reference gauge based at g , subtract $\delta_h^* \delta_h g$ from the left-hand side. The contracted Bianchi identity and integration by parts remove this gauge term, as in Proposition 4.3. The augmented linearization is surjective of index N along $\mathcal{M} \times \{0\}$. The implicit-function theorem and the gauge-transition argument in Proposition 4.5 give a C^2 manifold $\mathcal{W}$ of dimension N near this compact set, with parameter projection

$$p : \mathcal{W} \longrightarrow \mathbb{R}^N, \quad p([h], u) = u, \quad p^{-1}(0) = \mathcal{M}.$$

The stabilizers are torus translations and act trivially on invariant tensors, so no additional quotient or isotropy weights occur.

Choose $V \Subset \mathcal{W}$ containing the zero fibre. There is $c > 0$ such that $|p| \geq c$ on the boundary of V . For a sufficiently small regular value u , define

$$(C.3) \quad \mathfrak{n}(\mathfrak{R}) = \deg_2(p, V, 0) = \#(p^{-1}(u) \cap V) \pmod{2}.$$

The fibre is finite because it is a compact zero-dimensional manifold. Compact closure is taken in the finite-dimensional space $\mathcal{W}$; no compactness theorem for arbitrary perturbed Einstein metrics is needed.

Homotopy invariance and excision give independence of u and V . To compare two stabilizations, take their union. Near the common zero fibre, the extra parameter directions are identity factors in the parameter map, so they do not change its degree. Equivariance identifies changes of reference gauge. The same argument applies to a C^2 homotopy of equivariant divergence-free sections, including sections depending on u , provided the augmented linearizations remain surjective along the common compact zero fibre: localization near this fibre and a small regular parameter give a compact one-dimensional cobordism. Finally, at a nondegenerate metric the local degree is one modulo two. Such points are isolated and hence finite in number when all metrics are nondegenerate. $\square$

Changing the starting index of the cyclic rod list or making a global change of lattice basis gives an equivalent moduli problem.: transporting the metrics, allowed diffeomorphisms, and perturbing sections identifies the perturbed moduli spaces and their parameter projections. Reversing the orientation does not change the Einstein equation or the allowed diffeomorphism group. Since the count modulo two does not depend on orientation choices, it is invariant under all three operations.

C.2. Stabilized gluing and the boundary count. Fix a replacement $\mathfrak{R} = \mathfrak{R}_0 \#_{\mathfrak{p}} W$ from (4.1), and write $M = X_{\mathfrak{R}_0}$ and $X = X_{\mathfrak{R}}$. Contracting the intermediate rods of W recovers the corner $\mathfrak{p}$ of $\mathfrak{R}_0$. The first and last rods of W correspond to the two rods incident to $\mathfrak{p}$, with directions B, A in boundary order. We identify every other rod with its corresponding rod in $\mathfrak{R}_0$, preserving the cyclic order and lattice directions. Throughout, we use the same torus lattice and the orientations fixed above. Denote by $p_0 \in M$ the torus fixed point lying over $\mathfrak{p}$.

Let (Y, h) be the AE toric conical Ricci-flat gravitational instanton associated with W in Section 5.4. Its two infinite rods correspond, in order, to the B - and A -rods at $\mathfrak{p}$, and its finite rods have common cone angle $2\pi\beta_*$, where $\beta_* > 1$. Fix a scale for h and set $A_* = A(h)$, the total area of its finite rod spheres.

Theorem C.6. *If the rod directions of $\mathfrak{R}_0$ belong to $\{A, B, C\}$, then*

$$(C.4) \quad \mathfrak{n}(\mathfrak{R}_0 \#_{\mathfrak{p}} W) \equiv \mathfrak{n}(\mathfrak{R}_0) \pmod{2}.$$

Lemma C.7. *The stabilizing sections for the compact moduli space of smooth Einstein metrics can be chosen equivariant, divergence-free, and C^2 , with their values spanning the cokernel at every Einstein metric. Moreover, there is a uniform radius $\rho > 0$ such that every section vanishes on $B_{g'}(p_0, \rho)$ for every nearby invariant metric g' on M in its domain.*

Proof. Start with the divergence-free extensions in the proof of Proposition C.4. Fix a reference Einstein metric g and work in its reference gauge. Write such an extension as $\tau(g')$, where g' is a nearby invariant metric and $\delta_{g'}\tau(g') = 0$. Choose an invariant cutoff ψ equal to zero on $B_g(p_0, r)$ and one outside $B_g(p_0, 2r)$, with its transition compactly contained in the fixed annulus

$$\Omega = B_g(p_0, 2r) \setminus \overline{B_g(p_0, r)}.$$

For r sufficiently small and g' sufficiently close to g , every $\mathbb{T}^2$ -invariant Killing field on Ω is a constant linear combination of the two torus generators.

The divergence error $\delta_{g'}(\psi\tau(g'))$ is supported in the transition region of ψ and is orthogonal to every Killing field on Ω . We average any Killing field on Ω over the torus. Since $\delta_{g'}(\psi\tau(g'))$ is invariant, this averaging does not change the integral pairing. The averaged field is a constant linear combination of the torus generators and therefore extends to a Killing field Z on $B_g(p_0, 2r)$. Since $\delta_{g'}\tau(g') = 0$, we have

$$\delta_{g'}(\psi\tau(g')) = \delta_{g'}((\psi - 1)\tau(g')).$$

The tensor $(\psi - 1)\tau(g')$ vanishes near the boundary of $B_g(p_0, 2r)$. We may therefore integrate by parts to obtain

$$\int_{\Omega} [\delta_{g'}(\psi\tau(g'))](Z) \, d\text{Vol}_{g'} = \frac{1}{2} \int_{B_g(p_0, 2r)} \langle (\psi - 1)\tau(g'), \mathcal{L}_Z g' \rangle_{g'} \, d\text{Vol}_{g'} = 0.$$

Delay's divergence inverse produces an invariant tensor $s(g')$, supported in the closure of Ω and extendible by zero with the required regularity, such that

$$\delta_{g'} s(g') = \delta_{g'}(\psi\tau(g'));$$

see [22, Theorems 1.3, 3.8, and 6.1 and Section 9.3]. Thus $\psi\tau(g') - s(g')$ is divergence-free and vanishes on the fixed ball $B_g(p_0, r)$. The rescaled estimate gives

$$(C.5) \quad \|\psi\tau(g') - s(g') - \tau(g')\|_{L^2(g')} \leq Cr^2 \|\tau(g')\|_{C^0(B_g(p_0, 2r), g')}.$$

Consequently the cokernel-spanning property remains to hold for small r .

Apply this construction on fixed weighted spaces on the reference annulus Ω , using a fixed boundary-defining function. Let Z_1, Z_2 denote the fixed torus generators. The invariant Killing kernel is

$$K(g') = \text{span}\{g'(Z_1, \cdot), g'(Z_2, \cdot)\}.$$

Its weighted orthogonal projection depends C^2 on g' . These projections identify the orthogonal complements with the reference complement. Under these identifications, Delay's projected second-order operator is a C^2 family of bounded operators between fixed spaces and is invertible at the reference metric. The inverse-operator theorem gives a uniformly bounded inverse depending C^2 on g' after shrinking the reference neighborhood. The weighted boundary estimates ensure that extension by zero preserves the required regularity. Thus the localized sections depend C^2 on g' . Extend them through the reference gauges and multiply by the metric-space cutoffs as in Proposition C.4. These operations preserve divergence-freeness and equivariance.

Choose finitely many reference neighborhoods covering the compact smooth moduli space. After shrinking these neighborhoods, the nearby metrics are uniformly equivalent to their reference metrics. A common radius $\rho > 0$ can therefore be chosen so that each $B_{g'}(p_0, \rho)$ lies in the corresponding fixed ball on which the sections vanish. No trace-free condition is imposed on the perturbing sections. $\square$

Choose these sections $\sigma_1, \dots, \sigma_N$. The compact universal solution space $\mathcal{W}$ is a C^2 manifold of dimension N , with parameter projection $p : \mathcal{W} \rightarrow \mathbb{R}^N$. In local representatives,

$$(C.6) \quad \mathcal{E}_3(g_x) = \sum_{\nu=1}^N u_x^\nu \sigma_\nu(g_x), \quad p(x) = u_x.$$

In particular, g_x is Einstein near p_0 . Choose $V \subseteq \mathcal{W}$ containing $p^{-1}(0)$, sufficiently close to this compact set that $|u_x|$ is small on $\bar{V}$. There is $c > 0$ such that $|p| \geq c$ on the frontier of V . The case $p^{-1}(0) = \emptyset$ is treated at the end of the proof.

We represent tangent vectors to $\mathcal{W}$ by pairs (k, v) with $\delta_{g_x} k = 0$. Explicitly, a tangent vector represented in a reference gauge by $(\dot{g}, \dot{u})$ is identified with $(H_{g_x} \dot{g}, \dot{u})$, where

$$H_{g_x} \dot{g} = \dot{g} - \delta_{g_x}^* (\delta_{g_x} \delta_{g_x}^*)_{\perp}^{-1} \delta_{g_x} \dot{g}.$$

The inverse is taken on invariant one-forms orthogonal to the torus Killing fields. Equivariance and the differentiated divergence identity for $\mathcal{E}_3(g) - \sum_{\nu} u^\nu \sigma_\nu(g)$ identify these pairs with the kernel of the compact gauged linearization. All tangent pairings and restrictions to $T_x \mathcal{W}$ below use this identification. We choose C^2 linear conditions

$$(C.7) \quad \ell_x : (k, v) \mapsto \mathcal{E}_x, \quad \ell_x|_{T_x \mathcal{W}} \text{ is an isomorphism,}$$

where $\mathcal{E}$ has rank N and the tensor pairings defining ℓ_x are supported away from p_0 . To obtain them, pair with tangent vectors $(\dot{g}, \dot{u})$ by

$$(k, v) \mapsto \int_M \zeta \langle k, \dot{g} \rangle_{g_x} \, d\text{vol}_{g_x} + v \cdot \dot{u},$$

where ζ vanishes on a small ball and is positive outside a slightly larger ball, both contained in the common vanishing region of the sections. This pairing is positive definite on $T_x\mathcal{W}$: a null vector has $\dot{u} = 0$ and vanishes outside the larger ball, and unique continuation for the gauged linearized Einstein equation makes it zero inside as well. Finitely many smooth pairings embed $T\mathcal{W}$ in a trivial bundle. Approximating its C^1 orthogonal projection by a C^2 self-adjoint field and taking the spectral projection near one gives (C.7) uniformly on $\bar{V}$. We choose the pairings and the approximation on $\mathcal{W}$ and transport them equivariantly to local representatives.

Construct $b_{\varepsilon,x,\beta}$ as in Section 5.4, with total area $\varepsilon^2 A_*$ at $\beta = \beta_*$. Use the same identifications J_β , weights, and angle variable

$$\sigma = \varepsilon(\beta - \beta_*), \quad g = b_{\varepsilon,x,\beta} + J_\beta k.$$

Work at a fixed higher regularity sufficient for C^2 parameter dependence; denote the resulting weighted spaces by $X_\varepsilon, Y_\varepsilon$, with weights $-1, -3$ as before.

We extend the perturbations without changing them near the gluing region. Choose a uniform small $r > 0$ and an invariant cutoff χ_x , defined in g_x -normal coordinates, which is zero on $B_r^{g_x}(p_0)$ and one outside $B_{2r}^{g_x}(p_0)$. Choose r so small that $B_{2r}^{g_x}(p_0)$ lies in the vanishing region of every $\sigma_\nu(g')$ for all compact metrics g' under consideration. For small ε , the splice and angle variation lie inside $B_r^{g_x}(p_0)$. On the compact manifold define

$$(C.8) \quad \hat{g}_x(g) = g_x + \chi_x(g - g_x),$$

using the common exterior identification and setting the right-hand side equal to g_x on the inner ball. Transfer $\sigma_\nu(\hat{g}_x(g))$ to X and extend it by zero over the gluing region; denote the resulting tensor by $\hat{\sigma}_{\nu,\varepsilon,x}(g)$. Since $\hat{g}_x(g) = g$ wherever this tensor can be nonzero,

$$(C.9) \quad \delta_g \hat{\sigma}_{\nu,\varepsilon,x}(g) = 0, \quad \hat{\sigma}_{\nu,\varepsilon,x}(b_{\varepsilon,x,\beta}) = \sigma_\nu(g_x)$$

on the common exterior. Their first two derivatives are uniformly controlled on a fixed smooth region. Their angle derivatives vanish, and all these tensors vanish near the rescaled copy of Y and the connecting annulus. The construction commutes with the allowed changes of representative.

For $v \in \mathbb{R}^N$, consider

$$(C.10) \quad \mathcal{F}_{\varepsilon,x}(k, \sigma, v) = \begin{pmatrix} J_\beta^{-1} \left\{ \mathcal{E}_3(g) - \delta_g^* \delta_g b_{\varepsilon,x,\beta} - \sum_{\nu=1}^N (u_x^\nu + v^\nu) \hat{\sigma}_{\nu,\varepsilon,x}(g) \right\} \\ \varepsilon^{-1} (A(g) - \varepsilon^2 A_*) \\ \ell_x(k, v) \end{pmatrix}.$$

The last row uses only the common exterior. Write

$$N_\varepsilon(k, \sigma, v) = \|k\|_{X_\varepsilon} + |\sigma| + |v|.$$

“latex

Proposition C.8. *After shrinking V , there exist $\varepsilon_0, C > 0$ such that, for every $x \in \bar{V}$ and $0 < \varepsilon < \varepsilon_0$, the equation $\mathcal{F}_{\varepsilon,x}(k, \sigma, v) = 0$ has a unique solution satisfying*

$$(C.11) \quad N_\varepsilon(k_\varepsilon, \sigma_\varepsilon, v_\varepsilon) \leq C\varepsilon^{4/3}.$$

These solutions depend C^2 on (ε, x) for $\varepsilon > 0$, and

$$(C.12) \quad p_\varepsilon(x) = u_x + v_\varepsilon(x), \quad \|p_\varepsilon - p\|_{C^0(\bar{V})} \leq C\varepsilon^{4/3}.$$

Near its zero fibre, this family gives a regular N -dimensional stabilization of the fixed-area equation. Its zero fibre is in bijection with all toric conical Einstein metrics on X with the prescribed cone-angle data, $\beta \geq 1$, and $A = \varepsilon^2 A_$, modulo the allowed diffeomorphisms.*

Proof. We explain the changes to the gluing and converse arguments in Section 5.4 and Proposition 5.11. We work with $X_\varepsilon = C_{-1}^{m+2,\alpha}$ and $Y_\varepsilon = C_{-3}^{m,\alpha}$, for a fixed sufficiently large integer

m . We choose the spatial regularity in the preceding constructions sufficiently high for the fixed-region section bounds and two parameter derivatives in these spaces. We also put

$$H_\varepsilon^{(m)}(k) = \sup_X |k|_{m+2,\alpha;w_\varepsilon},$$

so that $H_\varepsilon^{(0)}$ is the norm used in Lemma 5.12.

We first prove the uniform inverse estimate. We write $\mathcal{L}_{\varepsilon,x} = D\mathcal{F}_{\varepsilon,x}(0)$ and $b_{\varepsilon,x} = b_{\varepsilon,x,\beta_*}$, and define

$$\mathcal{L}_{\varepsilon,x}^0(k, \sigma, v) = \mathcal{L}_{\varepsilon,x}(k, \sigma, v) + \left(\sum_\nu u_x^\nu D\hat{\sigma}_{\nu,\varepsilon,x}(b_{\varepsilon,x})[k], 0, 0 \right).$$

Its compact limiting operator is

$$(k, v) \mapsto \left(\mathcal{P}_{g_x} k - \sum_\nu v^\nu \sigma_\nu(g_x), \ell_x(k, v) \right).$$

Along $p^{-1}(0)$, this operator is invertible by the cokernel-spanning property and (C.7), using the divergence-free identification of $T_x\mathcal{W}$. By compactness and continuity, it remains uniformly invertible on the closure of a relatively compact neighborhood V_0 of the zero fibre.

We apply the three-limit argument of Proposition 5.9 to $\mathcal{L}_{\varepsilon,x}^0$. On Y , the perturbing tensors vanish, so Proposition B.7 eliminates the limiting tensor and angle variable. On the compact region, the bound $|k| \leq Cr^{-1}$ gives removability at p_0 , and the compact inverse eliminates both the limiting tensor and v . The remaining flat-annulus limit is excluded by the same weight- -1 argument. This proves a uniform estimate. For each fixed ε , the operator has index zero: we have added $N+1$ finite-dimensional variables and $N+1$ equations to $\mathcal{P}_{b_{\varepsilon,x}}$. Consequently,

$$\|(\mathcal{L}_{\varepsilon,x}^0)^{-1}\| \leq C_0$$

uniformly for $x \in \bar{V}_0$ and sufficiently small ε .

The omitted terms have operator norm at most $C_1|u_x|$. Indeed, their metric argument is $\chi_x k$, supported in a fixed exterior region, and

$$\|\chi_x k\|_{C^{m+2,\alpha}(M)} \leq C\|k\|_{X_\varepsilon}$$

uniformly in ε and x . The fixed-region bounds for the section derivatives give the assertion. We now shrink $V \Subset V_0$ so that $C_0 C_1 \sup_V |u_x| < 1/2$. The Neumann series gives a uniformly bounded inverse for $\mathcal{L}_{\varepsilon,x}$.

At the reference metric, the error vanishes away from the gluing region by (C.6) and (C.9). Near p_0 , the metrics g_x are Einstein and have uniformly controlled normal-coordinate expansions. With $r_* = \varepsilon^{2/3}$, we therefore obtain

$$\|\mathcal{F}_{\varepsilon,x}(0)\| \leq C(\varepsilon^3 + \varepsilon^2 r_*^{-1} + r_*^3) \leq C\varepsilon^{4/3}.$$

We write $Q_{\varepsilon,x}(z) = \mathcal{F}_{\varepsilon,x}(z) - \mathcal{F}_{\varepsilon,x}(0) - \mathcal{L}_{\varepsilon,x}z$. The nonlinear estimate in Theorem 5.8 remains uniform:

$$\|Q_{\varepsilon,x}(z) - Q_{\varepsilon,x}(z')\| \leq C\varepsilon^{-1}(N_\varepsilon(z) + N_\varepsilon(z'))N_\varepsilon(z - z')$$

whenever $N_\varepsilon(z), N_\varepsilon(z') \leq c\varepsilon$. The additional perturbation terms satisfy the same estimate without the factor ε^{-1} , since they are supported in a fixed exterior region. The contraction argument gives the unique solution in (C.11). Divergence-freeness in (C.9) permits the same Bianchi argument to remove the gauge term. The implicit-function theorem gives C^2 dependence on (ε, x) for $\varepsilon > 0$, and (C.12) follows from the bound on v_ε . We also have $|\beta_\varepsilon - \beta_*| \leq C\varepsilon^{1/3}$, so $\beta_\varepsilon > 1$ for sufficiently small ε .

We next identify all small-area Einstein metrics. Let $\beta_j \geq 1$ and $A(g_j) \rightarrow 0$. The angle bound, Theorem 5.7, and uniqueness in Appendix B give a compact limit g_0 , the area-normalized limit (Y, h) , and $\beta_j \rightarrow \beta_*$. We set

$$\varepsilon_j^2 = A(g_j)/A_*, \quad x_0 = ([g_0], 0) \in \mathcal{W}.$$

Lemma 5.12 applies uniformly for x near x_0 : its vector estimate uses the torus Killing fields and the sublinear kernel on Y , independently of the compact Einstein kernel. After x -dependent allowed pullbacks, we obtain representatives

$$g_j(x) = b_{\varepsilon_j, x, \beta_j} + J_{\beta_j} k_j(x), \quad \delta_{g_j(x)} b_{\varepsilon_j, x, \beta_j} = 0.$$

We justify the required parameter dependence of these representatives. In a fixed torus-linearizing chart, the compact metrics and their first two parameter derivatives have constant terms plus $O(r^2)$ remainders. The constant terms are included separately in the norms of Lemma 5.12, and the annular remainders of the splice are controlled by $r^2 + (\varepsilon/r)^2$, also after parameter differentiation. Thus the differentiated gauge equations have uniformly controlled right-hand sides. The uniform vector inverse gives

$$H_{\varepsilon_j}^{(0)}(k_j(x)) \leq o_j(1) + C|x - x_0|.$$

A second parameter differentiation and exterior elliptic estimates give C^1 convergence on the fixed exterior. The limiting gauge is identified by uniqueness of the compact gauge equation, modulo torus translations.

We put

$$\Phi_j(x) = \ell_x(k_j(x), -u_x).$$

These maps converge in C^1 to a map Φ_0 with $\Phi_0(x_0) = 0$. To compute its derivative, we let f_x be the limiting gauge diffeomorphism satisfying $\delta_{g_0}(f_x^* g_x) = 0$, with $f_{x_0} = \text{Id}$. For a tangent vector represented by $(\dot{g}, \dot{u})$, the derivative a of f_x , viewed as a one-form, satisfies

$$2\delta_{g_0}\delta_{g_0}^* a + \delta_{g_0}\dot{g} = 0.$$

The gauged representative of g_0 is $(f_x^{-1})^* g_0$. Hence the derivative of its difference from g_x is $-\dot{g} - 2\delta_{g_0}^* a = -H_{g_0}\dot{g}$, and

$$D\Phi_0(x_0)(\dot{g}, \dot{u}) = -\ell_{x_0}(H_{g_0}\dot{g}, \dot{u}).$$

This is invertible by (C.7). The implicit-function theorem therefore gives a unique $x_j \rightarrow x_0$ satisfying $\Phi_j(x_j) = 0$. The triple

$$z_j = (k_j(x_j), \varepsilon_j(\beta_j - \beta_*), -u_{x_j})$$

satisfies $\mathcal{F}_{\varepsilon_j, x_j}(z_j) = 0$.

The rescaled local estimates used in (5.17), applied at the chosen higher regularity, give

$$(C.13) \quad \|(D\mathcal{F}_{\varepsilon, x}(z) - D\mathcal{F}_{\varepsilon, x}(0))\dot{z}\| \leq C(H_{\varepsilon}^{(m)}(k) + |\beta - \beta_*| + N_{\varepsilon}(z))N_{\varepsilon}(\dot{z}).$$

We first establish the higher-order smallness needed here. We already have $H_{\varepsilon_j}^{(0)}(k_j(x_j)) \rightarrow 0$. Since $g_j(x_j)$ satisfies the Einstein equation and the chosen gauge condition, we can apply elliptic estimates in unit rescaled charts. The reference coefficients are uniformly controlled, and their errors in the rescaled Einstein equations tend to zero at every required finite order: the gluing error contributes $O(\varepsilon_j^{1/3})$, the angle variation contributes $O(|\beta_j - \beta_*|)$, and the fixed exterior contributes $O(|u_{x_j}|)$. Bootstrapping therefore gives

$$H_{\varepsilon_j}^{(m)}(k_j(x_j)) \rightarrow 0.$$

Since the weights are uniformly bounded above, this also gives $N_{\varepsilon_j}(z_j) \rightarrow 0$. We now apply Taylor's formula and the uniform inverse to obtain

$$N_{\varepsilon_j}(z_j) \leq C\varepsilon_j^{4/3} + o(1)N_{\varepsilon_j}(z_j).$$

Absorption places z_j in the contraction neighborhood, where uniqueness identifies it with the constructed solution.

We now verify regularity of the geometric stabilization. In local quotient coordinates, we put

$$R_{\varepsilon}(x, g, u) = \ell_x(k_x(g), u - u_x).$$

The preceding C^1 convergence and $p_{\varepsilon} \rightarrow p$ show that $D_x R_{\varepsilon}$ is invertible at every zero for sufficiently small ε : along any sequence of zeros with $\varepsilon \rightarrow 0$, its limit is $-\ell_{x_0}|_{T_{x_0}\mathcal{W}}$ in the

divergence-free identification. We can therefore solve $R_\varepsilon = 0$ for $x = x_\varepsilon(g, u)$ near the zero fibre. By eliminating this equation from the augmented system, the augmented inverse gives a stabilized fixed-area equation with surjective linearization and an N -dimensional kernel. Its universal solution space is locally parametrized by the constructed family, with projection p_ε .

Finally, the constructions commute with the allowed changes of representative. The rotations relating equivariant orthonormal frames at p_0 extend as torus isometries of (Y, h) , so uniqueness identifies the local families on overlaps. If injectivity on the zero fibre failed for arbitrarily small ε , we could choose two sequences of parameter zeros representing equivalent metrics. Compactness of the corresponding isometries would identify their compact limits, placing both sequences near the same point of $p^{-1}(0)$. The parameter-selection argument would then identify the parameters, a contradiction. Compactness of $p^{-1}(0)$ makes all area thresholds uniform. $\square$

““

Proof of Theorem C.6. We first identify the small-area degree. For fixed ε , solve

$$\ell_x(k_x(g), u - u_x) = 0$$

for $x = x_\varepsilon(g, u)$ near the compact zero fibre, using the parameter-selection argument above. The stabilized equation then reads

$$(C.14) \quad \mathcal{E}_3(g) = \sum_{\nu} u_{\nu} \hat{\sigma}_{\nu, \varepsilon, x_\varepsilon(g, u)}(g), \quad A(g) = \varepsilon^2 A_*.$$

Near the zero fibre, its regular universal solution space is identified with a neighborhood of $p_\varepsilon^{-1}(0)$ in V , with projection p_ε .

Replace $x_\varepsilon(g, u)$ in the section arguments by $x_\varepsilon(g, tu)$, $0 \leq t \leq 1$. At $u = 0$ the equation and its full linearization are independent of t . The degree is therefore unchanged, by the comparison of stabilizations in Proposition C.4. At $t = 0$ the sections depend only on g and remain equivariant and divergence-free. They define the required fixed-area count. Since $p_\varepsilon \rightarrow p$ uniformly and $|p| \geq c$ on the frontier of V , this count is

$$(C.15) \quad \deg_2(p_\varepsilon, V, 0) = \deg_2(p, V, 0) = \mathfrak{n}(\mathfrak{R}_0) \pmod{2}.$$

Choose $\eta = \varepsilon^2 A_*$ below the minimum of A on the smooth moduli space for $\mathfrak{R}$, if that space is nonempty. Its compactness follows from Lemma C.3. The conical solution space with $\beta \geq 1$ and $A \geq \eta$ is compact by Theorems 5.7 and 6.1. Its boundary neighborhoods at $A = \eta$ and $\beta = 1$ are disjoint. At the first boundary use the $t = 0$ sections just constructed, with area allowed to vary. At the second use the smooth count stabilization, with cone angle allowed to vary. Joint dependence on (ε, x) in Proposition C.8 makes area a free coordinate on the first universal solution space. This property persists for the $t = 0$ sections because the full linearization along the zero fibre is unchanged. Surjectivity of the fixed-angle stabilized equation makes β a free coordinate on the second.

Extend these sections and stabilize the remaining cokernels as in Proposition 4.5. Extra parameters at either boundary contribute identity factors to the degree. Choose a relatively compact neighborhood of the zero fibre in this finite-dimensional universal solution space, with parameter norm bounded away from zero on its artificial frontier. A sufficiently small common regular parameter then gives a compact one-dimensional manifold whose boundary counts are $\mathfrak{n}(\mathfrak{R}_0)$ at $A = \eta$ and $\mathfrak{n}(\mathfrak{R})$ at $\beta = 1$, modulo two. Their sum is zero, proving (C.4).

If the compact zero fibre is empty, Theorem 5.7 excludes sequences with $A \rightarrow 0$. The same argument has only the boundary $\beta = 1$ and gives $\mathfrak{n}(\mathfrak{R}) \equiv 0 \pmod{2}$, as required. $\square$

C.3. Reduction of the rod structures and proof of existence.

Proof of Theorem C.2. For two or three rods, Corollary 7.3 and Theorem 7.6 give that the invariant is 1 modulo 2. Every symmetry of these rod structures is realized by an isometry normalizing the corresponding standard torus. We may therefore adjust the isometry given by uniqueness so that it commutes with the fixed torus action and preserves each rod sphere. Thus the normalized moduli space consists of a single point in each case.

Suppose there are at least four rods. If two directions at distance two in the cyclic list agree, relabel so that three consecutive labels are A, B, A . The next label is B or C , giving a reduction

$$ABAB \mapsto AB, \quad ABAC \mapsto AC.$$

If no such pair agrees, each successive label is the third direction, so the cyclic list is $(ABC)^d$ after relabeling. For $d \geq 2$ it admits the reduction

$$ABCABC \mapsto AC.$$

These reductions preserve distinctness of adjacent labels and decrease the number of rods. Their inverses are the replacements (B, A, B, A) , (B, C, B, A) , and (B, C, A, B, C, A) , respectively, after a global lattice change. Every permutation of the three unsigned directions is realized by such a change. Theorem C.6 preserves the count modulo two at every step. Induction terminates at two or three rods and proves the theorem. $\square$

Proof of Theorem C.1. The necessity part is Theorem 7.12. For the sufficiency part, the connected-sum classification recalled in Section 2 reduces the manifold to

$$\#_r(\mathbb{S}^2 \times \mathbb{S}^2), \quad r \geq 0, \quad \text{or} \quad p\mathbb{CP}^2 \# q\overline{\mathbb{CP}^2}, \quad p, q \geq 0, \quad p + q \geq 1.$$

Here $r = 0$ denotes $\mathbb{S}^4$. We construct a rod structure with directions in $\{A, B, C\}$ on each manifold satisfying (C.1).

The alternating list $(AB)^{r+1}$ gives the first family by the topological calculation in Section 4. For the second family, reverse orientation if necessary so that $q \geq p$, and set $d = q - p$. If $d \geq 1$, the inequality is equivalent to $p \geq d - 1$. The cyclic list $(ABC)^d$ has every rod sphere of square -1 , so

$$\chi = 3d, \quad \tau = -d, \quad (b_2^+, b_2^-) = (d - 1, 2d - 1).$$

Its intersection form is odd. The connected-sum classification in Section 2 therefore identifies its oriented smooth manifold as

$$(C.16) \quad (d - 1)\mathbb{CP}^2 \# (2d - 1)\overline{\mathbb{CP}^2}.$$

Apply the alternating replacement $p - d + 1$ times. Each adds $\mathbb{S}^2 \times \mathbb{S}^2$, increasing both b_2^+ and b_2^- by one. The intersection form remains odd, so the resulting smooth manifold is $p\mathbb{CP}^2 \# q\overline{\mathbb{CP}^2}$. All rod directions remain in $\{A, B, C\}$.

If $d = 0$, then $p = q \geq 1$. Start with $ABAC$, whose underlying manifold is $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$, and apply the alternating replacement $p - 1$ times. Theorem C.2 gives the desired toric Einstein metric in every case. Reversing orientation does not change the Einstein equation. $\square$

Remark C.9. *Theorem C.1 concerns existence for some torus action on M . The inequality alone does not give existence for every prescribed action. For example, the action on the second Hirzebruch surface has rod spheres of square 2 and -2 , so it is excluded by Lemma 7.13. Its underlying manifold is $\mathbb{S}^2 \times \mathbb{S}^2$, which admits the round product metric with a different action.*

APPENDIX D. DISCUSSION ON THE VARIATIONAL FRAMEWORK

In this appendix we discuss the variational framework for toric conical Einstein metrics using the Einstein-Hilbert functional. It gives rise to a weaker version of uniqueness statement, and we discuss the issues related to a direct minimization method to the existence problem.

Definition D.1 (Conformally equivalent quotients). Let g and $\hat{g}$ be toric conical metrics with quotient disks B and $\hat{B}$ and rod structures $\mathfrak{R}$ and $\hat{\mathfrak{R}}$. We say that they have *conformally equivalence* if there is an equivalence of rod structures, in the sense of Definition 2.1, whose underlying map $f : B \rightarrow \hat{B}$ is conformal on the interiors with respect to the induced quotient metrics. Thus, writing $\underline{g}$ and $\underline{\hat{g}}$ for these metrics,

$$f^*\underline{\hat{g}} = e^{2u}\underline{g} \quad \text{on } B^\circ$$

for some smooth function u . The map f matches the corresponding rods, and their lattice directions are identified by a lattice automorphism.

D.1. The reduced Einstein–Hilbert functional. Recall the Einstein–Hilbert functional

$$\text{EH}_\lambda(g) = \int_X (\text{Scal}_g - 2\lambda) \, \text{dvol}_g, \quad \lambda > 0,$$

introduced in (2.3).

Fix a rod structure $\mathfrak{R}$ and a conformal class on its quotient disk B , compatible with the rod and corner regularity. Choose a representative $\underline{g}$ and write

$$g = e^{2\nu} \underline{g} + \rho \Phi_{\alpha\beta} \text{d}\varphi_\alpha \text{d}\varphi_\beta.$$

Thus $e^{2\nu} \underline{g}$ is the quotient metric, $G = \rho \Phi$, and $\det \Phi = 1$. Integrating the scalar-curvature formula (2.4) over the torus gives

$$\frac{\text{EH}_\lambda(g)}{(2\pi)^2} = \int_B \left[\rho \text{Scal}_{\underline{g}} - 2\rho \Delta_{\underline{g}} \nu - 2\Delta_{\underline{g}} \rho + \frac{|\text{d}\rho|_{\underline{g}}^2}{2\rho} - \rho |\text{d}\Phi|_{\underline{g}}^2 - 2\lambda \rho e^{2\nu} \right] \text{d}A_{\underline{g}}.$$

The two Dirichlet terms are integrated together: the expansions of Lemma 2.7 make their leading singularities cancel along the rods and at the turning points.

Suppose the cone angle along Σ_i is $2\pi\beta_i$, and set $\beta = (\beta_1, \dots, \beta_m)$. Let n and ξ be the outward unit normal and inward distance for $\underline{g}$. Set

$$\kappa = -\partial_n \rho, \quad c_i(\Phi) = \frac{1}{(2\pi)^2} \lim_{\xi \rightarrow 0} \frac{\Phi(\mathbf{v}_i, \mathbf{v}_i)}{\xi}.$$

The density $\kappa \, \text{d}s_{\underline{g}}$ is conformally invariant. By (2.12), its integrals determine the areas of the rod spheres:

$$(D.1) \quad M_i = \int_{\mathfrak{I}_i} \kappa \, \text{d}s_{\underline{g}} = \frac{\beta_i}{2\pi} \text{Area}_g(\Sigma_i), \quad M = \sum_i M_i.$$

The boundary expansions also give

$$(D.2) \quad \kappa c_i(\Phi) = \beta_i^2 e^{2\nu} \quad \text{on } \mathfrak{I}_i.$$

Thus the prescribed cone angles determine the boundary values of ν .

Integration by parts yields

$$-2 \int_B \Delta_{\underline{g}} \rho \, \text{d}A_{\underline{g}} = 2M,$$

and

$$-2 \int_B \rho \Delta_{\underline{g}} \nu \, \text{d}A_{\underline{g}} = -2 \int_B \nu \Delta_{\underline{g}} \rho \, \text{d}A_{\underline{g}} - 2 \sum_i \int_{\mathfrak{I}_i} \kappa \nu \, \text{d}s_{\underline{g}}.$$

These identities are justified by exhaustion using the rod and corner expansions. By (D.2), the last boundary term becomes

$$- \sum_i \int_{\mathfrak{I}_i} \kappa \log \frac{\kappa c_i(\Phi)}{\beta_i^2} \, \text{d}s_{\underline{g}}.$$

The remaining dependence on ν is therefore through the interior integral

$$\int_B (-2\nu \Delta_{\underline{g}} \rho - 2\lambda \rho e^{2\nu}) \, \text{d}A_{\underline{g}}.$$

For fixed (ρ, Φ) , this is strictly concave in ν . Its interior critical point satisfies

$$(D.3) \quad e^{2\nu} = \frac{-\Delta_{\underline{g}} \rho}{2\lambda \rho}.$$

This motivates the following definition.

Definition D.2. For prescribed cone angles $2\pi\beta_i$, a pair (ρ, Φ) is admissible if, for some smooth ν , the data $(e^{2\nu} \underline{g}, \rho \Phi)$ satisfy the boundary conditions of Lemma 2.7 with these angles, and $-\rho^{-1} \Delta_{\underline{g}} \rho$ extends smoothly and positively to the closed quotient.

This definition is independent of the representative $\underline{g}$. The reduced data of a toric Einstein metric with positive Einstein constant are admissible, but the positivity condition need not hold for a general toric metric.

For admissible data, (D.3) defines a positive quotient conformal factor. It does not, however, automatically satisfy the prescribed boundary values (D.2). The interior elimination and the cone-angle condition must therefore be distinguished.

Substituting the interior critical value of the ν -dependent integrand, retaining the boundary term determined by the prescribed angles, and subtracting M gives the normalized reduced functional

$$(D.4) \quad \begin{aligned} \mathcal{J}_{\underline{g},\beta}(\rho, \Phi) = & \int_B (-\Delta_{\underline{g}}\rho) \log \frac{-\Delta_{\underline{g}}\rho}{2\lambda\rho} dA_{\underline{g}} \\ & + \int_B \rho \left(\text{Scal}_{\underline{g}} + \frac{1}{2} |\mathrm{d} \log \rho|_{\underline{g}}^2 - |\mathrm{d}\Phi|_{\underline{g}}^2 \right) dA_{\underline{g}} \\ & - \sum_i \int_{\mathcal{I}_i} \kappa \log \frac{\kappa c_i(\Phi)}{\beta_i^2} ds_{\underline{g}}. \end{aligned}$$

Write $\mathcal{J}_{\underline{g}}$ for the expression obtained by setting every $\beta_i = 1$ in (D.4). Then

$$(D.5) \quad \mathcal{J}_{\underline{g},\beta}(\rho, \Phi) = \mathcal{J}_{\underline{g}}(\rho, \Phi) + \sum_i M_i \log \beta_i^2.$$

Thus the angle correction comes directly from the boundary term in the Einstein–Hilbert functional. It is constant when the individual rod fluxes are fixed. The prescribed angles also enter the admissible class: their relative values determine the corner asymptotics of Φ .

The relation to the original functional can be expressed without assuming (D.3). For any metric g realizing admissible data with the prescribed angles,

$$(D.6) \quad \mathcal{J}_{\underline{g},\beta}(\rho, \Phi) = \frac{\text{EH}_{\lambda}(g)}{(2\pi)^2} - M + \int_B f(-\Delta_{\underline{g}}\rho, 2\lambda\rho e^{2\nu}) dA_{\underline{g}},$$

where

$$(D.7) \quad f(a, b) = a \log \frac{a}{b} - a + b.$$

For $a, b > 0$, one has $f(a, b) \geq 0$, with equality exactly when $a = b$. Consequently, when (D.3) holds, the reduced functional equals the normalized Einstein–Hilbert functional minus M . This subtraction does not affect variations at fixed total flux.

Identity (D.6) also shows that the reduced functional is independent of the representative $\underline{g}$. Indeed, under $g \mapsto e^{2u}g$, both arguments of f are multiplied by e^{-2u} , while the area form is multiplied by e^{2u} ; the function f is homogeneous of degree one.

For Einstein data, $\text{Scal}_g = 4\lambda$ and (D.3) holds. Hence

$$(D.8) \quad \mathcal{J}_{\underline{g},\beta}(\rho, \Phi) = 0, \quad M = \frac{2\lambda}{(2\pi)^2} \text{Vol}_g(X).$$

The formula for M follows from (D.3) alone and therefore also holds for any metric reconstructed by that equation. Directly from (D.4),

$$(D.9) \quad \mathcal{J}_{\underline{g},\beta}(c\rho, \Phi) = c\mathcal{J}_{\underline{g},\beta}(\rho, \Phi) - cM \log c, \quad c > 0,$$

where M is the total flux of ρ .

We next isolate the dependence on the map Φ . For fixed ρ , define

$$(D.10) \quad \mathcal{E}_{\rho}(\Phi) = \int_B \left(\rho |\mathrm{d}\Phi|_{\underline{g}}^2 - \frac{|\mathrm{d}\rho|_{\underline{g}}^2}{2\rho} \right) dA_{\underline{g}} + \sum_i \int_{\mathcal{I}_i} \kappa \log c_i(\Phi) ds_{\underline{g}}.$$

The subtraction makes the integral finite and is independent of Φ . Thus

$$(D.11) \quad \begin{aligned} \mathcal{E}_\rho(\Phi_1) - \mathcal{E}_\rho(\Phi_0) &= \int_B \rho (|\mathrm{d}\Phi_1|_{\underline{g}}^2 - |\mathrm{d}\Phi_0|_{\underline{g}}^2) \mathrm{d}A_{\underline{g}} \\ &\quad + \sum_i \int_{\mathfrak{I}_i} \kappa \log \frac{c_i(\Phi_1)}{c_i(\Phi_0)} \mathrm{d}s_{\underline{g}}. \end{aligned}$$

The boundary term cancels the rod contribution in the first variation. In this notation,

$$(D.12) \quad \begin{aligned} \mathcal{J}_{\underline{g}, \beta}(\rho, \Phi) &= -\mathcal{E}_\rho(\Phi) + \int_B (-\Delta_{\underline{g}} \rho) \log \frac{-\Delta_{\underline{g}} \rho}{2\lambda \rho} \mathrm{d}A_{\underline{g}} \\ &\quad + \int_B \rho \mathrm{Scal}_{\underline{g}} \mathrm{d}A_{\underline{g}} - \sum_i \int_{\mathfrak{I}_i} \kappa \log \frac{\kappa}{\beta_i^2} \mathrm{d}s_{\underline{g}}. \end{aligned}$$

For fixed ρ , all terms other than $-\mathcal{E}_\rho(\Phi)$ are independent of Φ . Minimizing $\mathcal{E}_\rho$ therefore maximizes the reduced functional in the map variable.

Definition D.3. For a positive smooth function ρ on B° , a smooth map $\Phi : B^\circ \rightarrow \mathcal{H}$ is called ρ -weighted harmonic if it satisfies (2.6), namely

$$\mathrm{d}_{\underline{g}}^*(\rho \Phi^{-1} \mathrm{d}\Phi) = 0.$$

This condition is independent of the representative $\underline{g}$ of the quotient conformal class.

Lemma D.4. For fixed ρ and prescribed cone angles, if an admissible ρ -weighted harmonic map exists, it uniquely minimizes $\mathcal{E}_\rho$ among admissible maps.

Proof. Join two admissible maps by pointwise constant-speed hyperbolic geodesics Φ_t . The interpolation preserves the expansions of Lemma 2.7, and

$$\log c_i(\Phi_t) = (1-t) \log c_i(\Phi_0) + t \log c_i(\Phi_1).$$

Affine interpolation of the realizing conformal factors therefore preserves admissibility. The rod terms cancel in the first variation, and the corner-cap contributions are $O(\varepsilon^2)$.

Since in our convention the curvature of the target space $\mathcal{H}$ is -2 ,

$$(D.13) \quad \frac{\mathrm{d}^2}{\mathrm{d}t^2} \mathcal{E}_\rho(\Phi_t) = 2 \int_B \rho \left(|\nabla \partial_t \Phi_t|_{\underline{g}}^2 + 2 |\partial_t \Phi_t \wedge \mathrm{d}\Phi_t|_{\underline{g}}^2 \right) \mathrm{d}A_{\underline{g}} \geq 0.$$

Here ∇ is the pullback of the target connection. If Φ_0 is weighted harmonic, the first derivative vanishes at $t = 0$, proving minimality. Equality makes the distance between the two maps constant. Their common corner asymptotics make this distance tend to zero at a turning point, proving uniqueness. $\square$

Remark D.5. This is similar to related uniqueness results in the Ricci-flat setting, see [49, 31, 34].

D.2. Uniqueness for a prescribed conformal quotient. We begin with the case of prescribed boundary flux density. The elementary convexity underlying the argument is the following.

Lemma D.6. For positive affine functions τ, σ on an interval,

$$\left(-\tau \log \frac{\sigma}{\tau} \right)'' = \tau \left(\frac{\sigma'}{\sigma} - \frac{\tau'}{\tau} \right)^2 \geq 0.$$

Proof. Differentiate twice and use $\tau'' = \sigma'' = 0$. $\square$

Remark D.7. This is the perspective transform of the convex function $-\log x$: the function $af(x/a)$ is convex when f is convex and $a > 0$. Applying the same construction to $f(x) = |x|^2/2$ proves convexity of $|\mathrm{d}\rho|_{\underline{g}}^2/(2\rho)$ in $(\rho, \mathrm{d}\rho)$.

Thus $\mathcal{E}_\rho(\Phi)$ is concave in ρ for fixed Φ , and the logarithmic integral in (D.12) is convex. At fixed boundary flux density the last term is constant, and the full functional is strictly convex in ρ . More precisely,

$$(D.14) \quad \begin{aligned} D_\rho^2 \mathcal{J}_g(\rho, \Phi)[\dot{\rho}, \dot{\rho}] &= \int_B (-\Delta_g \rho) \left(\frac{\Delta_g \dot{\rho}}{\Delta_g \rho} - \frac{\dot{\rho}}{\rho} \right)^2 dA_g \\ &\quad + \int_B \rho \left| d \left(\frac{\dot{\rho}}{\rho} \right) \right|_g^2 dA_g. \end{aligned}$$

Equality makes $\dot{\rho}/\rho$ constant. Since $\partial_n \dot{\rho} = 0$ and $\kappa > 0$ on open rods, that constant is zero. The admissible map class is independent of the weight: multiplying a realizing metric by the ratio of two weights changes ρ while fixing Φ . Consequently, taking the supremum over Φ preserves convexity.

Proposition D.8. *Two toric Einstein metrics with the same positive Einstein constant and conformally equivalent quotients preserving the boundary flux density are equivariantly isometric.*

Proof. Use Lemma 2.8 to identify the quotients. At Einstein data, the first variation in ρ vanishes for variations preserving the boundary flux density. Strict convexity and Lemma D.4 therefore give

$$(D.15) \quad 0 = \mathcal{J}_g(\hat{\rho}, \hat{\Phi}) \geq \mathcal{J}_g(\hat{\rho}, \Phi) \geq \mathcal{J}_g(\rho, \Phi) = 0.$$

Equality gives $\hat{\rho} = \rho$ and $\hat{\Phi} = \Phi$. Equation (D.3) then identifies the quotient metrics. $\square$

To remove the boundary-density condition, take $\underline{g}$ to be the quotient metric of a toric Einstein metric g with $\text{Ric}(g) = \lambda g$. Compare its data (ρ, Φ) with admissible data $(\hat{\rho}, \hat{\Phi})$ having the same total boundary flux. The function f in (D.6) is convex, homogeneous of degree one, and nonnegative.

Proposition D.9. *If $\hat{\Phi}$ is $\hat{\rho}$ -weighted harmonic, then*

$$(D.16) \quad \mathcal{J}_g(\hat{\rho}, \hat{\Phi}) \geq 0.$$

If g is not conformally round, equality holds only for $(\hat{\rho}, \hat{\Phi}) = (\rho, \Phi)$.

Proof. The function $w = \sqrt{\hat{\rho}/\rho}$ extends to a positive smooth $\mathbb{T}^2$ -invariant function on X . The pair $(\hat{\rho}, \Phi)$ is realized by $w^2 g$, and Lemma D.4 gives

$$\mathcal{J}_g(\hat{\rho}, \hat{\Phi}) \geq \mathcal{J}_g(\hat{\rho}, \Phi).$$

Along the rods, $\hat{\kappa} = \kappa w^2$ and $\kappa c_i(\Phi) = 1$. Green's identity gives

$$(D.17) \quad \begin{aligned} \int_{\partial B} \hat{\kappa} \log w^2 ds_g &= \int_B (-\Delta_g \hat{\rho}) \log w^2 dA_g \\ &\quad - \int_B \langle d\hat{\rho}, d \log w^2 \rangle_g dA_g. \end{aligned}$$

Substituting the reduced equations for g and the equality of total flux yields

$$(D.18) \quad \begin{aligned} \mathcal{J}_g(\hat{\rho}, \Phi) &= \int_B f(-\Delta_g \hat{\rho}, 2\lambda \rho w^4) dA_g \\ &\quad - 2\lambda \int_B \rho (w^2 - 1)^2 dA_g + 6 \int_B \rho |dw|_g^2 dA_g. \end{aligned}$$

Jensen's inequality and homogeneity give

$$\begin{aligned} \int_B f(-\Delta_g \hat{\rho}, 2\lambda \rho w^4) dA_g &\geq f \left(\int_B (-\Delta_g \hat{\rho}) dA_g, 2\lambda \int_B \rho w^4 dA_g \right) \\ &= 2\lambda f \left(\int_B \rho dA_g, \int_B \rho w^4 dA_g \right). \end{aligned}$$

Expanding f in this inequality and combining the polynomial terms in (D.18), we obtain

$$\begin{aligned} \mathcal{J}_g(\hat{\rho}, \Phi) &\geq 4\lambda \left[\int_B \rho \left(w^2 + \frac{3}{2\lambda} |dw|_g^2 \right) dA_g - \int_B \rho dA_g \right. \\ &\quad \left. - \frac{1}{2} \left(\int_B \rho dA_g \right) \log \frac{\int_B \rho w^4 dA_g}{\int_B \rho dA_g} \right]. \end{aligned}$$

The sharp Sobolev inequality for the Einstein metric g is

$$\left(\frac{\int_X w^4 dV_g}{\text{Vol}_g(X)} \right)^{1/2} \leq \frac{1}{\text{Vol}_g(X)} \int_X \left(w^2 + \frac{3}{2\lambda} |dw|_g^2 \right) dV_g.$$

It follows by taking a Yamabe minimizer and applying Obata's theorem [40]. For a $\mathbb{T}^2$ -invariant function, angular integration gives

$$(D.19) \quad \left(\int_B \rho dA_g \int_B \rho w^4 dA_g \right)^{1/2} \leq \int_B \rho \left(w^2 + \frac{3}{2\lambda} |dw|_g^2 \right) dA_g.$$

Since $f(a, \sqrt{ab}) = \sqrt{ab} - a - \frac{a}{2} \log(b/a)$, the preceding two inequalities imply

$$\mathcal{J}_g(\hat{\rho}, \Phi) \geq 4\lambda f \left(\int_B \rho dA_g, \left(\int_B \rho dA_g \int_B \rho w^4 dA_g \right)^{1/2} \right) \geq 0.$$

This proves (D.16). Equality gives equality in the Sobolev inequality. If g is not conformally round, Obata's theorem makes w constant. The total flux then gives $w = 1$, and equality in the map comparison gives $\hat{\Phi} = \Phi$. $\square$

Proposition D.10. *Two smooth toric Einstein metrics with conformally equivalent quotients are equivariantly isometric up to scaling. In particular, if we fix Einstein constant, they have equal volume. If the metrics are not conformally round, the given conformal equivalence of their quotients lifts to an equivariant isometry.*

Proof. WLOG we assume the Einstein constant to be the same. Let their data be (ρ, Φ) and $(\hat{\rho}, \hat{\Phi})$, and set $c = \text{Vol}_g(X) / \text{Vol}_{\hat{g}}(X)$. The pair $(c\hat{\rho}, \hat{\Phi})$ is admissible, being realized by $c\hat{g}$, and its total flux equals that of g . Proposition D.9 and (D.9) give

$$(D.20) \quad 0 \leq \mathcal{J}_g(c\hat{\rho}, \hat{\Phi}) = -M \log c.$$

Thus $\text{Vol}_{\hat{g}}(X) \geq \text{Vol}_g(X)$. Interchanging the metrics gives equality. If g is not conformally round, the equality statement in Proposition D.9 identifies ρ and Φ ; (D.3) then identifies the quotient metrics.

It remains to consider the case where g is conformally round. Since g is Einstein, it is round, and the volume equality gives

$$\text{Vol}_{\hat{g}}(X) = \text{Vol}_g(X) = \frac{24\pi^2}{\lambda^2}.$$

Equality in Bishop's volume comparison therefore implies that $\hat{g}$ is round as well. The images of the effective isometric $\mathbb{T}^2$ -actions are maximal tori in $\text{SO}(5)$. Conjugacy of maximal tori gives an equivariant isometry, with equivariance understood up to a lattice automorphism. $\square$

Remark D.11. *If we fix the quotient metric instead of the conformal structure, the uniqueness statement is already proved by Liu [35, Theorem 1.2].*

Remark D.12. *Similar uniqueness result also holds for toric conical Einstein metrics with the same positive Einstein constant, the same cone angles, and conformally equivalent quotients. Indeed, a common reduction of the angular periods makes all cone angles less than 2π , so the key inequality follows from [37, Theorem 3.1] for $\mathbb{T}^2$ -invariant functions.*

Using $\mathcal{J}_{g,\beta}$, the same comparison gives equal volumes, and equality yields $\hat{g} = w^2 g$ for a positive $\mathbb{T}^2$ -invariant function w . If w is constant, the common Einstein constant gives $w = 1$.

Otherwise, the conformal Einstein equation shows that a suitable time map ψ of the flow of $\nabla_g(w^{-1})$ satisfies $\psi^*g = \hat{g}$. This flow is complete and equivariant: the rod and corner regularity makes its generator smooth in the local developments and tangent to every singular stratum.

D.3. Infinitesimal rigidity. We prove an infinitesimal counterpart of Proposition D.10. The argument combines the Lichnerowicz–Obata spectral gap in the nonround case with the non-negative second variation of the weighted harmonic-map energy. Together, these show that infinitesimal Einstein deformations preserving the quotient conformal structure are induced by equivariant diffeomorphisms.

Let k be a $\mathbb{T}^2$ -invariant infinitesimal Einstein deformation of g , with the Einstein constant fixed.

Lemma D.13. *There is a smooth $\mathbb{T}^2$ -invariant vector field Y , tangent to the torus orbits, such that $k + \mathcal{L}_Y g$ has no mixed quotient-angular components.*

Proof. Write $X_i = \partial_{\varphi_i}$. The mixed components are encoded by one-forms a^i on B° through

$$a^i(V) = \sum_j G^{ij} k(\tilde{V}, X_j), \quad k_{\text{mix}} = \sum_{i,j} G_{ij} (a^i \otimes d\varphi_j + d\varphi_j \otimes a^i),$$

where $\tilde{V}$ is the horizontal lift. Apply the Killing identity in the proof of Lemma 2.4 to $g + \varepsilon k$. Its Einstein error is $O(\varepsilon^2)$, so the first variations of the twist functions are constant. They vanish on the rods, and their expressions on the interior are

$$\dot{\Theta}_i = \rho \sum_j G_{ij} *_g da^j.$$

Hence $da^j = 0$. Since B° is simply connected, $a^j = df^j$. The expansions of Lemma 2.7 make these primitives smooth in squared normal coordinates, both on rods and at corners. Thus $Y = -\sum_j f^j X_j$ is smooth and $\mathcal{L}_Y g = -k_{\text{mix}}$. $\square$

Suppose now that k preserves the quotient conformal structure and corresponding rods. After the preceding gauge change and a quotient diffeomorphism, its quotient component is $2\dot{\nu}g$. Denote the induced variations by $\dot{\rho}, \dot{\Phi}$.

Lemma D.14. *The deformation k is tangent to a smooth family of admissible metrics with fixed quotient conformal structure and constant total boundary flux. Along this family,*

$$\left. \frac{d^2}{d\varepsilon^2} \right|_{\varepsilon=0} \mathcal{J}_g(\rho_\varepsilon, \Phi_\varepsilon) = 0.$$

Proof. The linearized fiber trace equation is

$$(D.21) \quad \Delta_g \dot{\rho} + 2\lambda \dot{\rho} = -4\lambda \rho \dot{\nu}.$$

The integrated scalar-curvature variation gives $\int_X \text{tr}_g k dV_g = 0$, so the volume derivative vanishes. Integrating (D.21) therefore gives

$$\dot{M} = 2\lambda \int_B (\dot{\rho} + 2\rho \dot{\nu}) dA_g = \frac{2\lambda}{(2\pi)^2} \left. \frac{d}{d\varepsilon} \right|_0 \text{Vol}_{g+\varepsilon k}(X) = 0.$$

If M_ε denotes the total flux of $g + \varepsilon k$, take $g_\varepsilon = (M/M_\varepsilon)(g + \varepsilon k)$. This preserves the tangent k and makes the flux constant.

The fiber trace identity and $\text{Ric}(g_\varepsilon) - \lambda g_\varepsilon = O(\varepsilon^2)$ give

$$\frac{-\Delta_g \rho_\varepsilon}{2\lambda \rho_\varepsilon e^{2\nu_\varepsilon}} = 1 + O(\varepsilon^2).$$

The quotient and angular coefficients have the differentiated boundary expansions, so the data are admissible for small ε . Since $f(a, b)$ vanishes to second order at $a = b$, (D.6) gives

$$\mathcal{J}_g(\rho_\varepsilon, \Phi_\varepsilon) = \frac{\text{EH}_\lambda(g_\varepsilon)}{(2\pi)^2} - M + O(\varepsilon^4).$$

The Einstein–Hilbert Hessian vanishes on an infinitesimal Einstein deformation. Taking second derivatives proves the claim. $\square$

Proposition D.15. *For a nonround toric Einstein metric, every $\mathbb{T}^2$ -invariant infinitesimal Einstein deformation preserving the quotient conformal structure and corresponding rods is induced by an equivariant diffeomorphism.*

Proof. Use the constant-flux family from Lemma D.14. Differentiating the weighted harmonic-map equation at $\varepsilon = 0$ gives

$$D_\rho D_\Phi \mathcal{E}_\rho(\Phi)[\dot{\rho}, \dot{\Phi}] + D_\Phi^2 \mathcal{E}_\rho(\Phi)[\dot{\Phi}, \dot{\Phi}] = 0.$$

The same rod and corner estimates as in Lemma D.4 justify this identity for the admissible variation. Since $D_\Phi \mathcal{E}_\rho(\Phi) = 0$, the chain rule consequently gives

$$\left. \frac{d^2}{d\varepsilon^2} \right|_0 (\mathcal{E}_{\rho_\varepsilon}(\Phi) - \mathcal{E}_{\rho_\varepsilon}(\Phi_\varepsilon)) = D_\Phi^2 \mathcal{E}_\rho(\Phi)[\dot{\Phi}, \dot{\Phi}].$$

This is the nonnegative map Hessian in (D.13).

With $u = \log \rho$, write $\dot{u} = \dot{\rho}/\rho$ and

$$\bar{u} = \frac{\int_B \rho \dot{u} dA_g}{\int_B \rho dA_g}.$$

Taking second variations in (D.18) and its Jensen estimate yields

$$(D.22) \quad \begin{aligned} 0 \geq & 4\lambda \bar{u}^2 \int_B \rho dA_g + 3 \int_B \rho |d\dot{u}|_g^2 dA_g - 4\lambda \int_B \rho (\dot{u} - \bar{u})^2 dA_g \\ & + 2 \int_B \rho (|\nabla \dot{\Phi}|_g^2 + 2|\dot{\Phi} \wedge d\Phi|_g^2) dA_g. \end{aligned}$$

By the Lichnerowicz–Obata estimate, the first positive scalar eigenvalue of a nonround Einstein metric is strictly greater than $4\lambda/3$. The right-hand side is therefore nonnegative and vanishes only if $\dot{u} = 0$ and the map Hessian vanishes. Equation (D.21) then gives $\dot{\nu} = 0$, while the corner argument in Lemma D.4 gives $\dot{\Phi} = 0$. Thus the deformation vanishes after the chosen diffeomorphisms. $\square$

Remark D.16. *The round metric is also infinitesimally rigid: its Einstein operator on trace-free tensors is $\nabla_g^* \nabla_g + 2\lambda/3$. For a general rod structure, the proposition concerns deformations fixing the quotient conformal structure. A disk with $m \geq 3$ ordered boundary vertices has $m - 3$ real conformal parameters. Hence the space of $\mathbb{T}^2$ -invariant infinitesimal Einstein deformations, modulo diffeomorphisms and with scale fixed, has dimension at most $m - 3$.*

D.4. The case of $\mathbb{S}^4$ and $\mathbb{CP}^2$. We now apply the preceding uniqueness result to $\mathbb{S}^4$ and $\mathbb{CP}^2$. The key simplification is that their quotients have no conformal moduli, as expressed by the following elementary observation.

Lemma D.17. *Quotient disks with the same two-rod or three-rod structure are conformally equivalent.*

Proof. Uniformization identifies each quotient with the unit disk. A Möbius automorphism then matches their two or three cyclically ordered boundary vertices. $\square$

It therefore remains to identify the rod structures. The classification of torus actions [41] allows us to compare with the round and Fubini–Study metrics, giving the following result. Notice that this is already implied by Corollary 7.3. But the proof is here and may be of independent interest.

Theorem D.18. *Every smooth toric Einstein metric on $\mathbb{S}^4$ or $\mathbb{CP}^2$ is, up to scaling and equivariant diffeomorphism, the round or Fubini–Study metric, respectively.*

Proof. We normalize the Einstein constant to be 3. By [41], the rod structure agrees with that of the corresponding standard metric, up to a change of lattice basis. Their quotients are conformally equivalent by Lemma D.17, so Proposition D.10 gives the required equivariant isometry. $\square$

Remark D.19. *The second author was told in a Brisbane winter school in July 2026 that the uniqueness in the case of $\mathbb{S}^4$ was proved by Kyle Broder and Ramiro Lafuente.*

D.5. The case of $\mathbb{S}^2 \times \mathbb{S}^2$. Again, this case is already contained in Corollary 7.3. Here we describe explicitly how the equality case of the Weyl pinching gives a conformal product decomposition for the alternating four-rod structure. Smoothness at the poles then forces the conformal factor to be constant.

Let g be a toric Einstein metric on $\mathbb{S}^2 \times \mathbb{S}^2$ with $\text{Ric}(g) = \lambda g$, $\lambda > 0$, and rod structure (A, B, A, B) . By [35, Proposition 5] and Lemma 2.4, we may write

$$g = \underline{g} + a^2 d\varphi_1^2 + b^2 d\varphi_2^2,$$

where both angular variables have period 2π . The functions a, b are positive in B° and vanish on their respective pairs of opposite rods. The rods are geodesic and the four corner angles are $\pi/2$, so

$$(D.23) \quad \int_B K dA_{\underline{g}} = 0.$$

Set

$$P_a = a^{-1} \text{Hess}_{\underline{g}} a, \quad P_b = b^{-1} \text{Hess}_{\underline{g}} b, \quad u = \log \rho = \log(ab), \quad v = \log(a/b).$$

The diagonal Einstein equations are

$$(D.24) \quad \begin{aligned} P_a + P_b &= (K - \lambda)\underline{g}, \\ \Delta_{\underline{g}} a &= (K - \lambda)a, \\ \Delta_{\underline{g}} b &= (K - \lambda)b, \\ \langle da, db \rangle_{\underline{g}} &= -Kab. \end{aligned}$$

In particular, $P_a - P_b$ is trace free.

Lemma D.20. *We have $K < \lambda/3$ and*

$$(D.25) \quad |P_a - P_b|_{\underline{g}}^2 = 2(\lambda - 3K)^2.$$

Moreover, setting

$$W = (\lambda - 3K)^{1/3}, \quad \tilde{g} = W^2 \underline{g}, \quad T = \frac{P_b - P_a}{W},$$

the metric $\tilde{g}$ is flat and T is parallel, with eigenvalues $1, -1$ relative to $\tilde{g}$.

Proof. Diagonality implies that Φ takes values in a geodesic, so its Jacobian J vanishes. The diagonal curvature formulas and (7.5) give

$$w = \frac{\lambda}{3} - K, \quad |q|^2 = \frac{1}{8} |P_a - P_b|_{\underline{g}}^2.$$

By (D.23), w cannot vanish identically. Proposition 7.10 therefore gives $w > 0$ on the closed quotient. Thus $K < \lambda/3$, and W is smooth and positive there.

Define

$$\chi = \frac{1}{2} |T|_{\tilde{g}}^2 = \frac{|P_a - P_b|_{\underline{g}}^2}{2(\lambda - 3K)^2} = \frac{4|q|^2}{9w^2}.$$

Lemma 7.11 gives $\chi \leq 1$. Since $W = (3w)^{1/3}$, equation (7.17) and the conformal curvature formula yield

$$(D.26) \quad \widetilde{K} = W^{-2}(K - \Delta_{\underline{g}} \log W) = -W^{-2} \left(\lambda + \frac{3}{4} |du|_{\underline{g}}^2 \right) (1 - \chi) \leq 0.$$

The evenness of W preserves the geodesic boundary, and the corner angles remain $\pi/2$. Gauss–Bonnet therefore gives

$$\int_B \widetilde{K} dA_{\widetilde{g}} = 0.$$

It follows that $\widetilde{K} = 0$ and $\chi = 1$, proving (D.25).

To prove parallelness, differentiating (D.24) and commuting Hessian derivatives gives

$$(D.27) \quad \begin{aligned} \operatorname{div}_{\underline{g}}(P_a - P_b) + \frac{1}{2}(P_a - P_b)(\nabla_{\underline{g}}u, \cdot) &= \frac{1}{2}(3K - \lambda)dv, \\ 2dK + (3K - \lambda)du &= (P_a - P_b)(\nabla_{\underline{g}}v, \cdot). \end{aligned}$$

Since $d \log W = dK/(3K - \lambda)$, their conformal form is

$$(D.28) \quad d(u + 2 \log W) = T(\widetilde{\nabla}v, \cdot), \quad \operatorname{div}_{\widetilde{g}}T = \frac{1}{2}(1 - \chi)dv.$$

Here $\widetilde{\nabla}$ denotes the gradient for $\widetilde{g}$; in deriving the second identity we use $T^2 = \chi \operatorname{Id}$ after raising an index with $\widetilde{g}$.

Thus T has constant eigenvalues 1, -1 and vanishing divergence. In a local orthonormal eigenframe, the divergence equation forces the connection one-form to vanish. Hence T is parallel. All these identities extend to the boundary by the rod and corner regularity. $\square$

The parallel eigenlines provide coordinates in which the conformally rescaled metric separates.

Corollary D.21. *There are global quotient coordinates on a rectangle such that*

$$(D.29) \quad g = W^{-2}(g_1 + g_2), \quad g_1 = dx^2 + A(x)^2 d\varphi_1^2, \quad g_2 = dy^2 + B(y)^2 d\varphi_2^2.$$

Both factors extend smoothly to metrics on $\mathbb{S}^2$.

Proof. Since B is simply connected, parallel unit covectors in the eigenlines of T integrate to global functions x, y satisfying

$$\widetilde{g} = dx^2 + dy^2, \quad T = dx^2 - dy^2.$$

On an a -rod, the rod expansions and (D.24) give, in the normal–tangent frame,

$$P_a - P_b = \operatorname{diag}(3K - \lambda, -(3K - \lambda)).$$

The signs reverse on a b -rod. Thus every rod is tangent to an eigenline of T . The developing map (x, y) takes the boundary to four coordinate segments forming a rectangle traced once. Its degree is one inside this rectangle. Since every interior local degree is positive, it identifies B with the rectangle.

Equation (D.28) now reads

$$\partial_x(u + 2 \log W) = \partial_x v, \quad \partial_y(u + 2 \log W) = -\partial_y v.$$

Consequently, $Wa = A(x)$ and $Wb = B(y)$, proving (D.29).

The positive function W lifts smoothly to $\mathbb{S}^2 \times \mathbb{S}^2$, so $W^2 g$ is smooth. The two factors therefore complete smoothly at their poles. Writing the rectangle as $[x_-, x_+] \times [y_-, y_+]$, we have

$$(D.30) \quad \begin{aligned} A(x_{\pm}) &= 0, \quad A'(x_-) = 1, \quad A'(x_+) = -1, \\ B(y_{\pm}) &= 0, \quad B'(y_-) = 1, \quad B'(y_+) = -1. \end{aligned}$$

$\square$

It remains to show that the conformal factor is constant. We give a direct proof using the smooth pole conditions; compare the conformally Einstein product equations in [30, Section 3].

Theorem D.22. *A toric Einstein metric on $\mathbb{S}^2 \times \mathbb{S}^2$ with $\operatorname{Ric}(g) = \lambda g$, $\lambda > 0$, and the alternating four-rod structure is equivariantly isometric to the product of two round spheres of curvature λ . In particular, its conformal quotient is a square.*

Proof. Write $k = g_1 + g_2$ in (D.29). The mixed and trace-free components of the conformal Einstein equation give

$$W_{xy} = 0, \quad W_{xx} = \frac{A'}{A}W_x, \quad W_{yy} = \frac{B'}{B}W_y.$$

Hence

$$W_x = cA, \quad W_y = dB$$

for constants c, d .

Let $K_1 = -A''/A$ and $K_2 = -B''/B$ be the curvatures of the two factors. The preceding equations give

$$\Delta_k W = 2(W_{xx} + W_{yy}), \quad K = W(W_{xx} + W_{yy}) - (W_x^2 + W_y^2).$$

The scalar curvature transformation therefore yields

$$4\lambda = 2W^2(K_1 + K_2) + 6W\Delta_k W - 12|dW|_k^2 = 2W^2(K_1 + K_2) + 12K.$$

Since $W^3 = \lambda - 3K$, we obtain $K_1 + K_2 = 2W$. Differentiating in x gives $K_1' = 2cA$, and hence

$$\begin{aligned} 2c \int_{x_-}^{x_+} A^3 dx &= \int_{x_-}^{x_+} K_1' A^2 dx \\ &= [-A''A + (A')^2]_{x_-}^{x_+} = 0. \end{aligned}$$

The last equality follows from (D.30). Since $A > 0$ in the interior, $c = 0$. The same argument for B gives $d = 0$. Thus W is constant.

The Einstein equation now gives

$$K_1 = K_2 = \lambda W^{-2}.$$

Together with the pole conditions, this makes both factors round, with quotient intervals of length $\pi W/\sqrt{\lambda}$. The quotient rectangle is therefore a square, and $g = W^{-2}(g_1 + g_2)$ is the product of round spheres of curvature λ . The angular coordinates are unchanged, so the isometry is equivariant. $\square$

D.6. A variational approach to existence. Motivated by the construction of toric gravitational instantons in [34], it is natural to seek compact toric Einstein metrics by first constructing conical metrics and then varying the cone angles in their moduli space until the singularities disappear. We discuss a possible approach using the reduced Einstein–Hilbert functional. The calculations below give a necessary condition on the rod fluxes, determine the cone angles at a regular critical point, and give a limited positivity statement for relaxed minimizers. Existence and regularity of minimizers remain open in this approach.

Fix a rod structure $\mathfrak{R}$, an Einstein constant $\lambda > 0$, and positive angle factors $\beta = (\beta_i)$. We use the admissible class of Definition D.2 and the functional $\mathcal{J}_{g,\beta}$ in (D.5). Recall that

$$\kappa = -\partial_n \rho, \quad M_i = \int_{\mathcal{J}_i} \kappa ds_{\underline{g}}, \quad M = \sum_i M_i.$$

At a fixed quotient conformal structure, the proposed variational problem is

$$(D.31) \quad \inf_{\rho: \int_{\mathcal{J}_i} \kappa ds_{\underline{g}} = M_i} \sup_{\Phi} \mathcal{J}_{g,\beta}(\rho, \Phi),$$

with both variables satisfying the admissibility conditions. One would then seek stationarity with respect to the quotient conformal structure.

For fixed ρ , Lemma D.4 shows that an admissible weighted harmonic map, if it exists, uniquely attains the inner supremum. The lemma does not establish existence with the differentiated rod and corner expansions. There is already a necessary restriction on the fluxes, analogous to the Hilbert–Mumford criterion for weighted points on $\mathbb{CP}^1 \simeq \mathbb{S}^2$: the rod directions determine points of $\mathbb{RP}^1 \subset \mathbb{CP}^1$, with weights given by the corresponding fluxes M_i .

Proposition D.23. *If an admissible weighted harmonic map exists, then for every rod direction $\mathbf{v}$,*

$$2 \sum_{\mathbf{v}_i \parallel \mathbf{v}} M_i \leq M.$$

The inequalities are strict if at least three distinct rod directions occur. If there are exactly two directions, the total flux belonging to each direction is $M/2$.

Proof. The target function $b_{\mathbf{v}} = 2^{-1/2} \log \Phi(\mathbf{v}, \mathbf{v})$ has unit gradient and Hessian $\sqrt{2}(g_{\mathcal{H}} - db_{\mathbf{v}} \otimes db_{\mathbf{v}})$. The weighted harmonic-map equation gives

$$\operatorname{div}_{\underline{g}}(\rho \nabla_{\underline{g}} b_{\mathbf{v}}) = \sqrt{2} \rho (|d\Phi|_{\underline{g}}^2 - |db_{\mathbf{v}}|_{\underline{g}}^2) \geq 0.$$

Its outward flux is $-\kappa/\sqrt{2}$ on rods parallel to $\mathbf{v}$ and $+\kappa/\sqrt{2}$ on the other rods. The corner-cap integrals tend to zero, so

$$(D.32) \quad M - 2 \sum_{\mathbf{v}_i \parallel \mathbf{v}} M_i = 2 \int_B \rho (|d\Phi|_{\underline{g}}^2 - |db_{\mathbf{v}}|_{\underline{g}}^2) dA_{\underline{g}}.$$

Equality forces the image into a target geodesic, whose ideal endpoints account for at most two rod directions. In the two-direction case, applying the inequality to both directions gives equality. $\square$

These conditions are necessary and we are not proving the sufficiency for the admissible map problem here. Even if the supremum is attained, the minimization in ρ requires further estimates. At fixed boundary flux density, (D.14) gives strict convexity in ρ . Fixing only the integrals M_i allows the density to vary, and along an affine positive variation,

$$(D.33) \quad \frac{d^2}{dt^2} \left(- \int_{\partial B} \kappa_t \log \kappa_t ds_{\underline{g}} \right) = - \int_{\partial B} \frac{\dot{\kappa}_t^2}{\kappa_t} ds_{\underline{g}}.$$

The flux constraints do not remove this negative term. Consequently, the fixed-density argument does not establish convexity or compactness for (D.31).

We can nevertheless identify the boundary equation at an admissible critical pair. This calculation concerns the functional itself and does not require differentiability of an optimized family of harmonic maps.

Proposition D.24. *Suppose that an admissible pair (ρ, Φ) is stationary for $\mathcal{J}_{g,\beta}$ under variations of ρ and Φ preserving the quotient conformal structure and the individual rod fluxes. Reconstruct the quotient metric by (D.3). Then its cone factors are constant on the rods and satisfy*

$$(D.34) \quad \widehat{\beta}_i = c\beta_i \quad \text{for every } i$$

for one constant $c > 0$.

Proof. Write

$$e^{2\nu} = \frac{-\Delta_g \rho}{2\lambda \rho}, \quad \widehat{\beta}_i^2 = \kappa c_i(\Phi) e^{-2\nu} \quad \text{on } \mathfrak{I}_i.$$

Interior variations give the interior Euler–Lagrange equations. For weight variations supported away from the vertices, integration by parts therefore gives

$$(D.35) \quad D_{\rho} \mathcal{J}_{g,\beta}(\rho, \Phi)[\dot{\rho}] = \sum_i \int_{\mathfrak{I}_i} \left(\log \frac{\beta_i^2}{\widehat{\beta}_i^2} - 1 \right) \dot{\kappa} ds_{\underline{g}}.$$

Indeed, the logarithmic bulk term and the explicit boundary term contribute

$$\log \frac{-\Delta_g \rho}{2\lambda \rho} + 1, \quad - \log \frac{\kappa c_i(\Phi)}{\beta_i^2} - 1,$$

respectively. The singular Dirichlet term contributes $-\dot{\kappa}$, since on a rod cutoff

$$\frac{\partial_n \rho}{\rho} \dot{\rho} \longrightarrow -\dot{\kappa}.$$

The constant in (D.35) disappears under the flux constraints. Variations with zero integral on each open rod thus force $\widehat{\beta}_i$ to be constant there.

Let $e^{2\nu_0}\underline{g}$ realize the admissible data with angle factors β_i . Along each rod,

$$\frac{\widehat{\beta}_i^2}{\beta_i^2} = e^{2\nu_0 - 2\nu}.$$

Both conformal factors extend smoothly and positively to the corners. The constants on incident rods therefore agree. Connectedness of the boundary proves (D.34). $\square$

In particular, the admissible class with $\beta_i = 1$ gives a common cone angle at a regular critical pair. Replacing ρ by $c^{-2}\rho$ normalizes the actual angles to the prescribed ones. This leaves the reconstructed quotient metric unchanged and multiplies every rod flux by c^{-2} . Thus angle normalization also changes the absolute fluxes.

There is a separate issue concerning the quotient conformal structure. For $m \geq 3$ rods, the quotient has $m - 3$ real conformal parameters. A complete variational characterization requires a first-variation argument showing that stationarity in these parameters, together with the other equations and boundary conditions, implies (2.8). We do not carry out that argument here. Finding a critical conformal structure is a further global problem; no convexity in these parameters has been established.

The main analytic difficulty in (D.31) is control of minimizing sequences. For metrics reconstructed by (D.3), one has

$$M = \frac{2\lambda}{(2\pi)^2} \text{Vol}_g(X).$$

Fixed total flux therefore fixes the volume, but gives no direct control of the distribution of the source $-\Delta_g \rho$. Indeed, if ω_i is the harmonic measure of $\mathcal{I}_i$, Green's identity gives

$$(D.36) \quad M_i = \int_B \omega_i(-\Delta_g \rho) dA_g.$$

The flux constraints prescribe only finitely many moments of this positive source. They do not by themselves exclude concentration near vertices. Moreover, the bulk and boundary logarithmic terms have opposing contributions under local dilations, so a concentration estimate must control their combination and the optimized map energy.

Before studying genuine concentration, one must remove degeneration caused by conformal reparametrization. The round sphere gives a simple example. In an upper-half-plane coordinate on its quotient, with $\lambda = 3$, write

$$(D.37) \quad \rho_\varepsilon(z) = \frac{2\varepsilon \text{Im } z}{(\varepsilon + |z|)^2}, \quad e^{2\nu_\varepsilon}(z) = \frac{\varepsilon}{|z|(\varepsilon + |z|)^2}.$$

Each half-axis has flux 2, and both coefficients tend to zero locally away from the origin as $\varepsilon \rightarrow 0$. Nevertheless, these data describe pullbacks of the same round metric by $z \mapsto z/\varepsilon$. Thus this example shows the need to fix conformal reparametrizations; it does not exhibit geometric degeneration.

The logarithmic term does give the following limited positivity result. A relaxed weight is positive on B° and satisfies the same positive rod and corner expansions as an admissible weight, but $-\Delta_g \rho/\rho$ is allowed to extend smoothly and nonnegatively to the closed quotient. The logarithmic integrand is interpreted continuously at zero.

Lemma D.25. *Fix the quotient conformal structure, angle data, and rod fluxes. Let ρ be a relaxed weight and $\widehat{\rho}$ an admissible competitor with the same rod fluxes, such that $\widehat{\rho}/\rho$ extends smoothly and positively to the closed quotient. Assume that the logarithmic integrals and the suprema of $-\mathcal{E}_\rho$ over the common map class are finite for both weights. If $-\Delta_g \rho$ vanishes on a set of positive area in B° , then ρ cannot minimize the relaxed version of (D.31).*

Proof. Set

$$\rho_t = (1 - t)\rho + t\widehat{\rho}, \quad Z = \{\Delta_g \rho = 0\} \cap B^\circ.$$

The rod fluxes remain fixed. On Z ,

$$\begin{aligned} & \int_Z (-\Delta_{\underline{g}} \rho_t) \log \frac{-\Delta_{\underline{g}} \rho_t}{2\lambda \rho_t} dA_{\underline{g}} \\ &= t \log t \int_Z (-\Delta_{\underline{g}} \hat{\rho}) dA_{\underline{g}} + O(t). \end{aligned}$$

On $B \setminus Z$, convexity of the logarithmic integrand bounds its increase above by $O(t)$. The same bound holds for $\sup_{\Phi}(-\mathcal{E}_{\rho_t}(\Phi))$, by convexity in ρ , and for the linear curvature term.

The positive smooth ratio $\hat{\rho}/\rho$ gives bounded boundary-density ratios, so the boundary entropy changes by $O(t)$. The conical correction is constant at fixed rod fluxes. Hence

$$\begin{aligned} & \sup_{\Phi} \mathcal{J}_{\underline{g},\beta}(\rho_t, \Phi) - \sup_{\Phi} \mathcal{J}_{\underline{g},\beta}(\rho, \Phi) \\ & \leq t \log t \int_Z (-\Delta_{\underline{g}} \hat{\rho}) dA_{\underline{g}} + O(t) < 0 \end{aligned}$$

for sufficiently small $t > 0$. □

The lemma excludes a zero set of positive area under its hypotheses. It assumes the boundary expansions and smoothness of the source ratio from the outset, and therefore does not establish these properties for a weak limit of a minimizing sequence. Nor does it exclude isolated zeros or give a positive lower bound for the reconstructed quotient factor.

An existence theorem would require compactness modulo reparametrization, lower semicontinuity of the optimized functional, and regularity preserving the rod and corner expansions and strict positivity. We will not discuss the details here and we leave for future exploration.

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